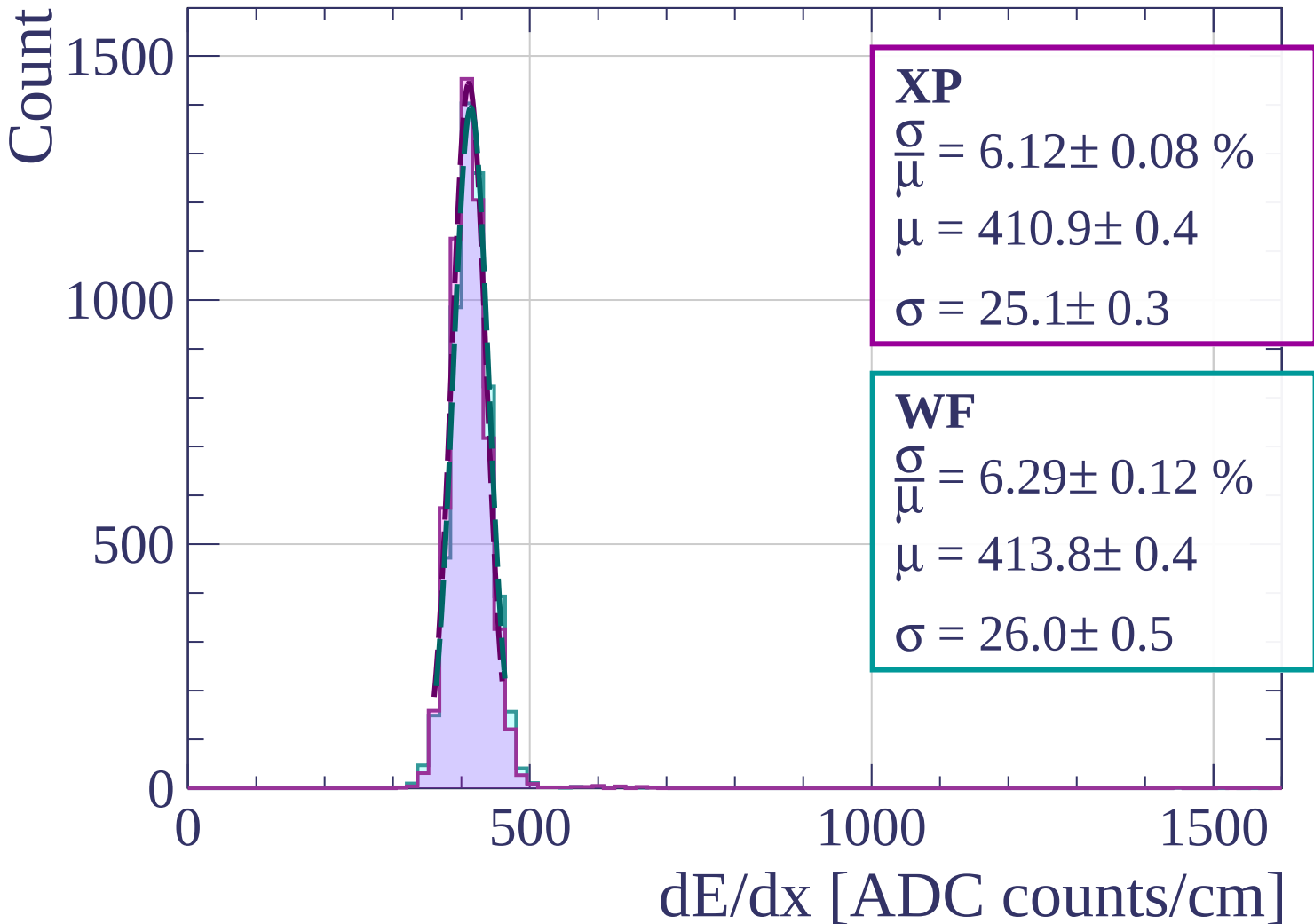
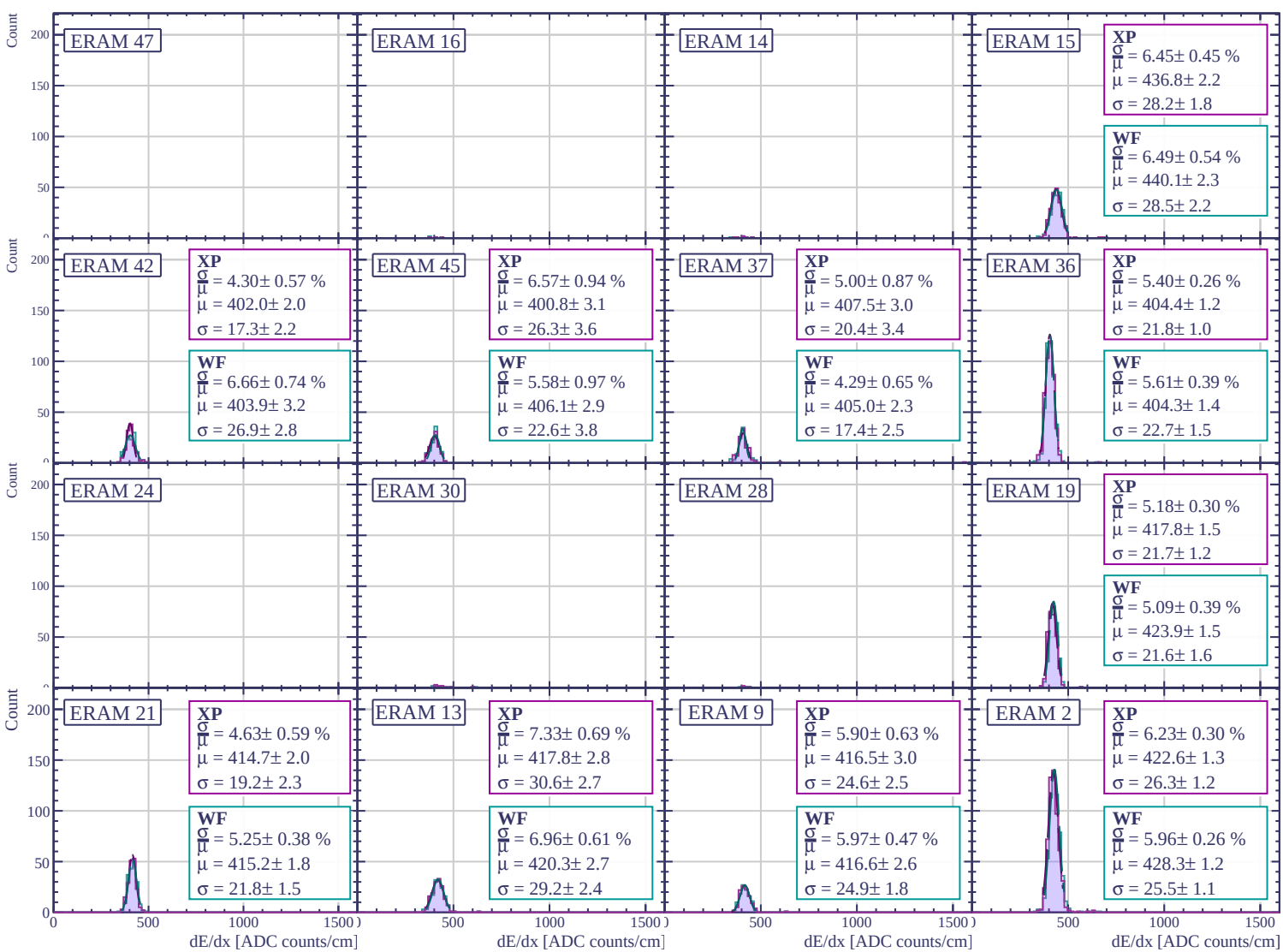
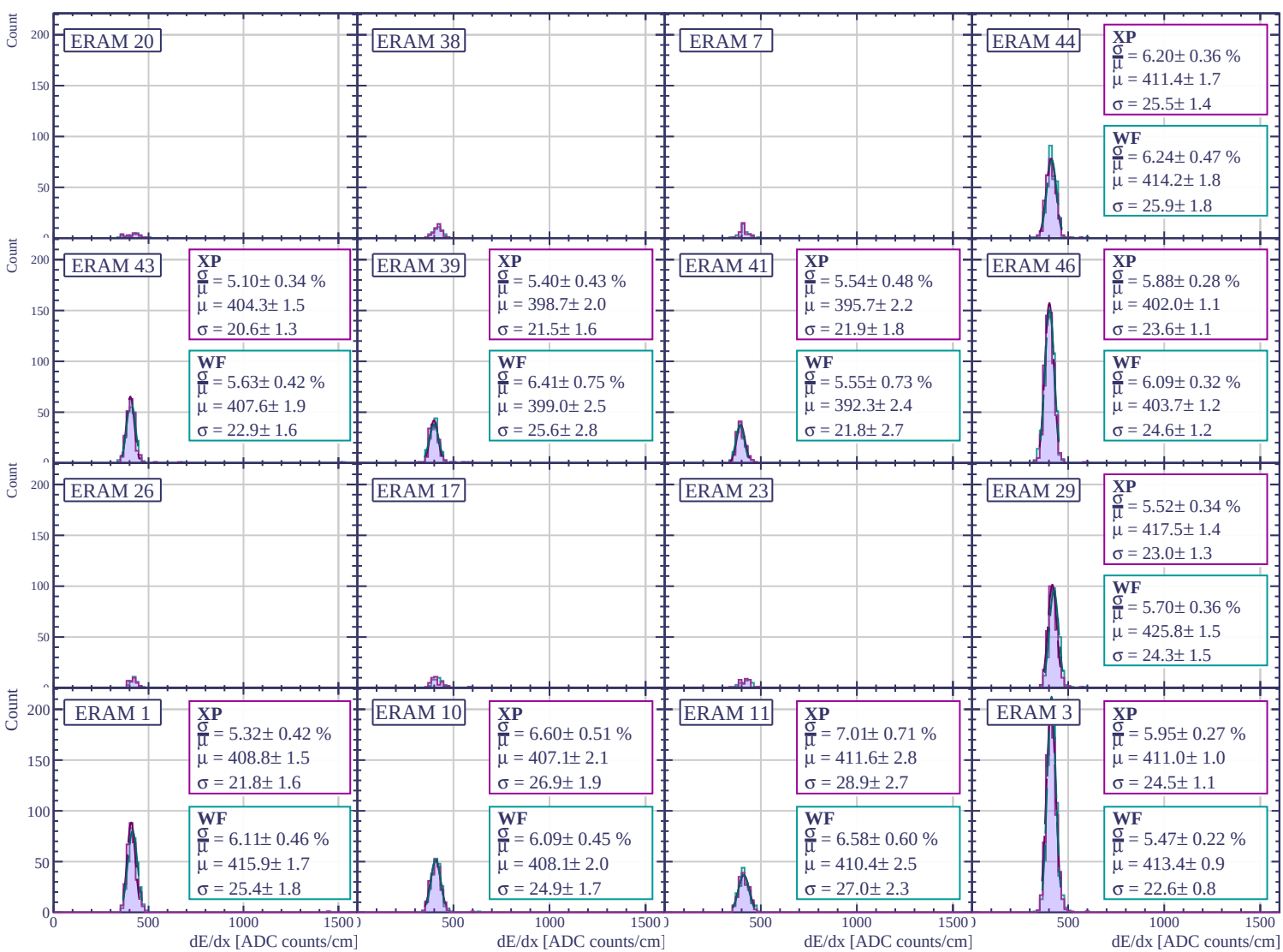
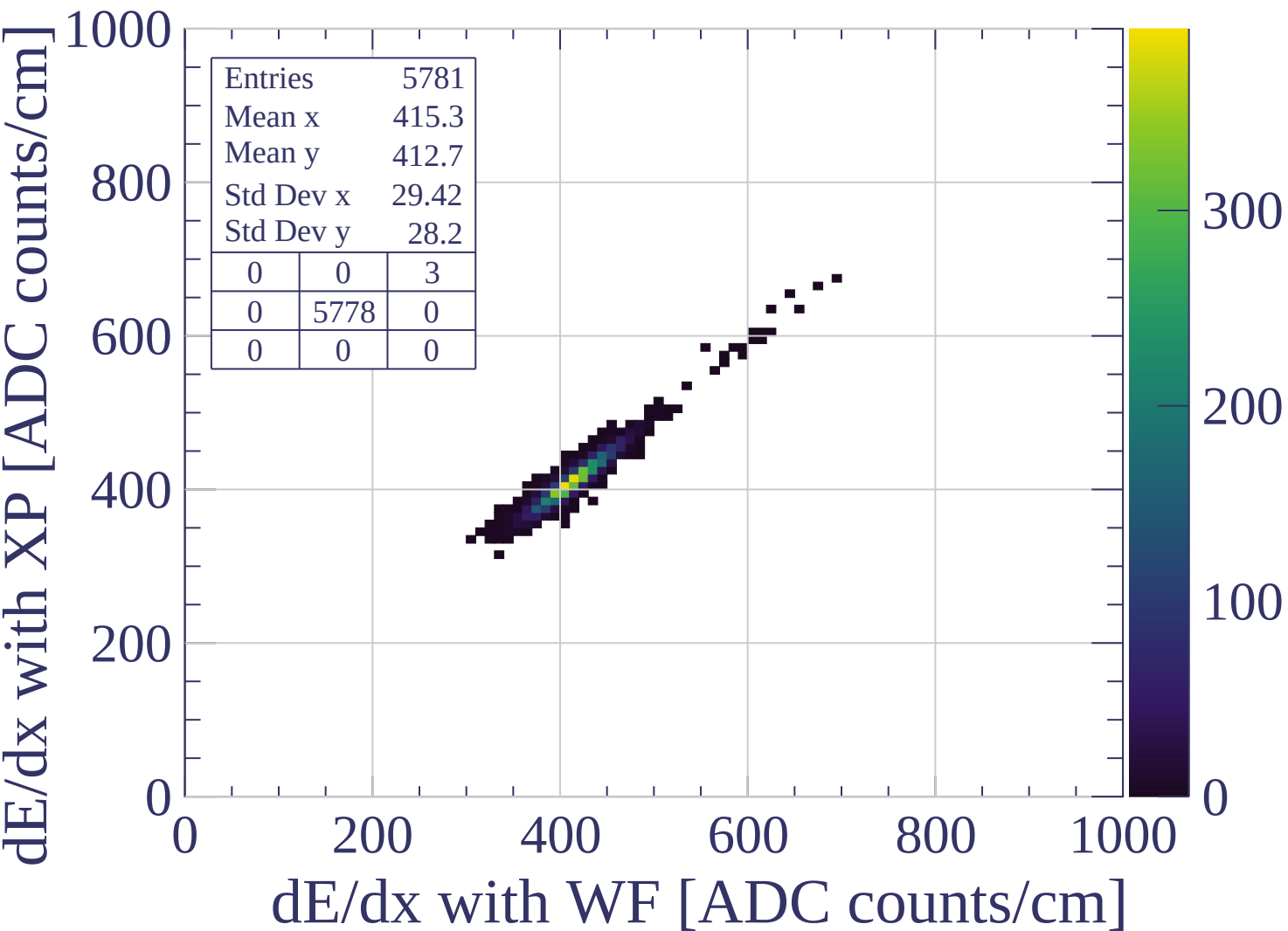
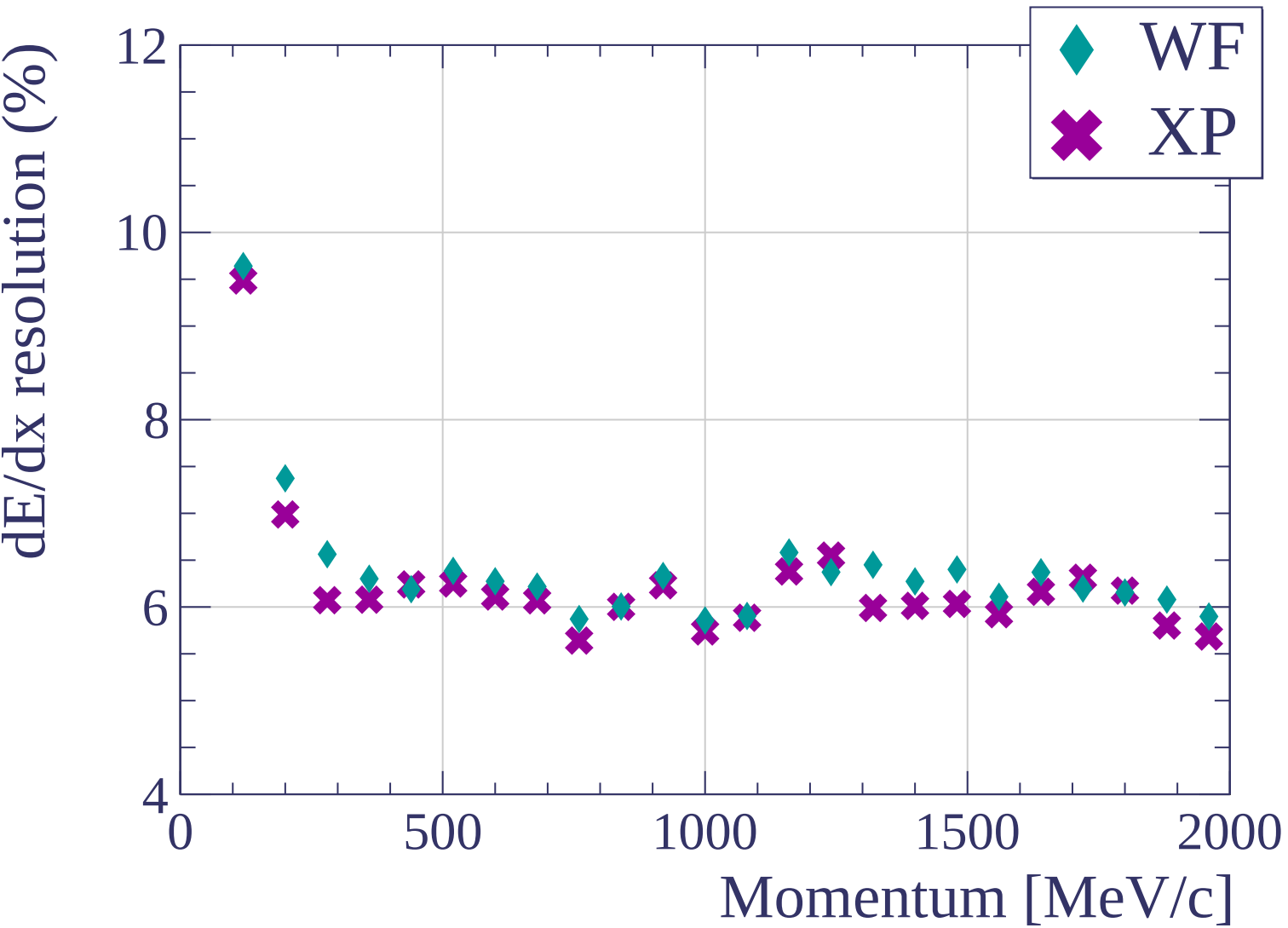


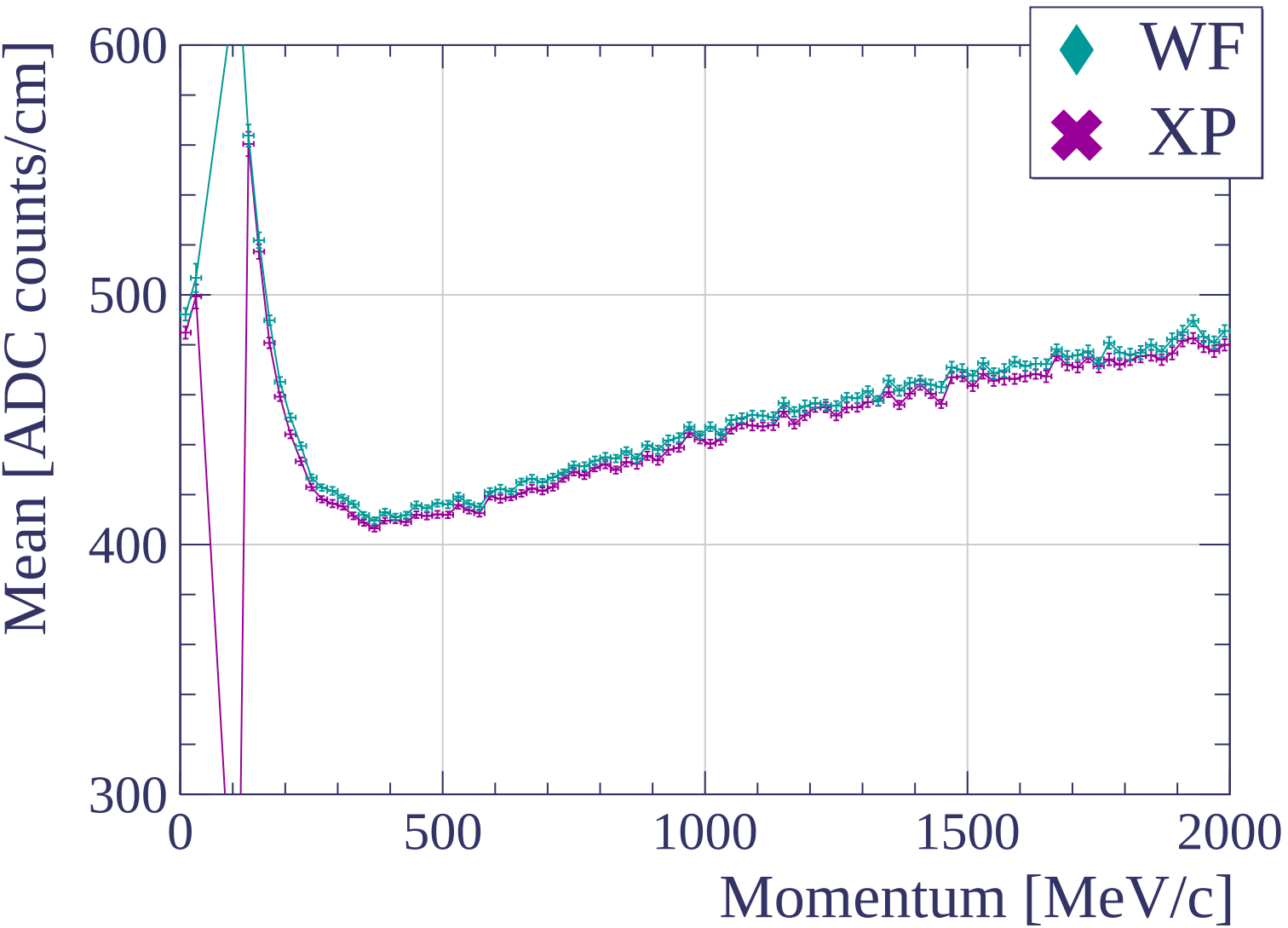


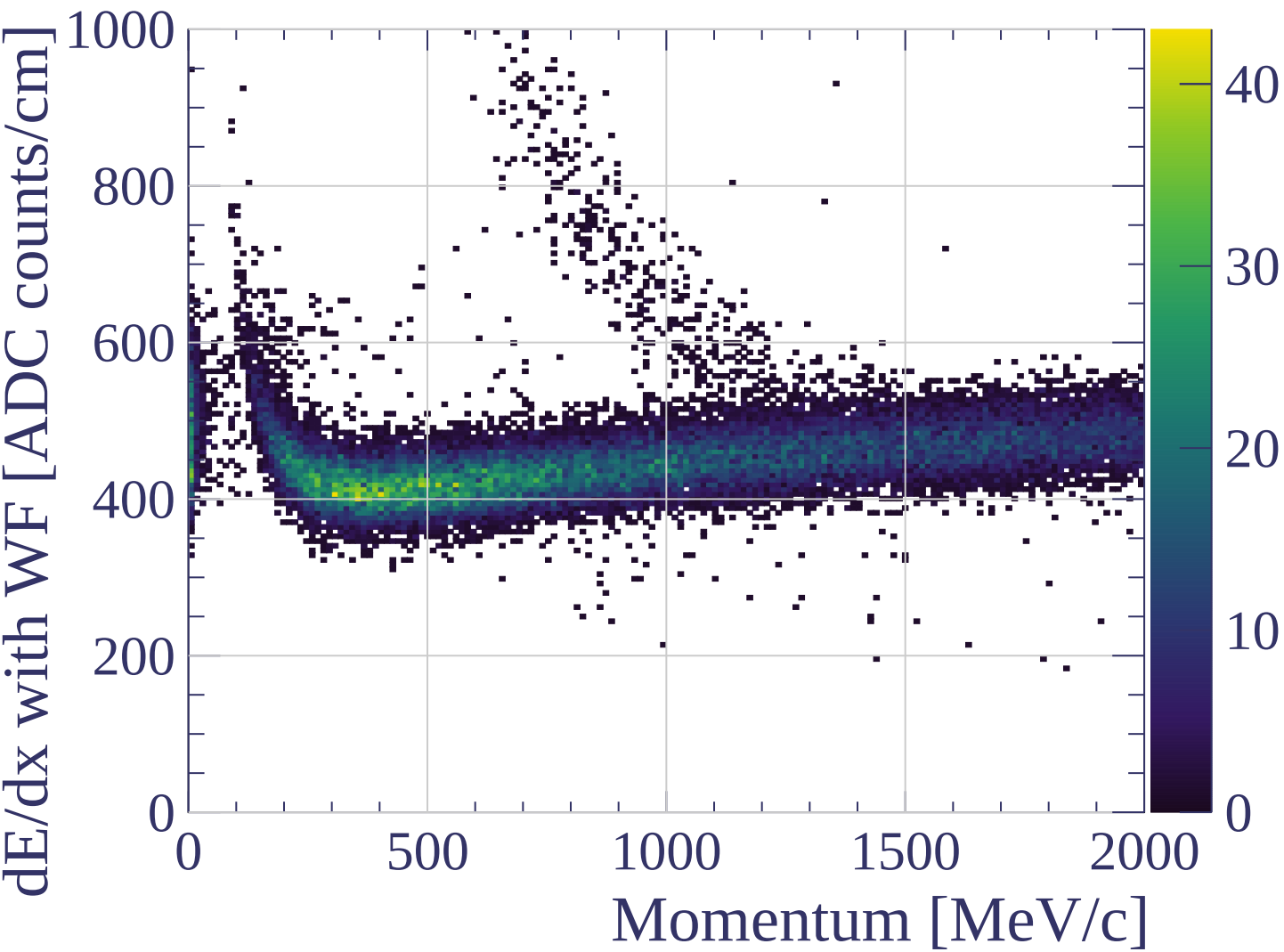


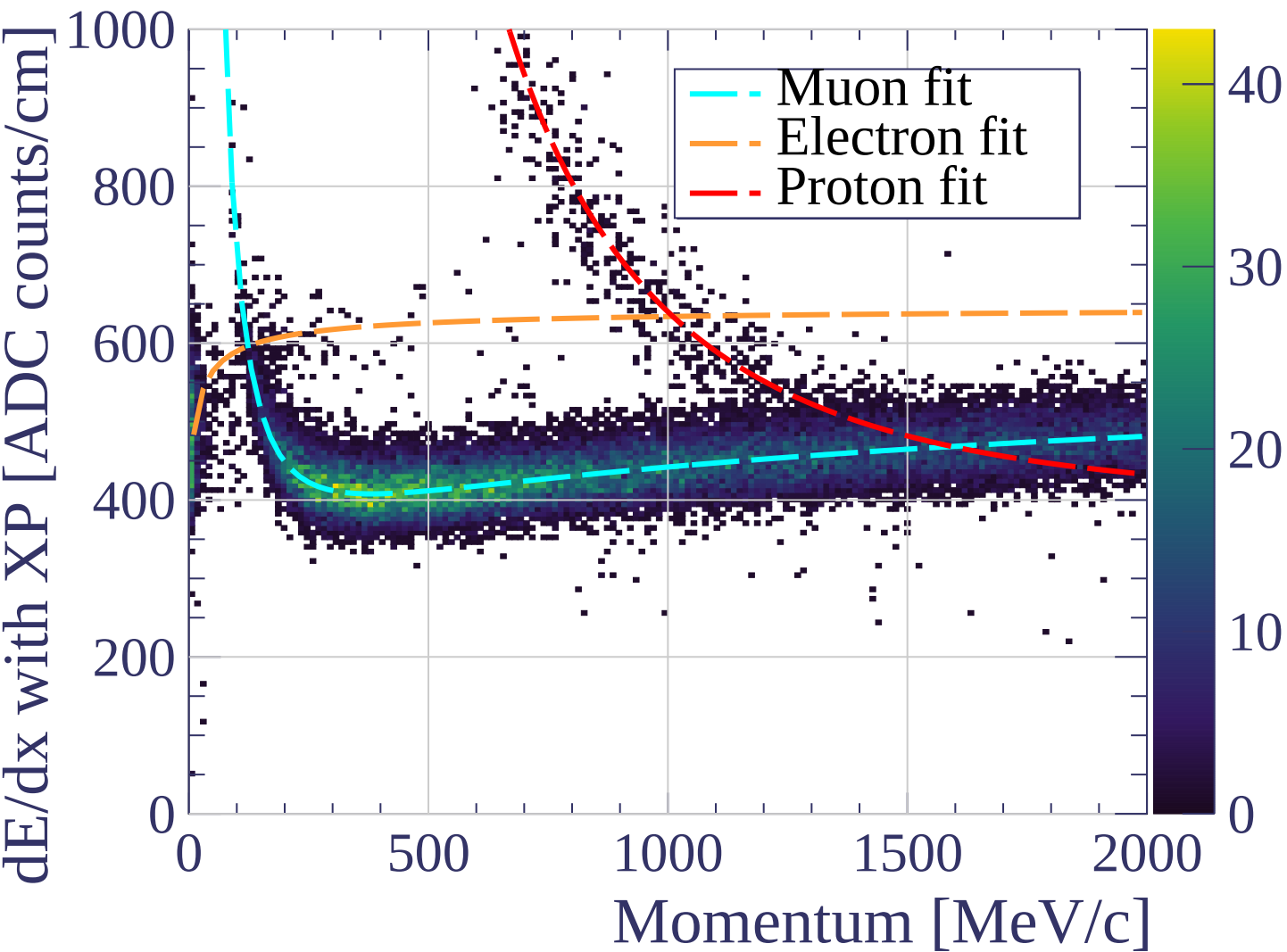




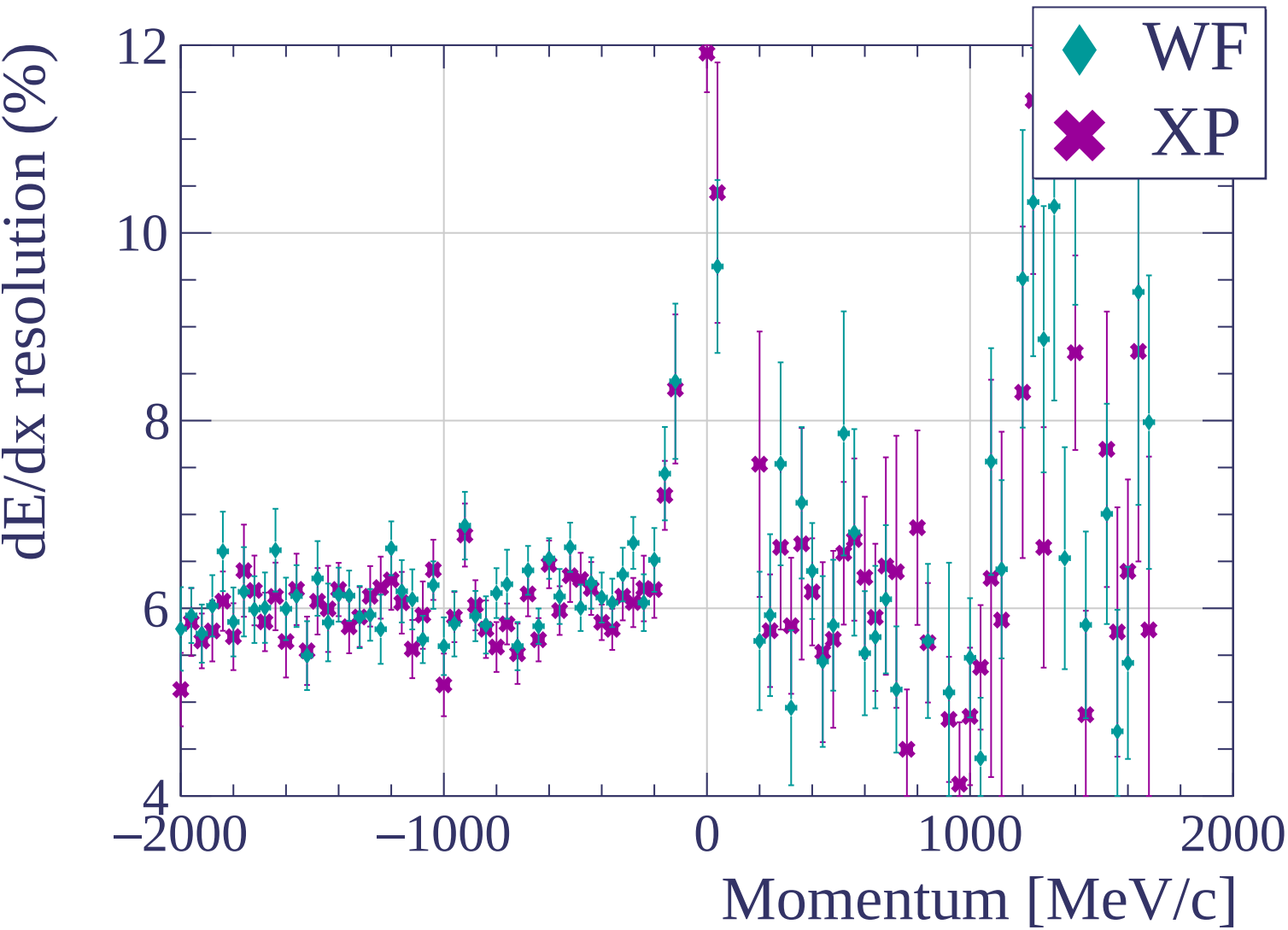


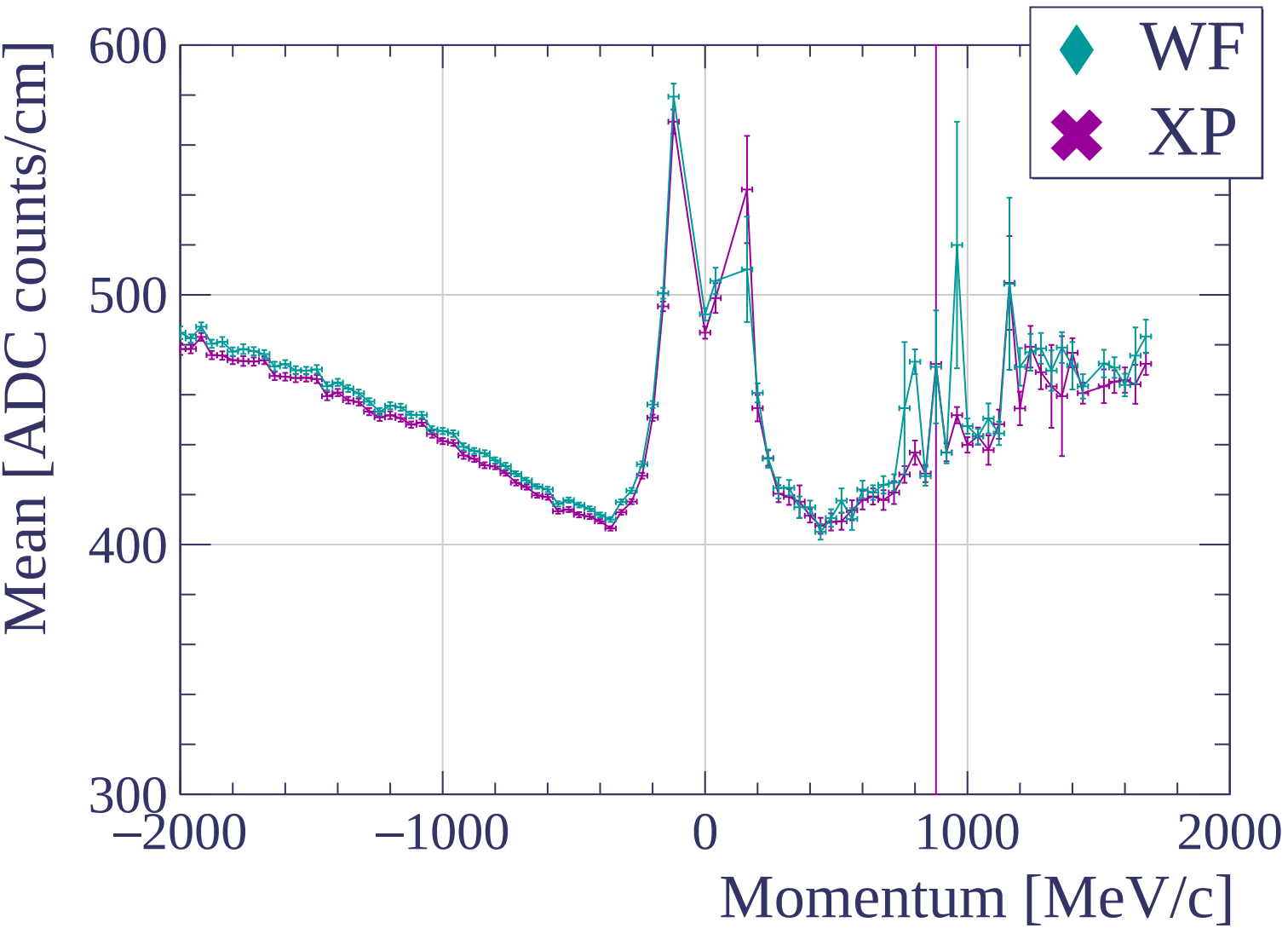


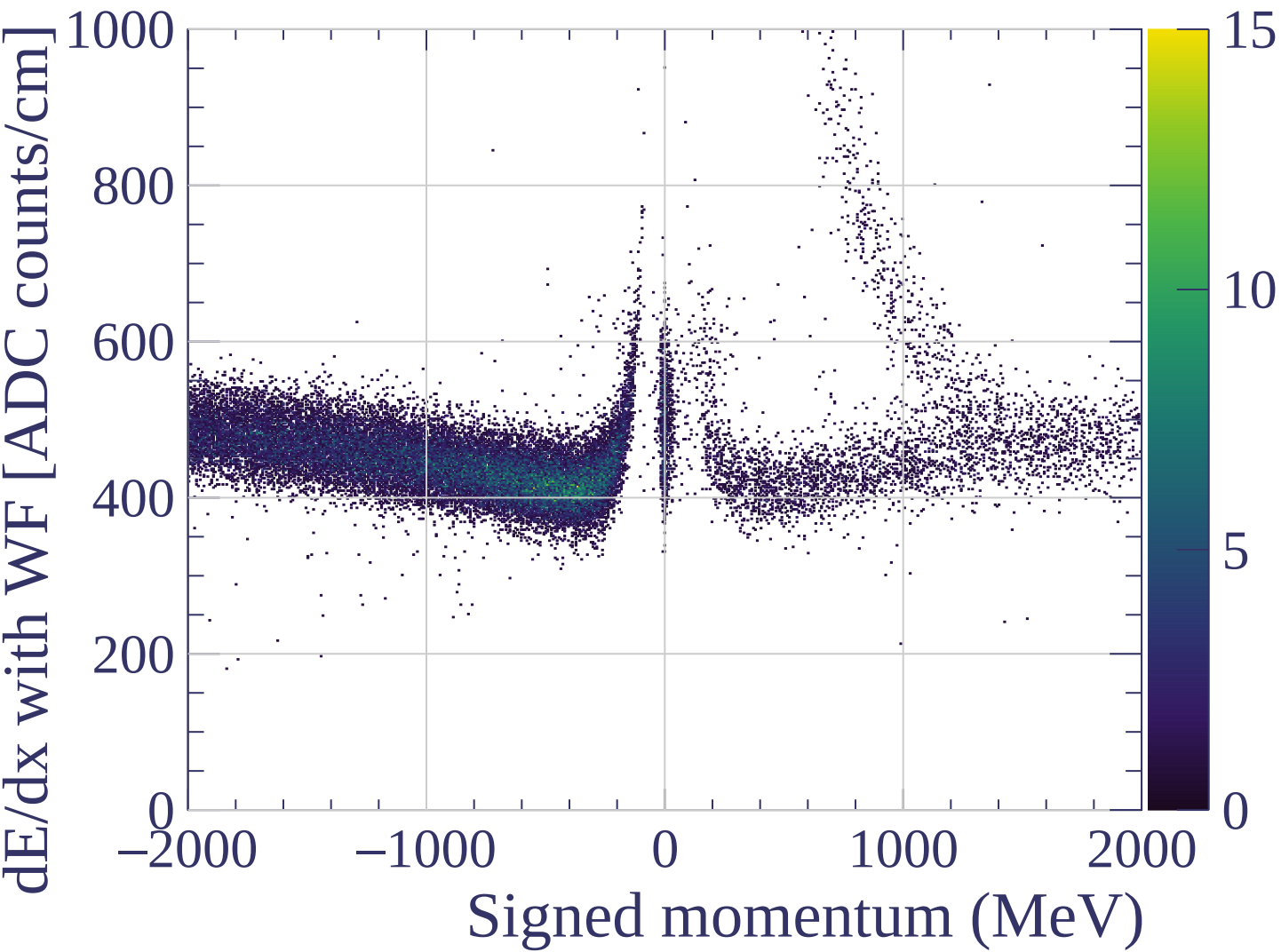


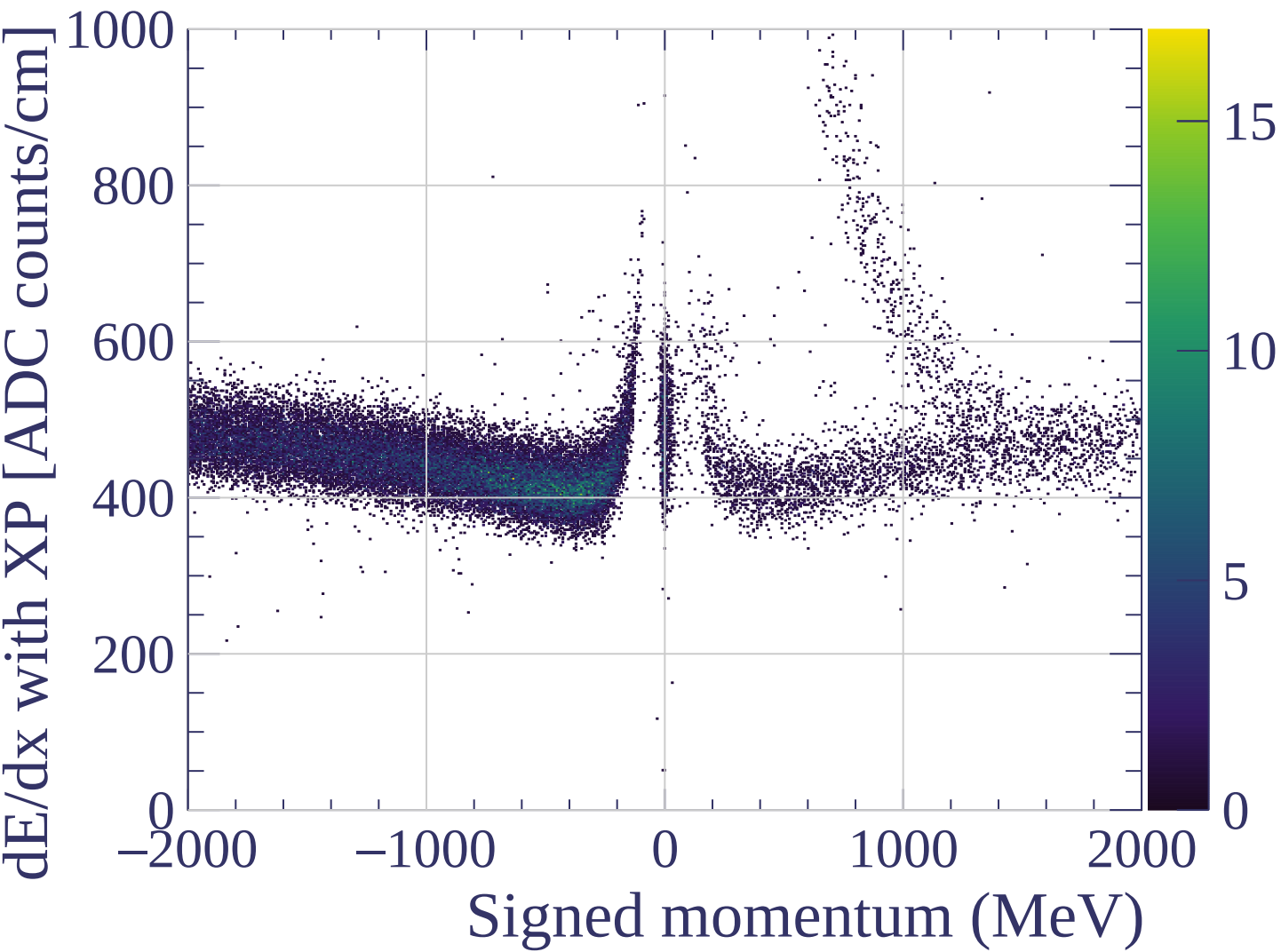




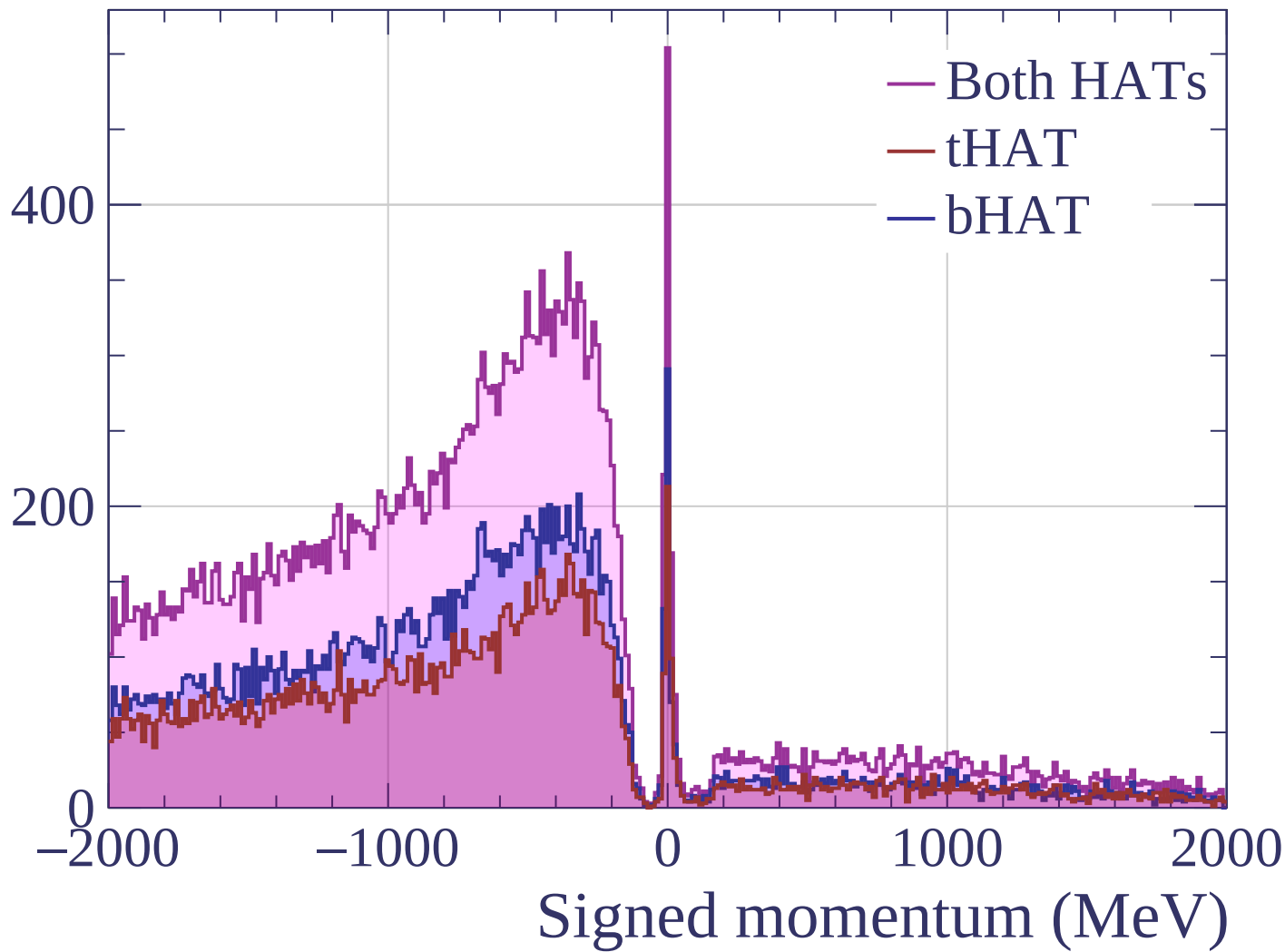


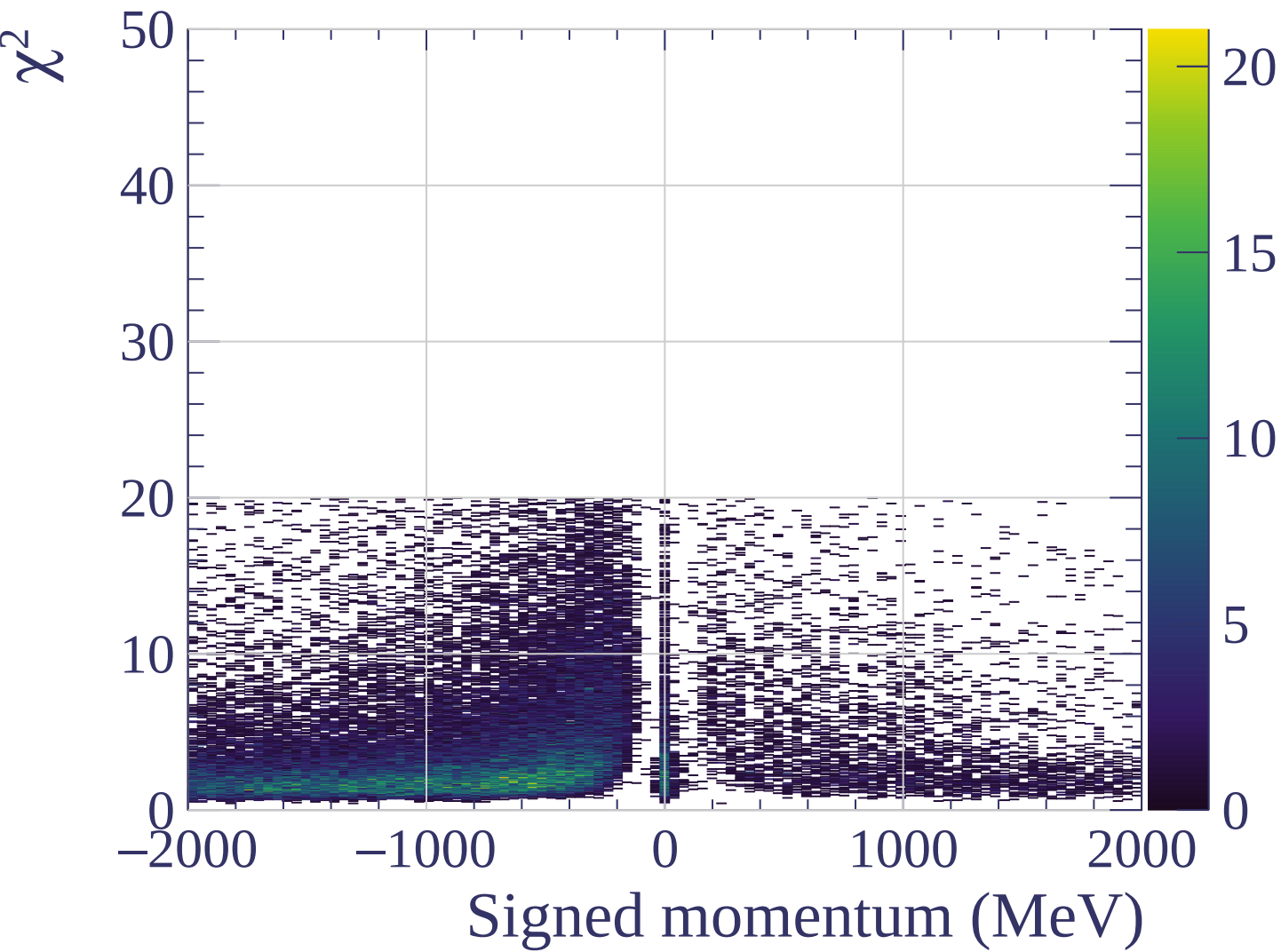


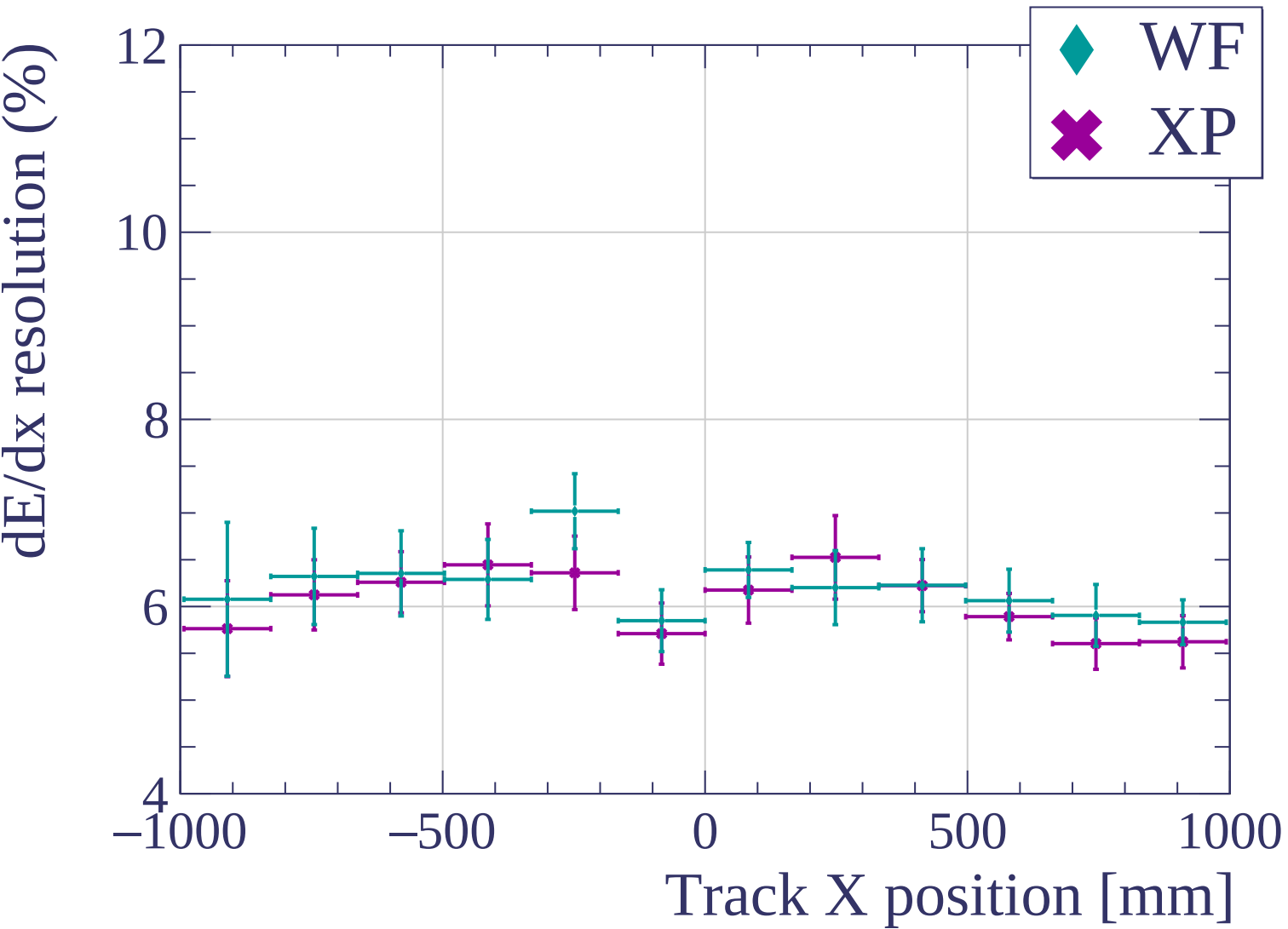


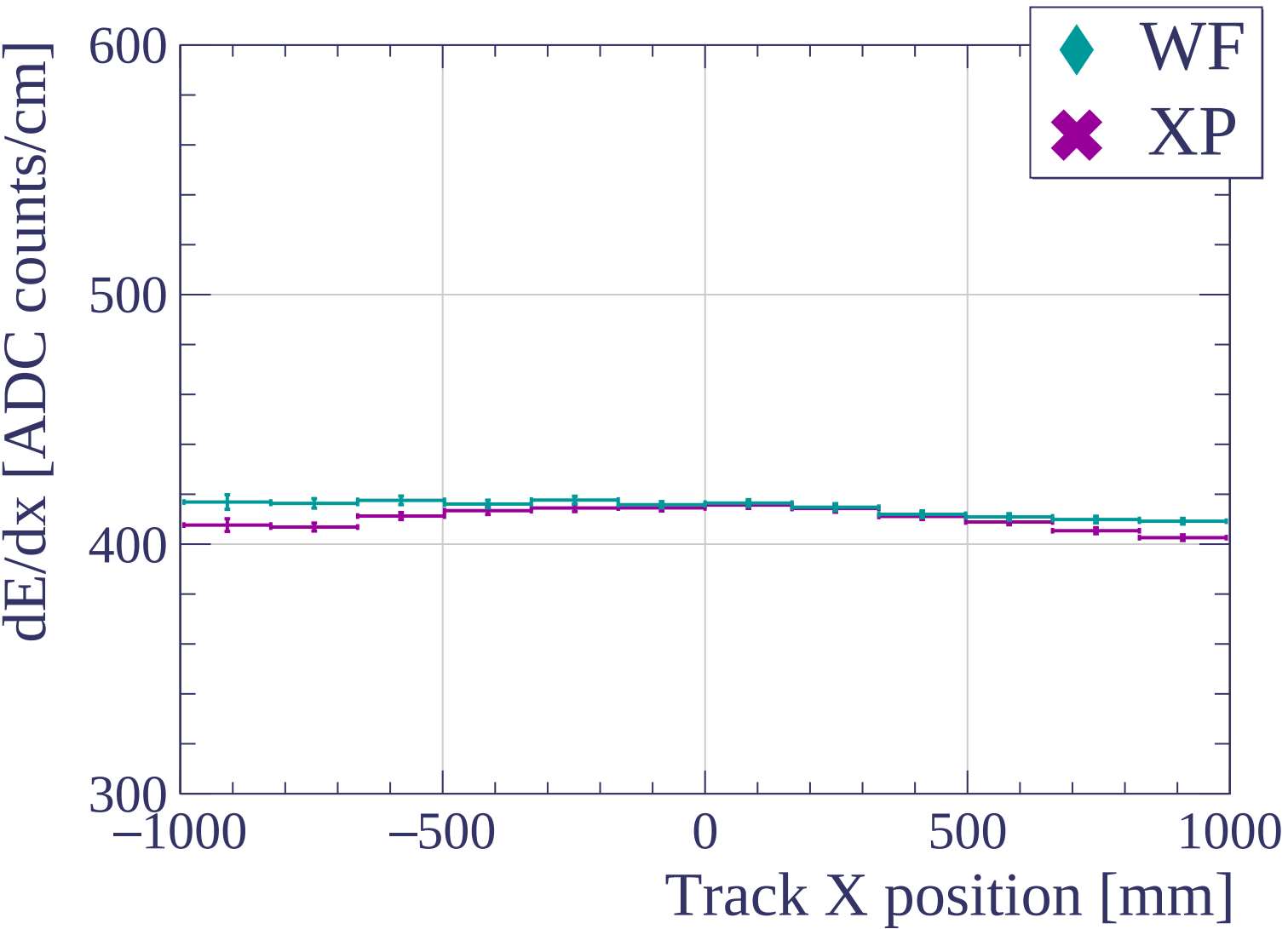


Count

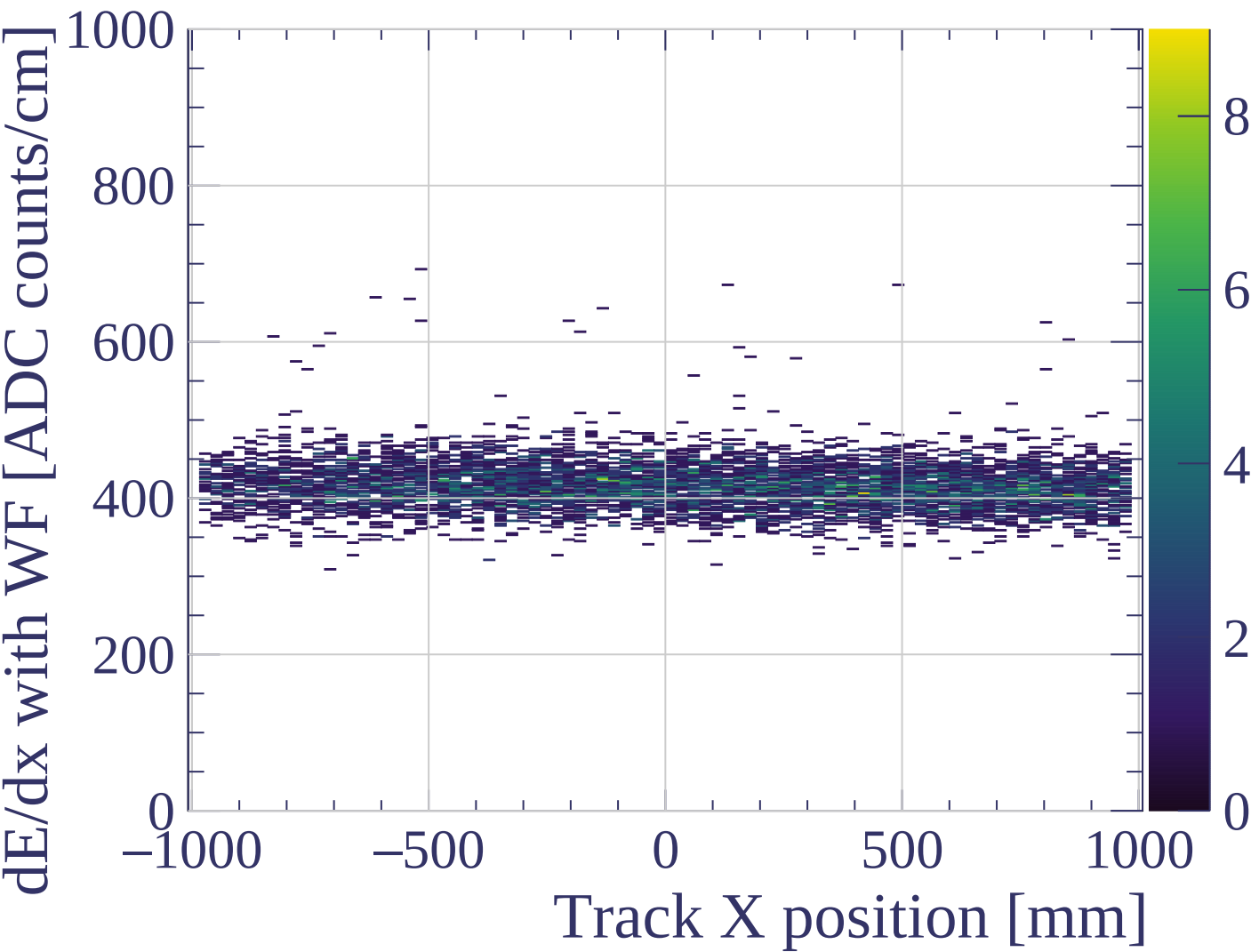


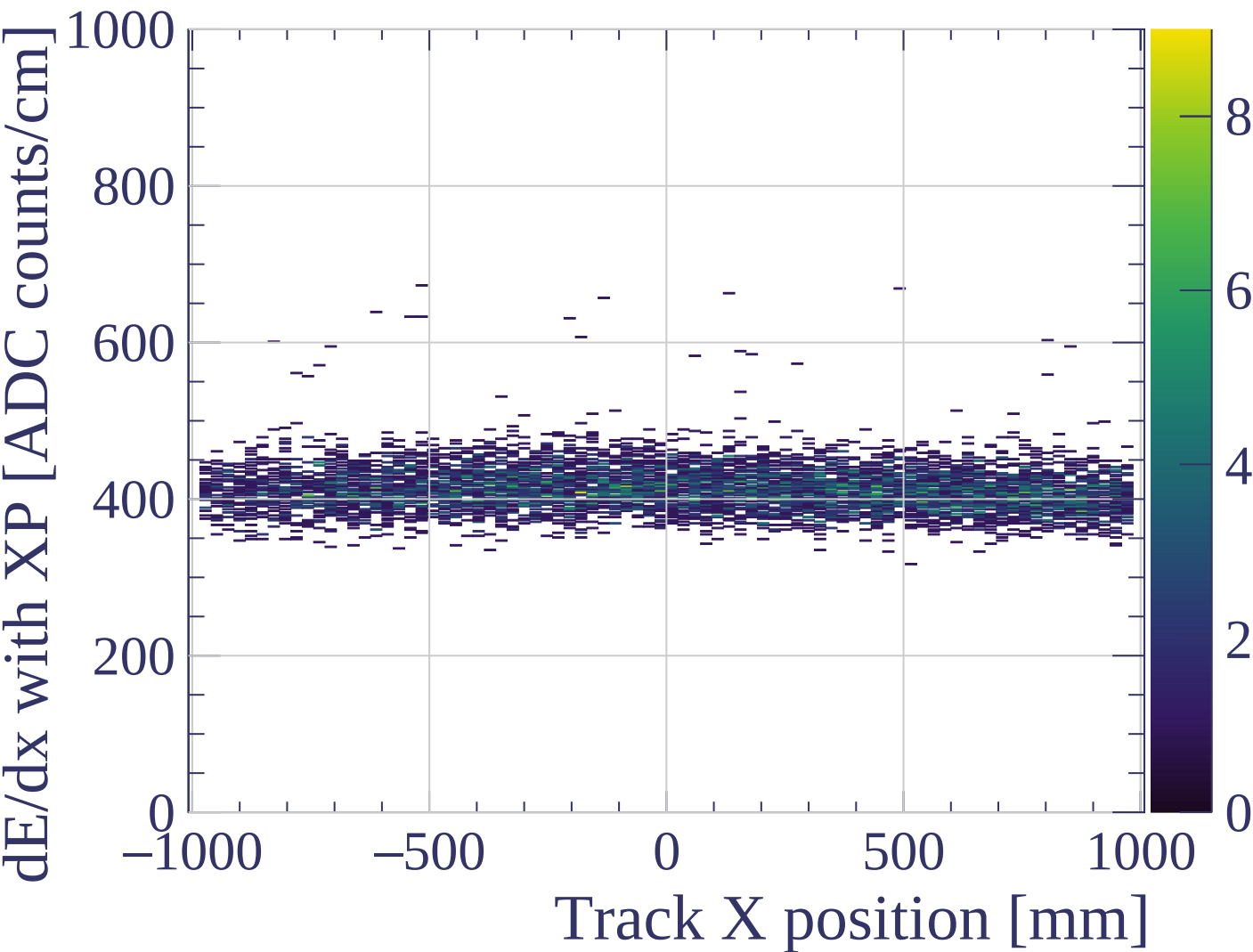


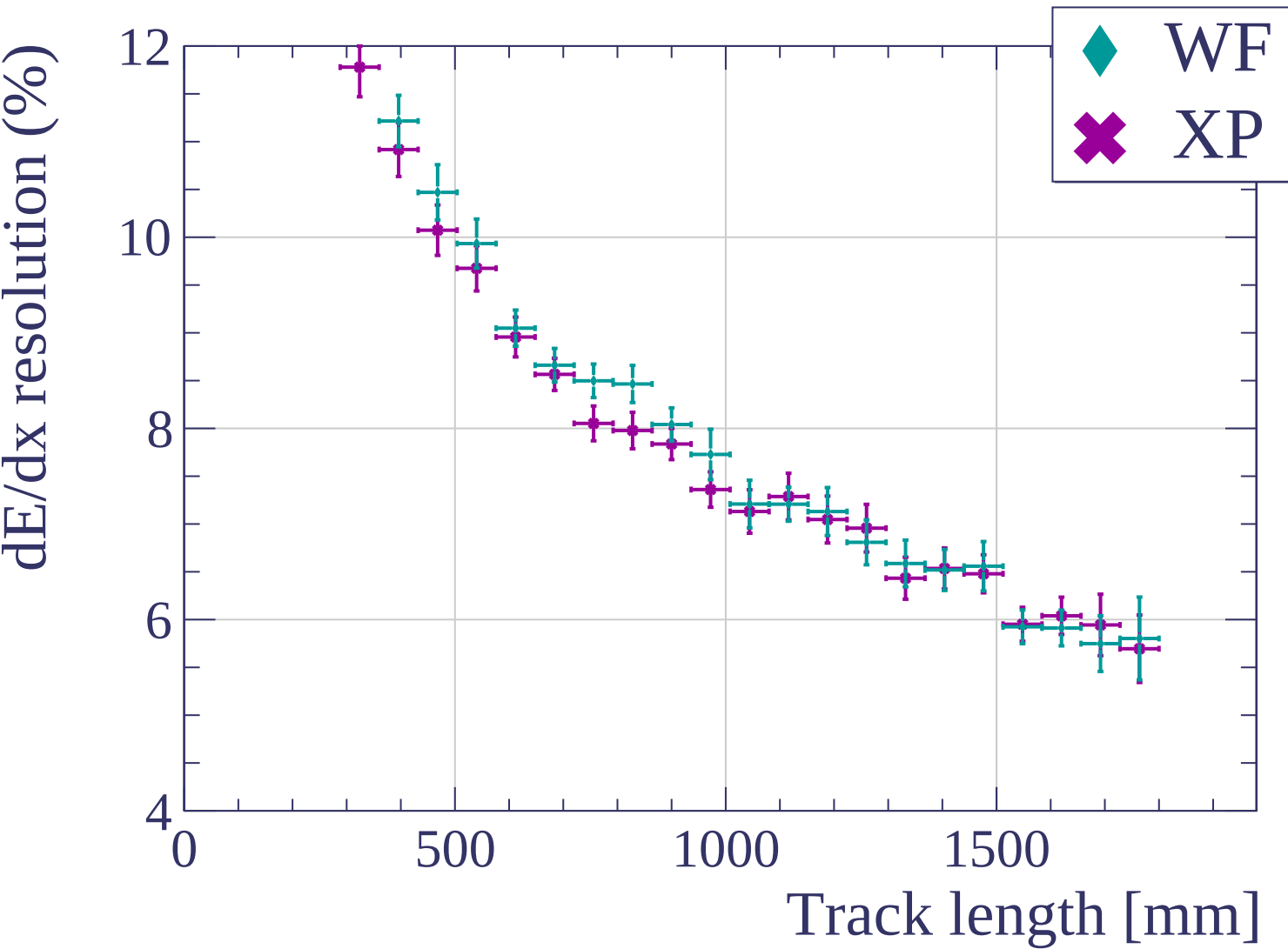


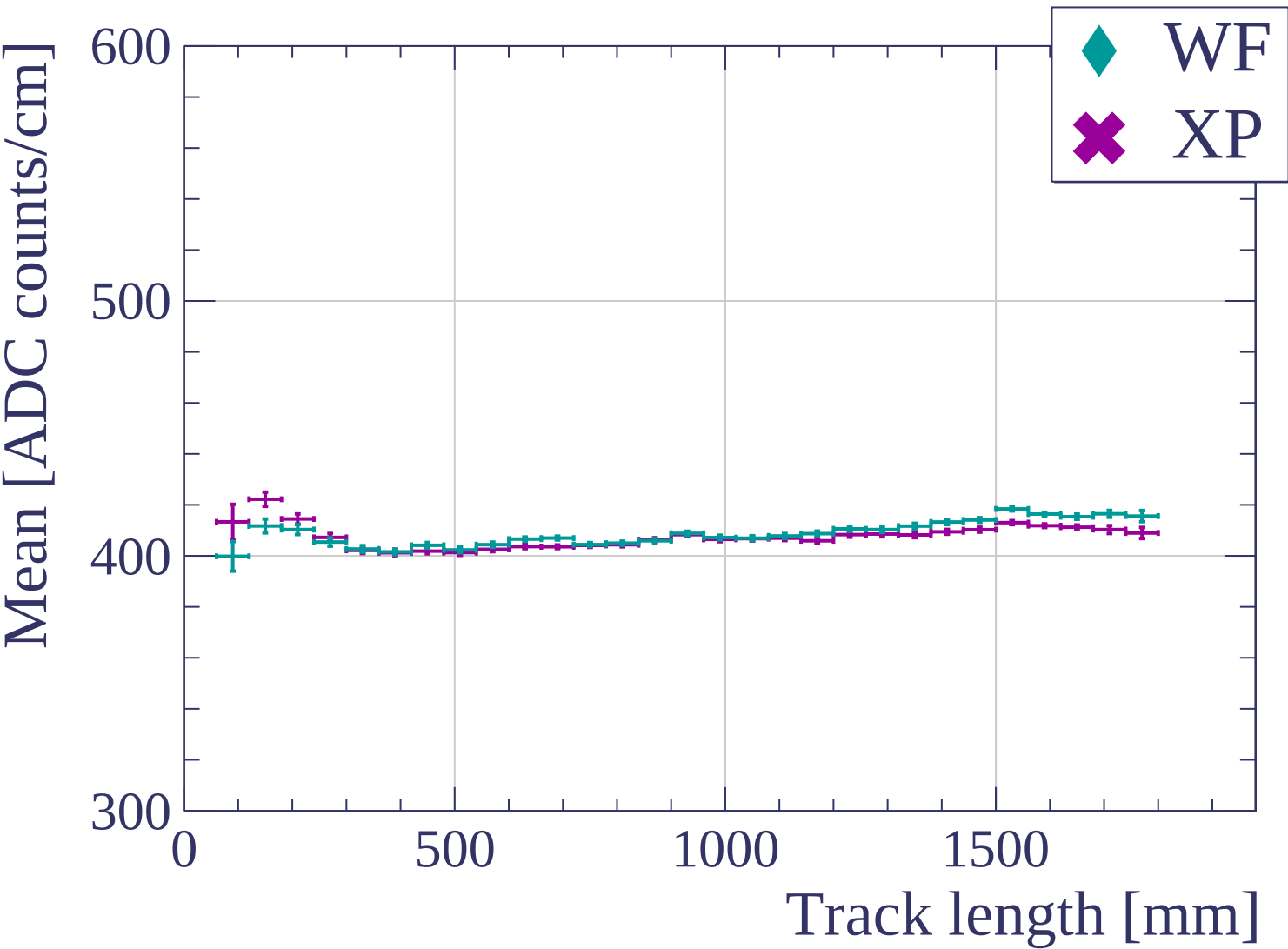


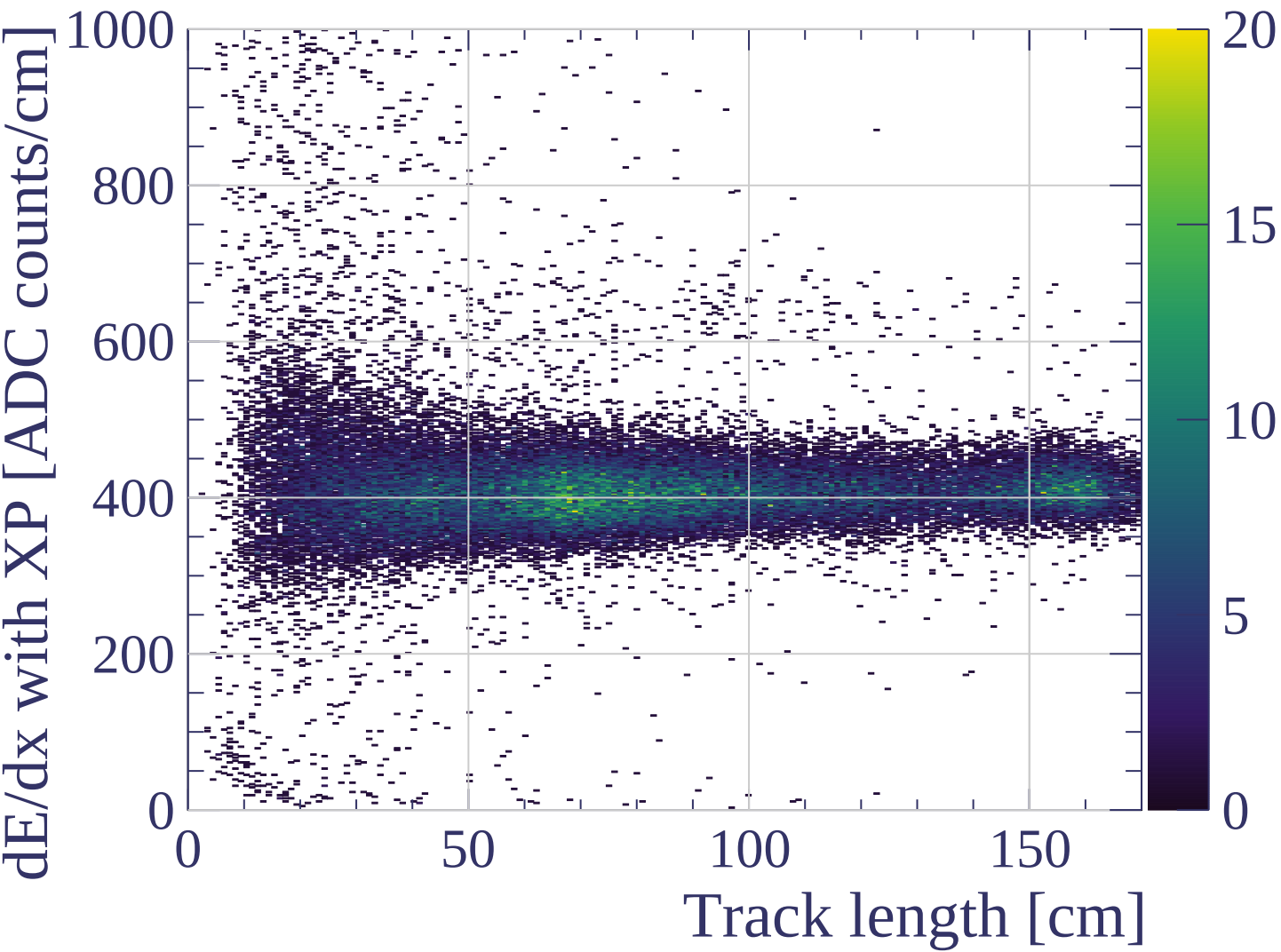


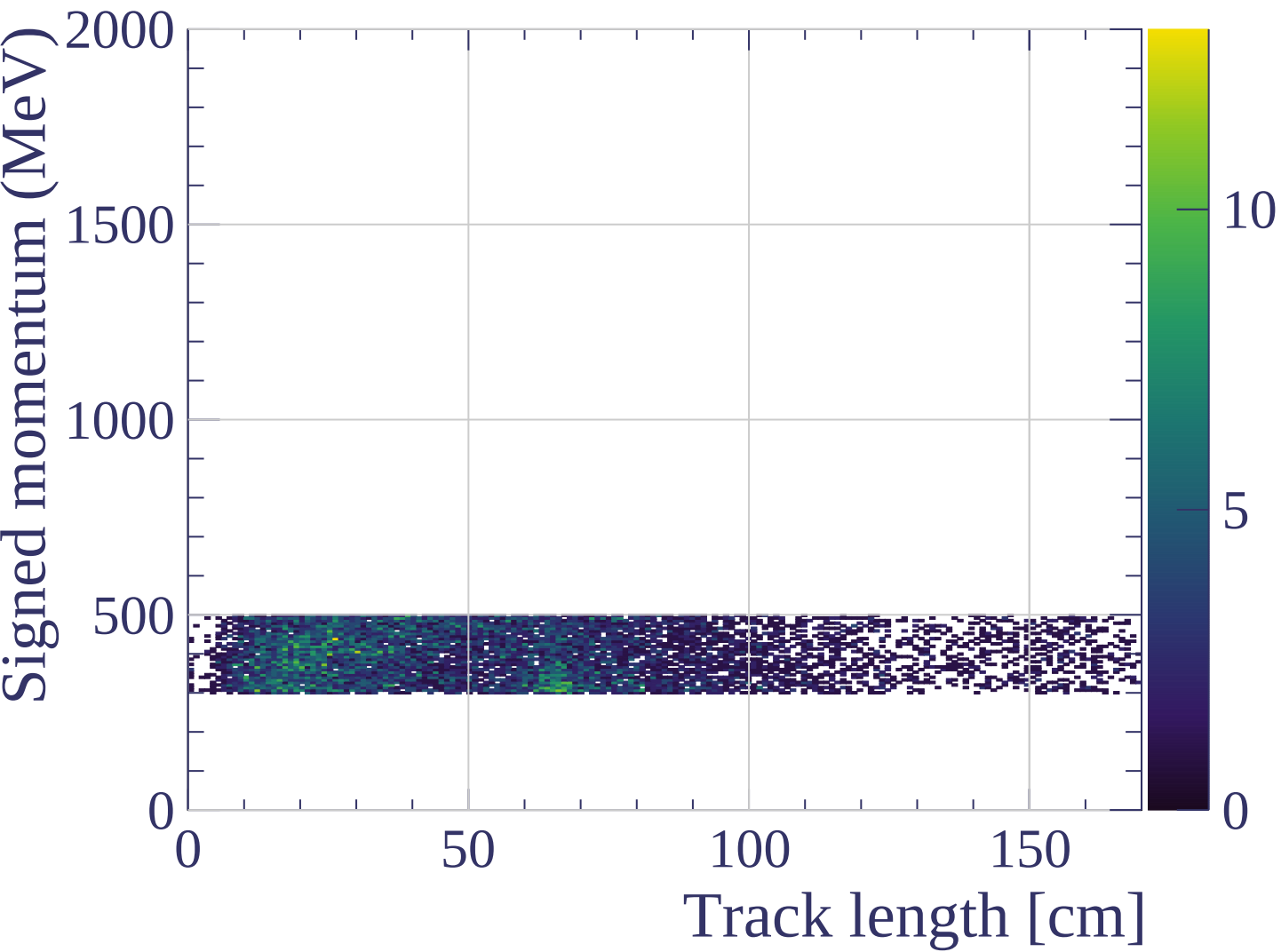


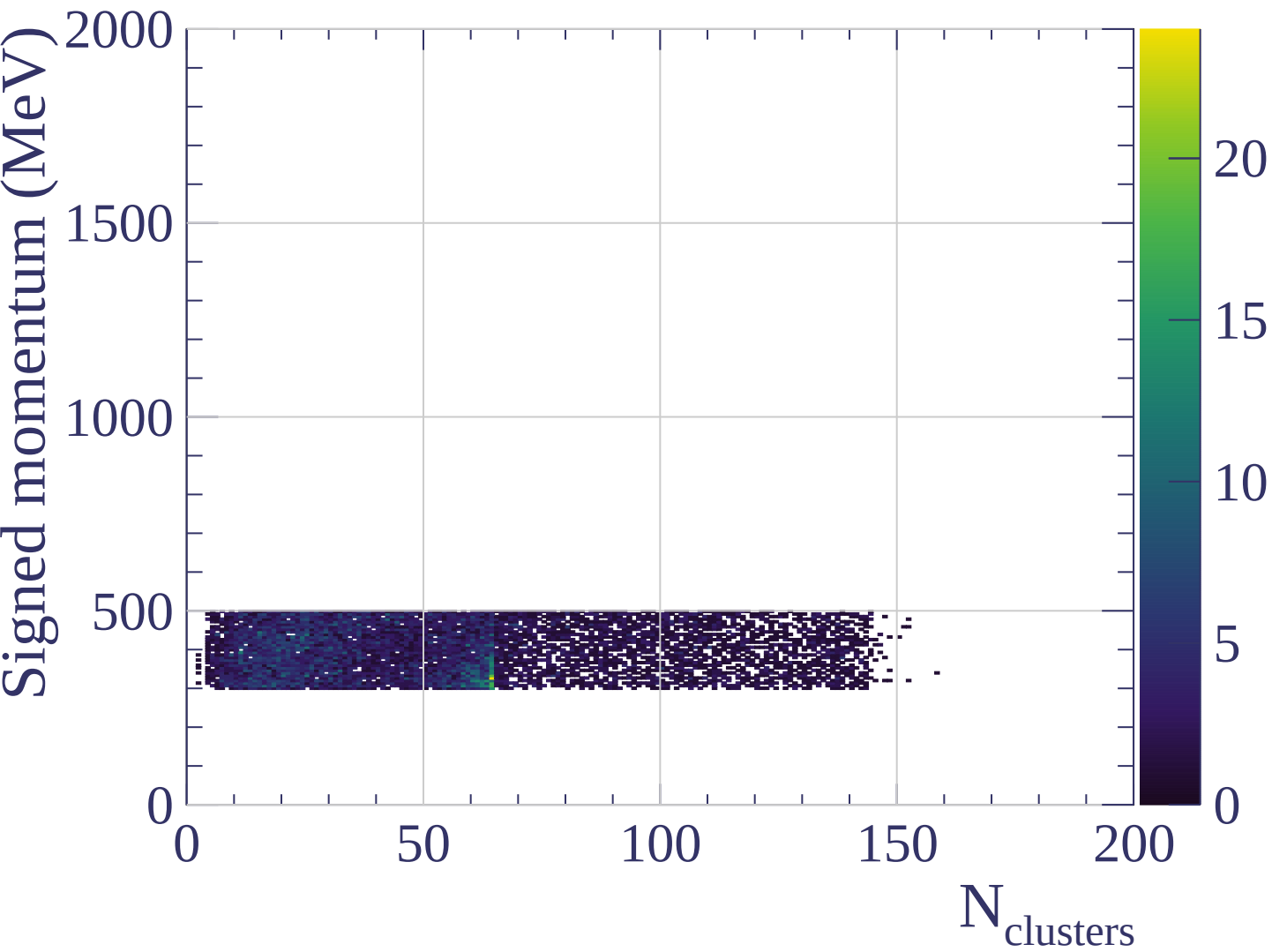


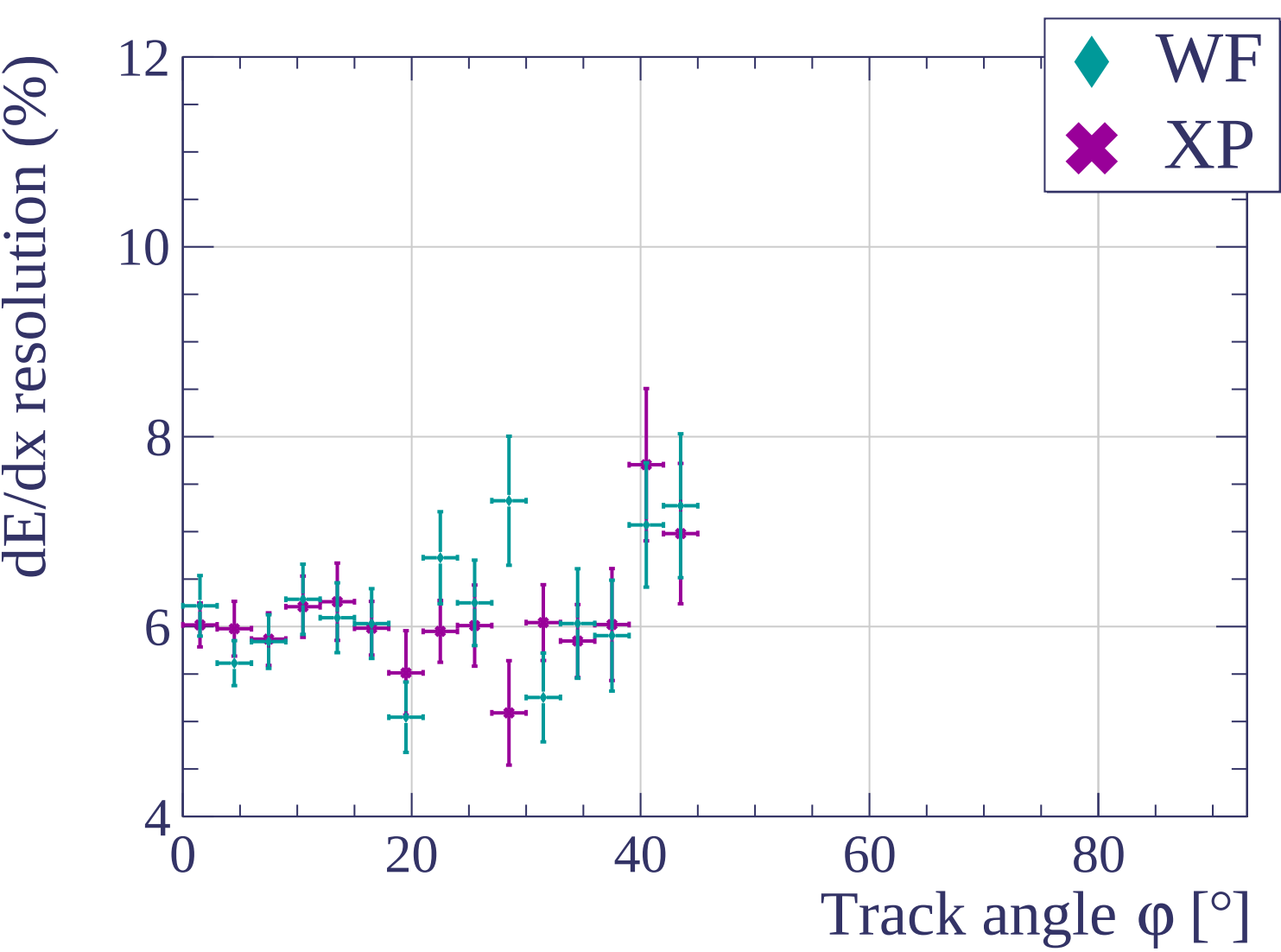




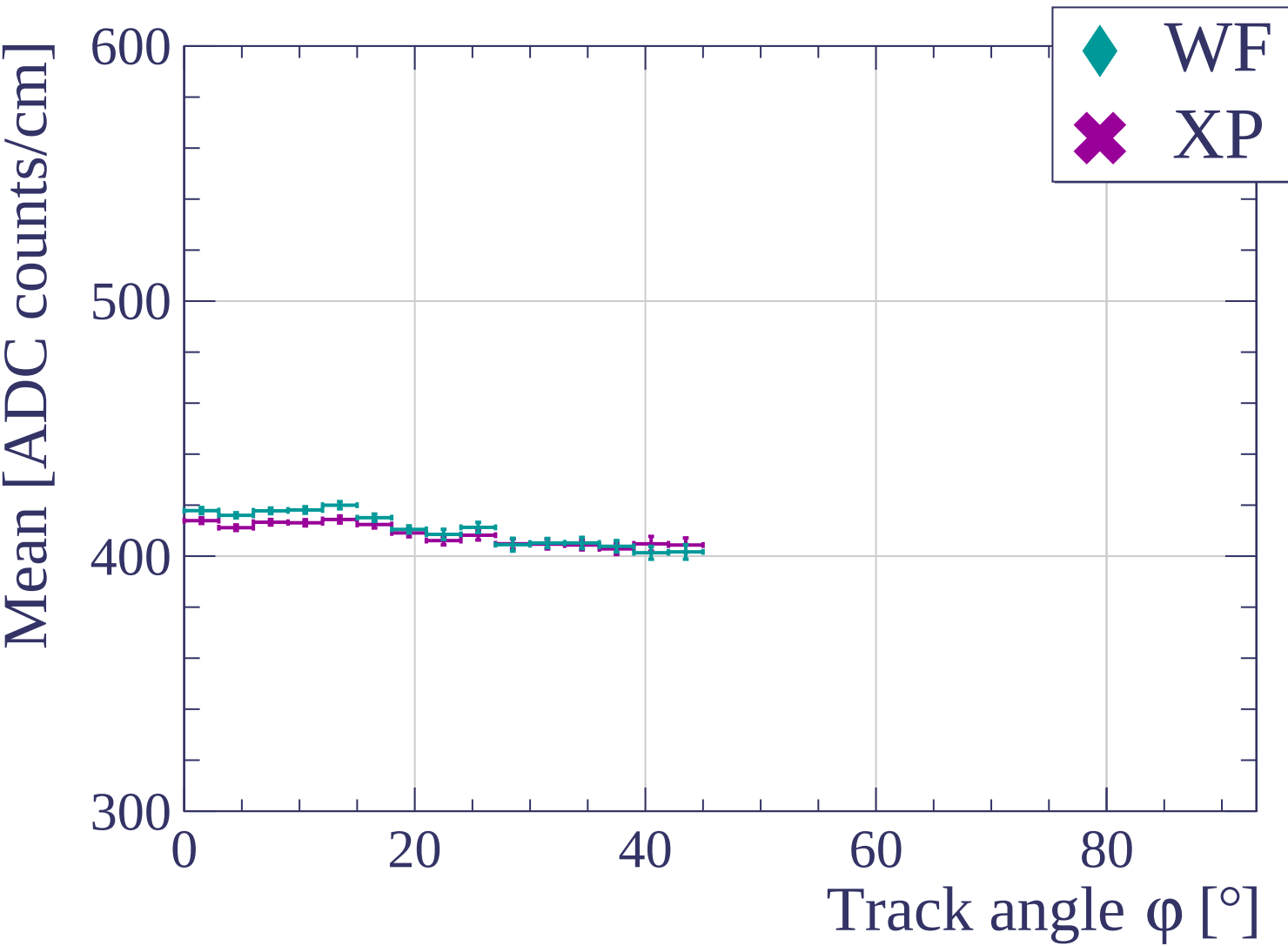


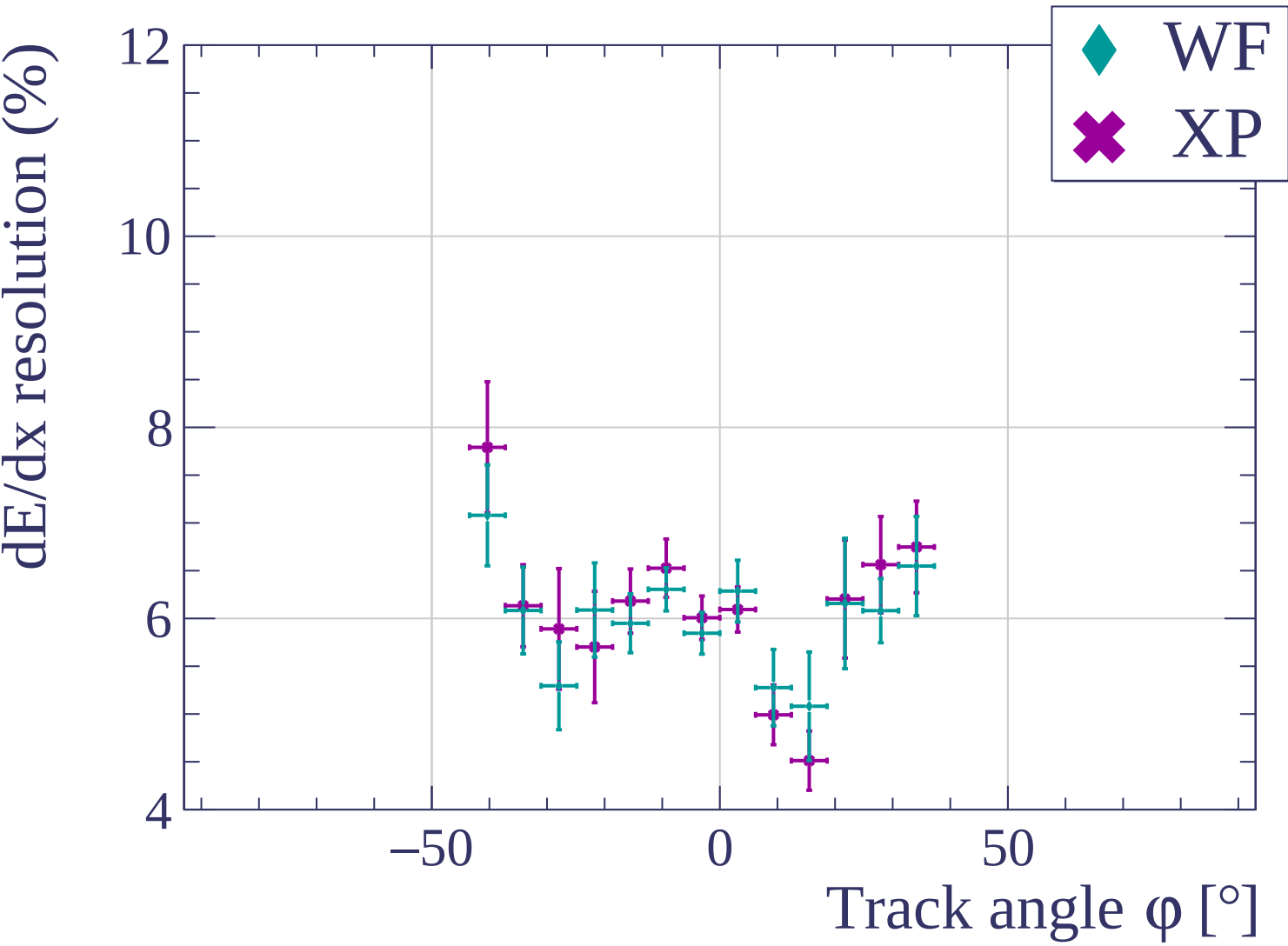


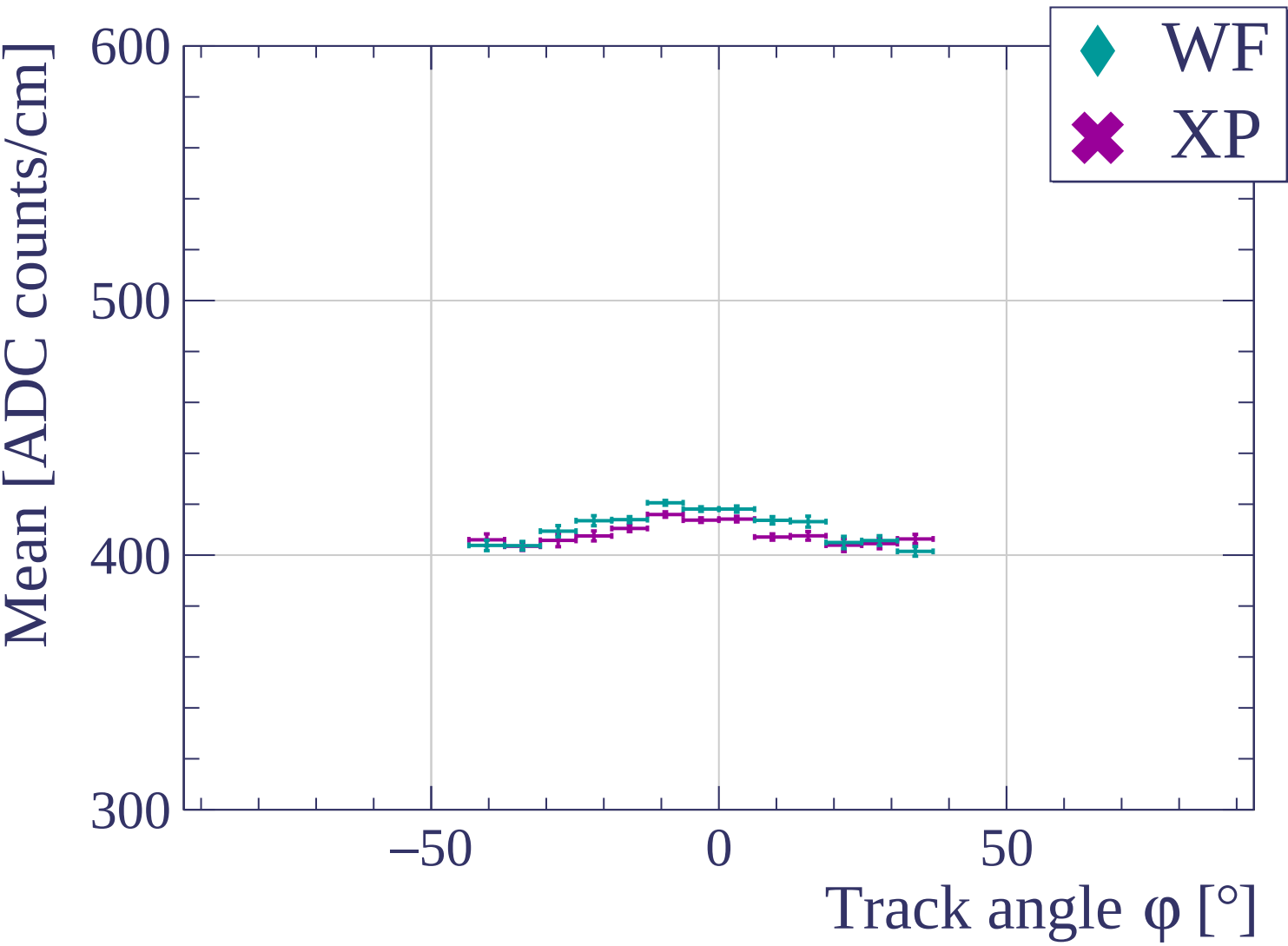


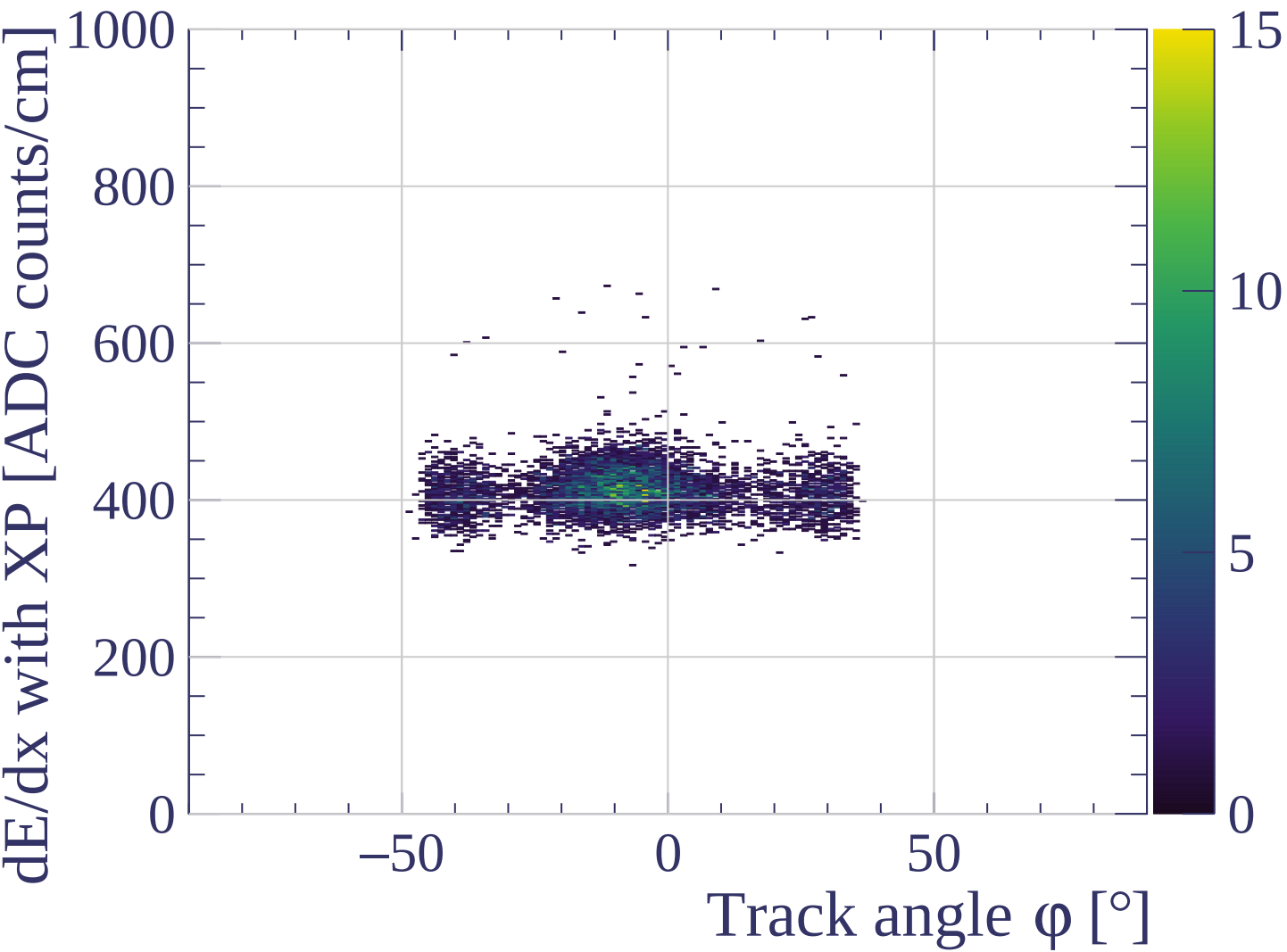


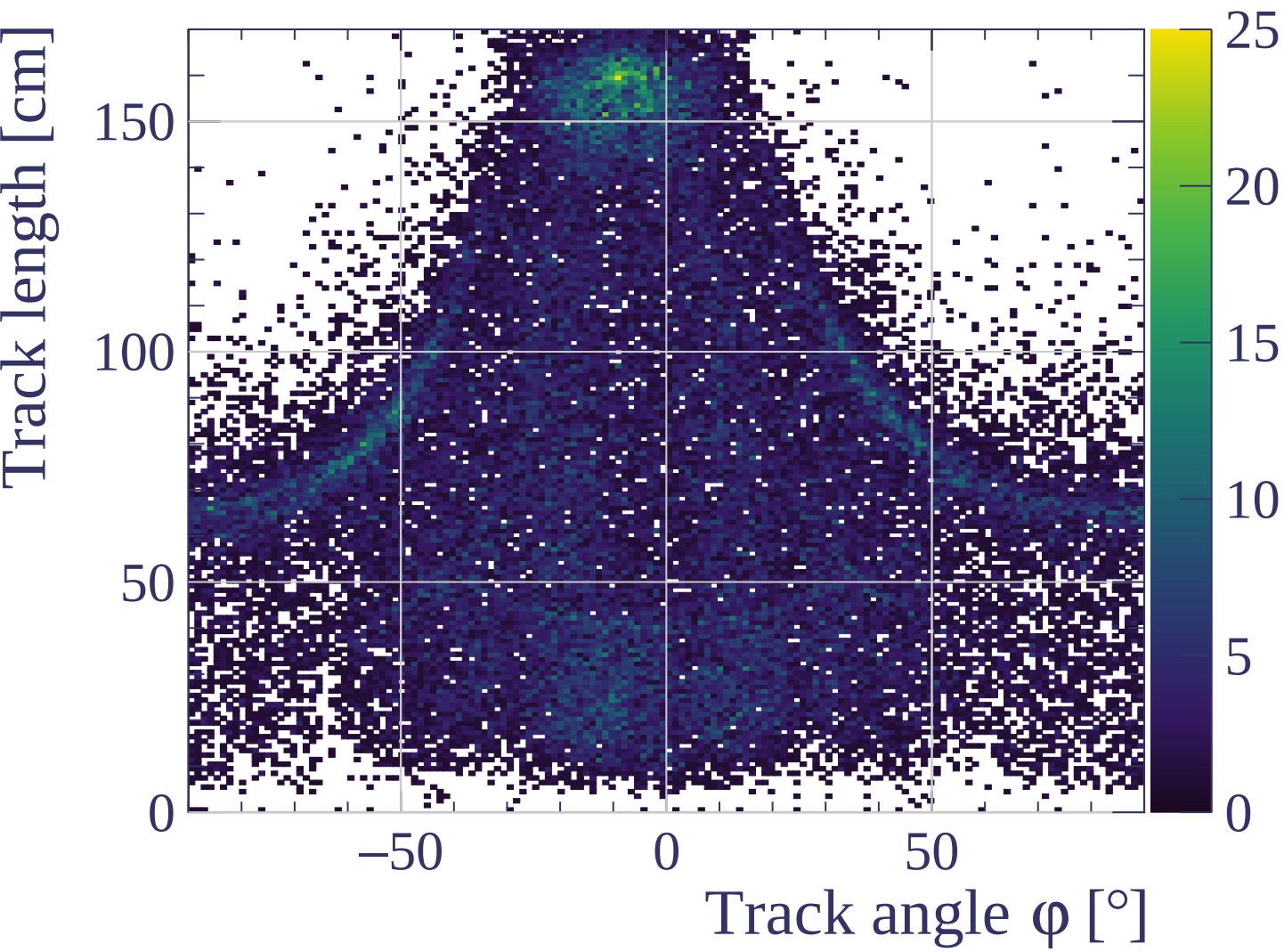


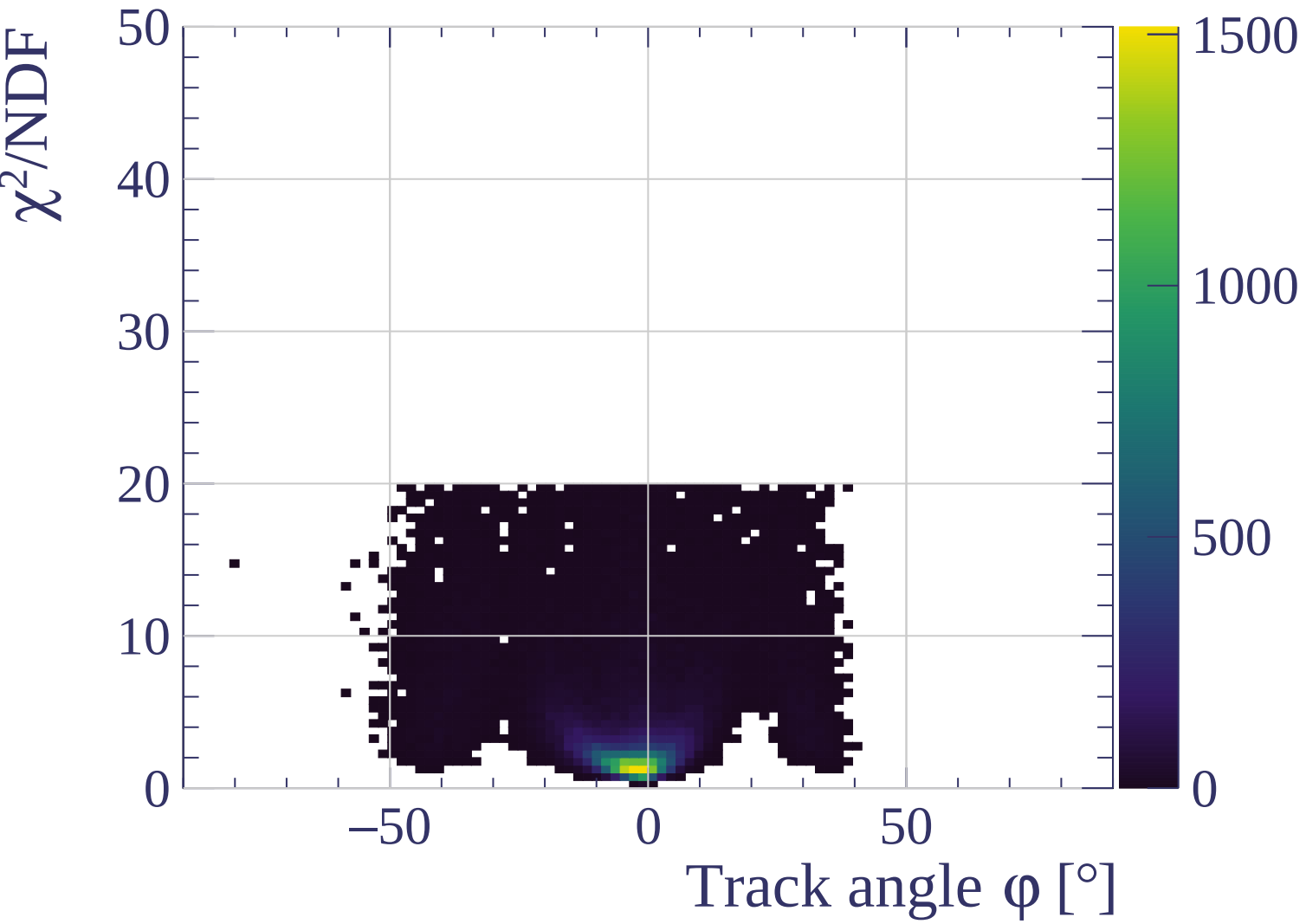


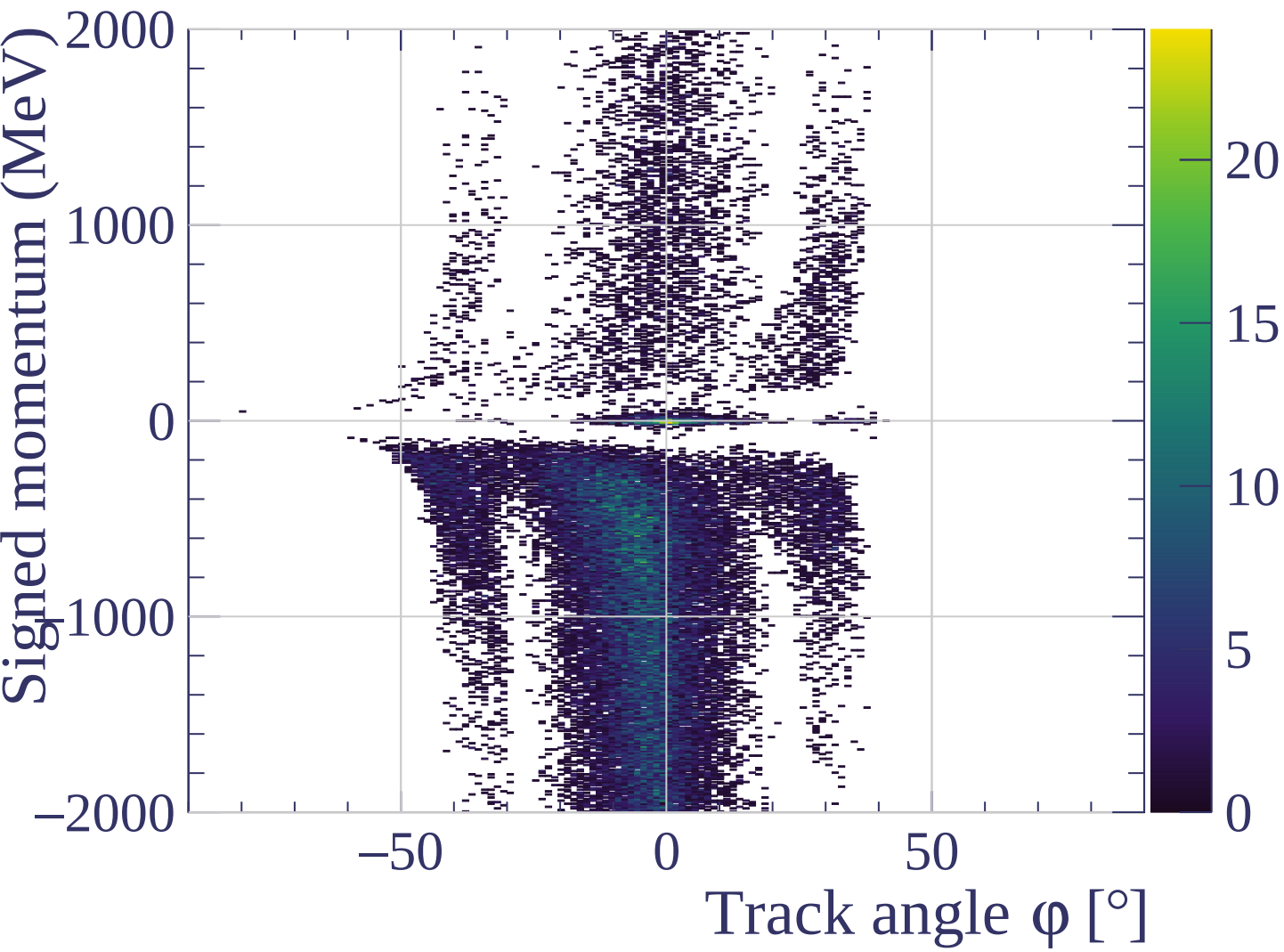




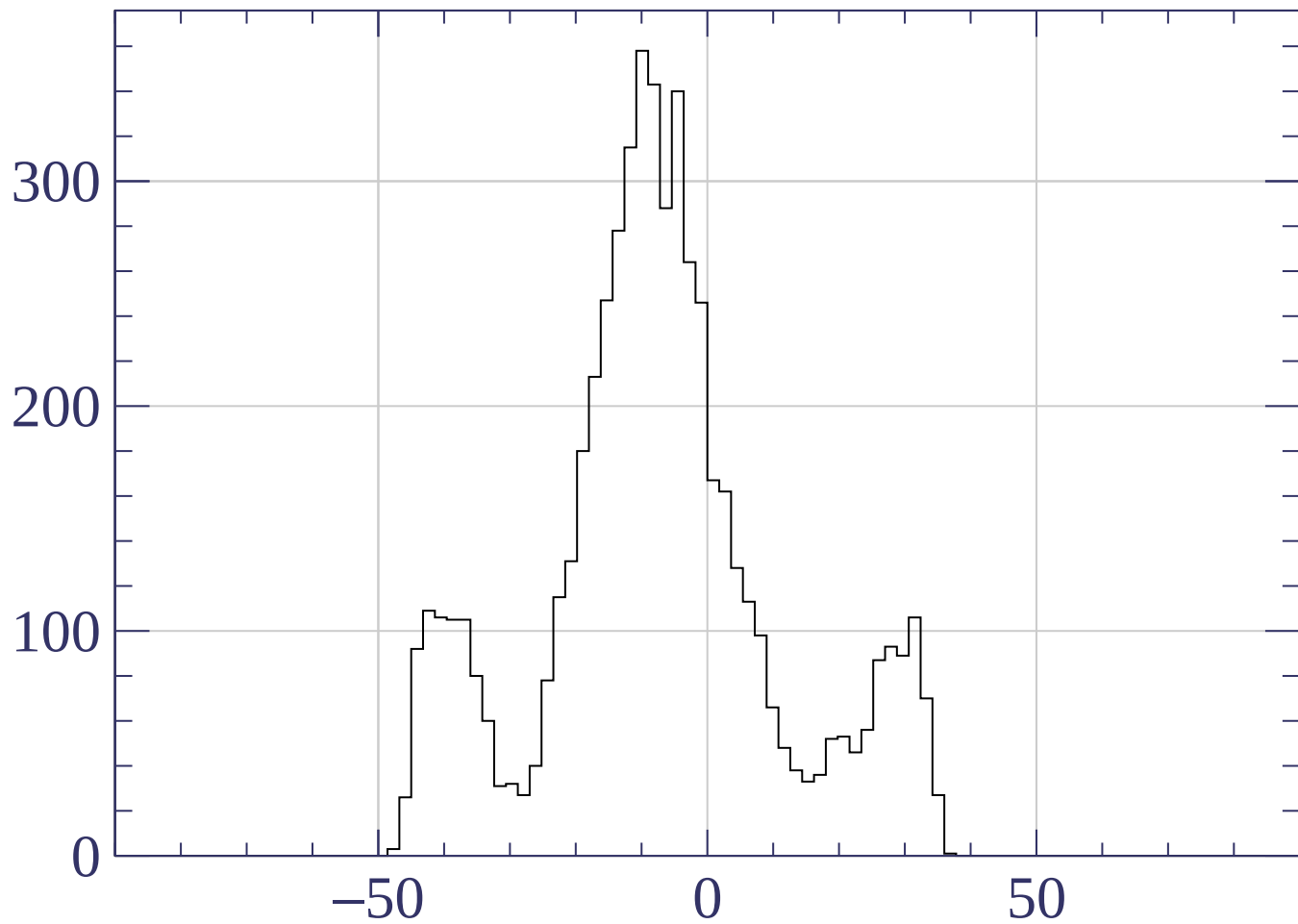






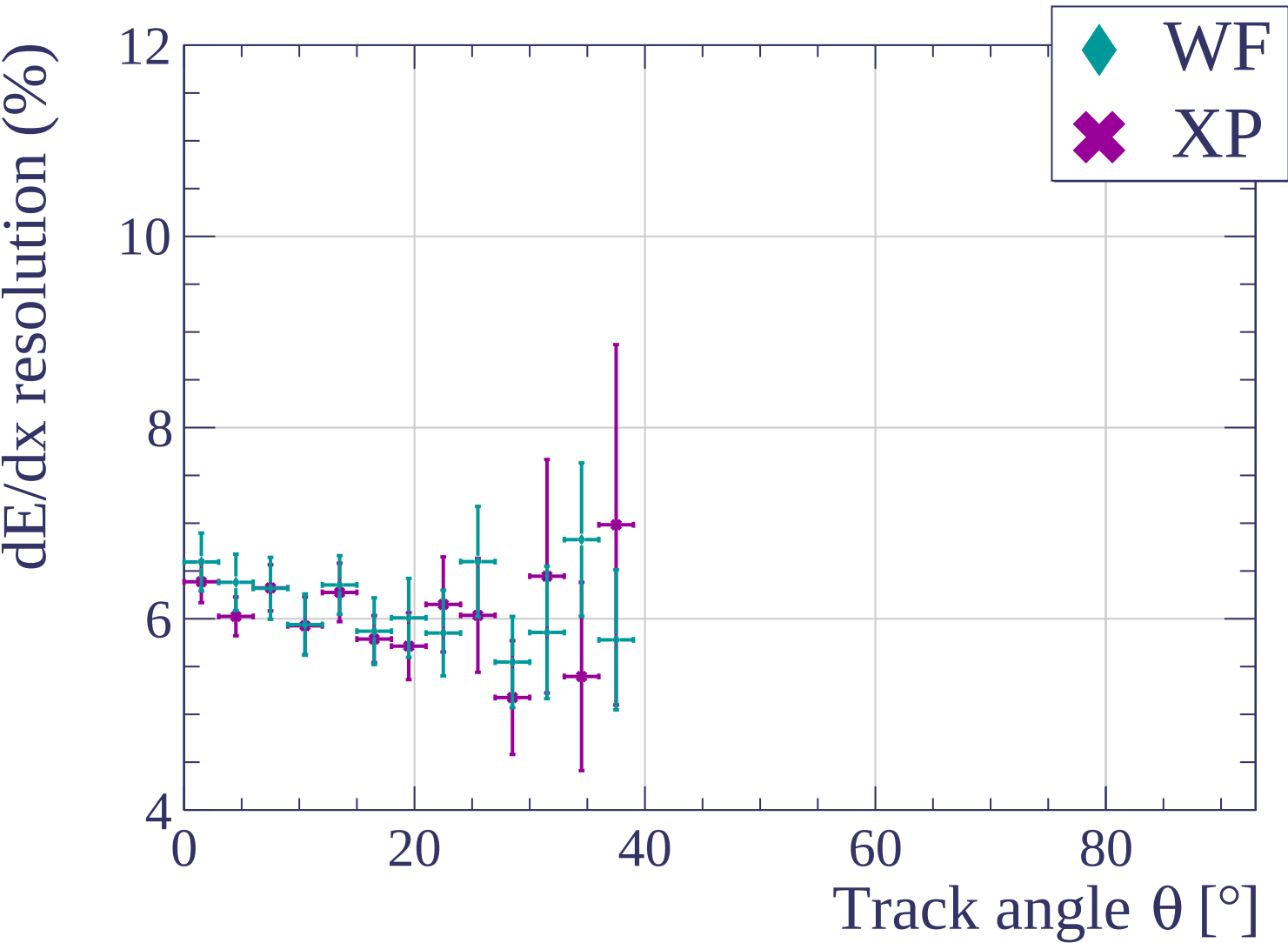


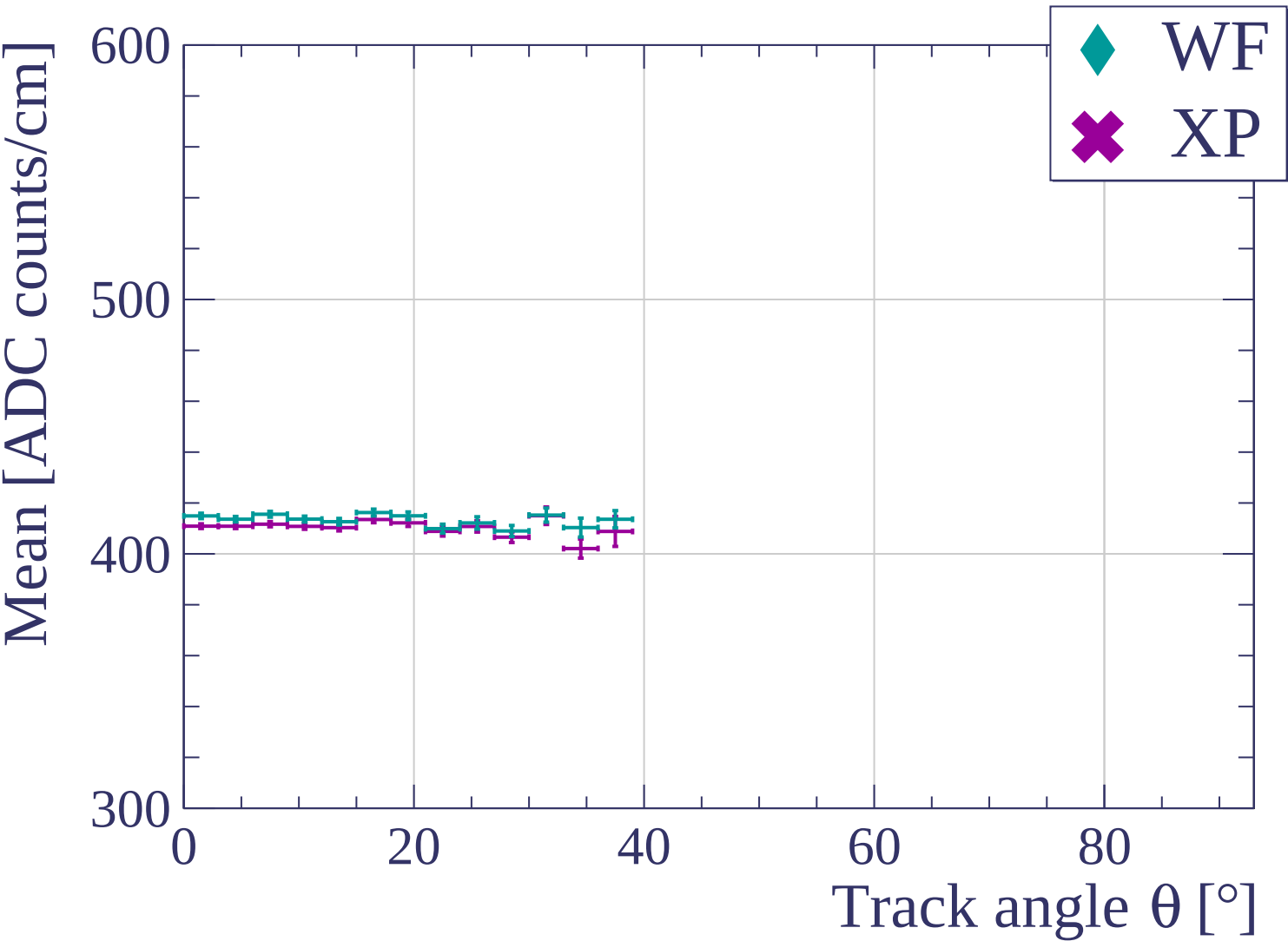
Count

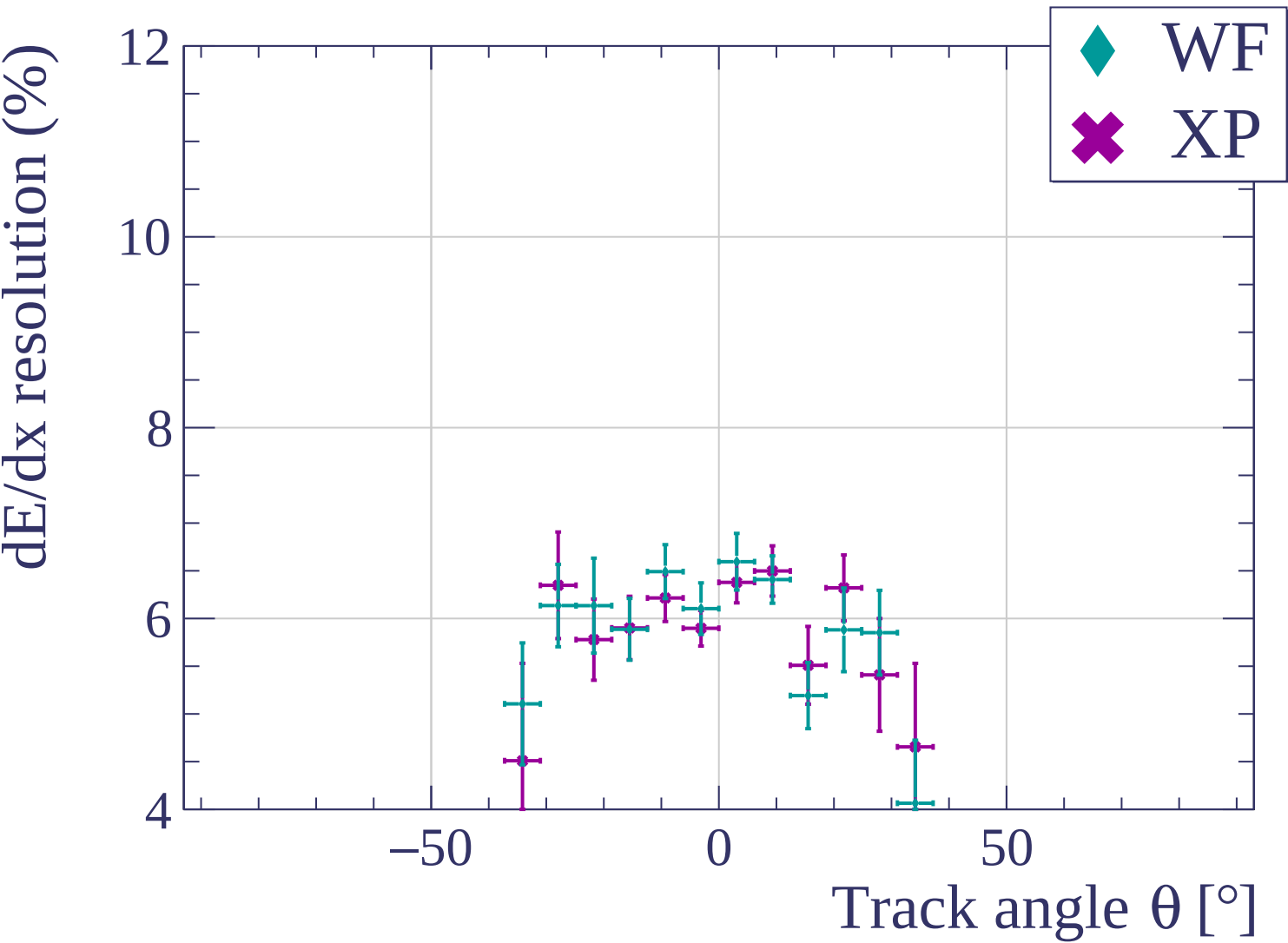


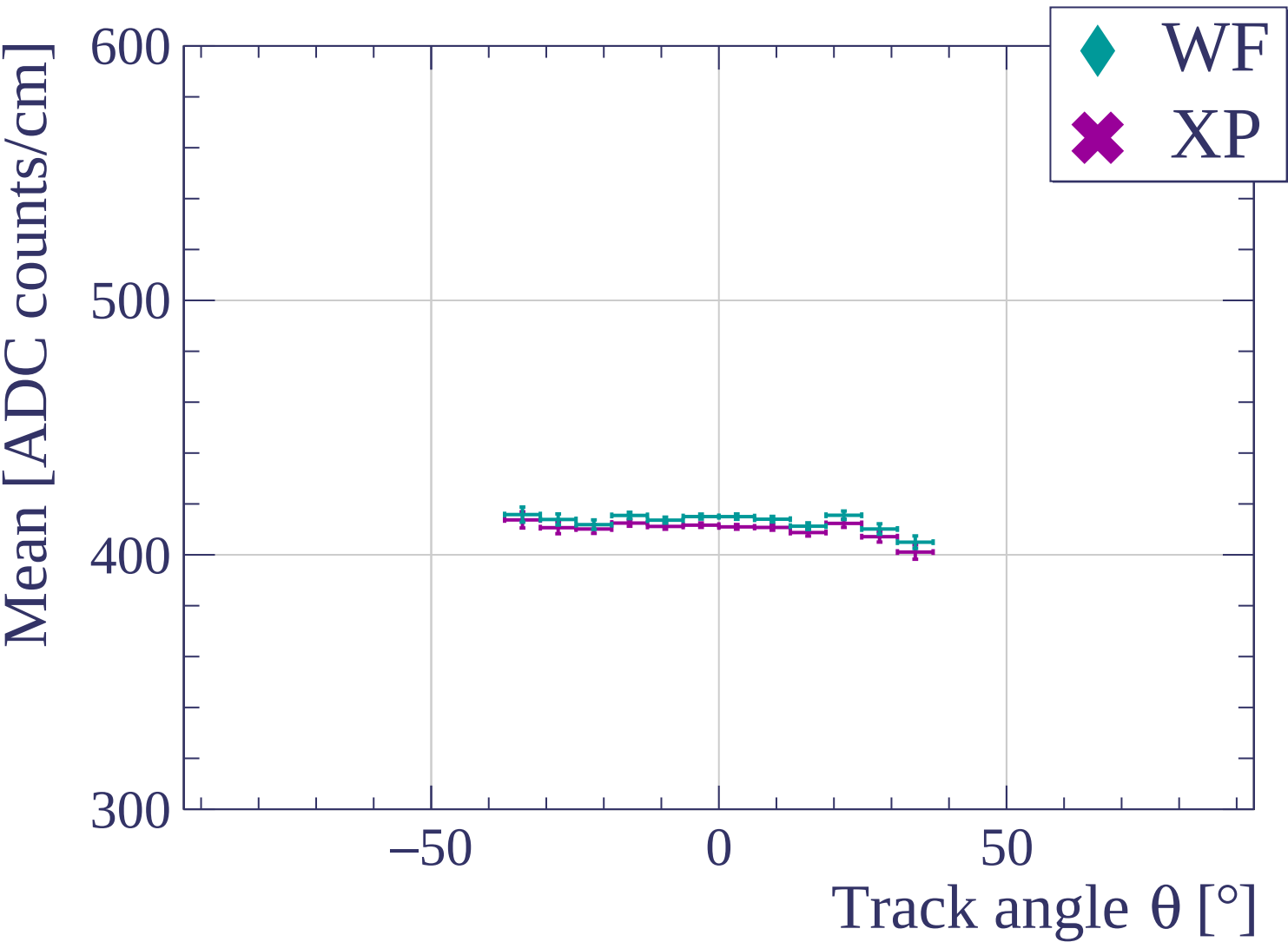
Track angle  $\phi$  [°] angle

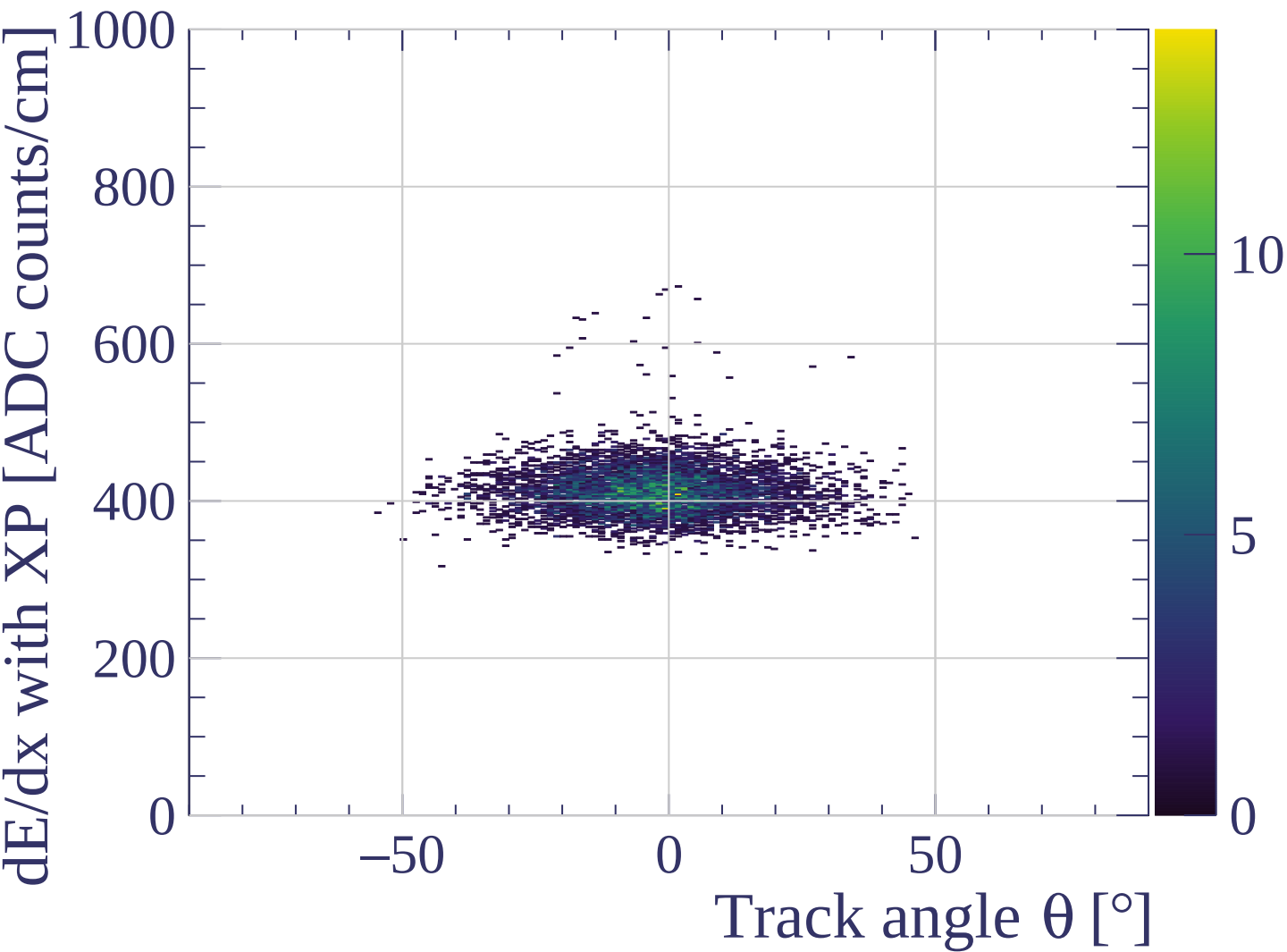




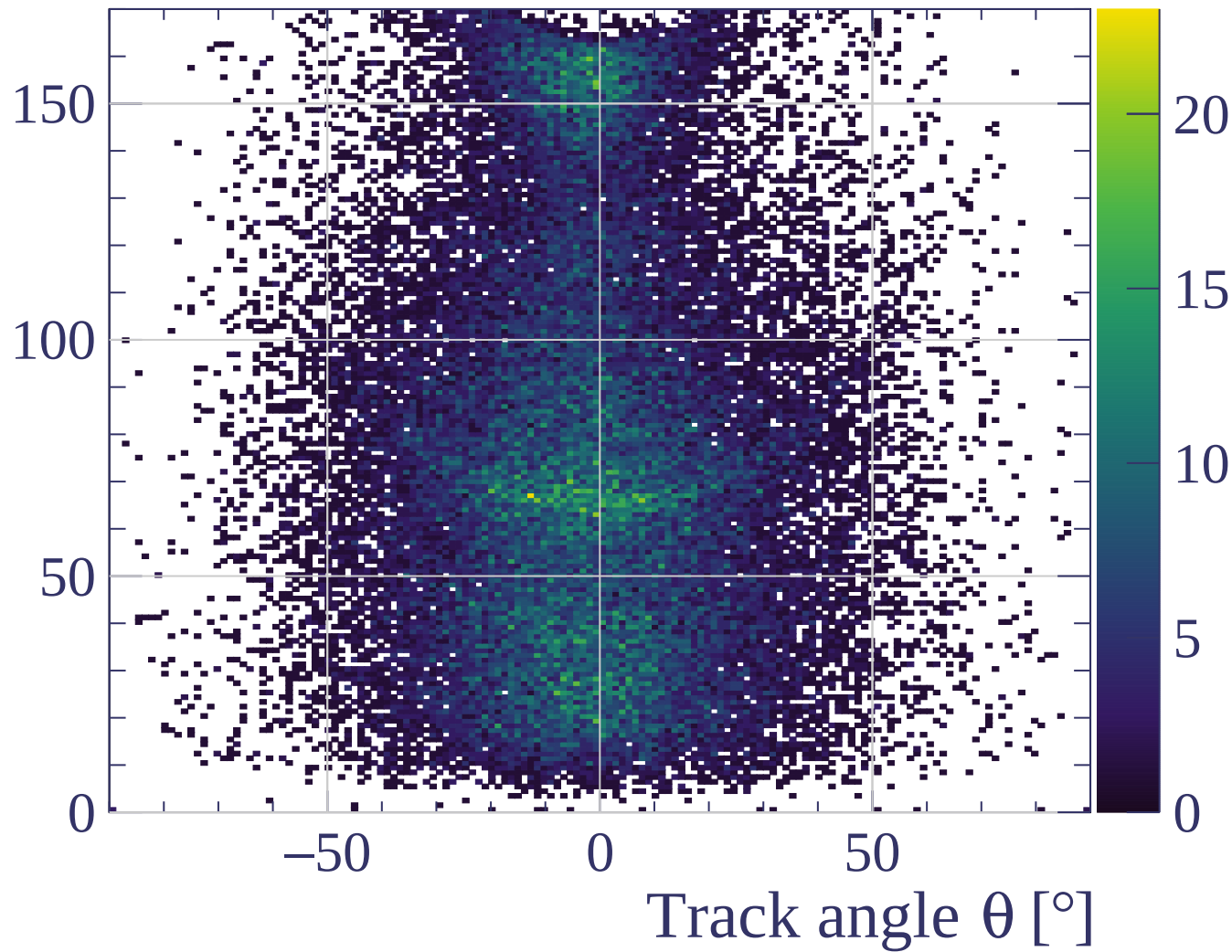


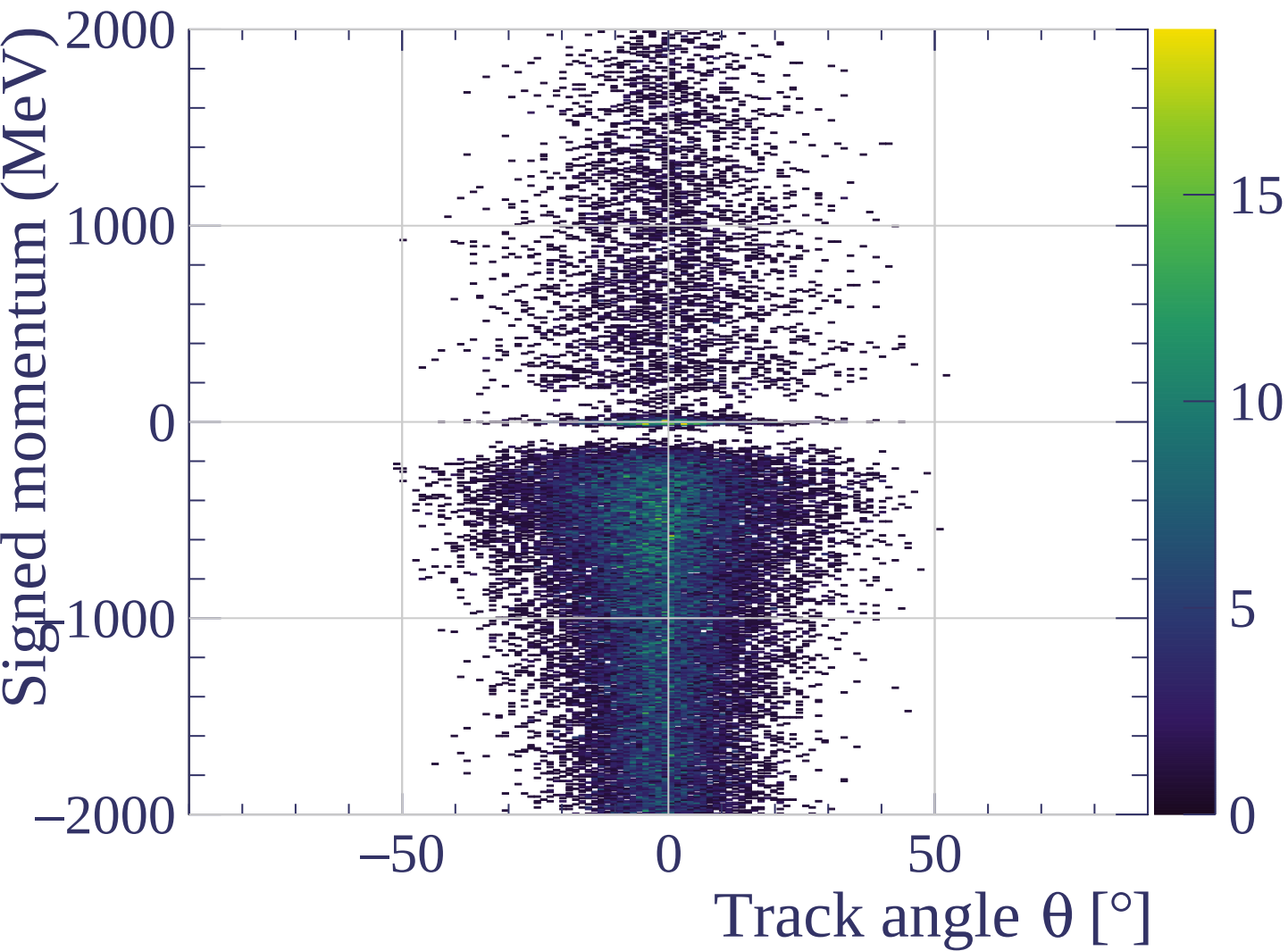




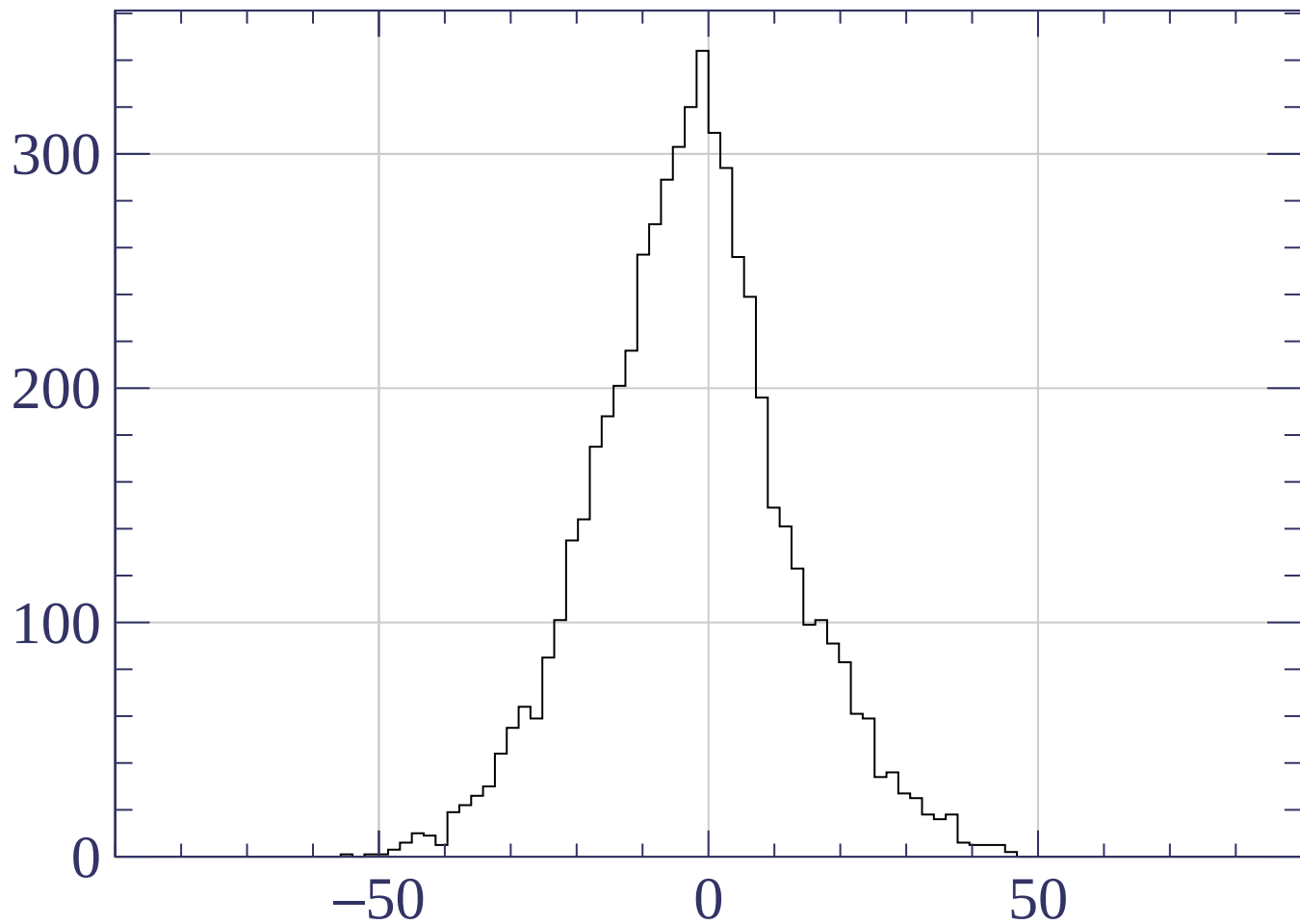


Track length [cm]





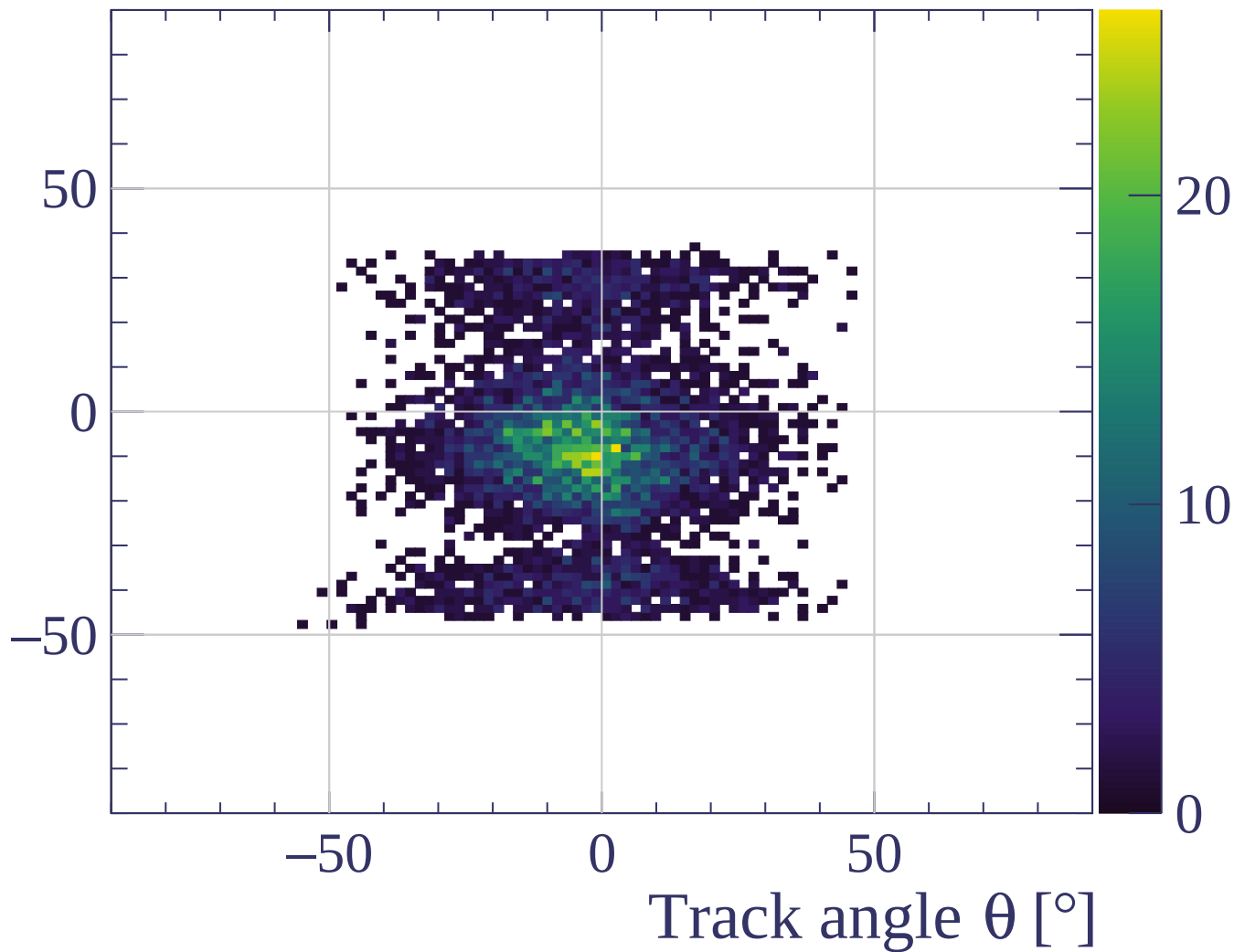
Count



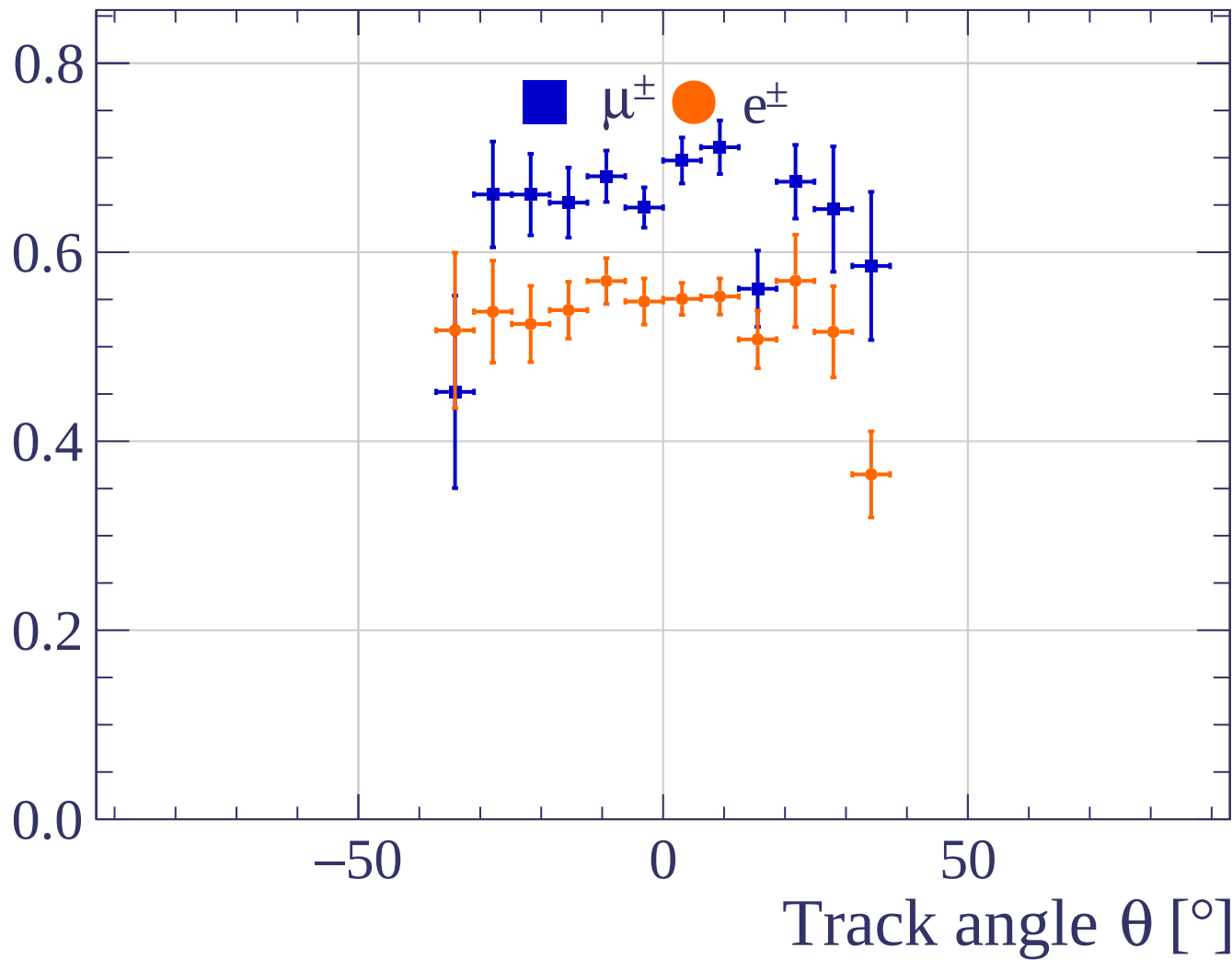
Track angle  $\theta$  [°]



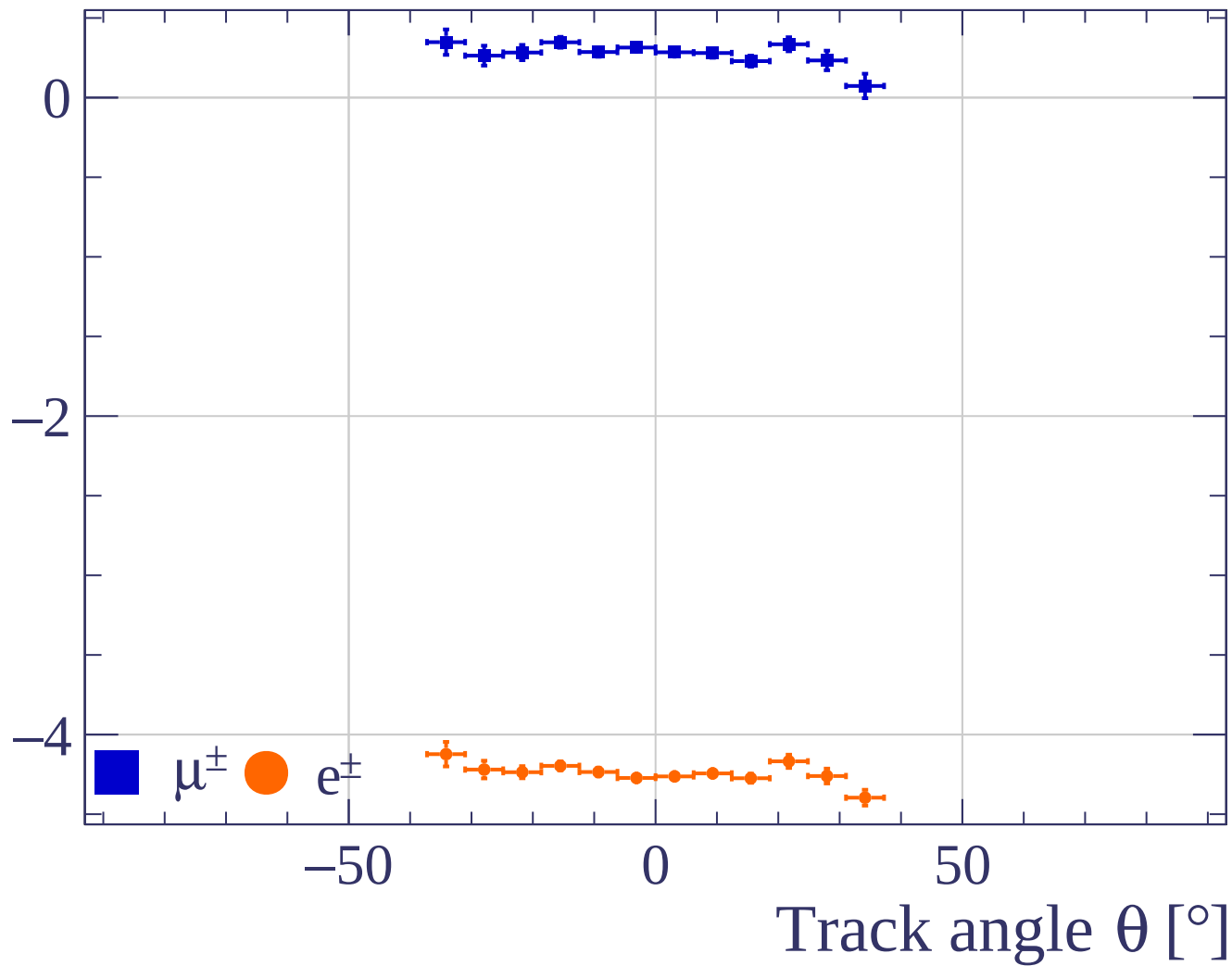
Track angle  $\phi$  [°]



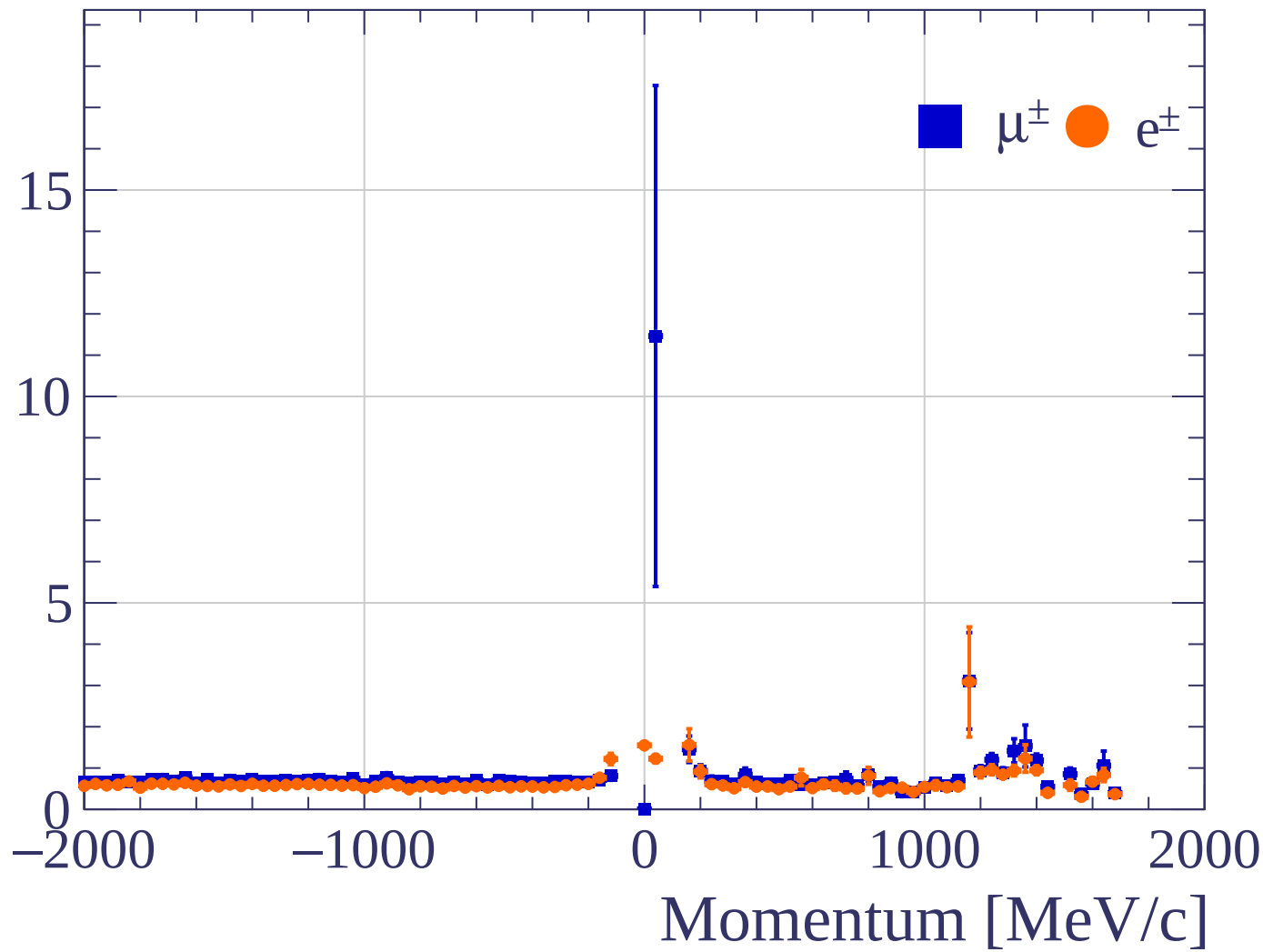
Pulls standard deviation



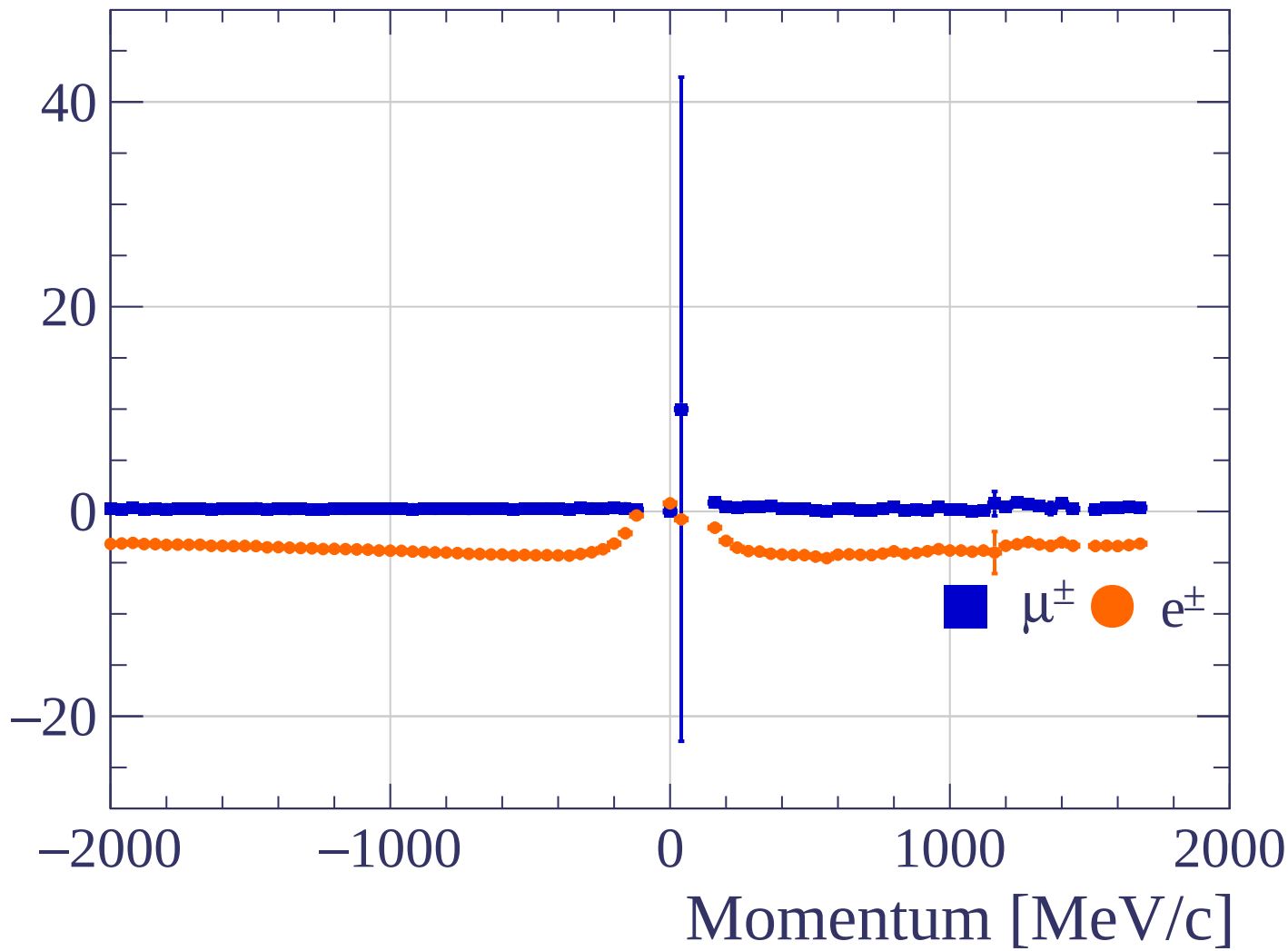
Pulls mean



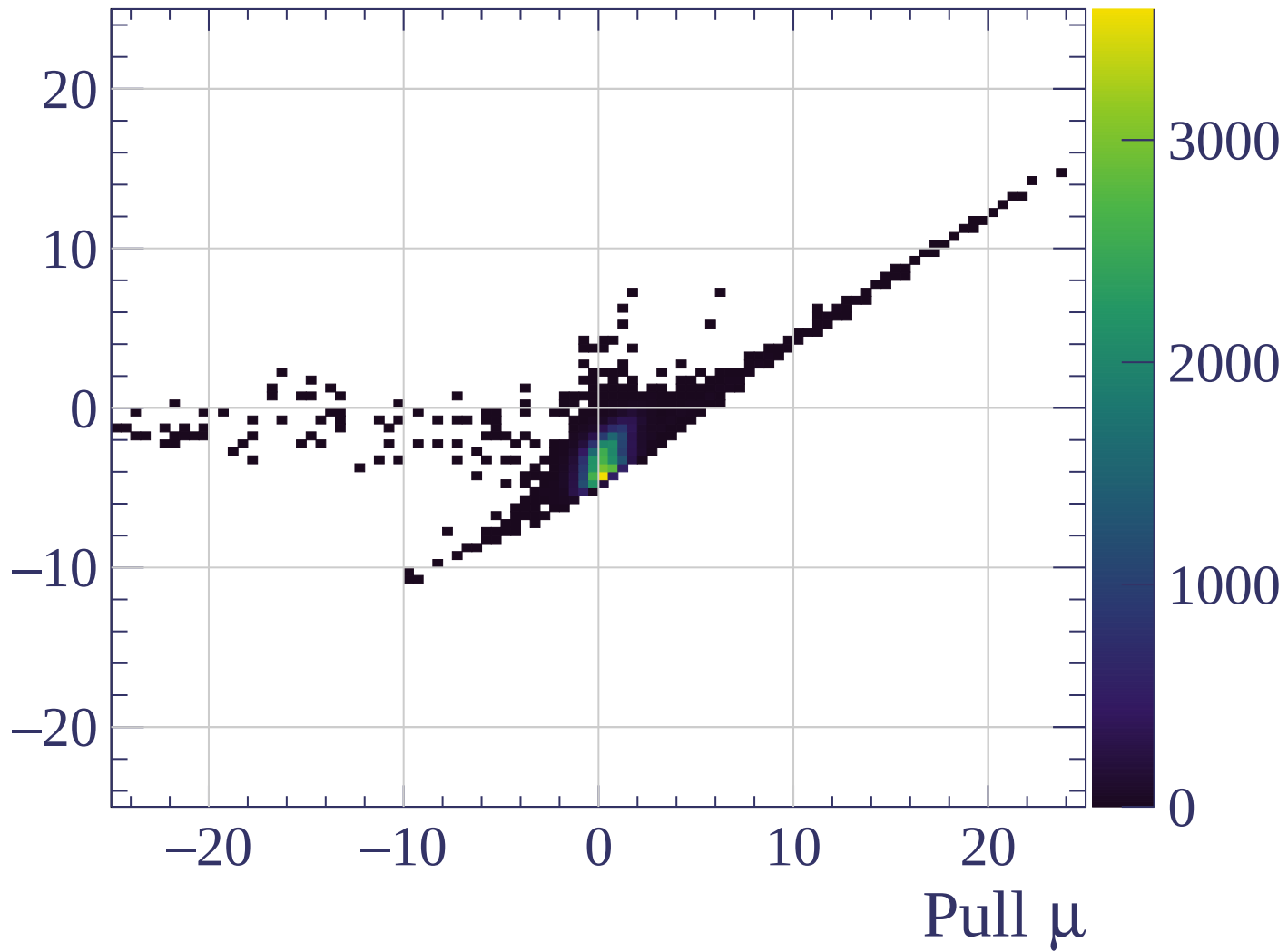
Pulls standard deviation

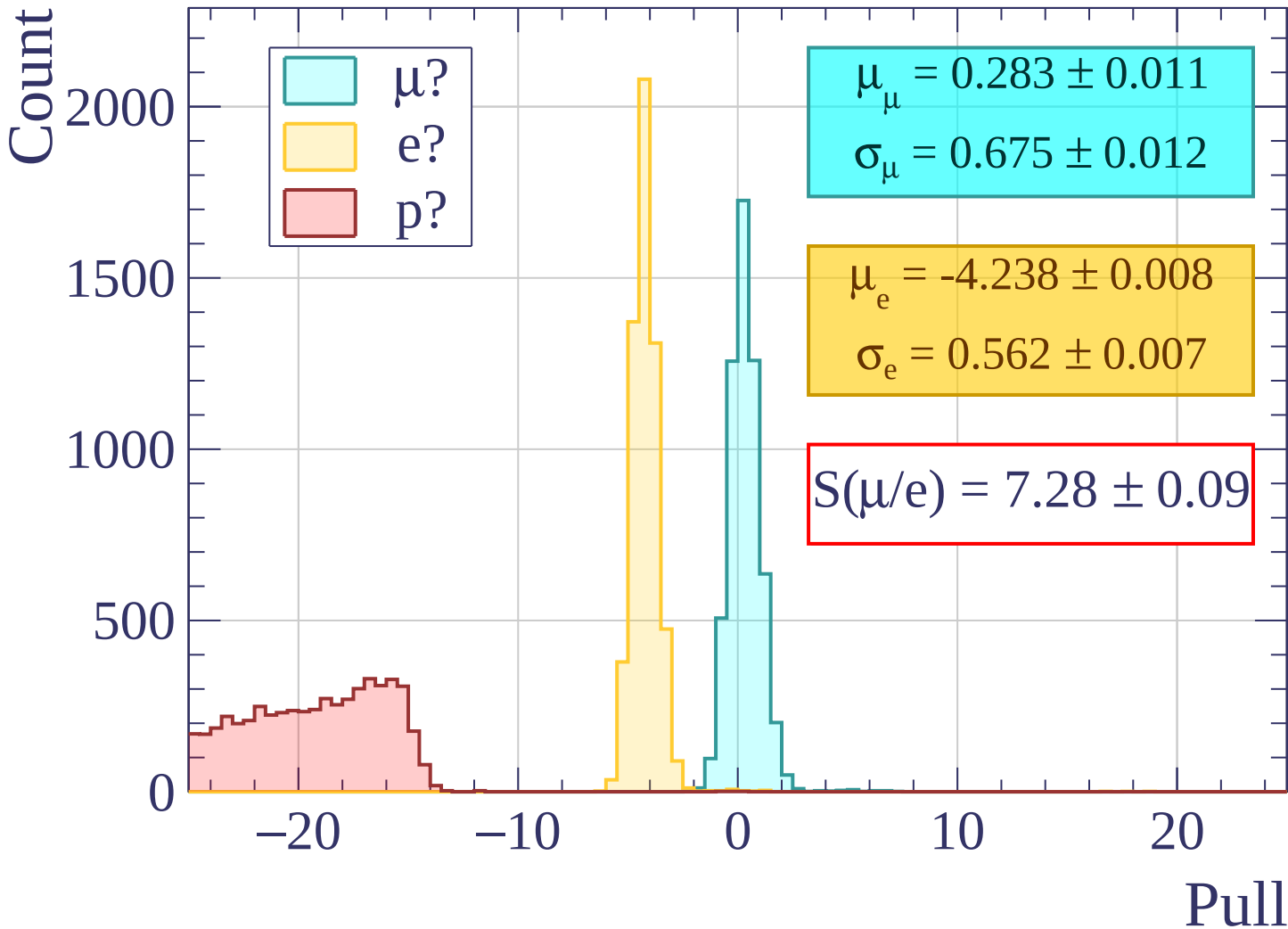


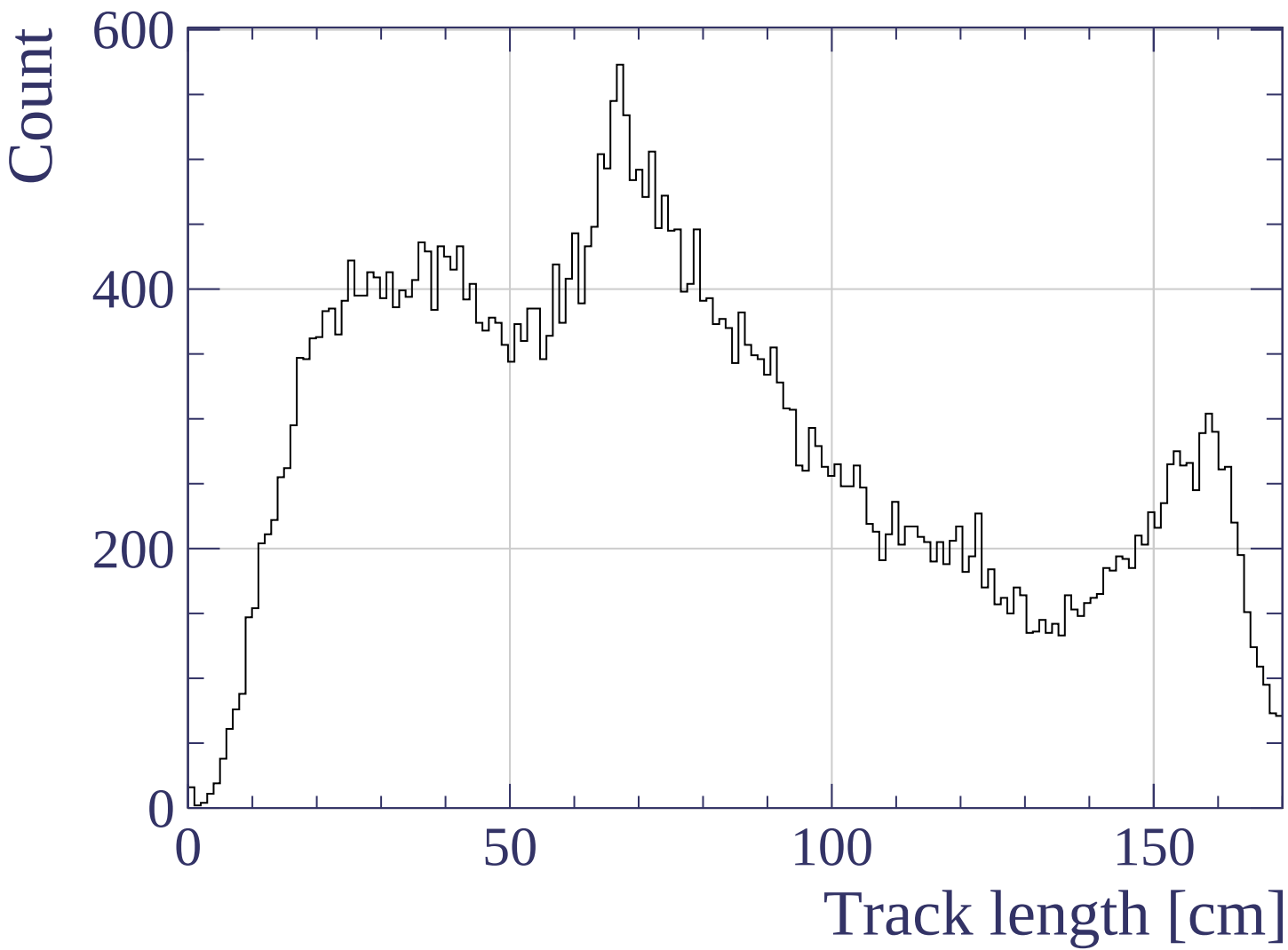
Pulls mean



Pull  $e$

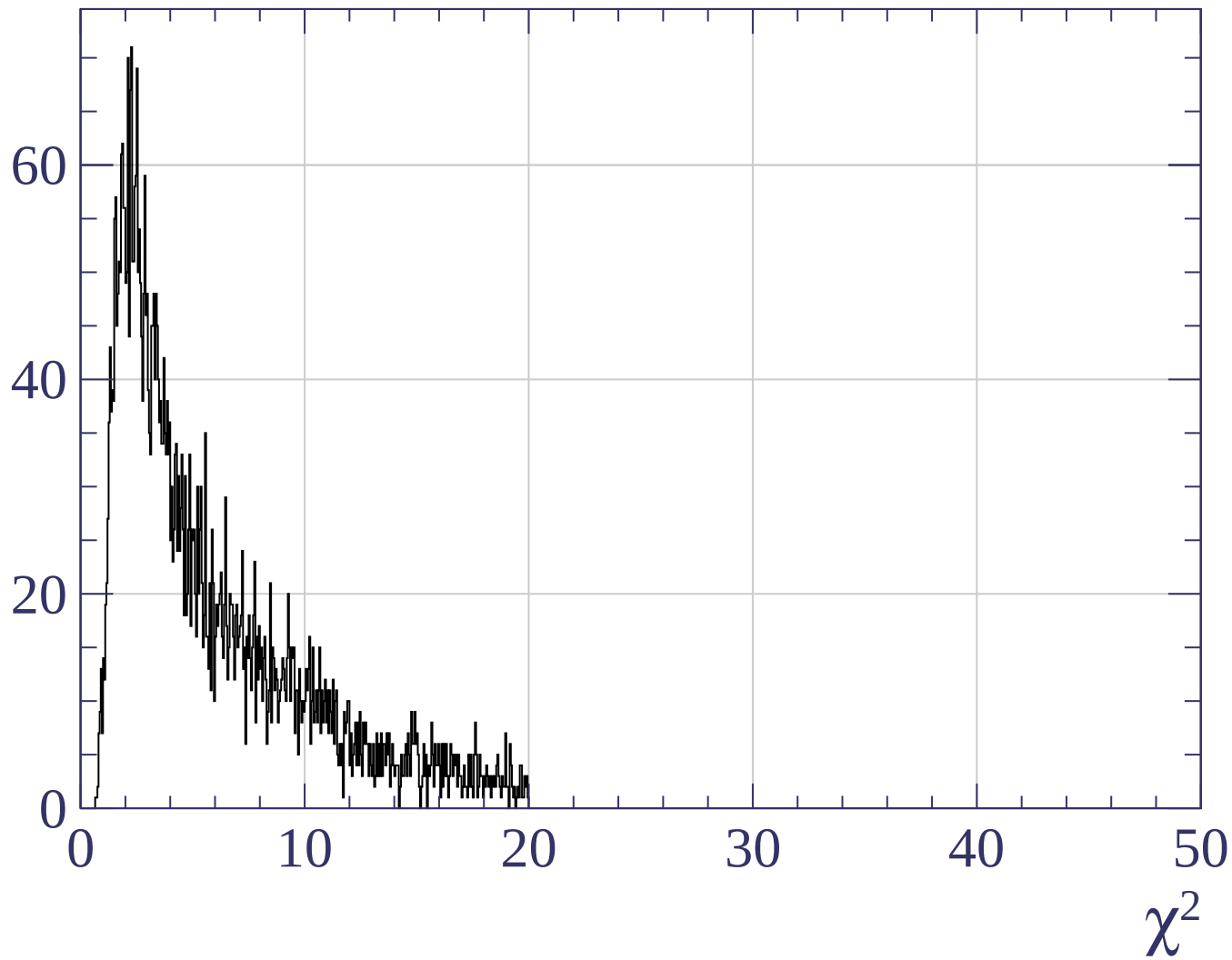




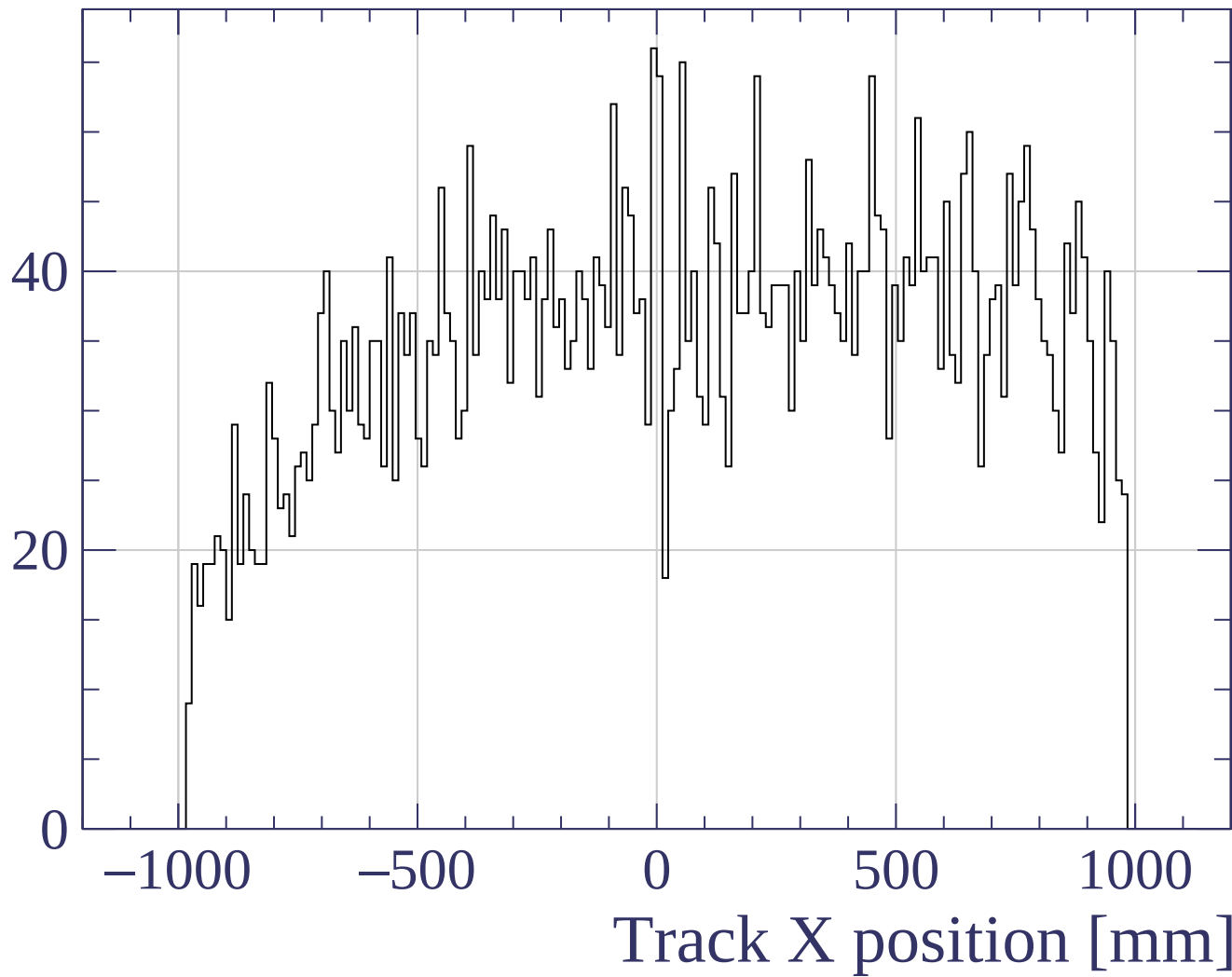




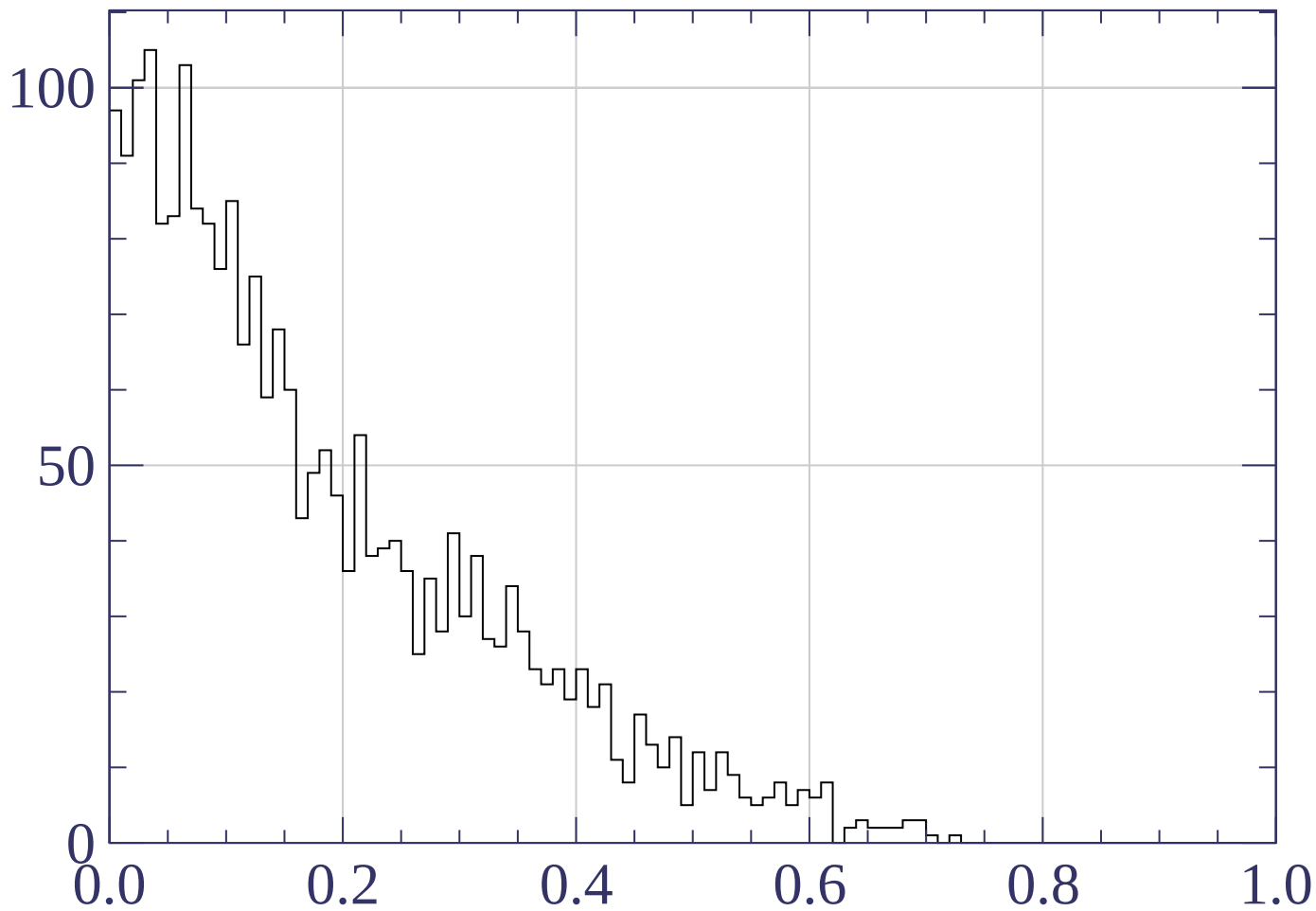
Count



Count

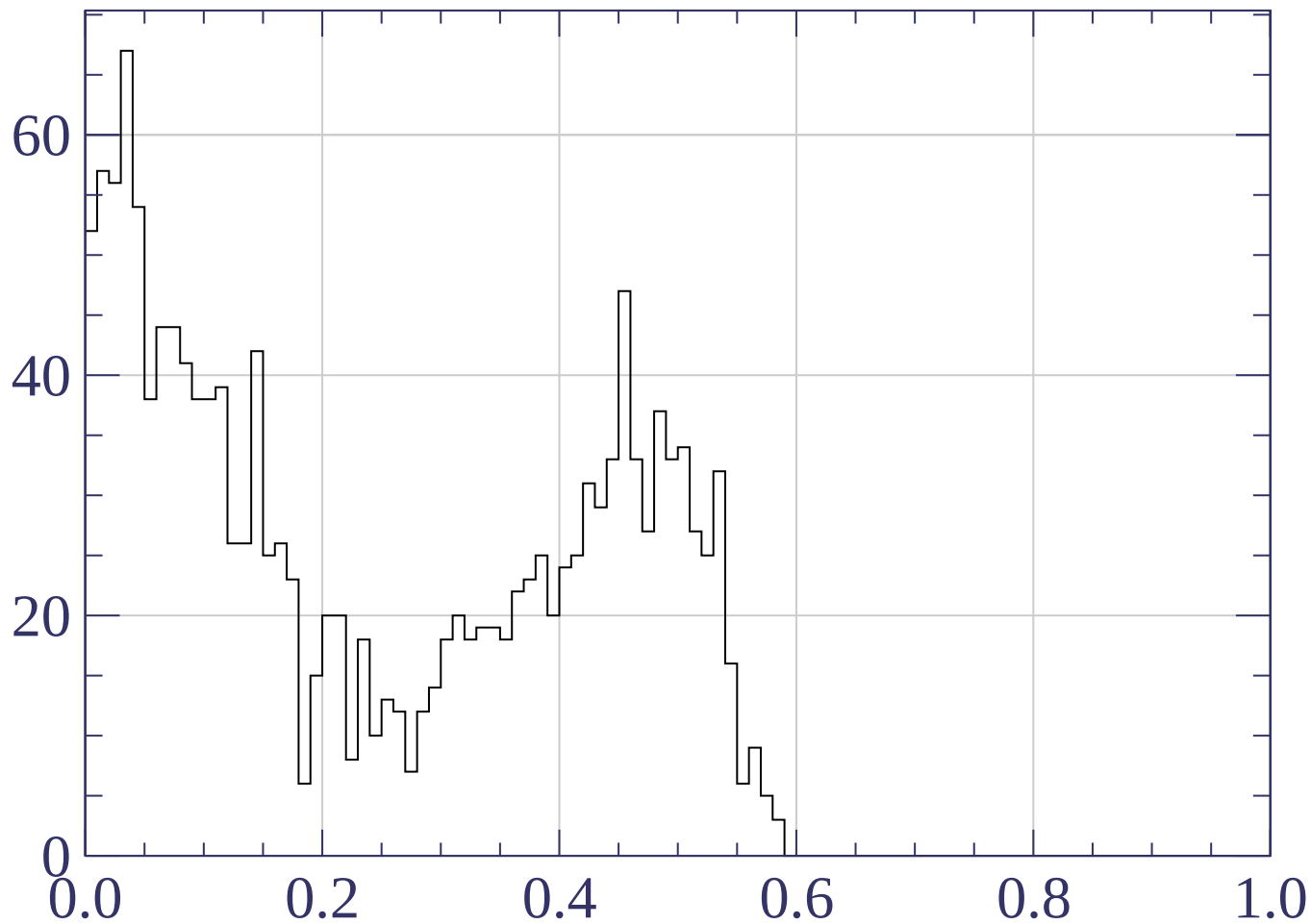


Counts



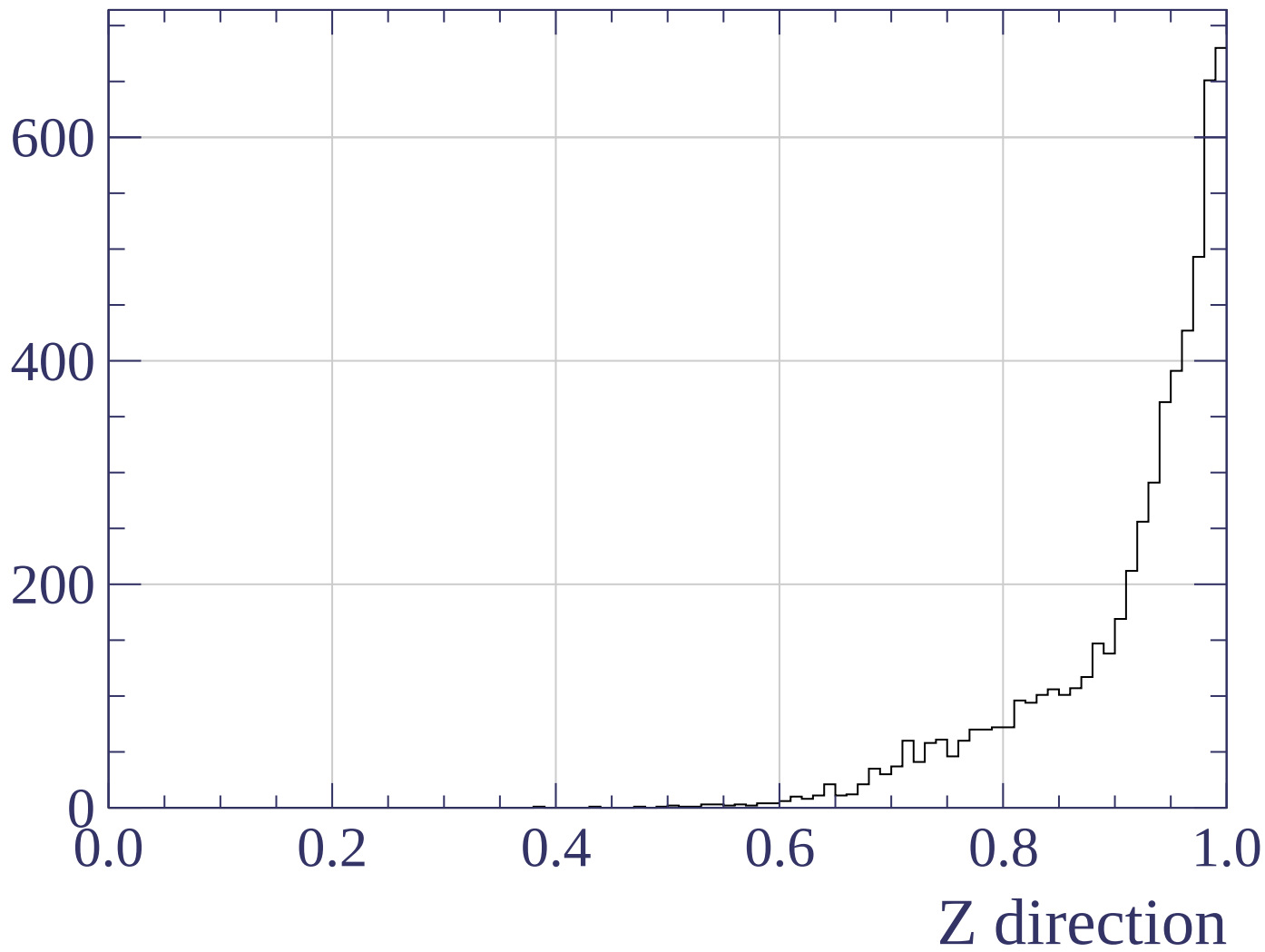
X direction

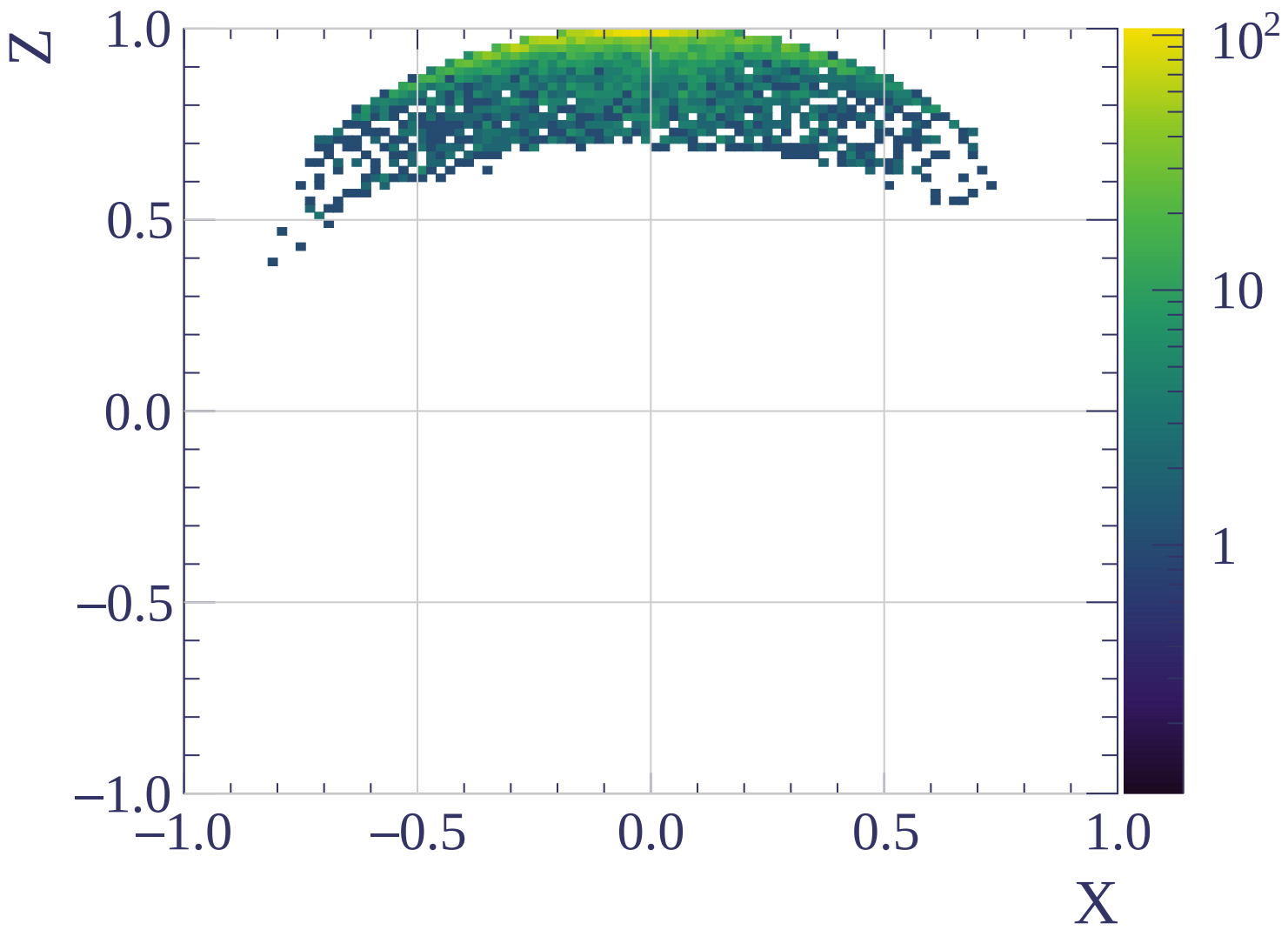
Counts

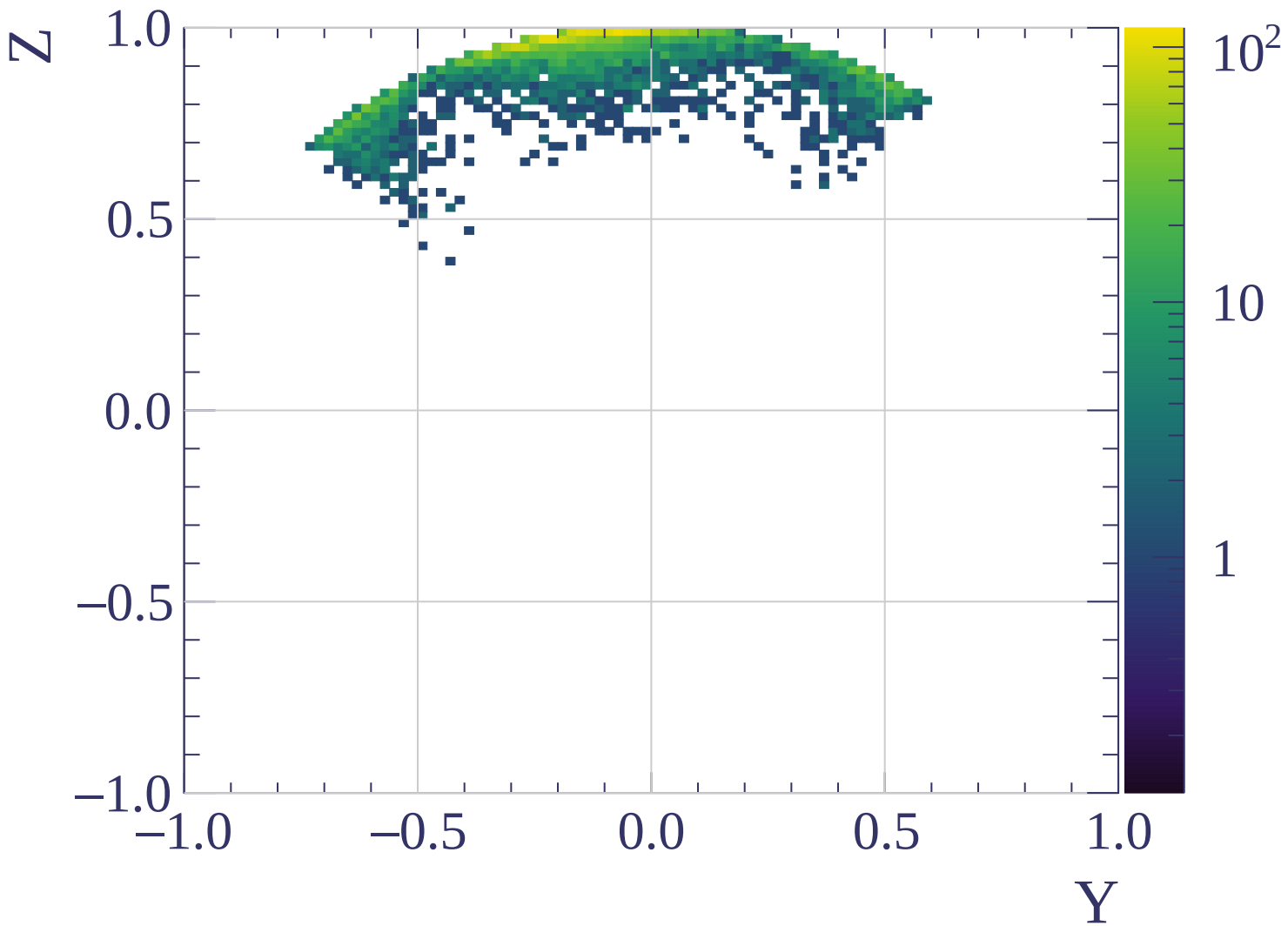


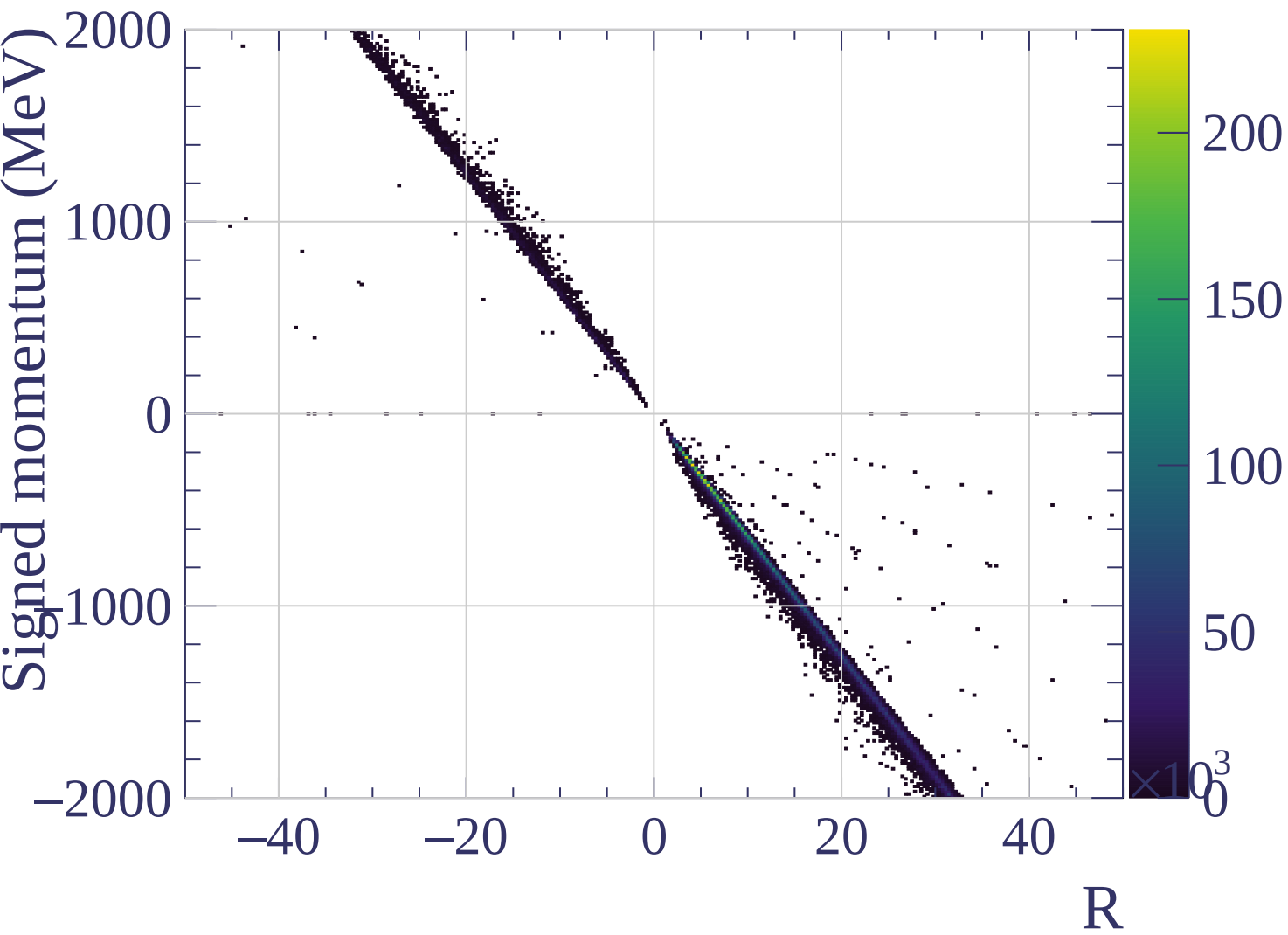
Y direction

Counts

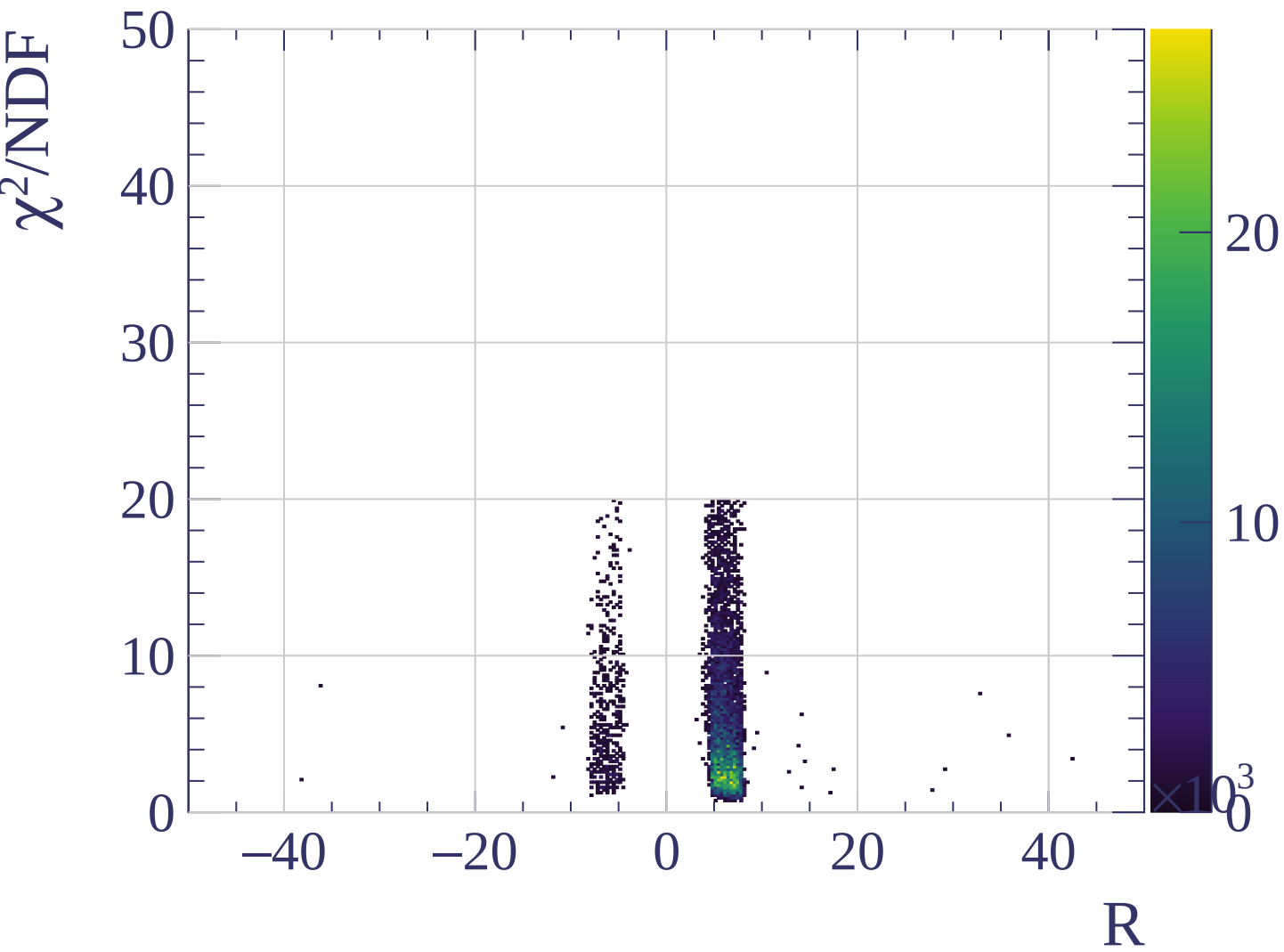






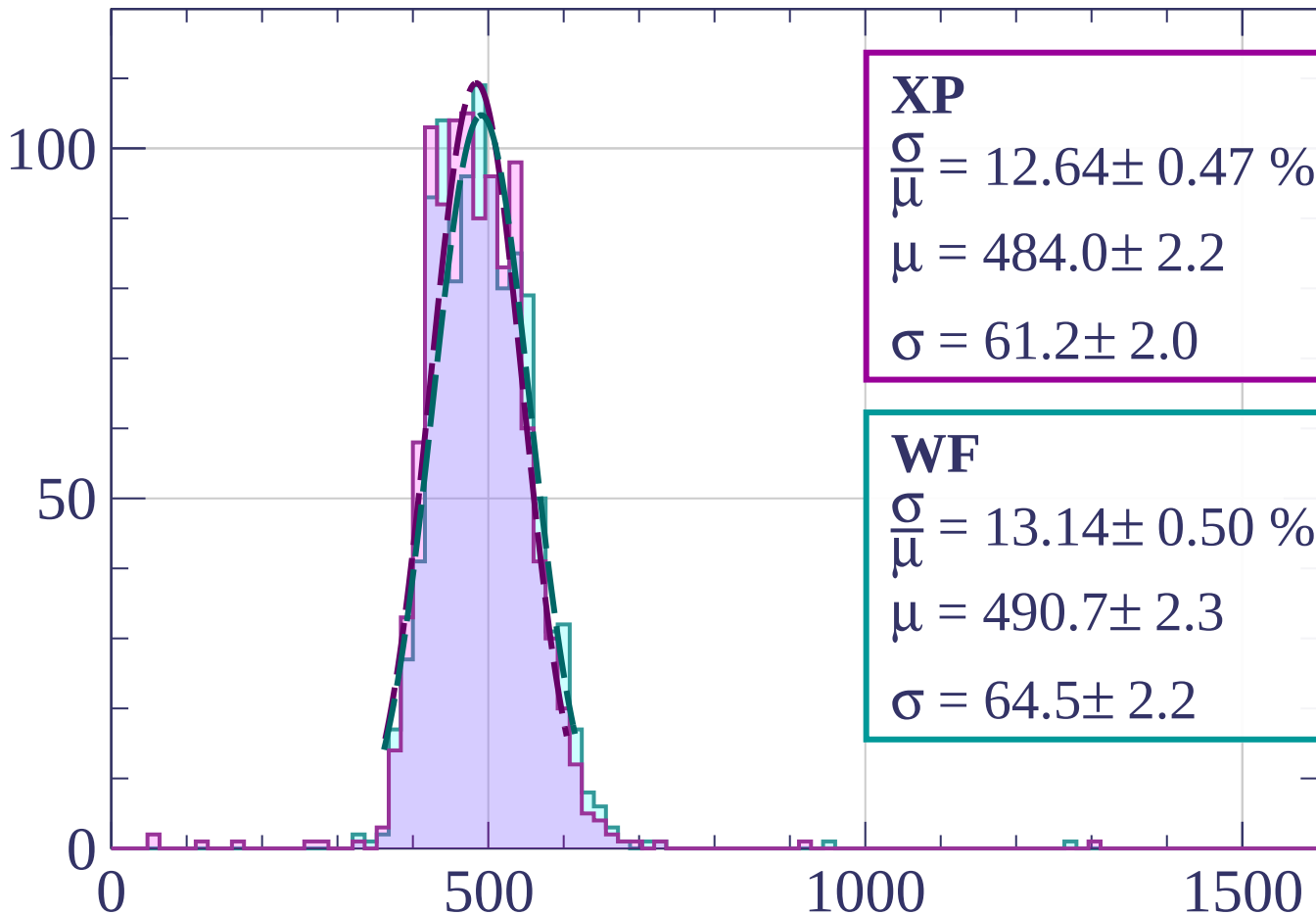






Energy loss |  $0 < p < 80$

Count



**XP**

$$\frac{\sigma}{\mu} = 12.64 \pm 0.47 \%$$

$$\mu = 484.0 \pm 2.2$$

$$\sigma = 61.2 \pm 2.0$$

**WF**

$$\frac{\sigma}{\mu} = 13.14 \pm 0.50 \%$$

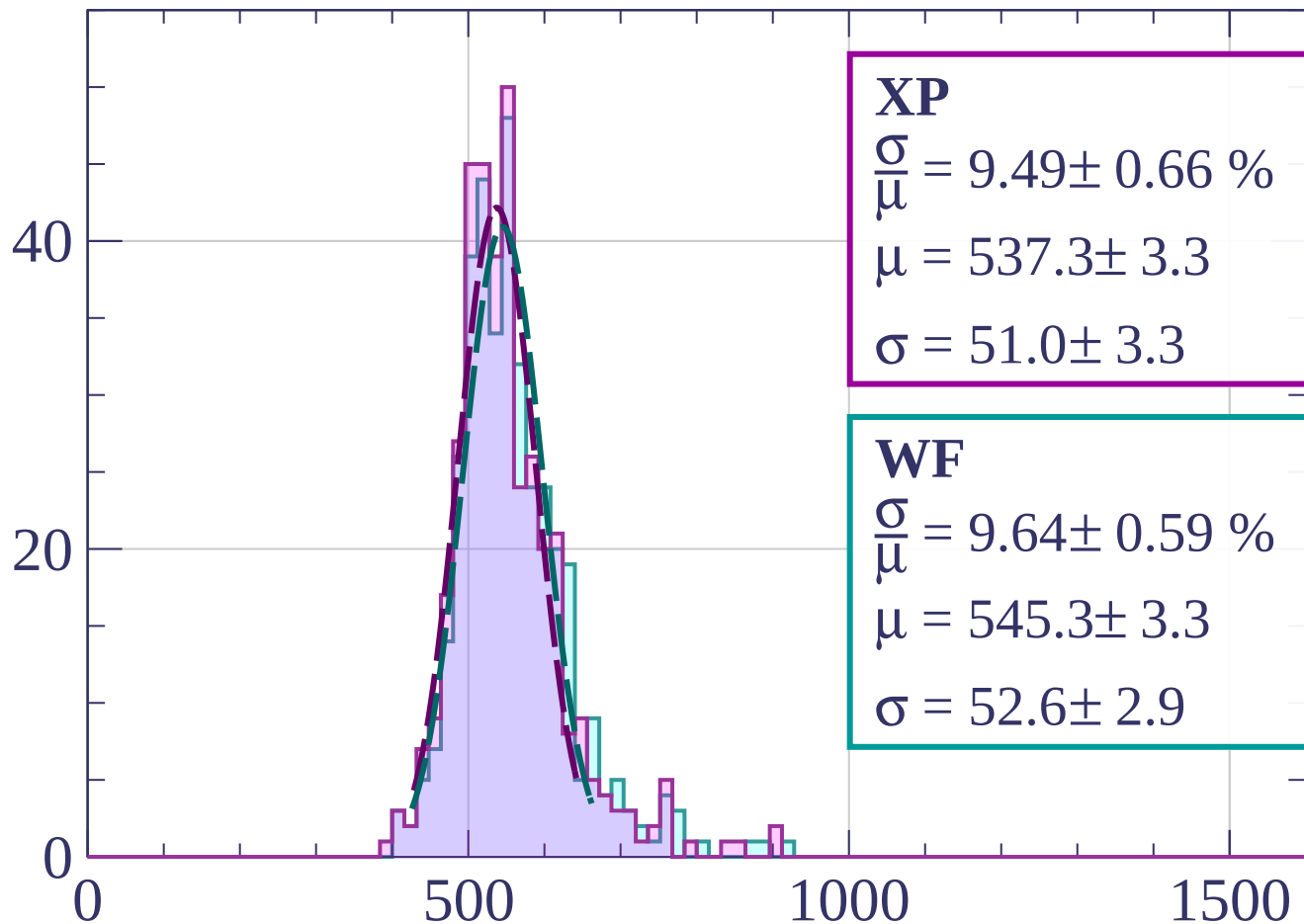
$$\mu = 490.7 \pm 2.3$$

$$\sigma = 64.5 \pm 2.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $80 < p < 160$

Count



**XP**

$$\frac{\sigma}{\mu} = 9.49 \pm 0.66 \%$$

$$\mu = 537.3 \pm 3.3$$

$$\sigma = 51.0 \pm 3.3$$

**WF**

$$\frac{\sigma}{\mu} = 9.64 \pm 0.59 \%$$

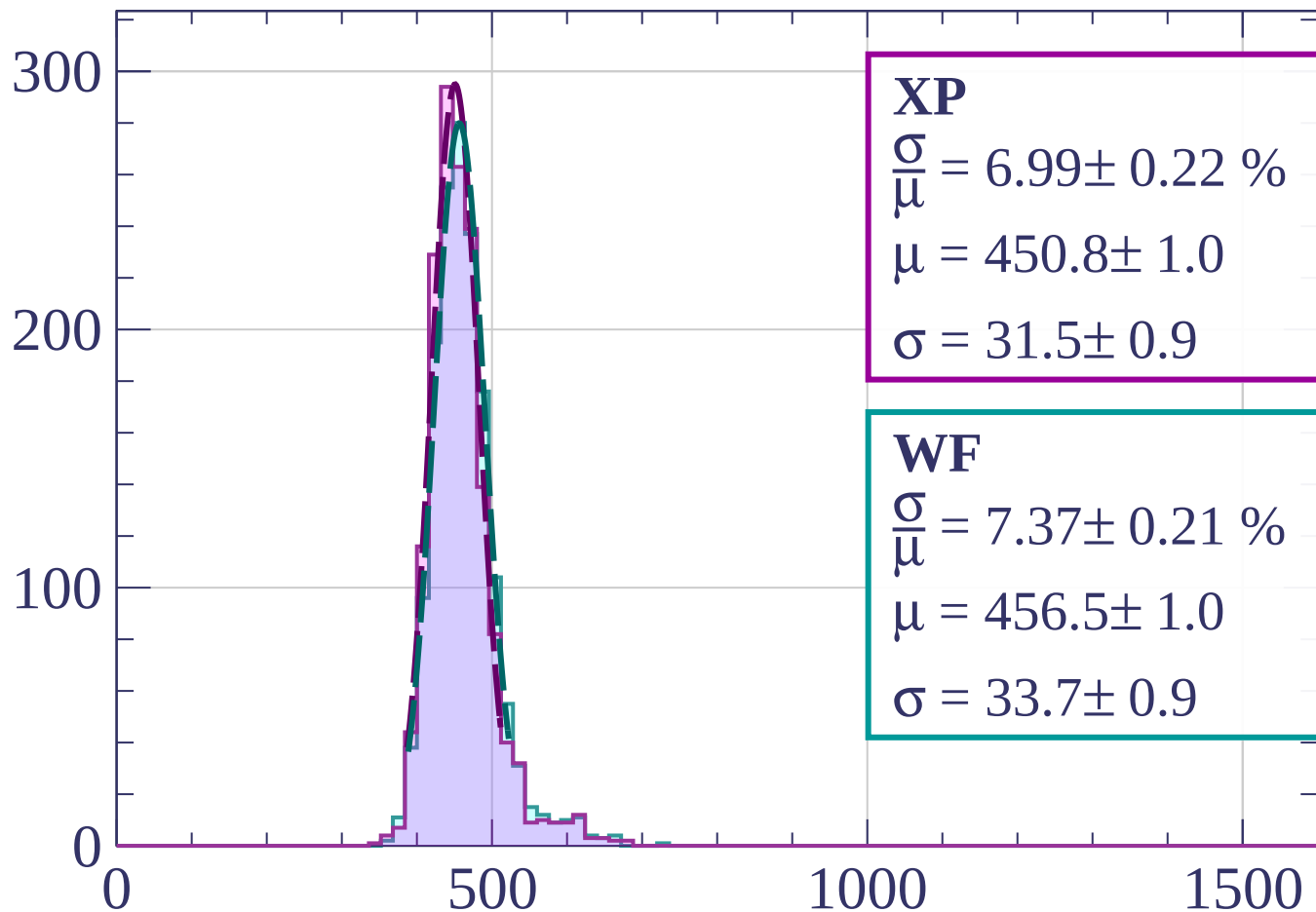
$$\mu = 545.3 \pm 3.3$$

$$\sigma = 52.6 \pm 2.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $160 < p < 240$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.99 \pm 0.22 \%$$

$$\mu = 450.8 \pm 1.0$$

$$\sigma = 31.5 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 7.37 \pm 0.21 \%$$

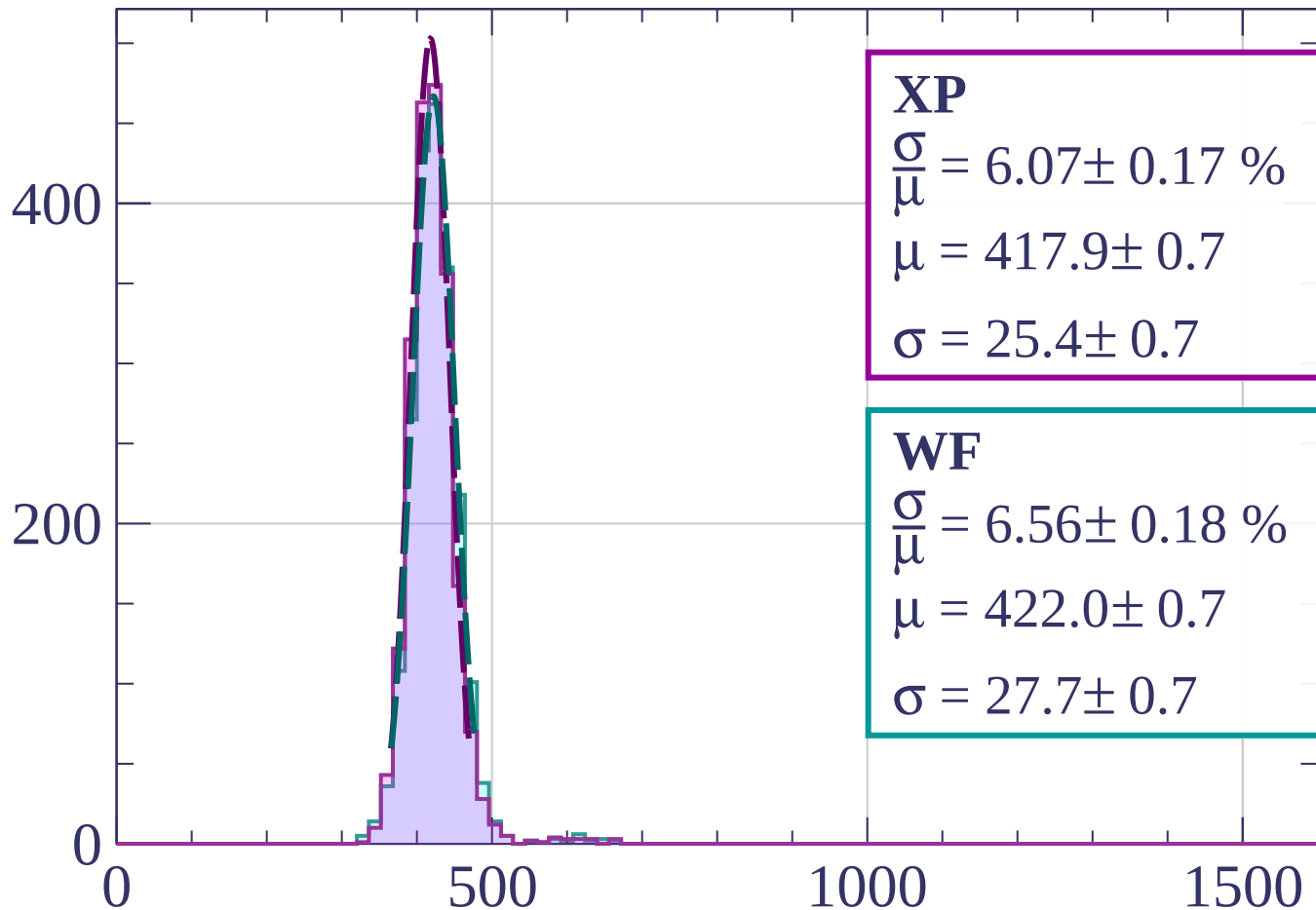
$$\mu = 456.5 \pm 1.0$$

$$\sigma = 33.7 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $240 < p < 320$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.07 \pm 0.17 \%$$

$$\mu = 417.9 \pm 0.7$$

$$\sigma = 25.4 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 6.56 \pm 0.18 \%$$

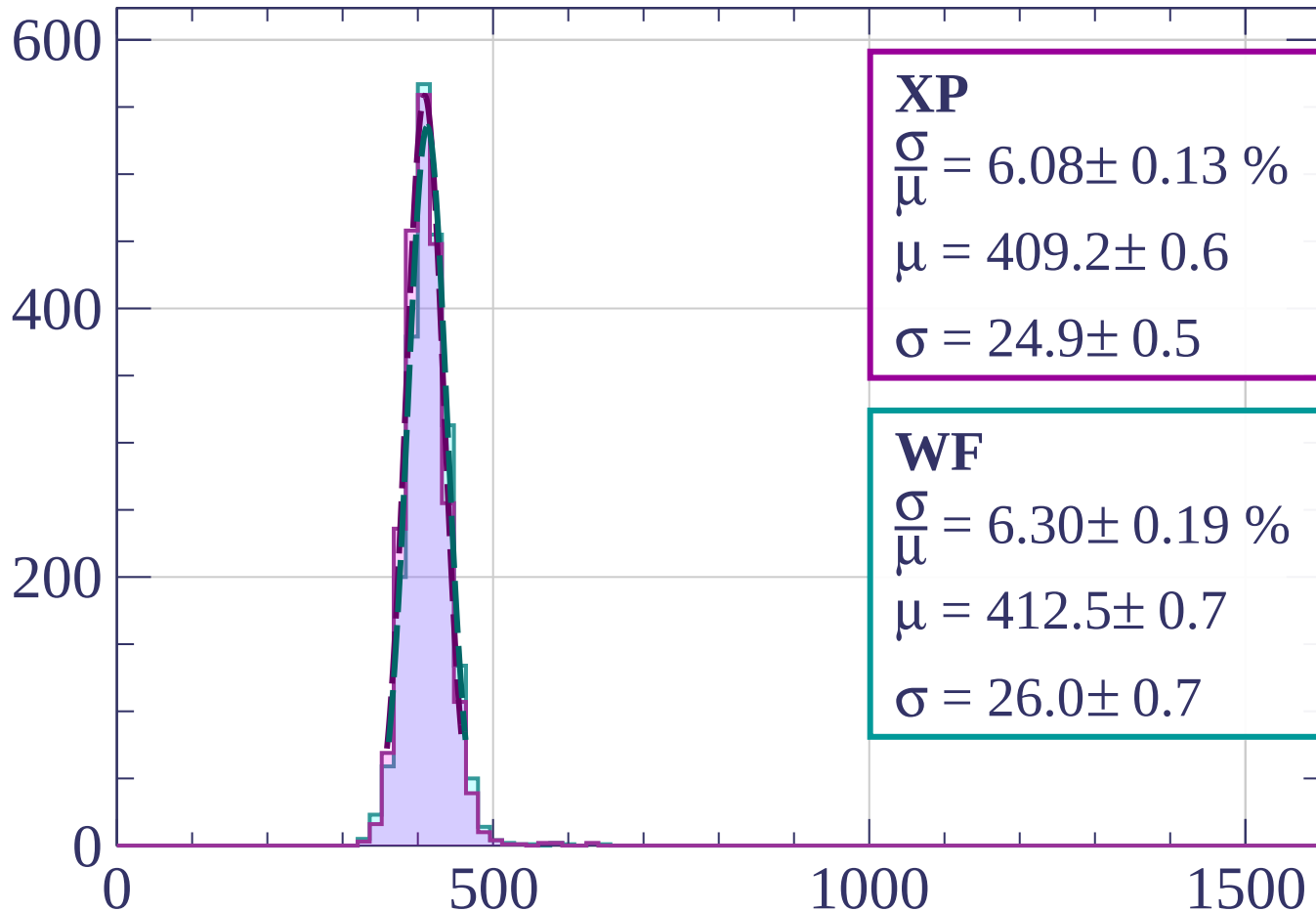
$$\mu = 422.0 \pm 0.7$$

$$\sigma = 27.7 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $320 < p < 400$

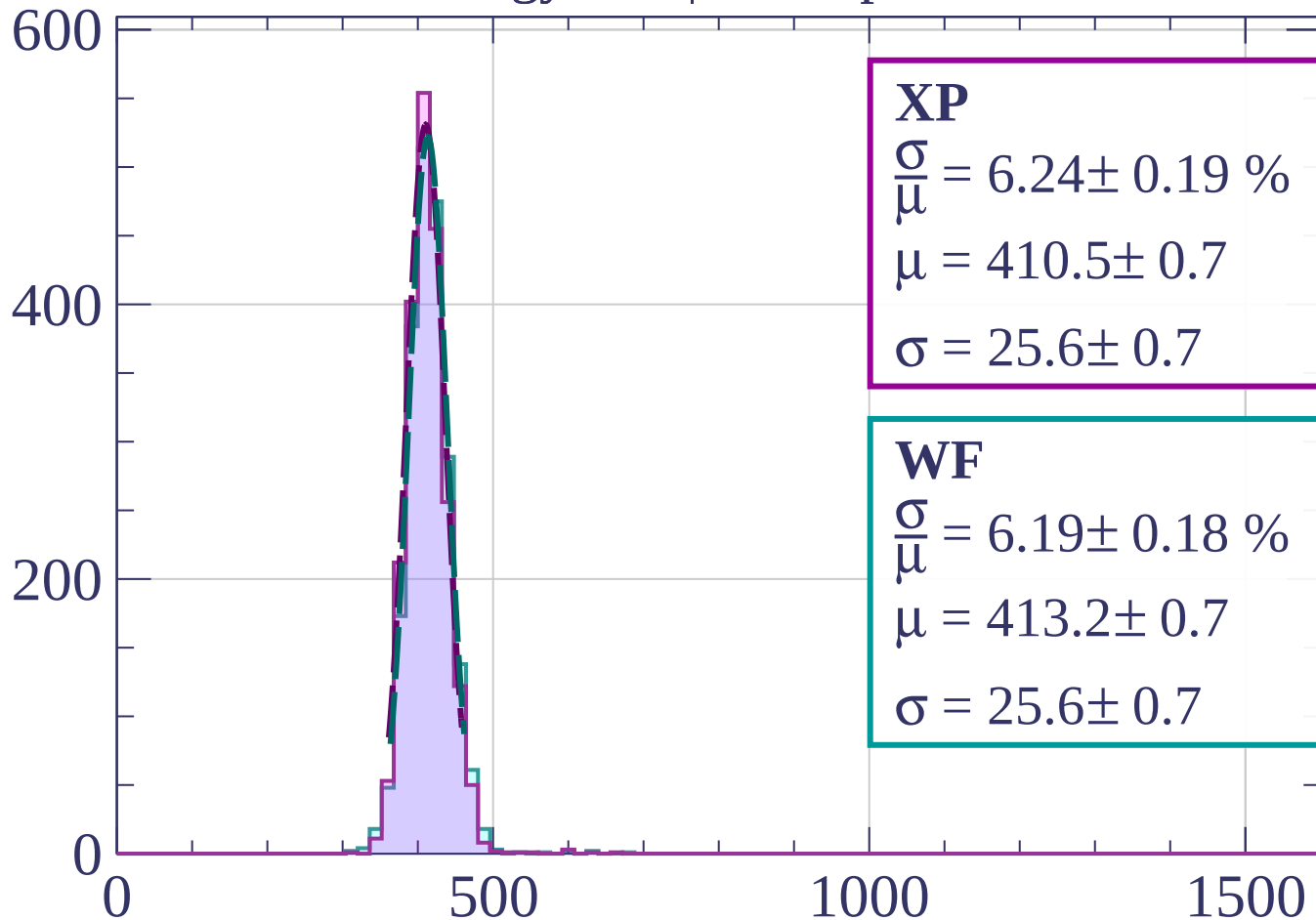
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $400 < p < 480$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.24 \pm 0.19 \%$$

$$\mu = 410.5 \pm 0.7$$

$$\sigma = 25.6 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 6.19 \pm 0.18 \%$$

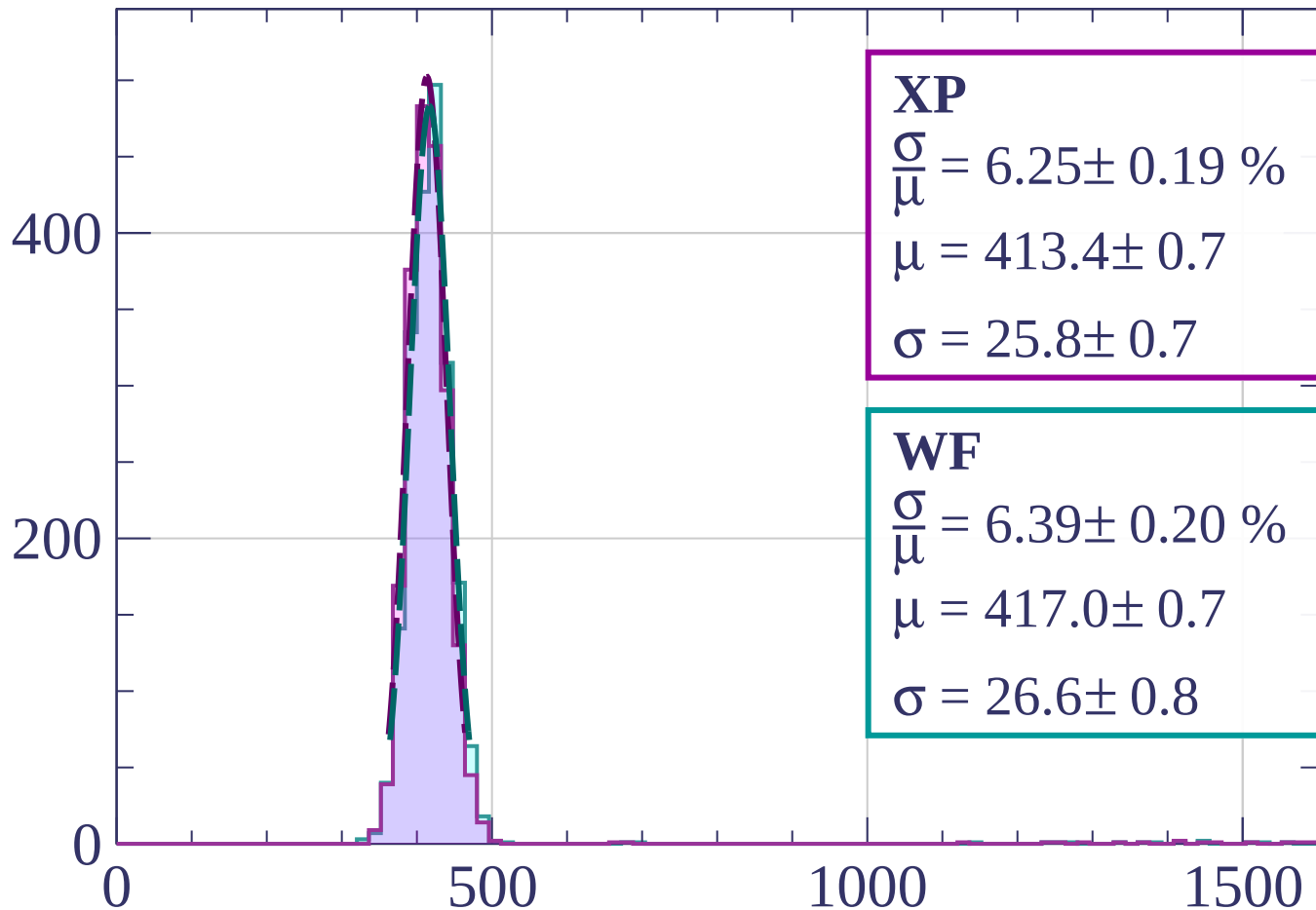
$$\mu = 413.2 \pm 0.7$$

$$\sigma = 25.6 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $480 < p < 560$

Count

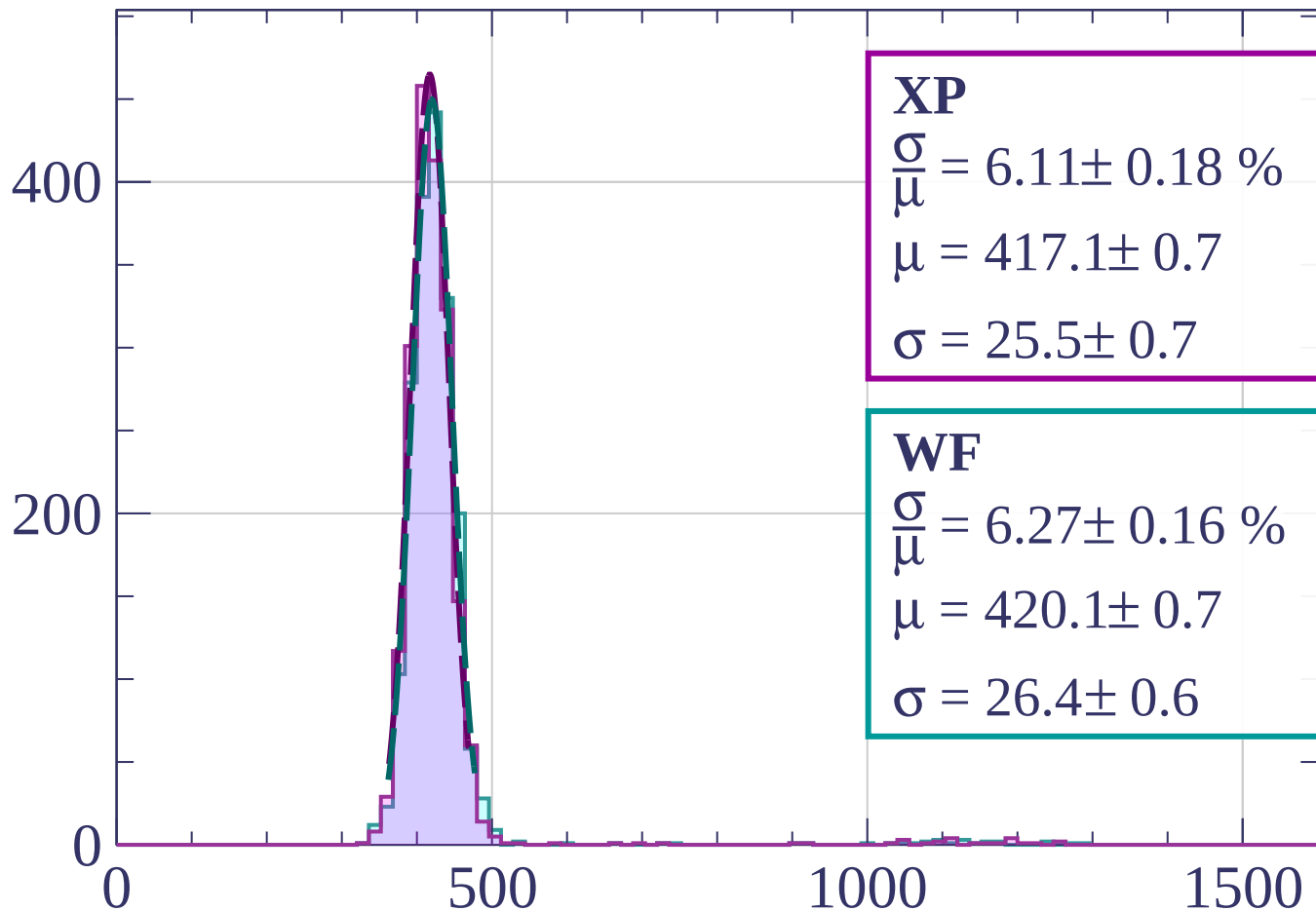


$dE/dx$  [ADC counts/cm]



Energy loss |  $560 < p < 640$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.11 \pm 0.18 \%$$

$$\mu = 417.1 \pm 0.7$$

$$\sigma = 25.5 \pm 0.7$$

**WF**

$$\frac{\sigma}{\mu} = 6.27 \pm 0.16 \%$$

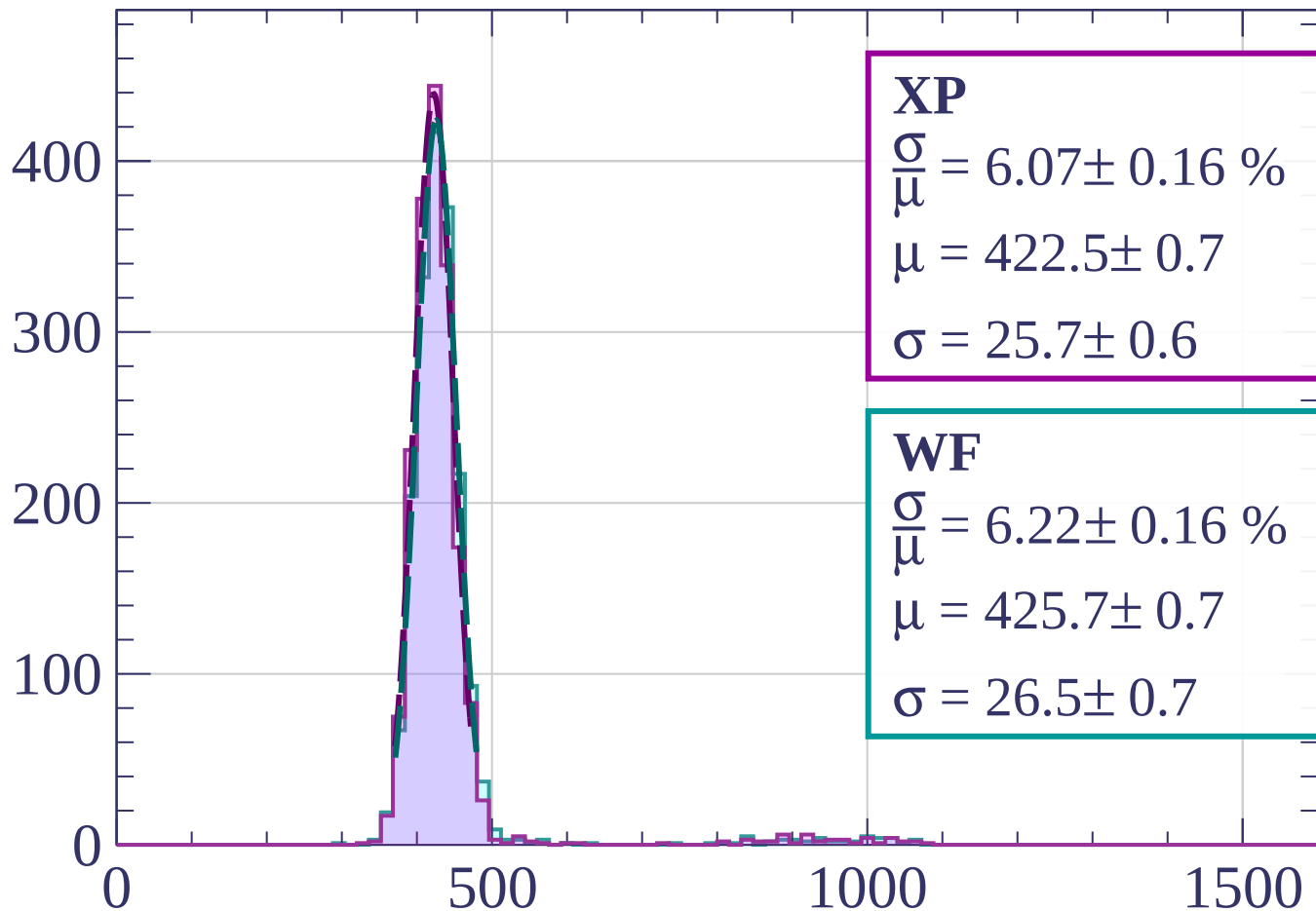
$$\mu = 420.1 \pm 0.7$$

$$\sigma = 26.4 \pm 0.6$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $640 < p < 720$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.07 \pm 0.16 \%$$

$$\mu = 422.5 \pm 0.7$$

$$\sigma = 25.7 \pm 0.6$$

**WF**

$$\frac{\sigma}{\mu} = 6.22 \pm 0.16 \%$$

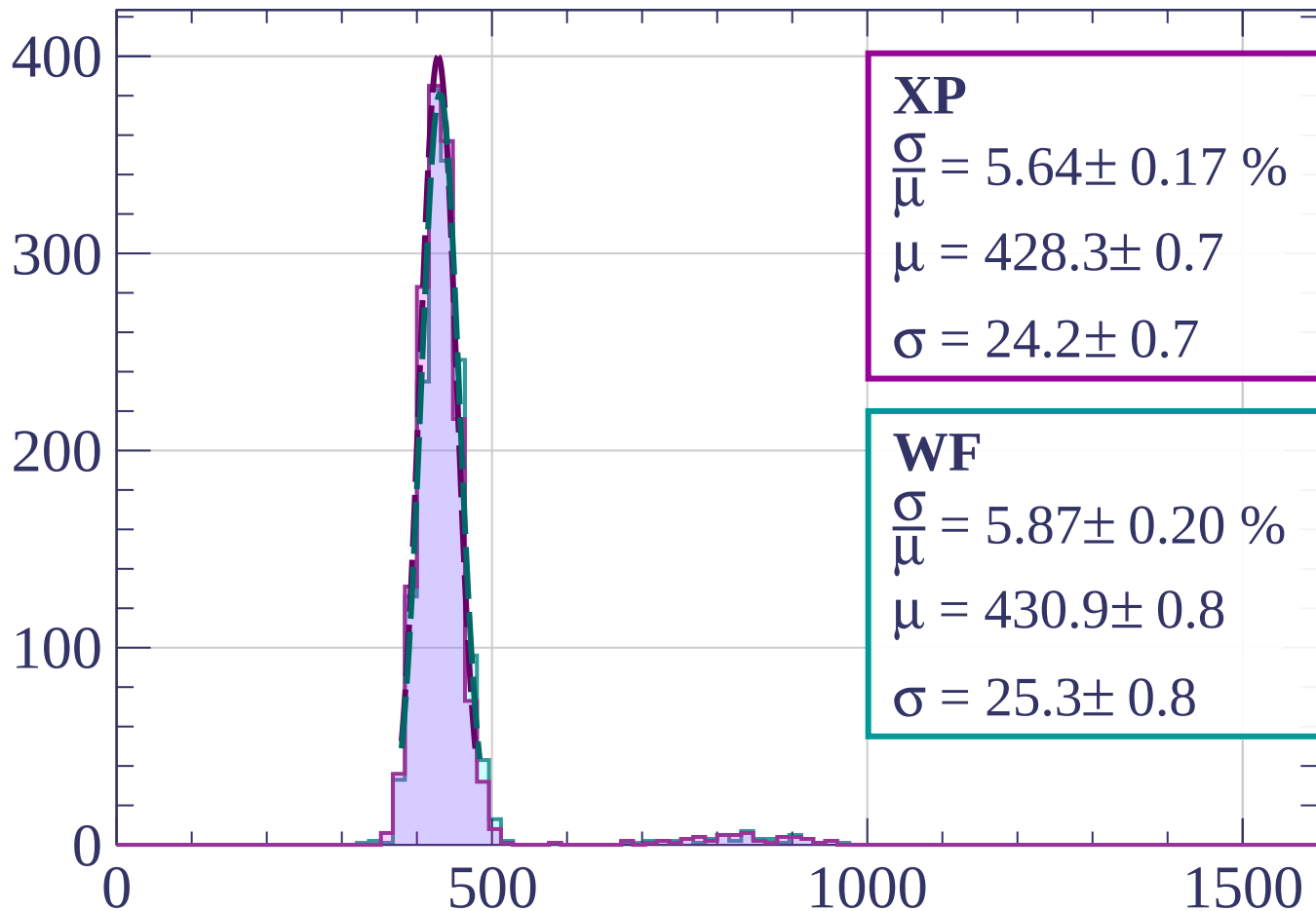
$$\mu = 425.7 \pm 0.7$$

$$\sigma = 26.5 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $720 < p < 800$

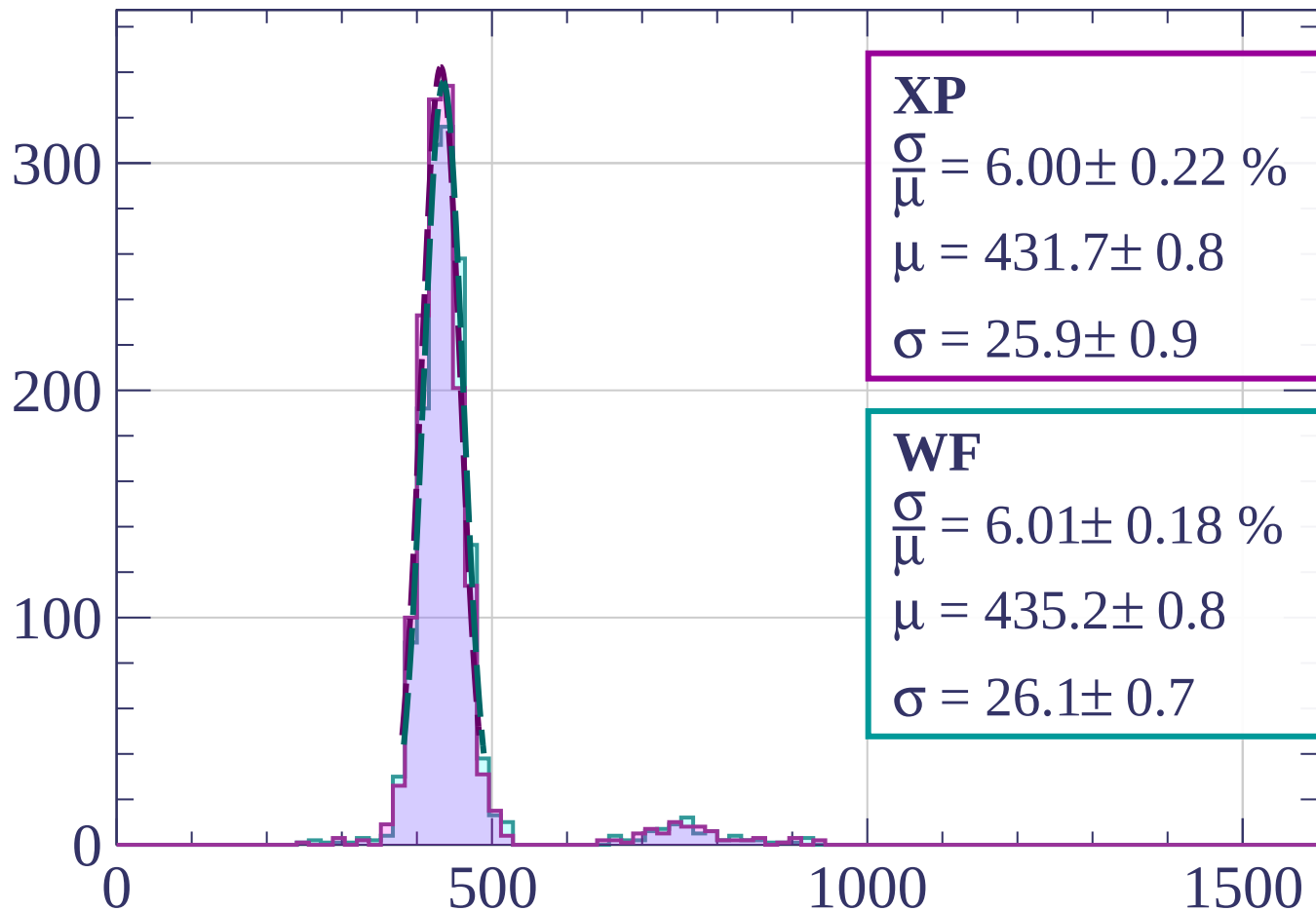
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $800 < p < 880$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.00 \pm 0.22 \%$$

$$\mu = 431.7 \pm 0.8$$

$$\sigma = 25.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.01 \pm 0.18 \%$$

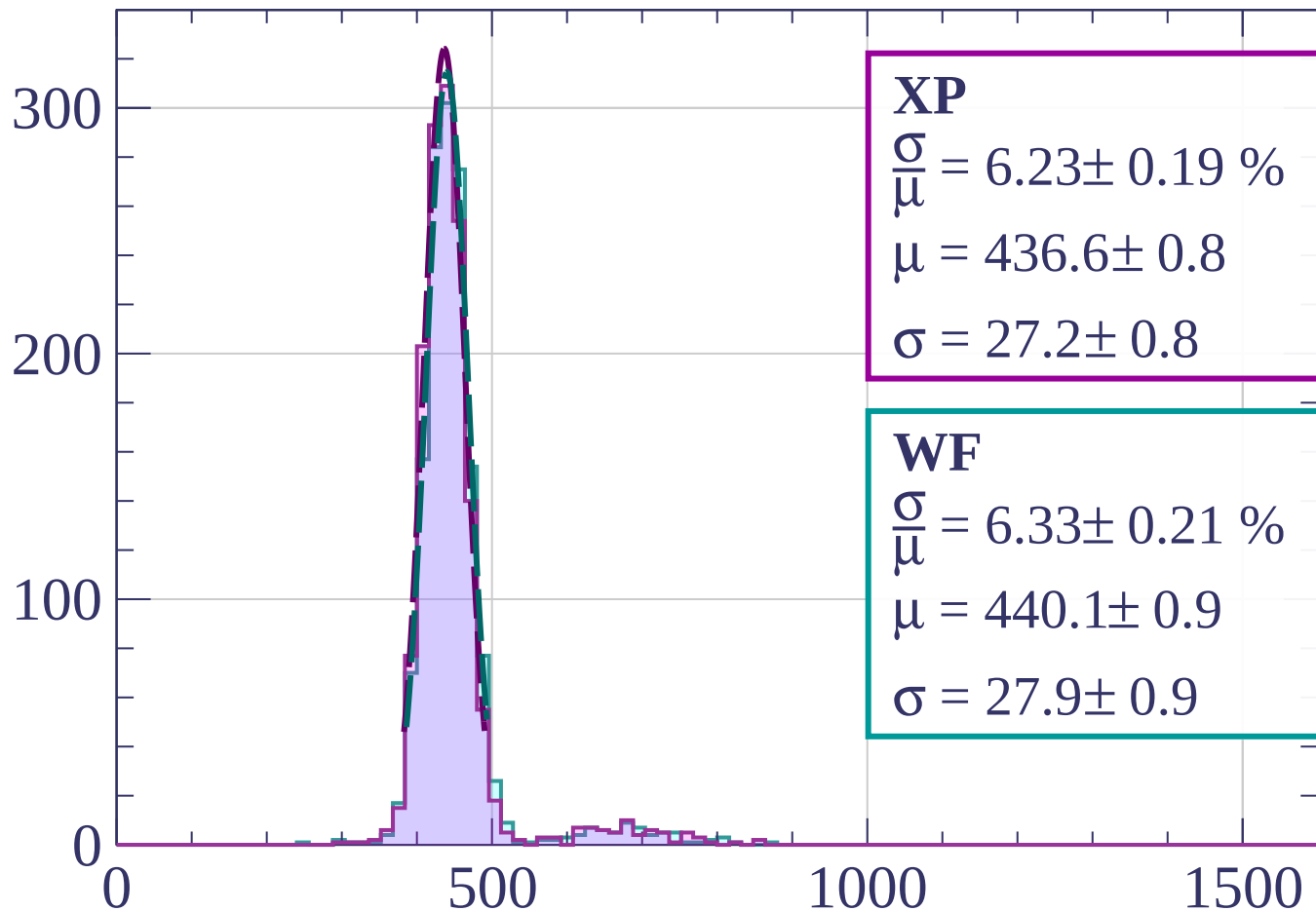
$$\mu = 435.2 \pm 0.8$$

$$\sigma = 26.1 \pm 0.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $880 < p < 960$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.23 \pm 0.19 \%$$

$$\mu = 436.6 \pm 0.8$$

$$\sigma = 27.2 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 6.33 \pm 0.21 \%$$

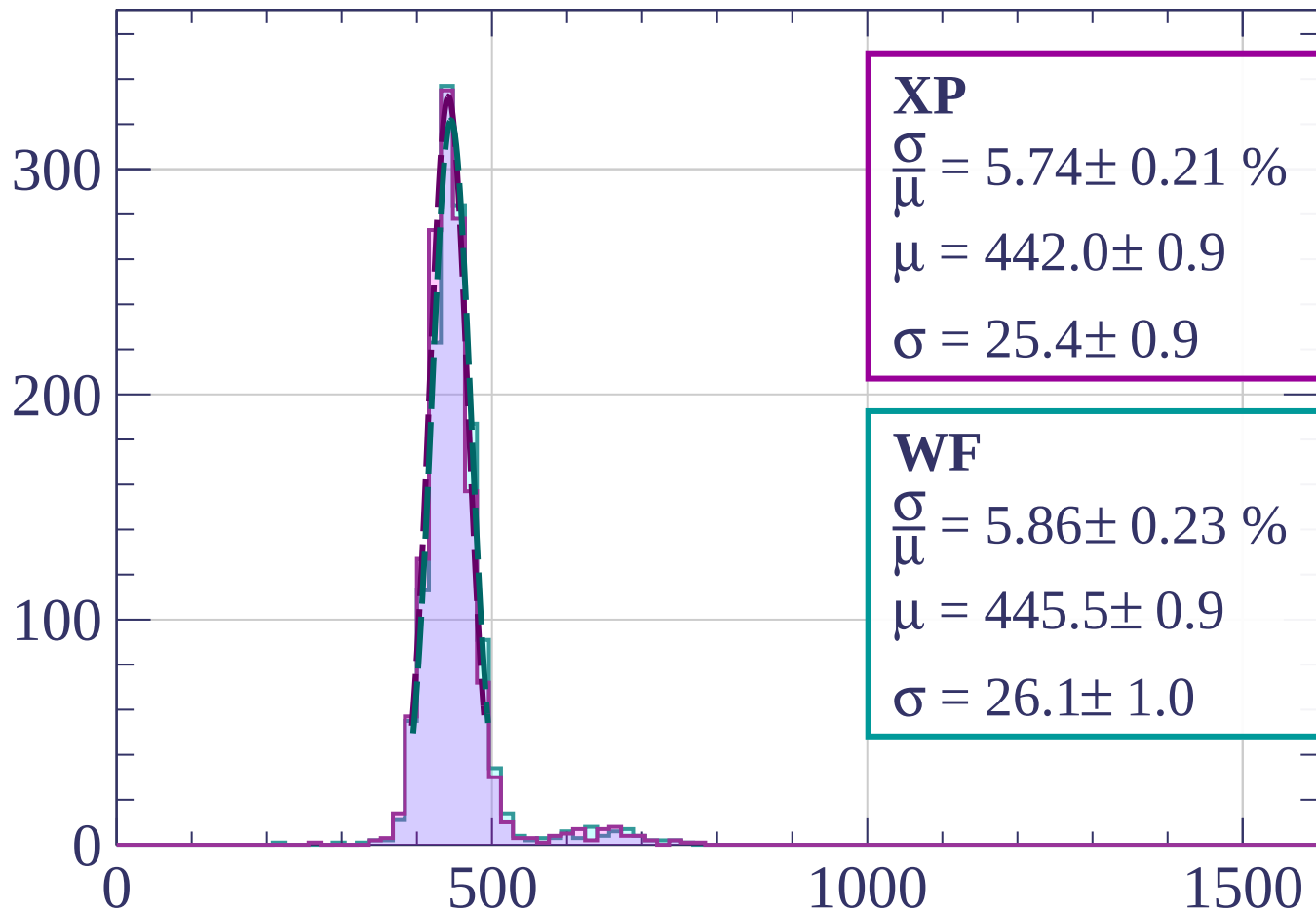
$$\mu = 440.1 \pm 0.9$$

$$\sigma = 27.9 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $960 < p < 1040$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.74 \pm 0.21 \%$$

$$\mu = 442.0 \pm 0.9$$

$$\sigma = 25.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 5.86 \pm 0.23 \%$$

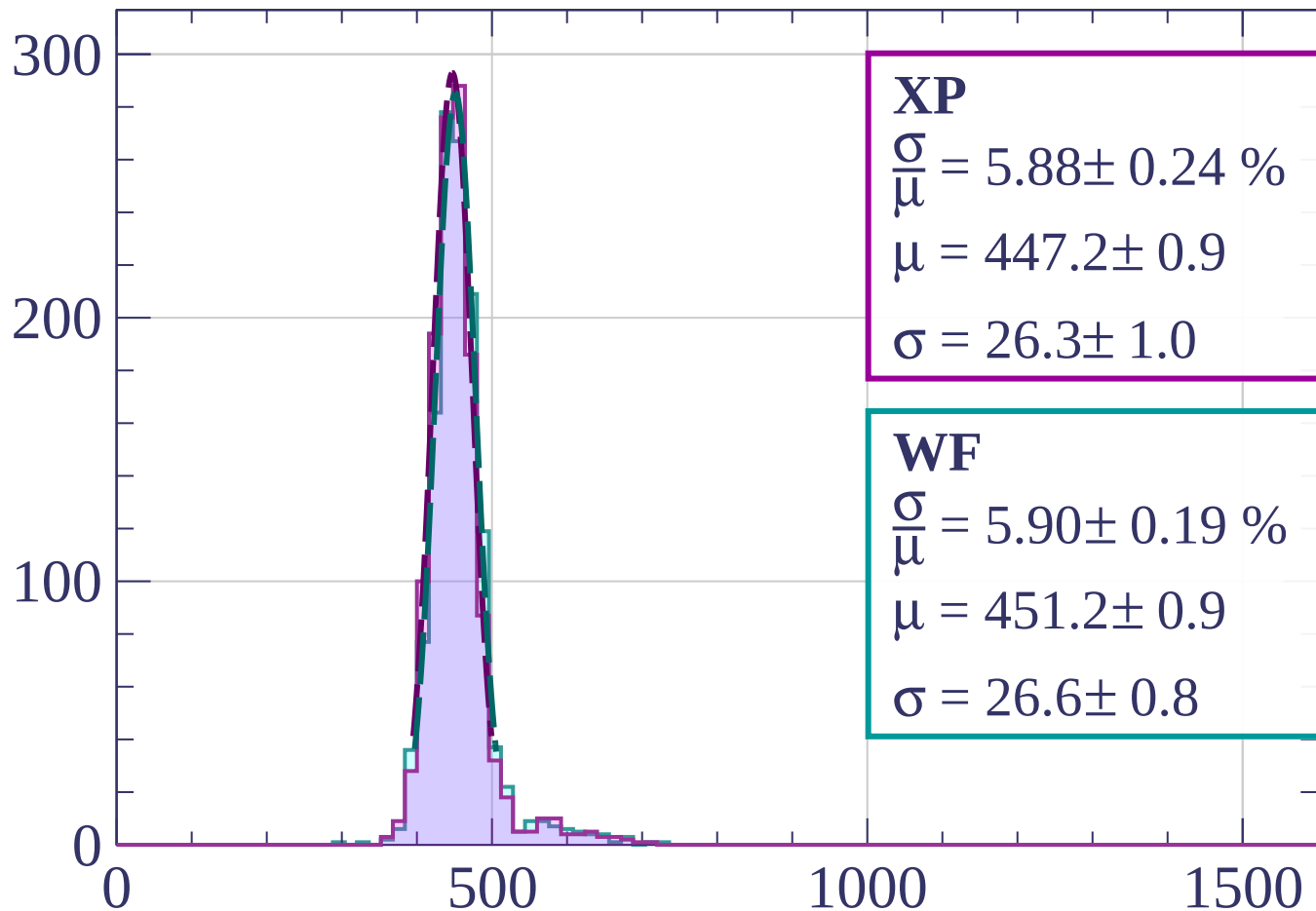
$$\mu = 445.5 \pm 0.9$$

$$\sigma = 26.1 \pm 1.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1040 < p < 1120$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.88 \pm 0.24 \%$$

$$\mu = 447.2 \pm 0.9$$

$$\sigma = 26.3 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 5.90 \pm 0.19 \%$$

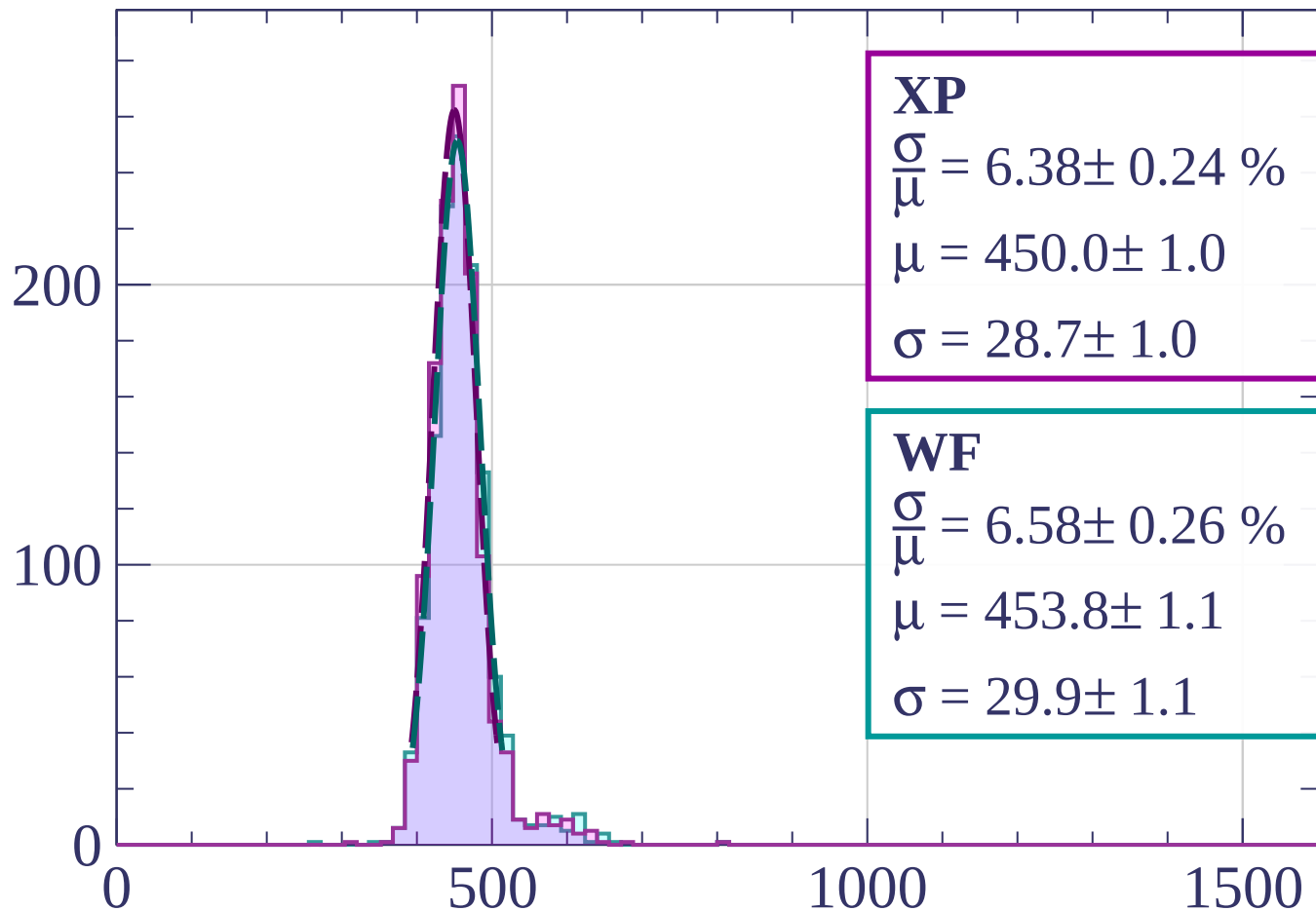
$$\mu = 451.2 \pm 0.9$$

$$\sigma = 26.6 \pm 0.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1120 < p < 1200$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.38 \pm 0.24 \%$$

$$\mu = 450.0 \pm 1.0$$

$$\sigma = 28.7 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 6.58 \pm 0.26 \%$$

$$\mu = 453.8 \pm 1.1$$

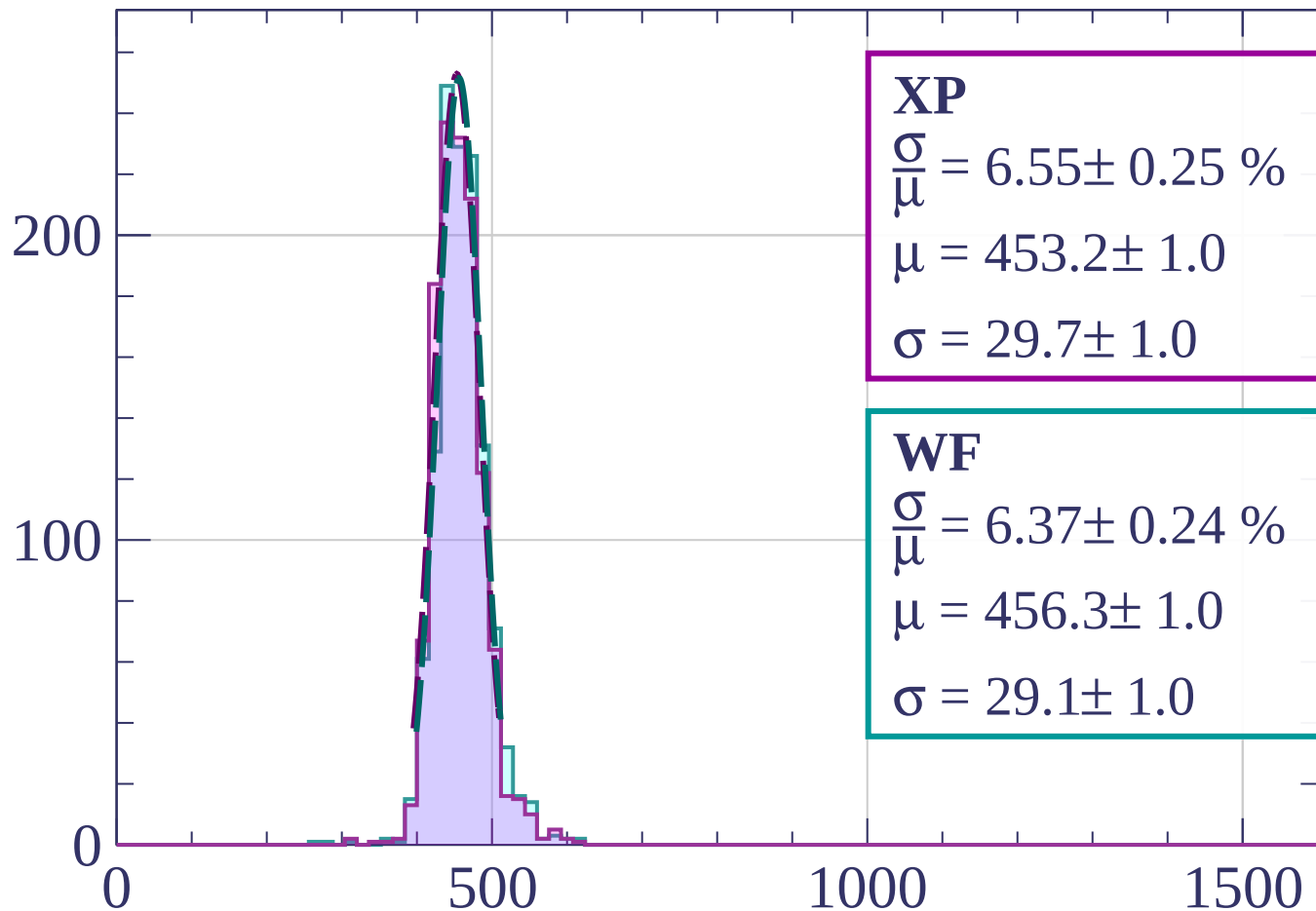
$$\sigma = 29.9 \pm 1.1$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $1200 < p < 1280$

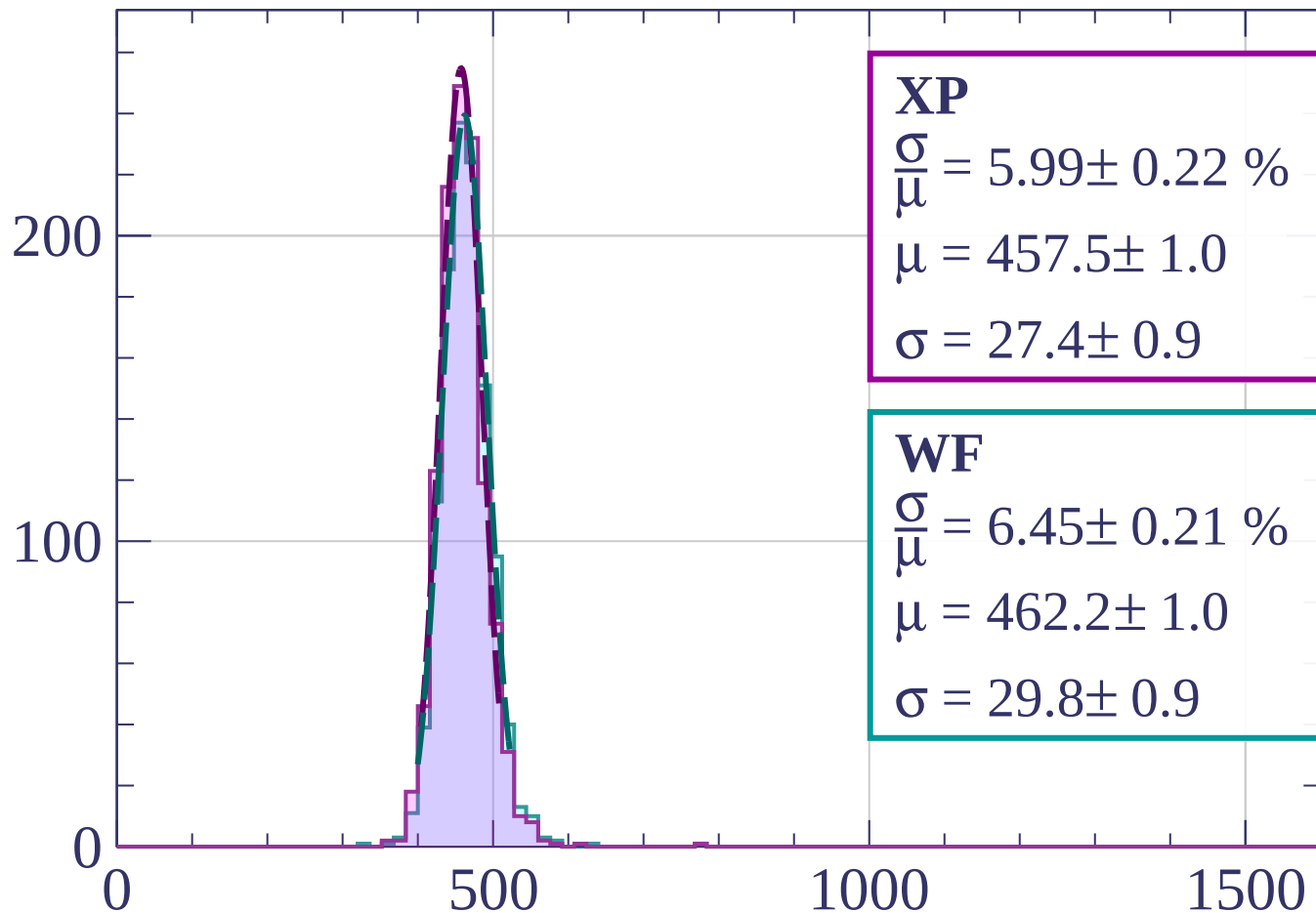
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $1280 < p < 1360$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.99 \pm 0.22 \%$$

$$\mu = 457.5 \pm 1.0$$

$$\sigma = 27.4 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.45 \pm 0.21 \%$$

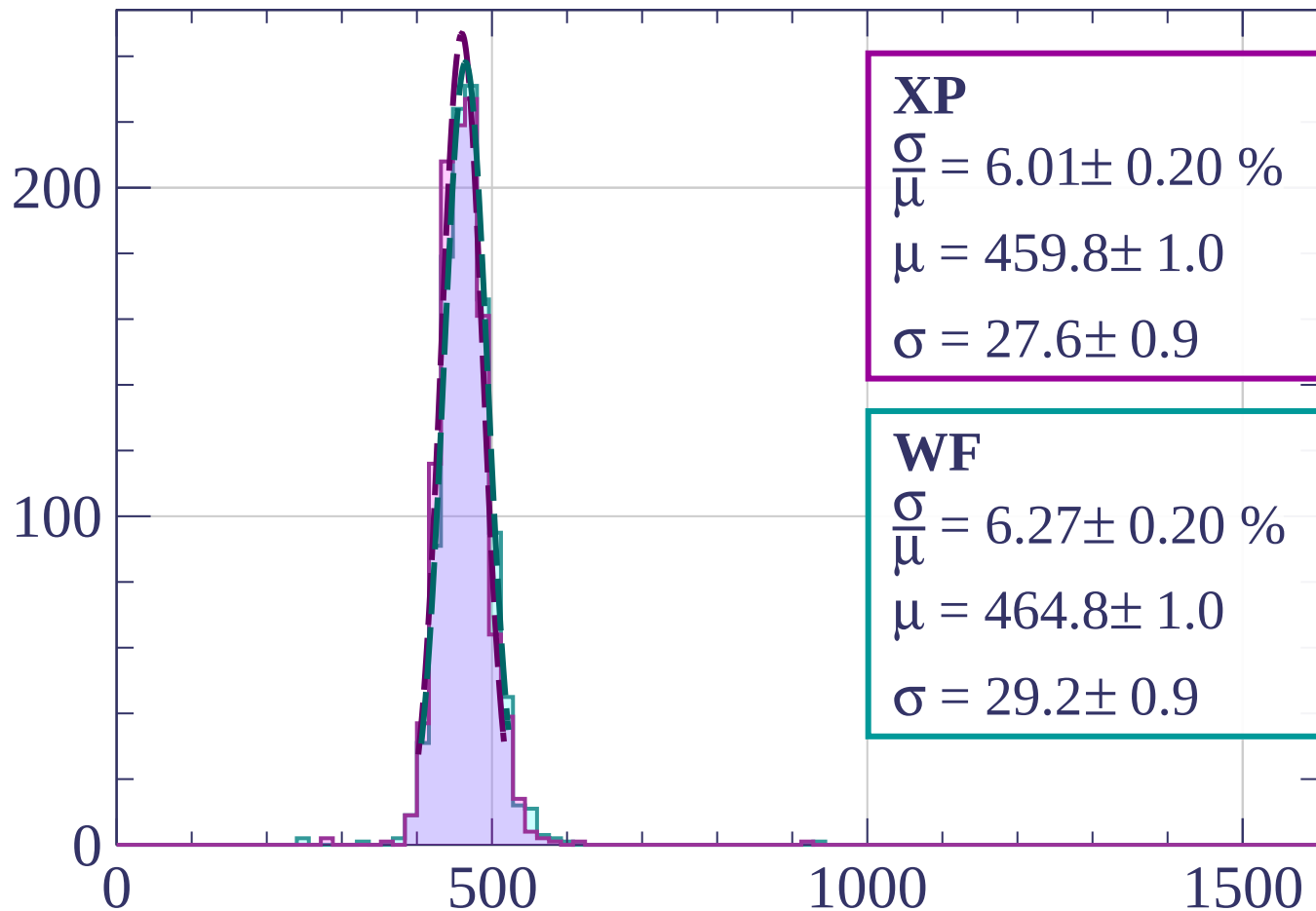
$$\mu = 462.2 \pm 1.0$$

$$\sigma = 29.8 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1360 < p < 1440$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.01 \pm 0.20 \%$$

$$\mu = 459.8 \pm 1.0$$

$$\sigma = 27.6 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.27 \pm 0.20 \%$$

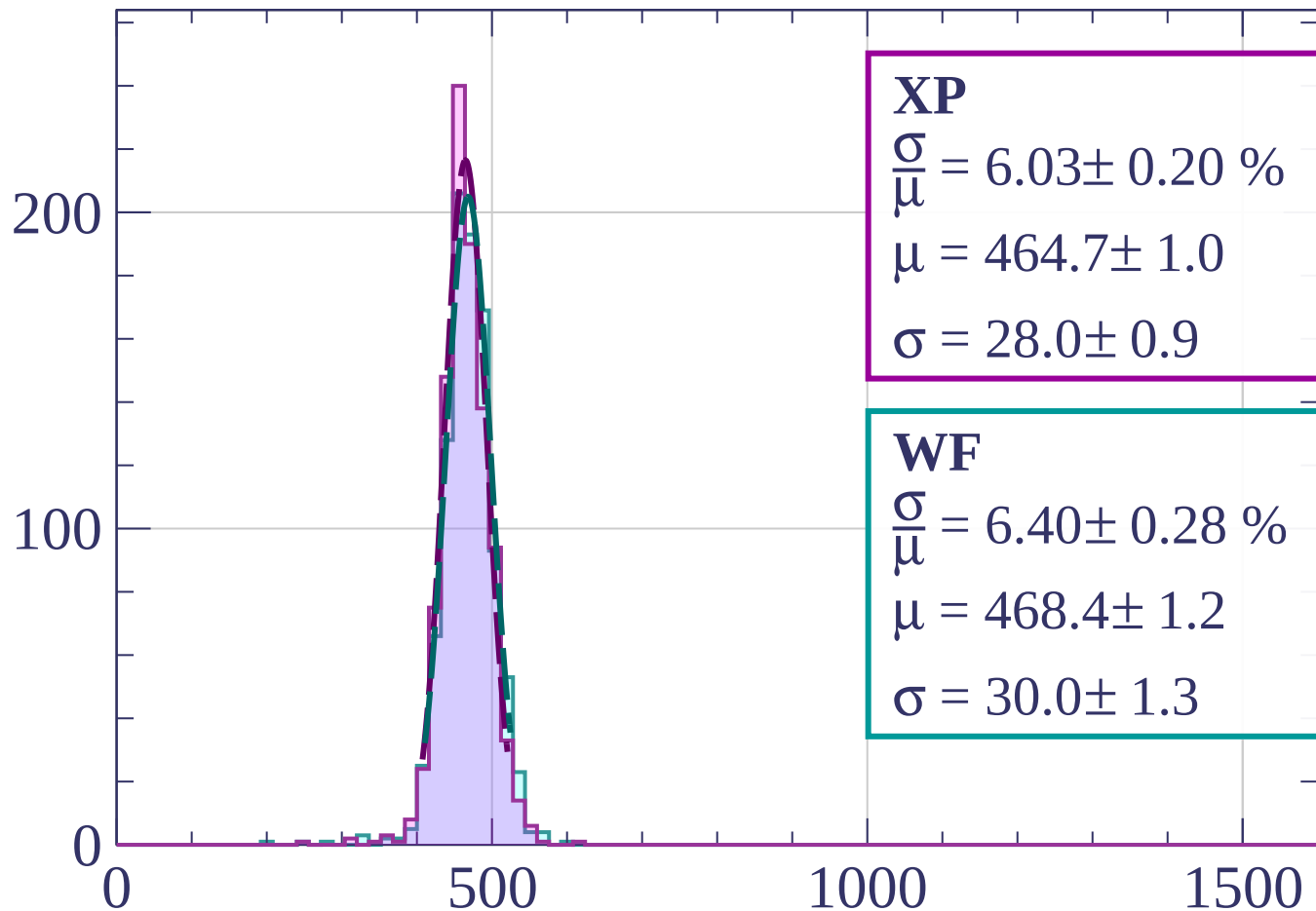
$$\mu = 464.8 \pm 1.0$$

$$\sigma = 29.2 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1440 < p < 1520$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.03 \pm 0.20 \%$$

$$\mu = 464.7 \pm 1.0$$

$$\sigma = 28.0 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.40 \pm 0.28 \%$$

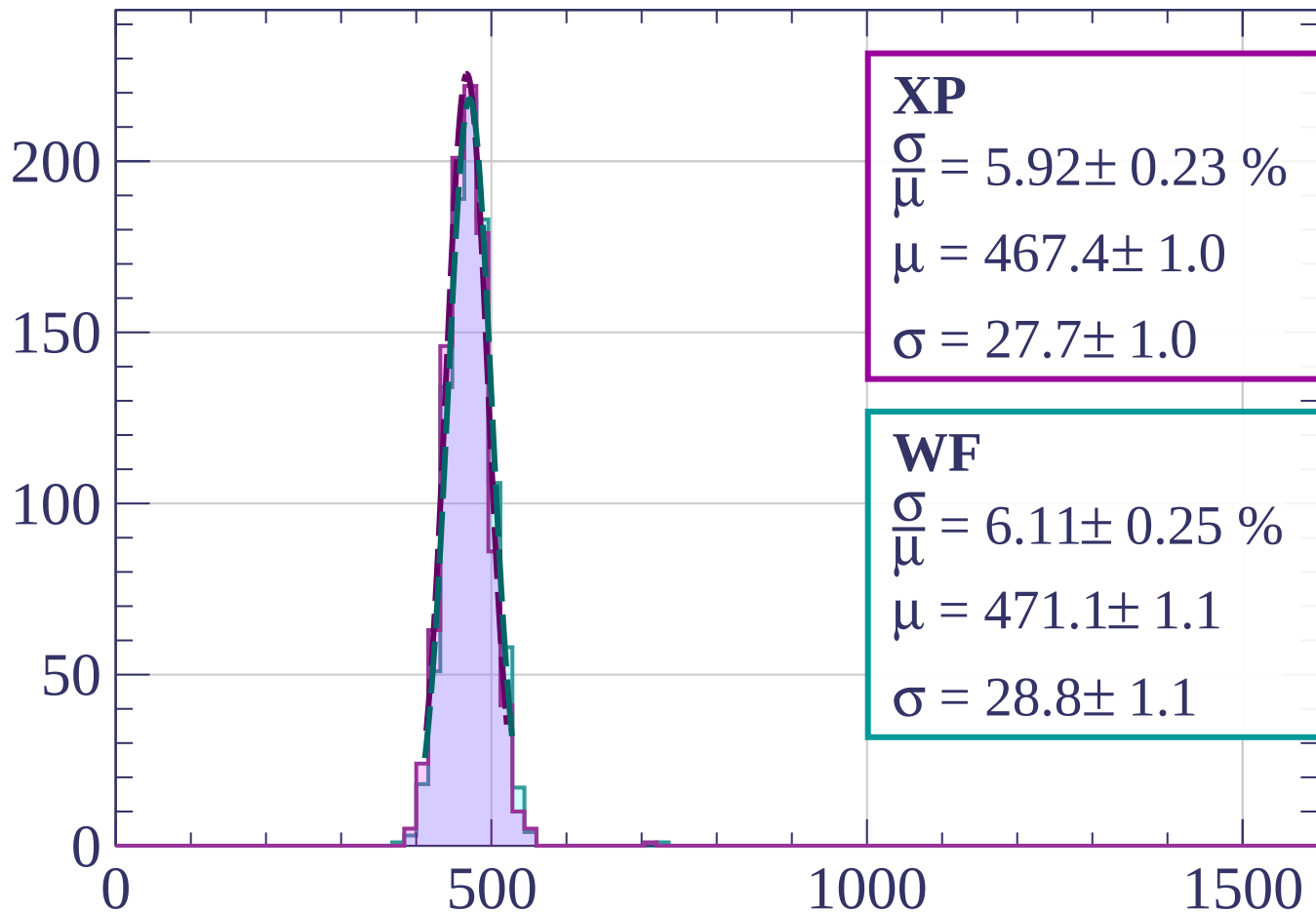
$$\mu = 468.4 \pm 1.2$$

$$\sigma = 30.0 \pm 1.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1520 < p < 1600$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.92 \pm 0.23 \%$$

$$\mu = 467.4 \pm 1.0$$

$$\sigma = 27.7 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 6.11 \pm 0.25 \%$$

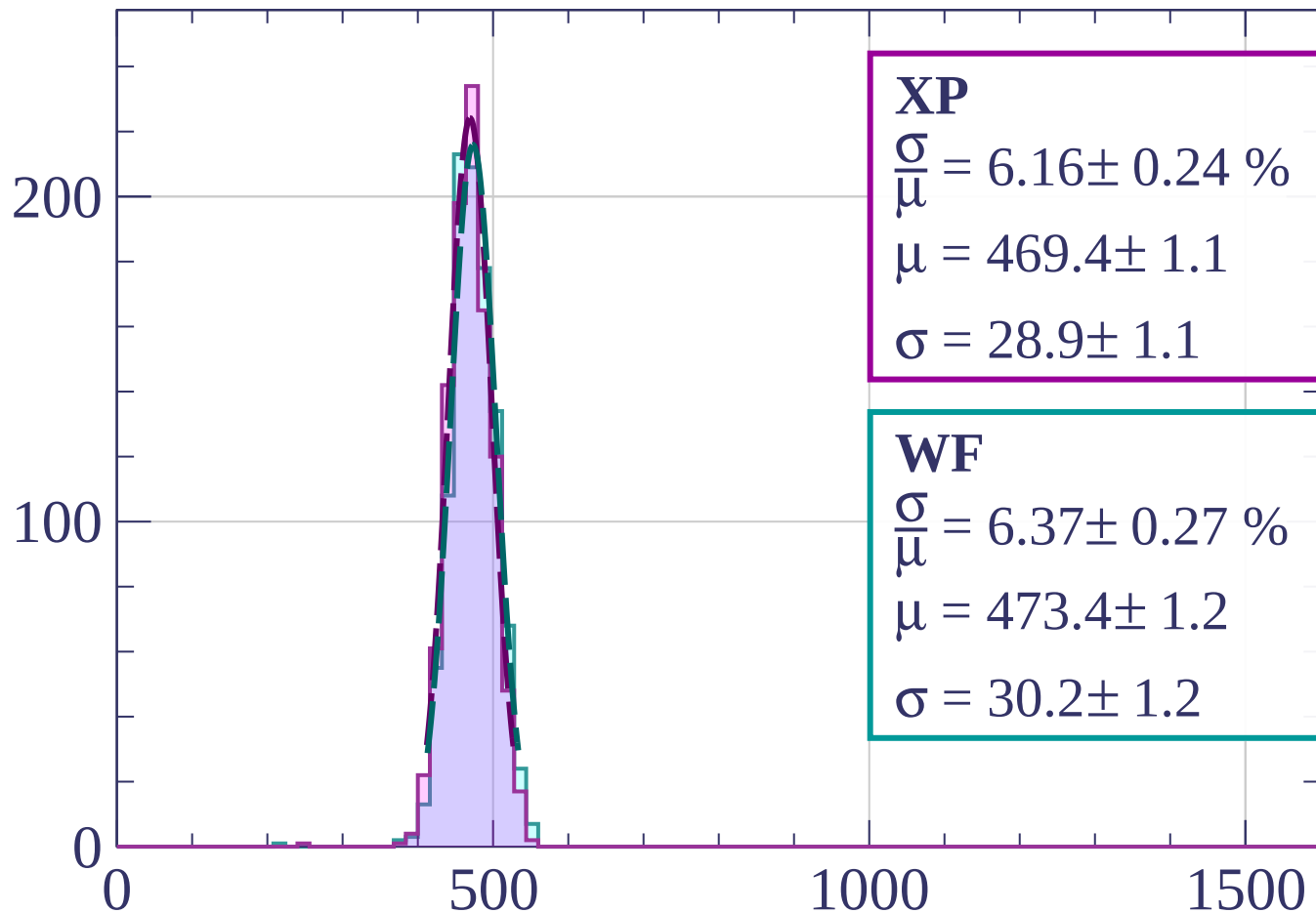
$$\mu = 471.1 \pm 1.1$$

$$\sigma = 28.8 \pm 1.1$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1600 < p < 1680$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.16 \pm 0.24 \%$$

$$\mu = 469.4 \pm 1.1$$

$$\sigma = 28.9 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 6.37 \pm 0.27 \%$$

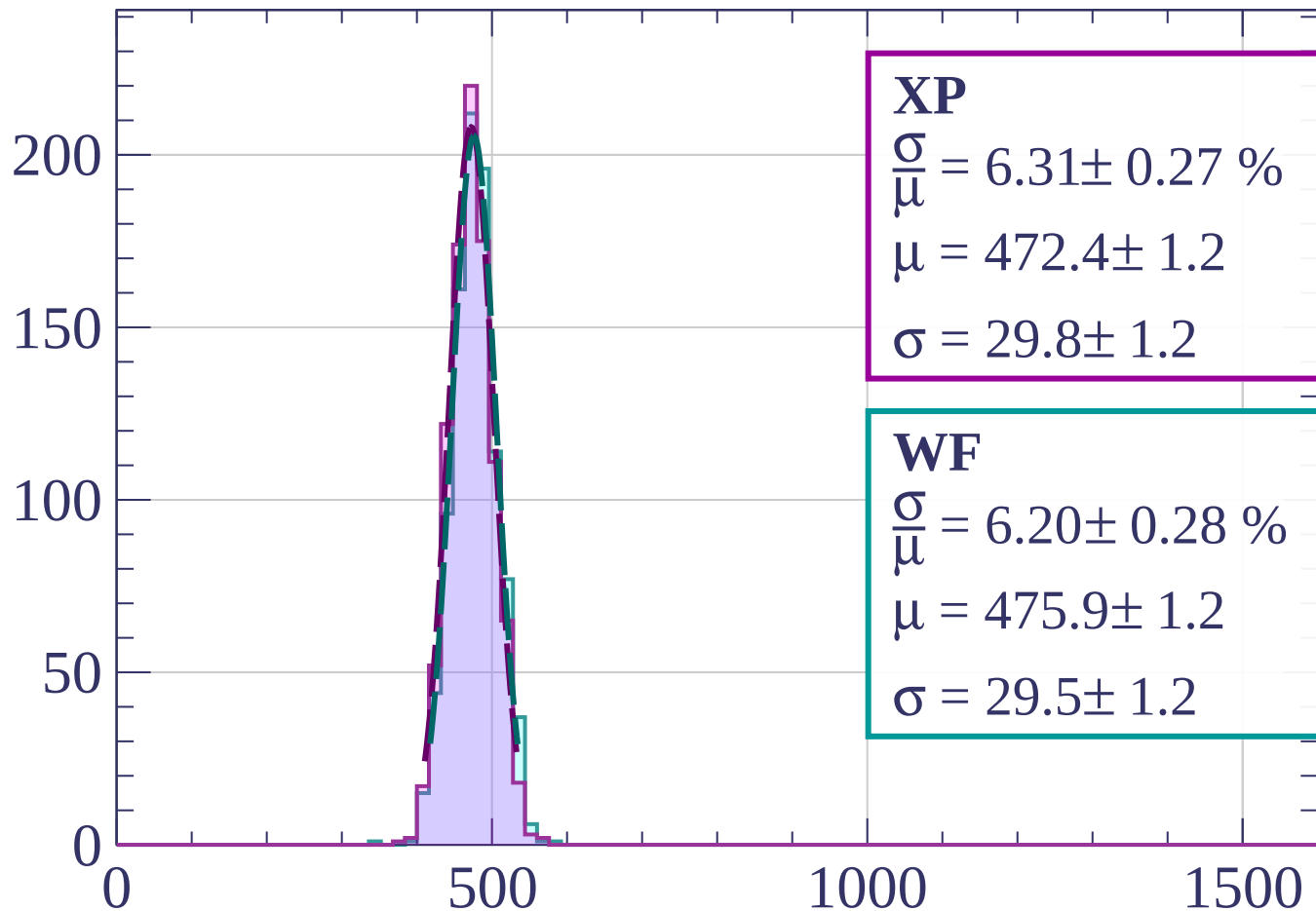
$$\mu = 473.4 \pm 1.2$$

$$\sigma = 30.2 \pm 1.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1680 < p < 1760$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.31 \pm 0.27 \%$$

$$\mu = 472.4 \pm 1.2$$

$$\sigma = 29.8 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.20 \pm 0.28 \%$$

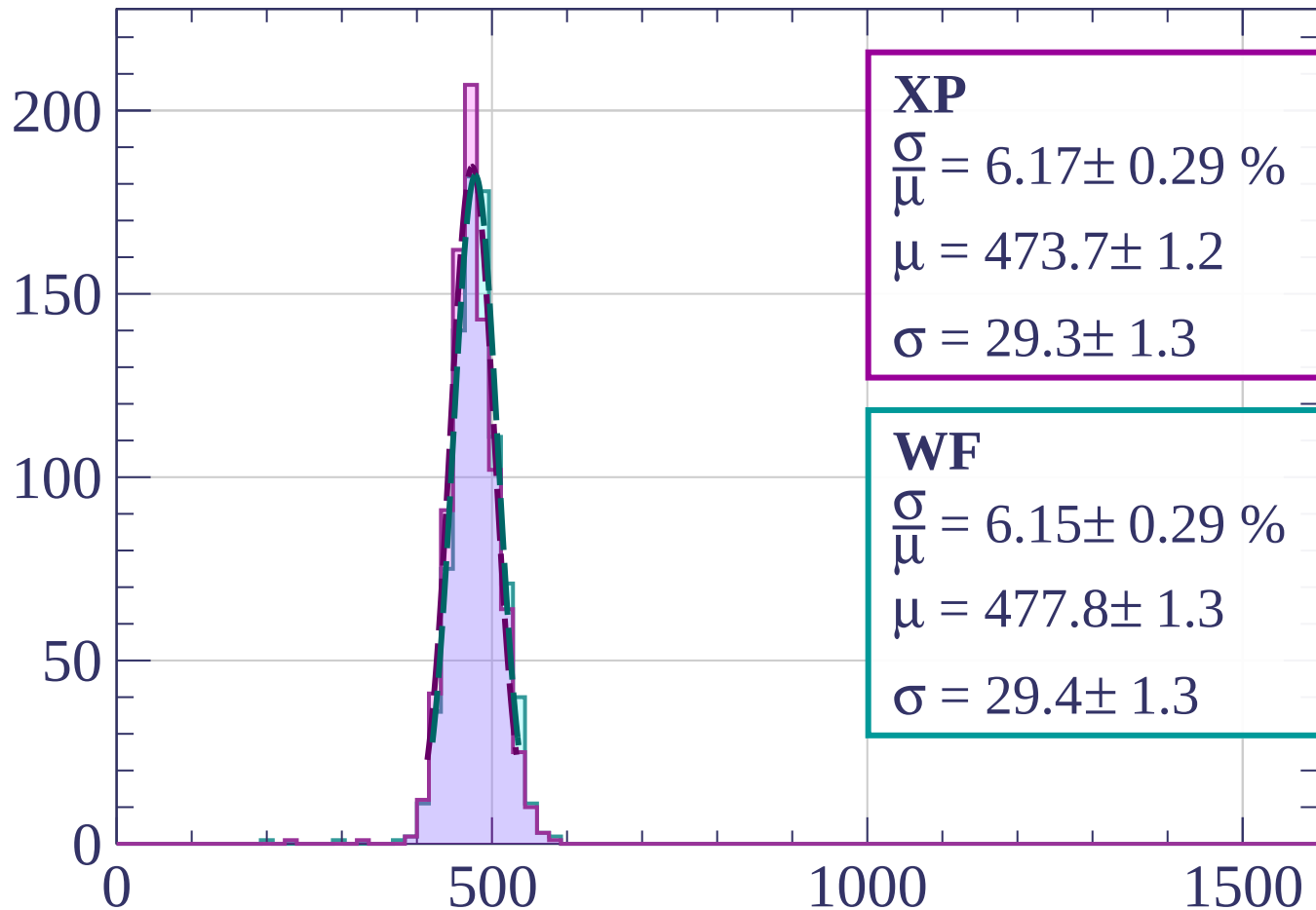
$$\mu = 475.9 \pm 1.2$$

$$\sigma = 29.5 \pm 1.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $1760 < p < 1840$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.17 \pm 0.29 \%$$

$$\mu = 473.7 \pm 1.2$$

$$\sigma = 29.3 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.15 \pm 0.29 \%$$

$$\mu = 477.8 \pm 1.3$$

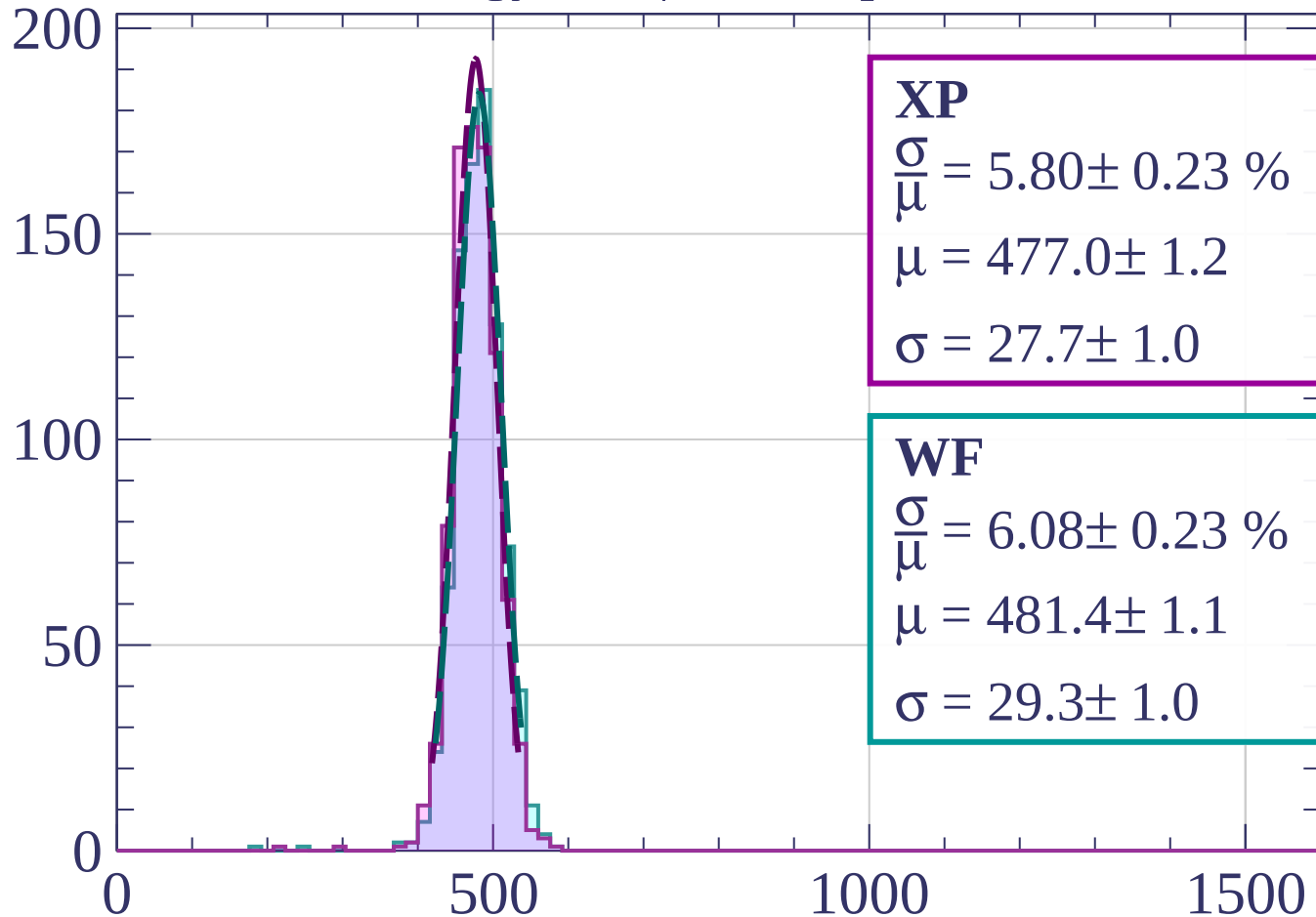
$$\sigma = 29.4 \pm 1.3$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $1840 < p < 1920$

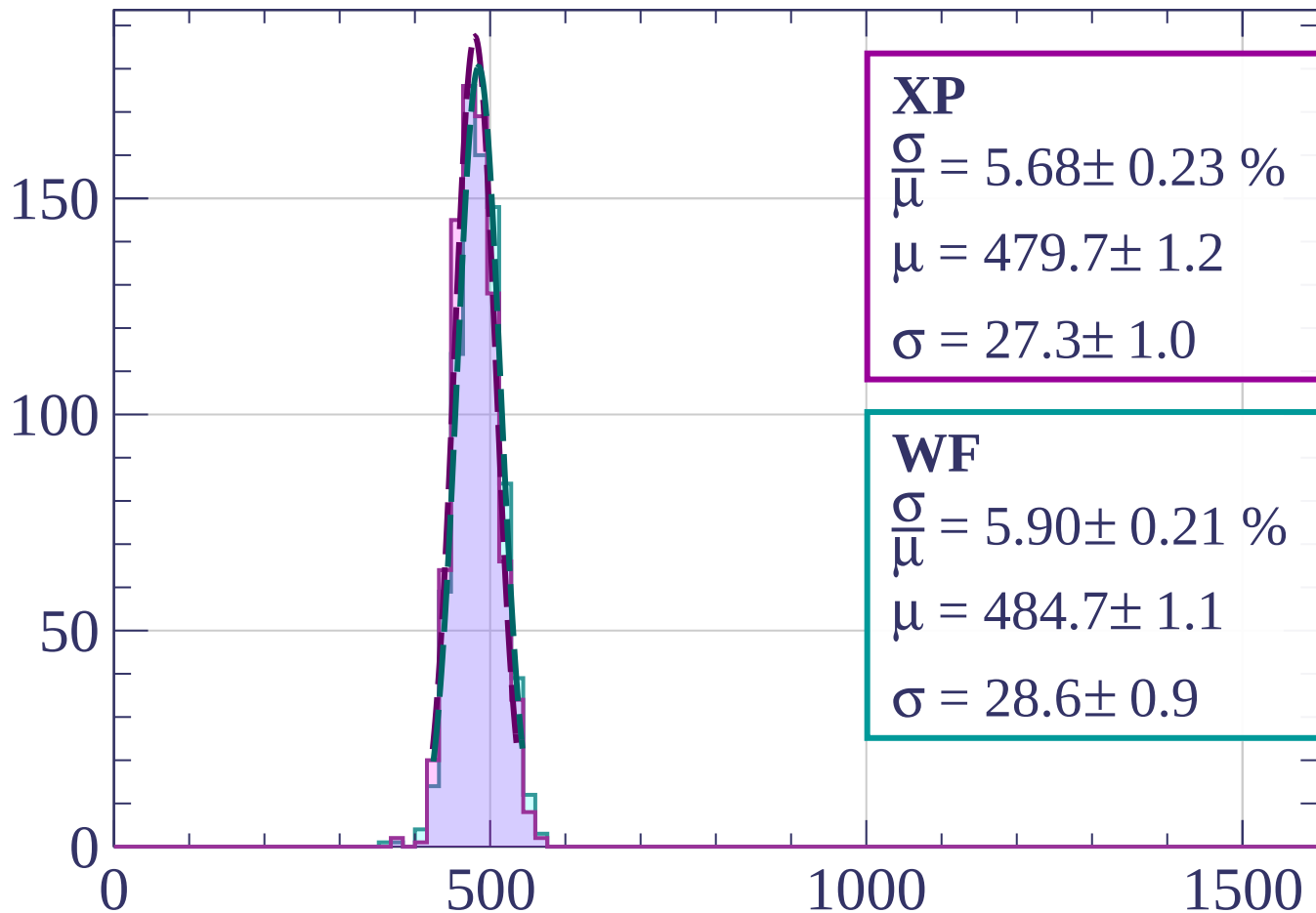
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $1920 < p < 2000$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.68 \pm 0.23 \%$$

$$\mu = 479.7 \pm 1.2$$

$$\sigma = 27.3 \pm 1.0$$

**WF**

$$\frac{\sigma}{\mu} = 5.90 \pm 0.21 \%$$

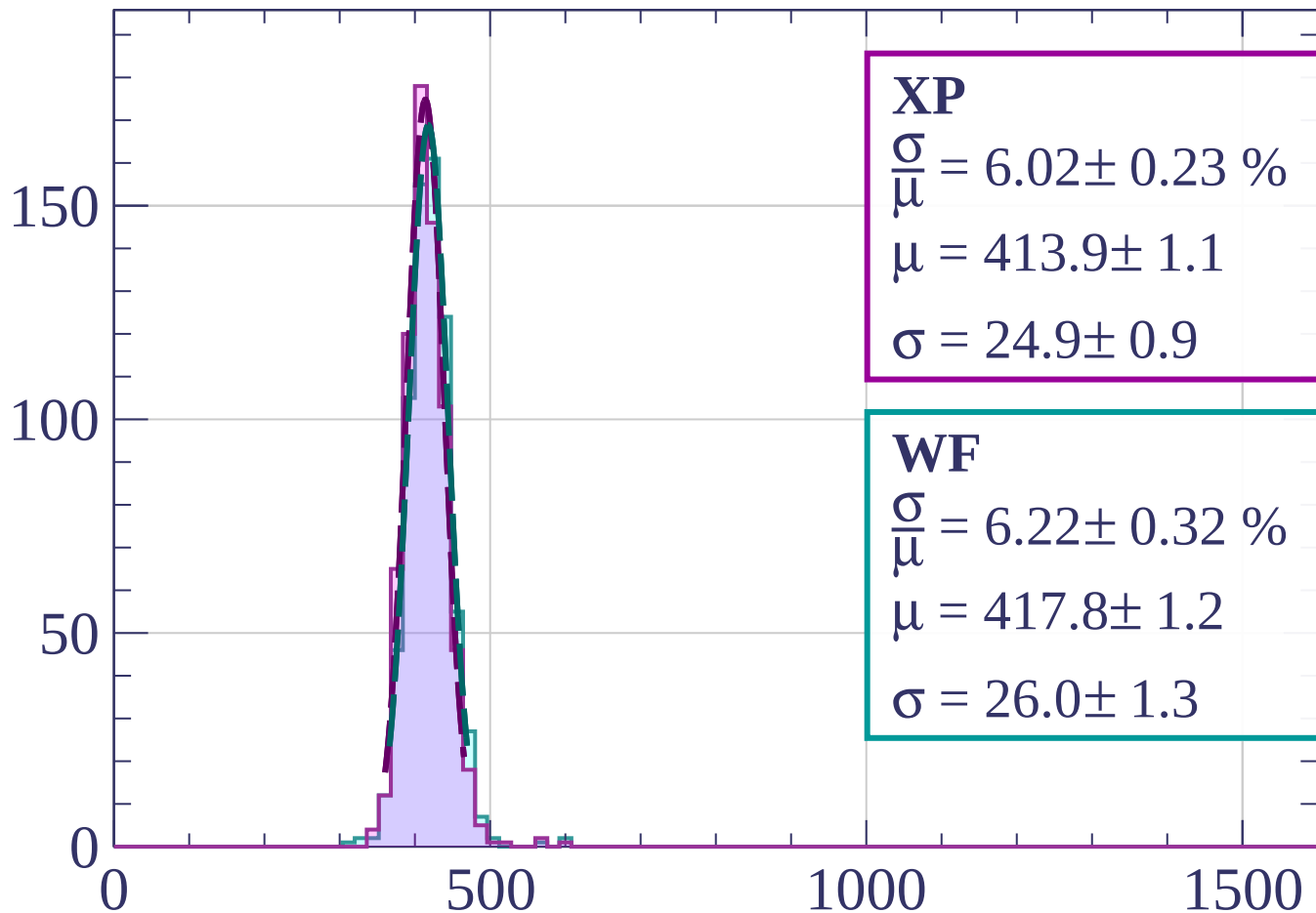
$$\mu = 484.7 \pm 1.1$$

$$\sigma = 28.6 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $0 < \phi < 3$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.02 \pm 0.23 \%$$

$$\mu = 413.9 \pm 1.1$$

$$\sigma = 24.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.22 \pm 0.32 \%$$

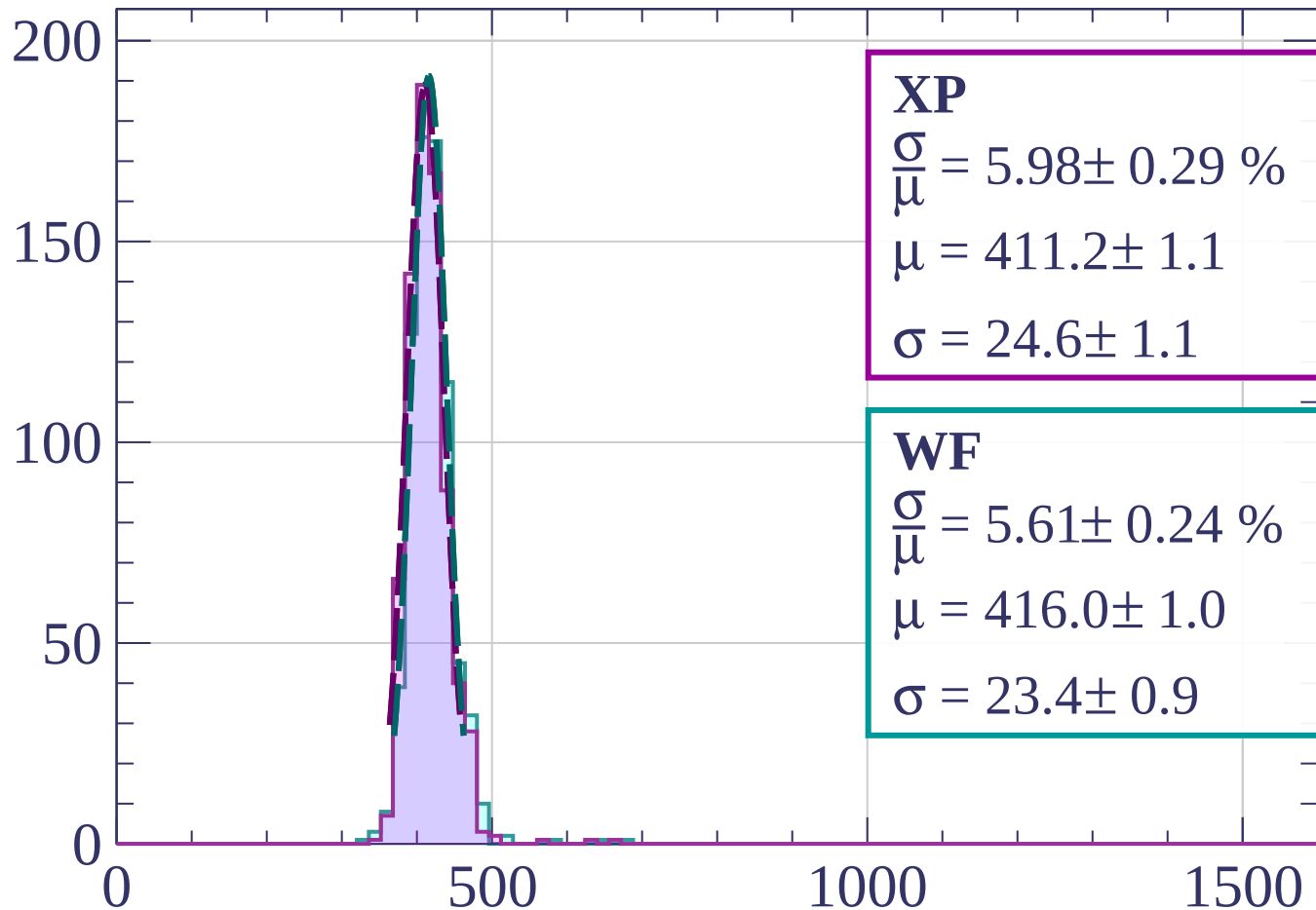
$$\mu = 417.8 \pm 1.2$$

$$\sigma = 26.0 \pm 1.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $3 < \phi < 6$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.98 \pm 0.29 \%$$

$$\mu = 411.2 \pm 1.1$$

$$\sigma = 24.6 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 5.61 \pm 0.24 \%$$

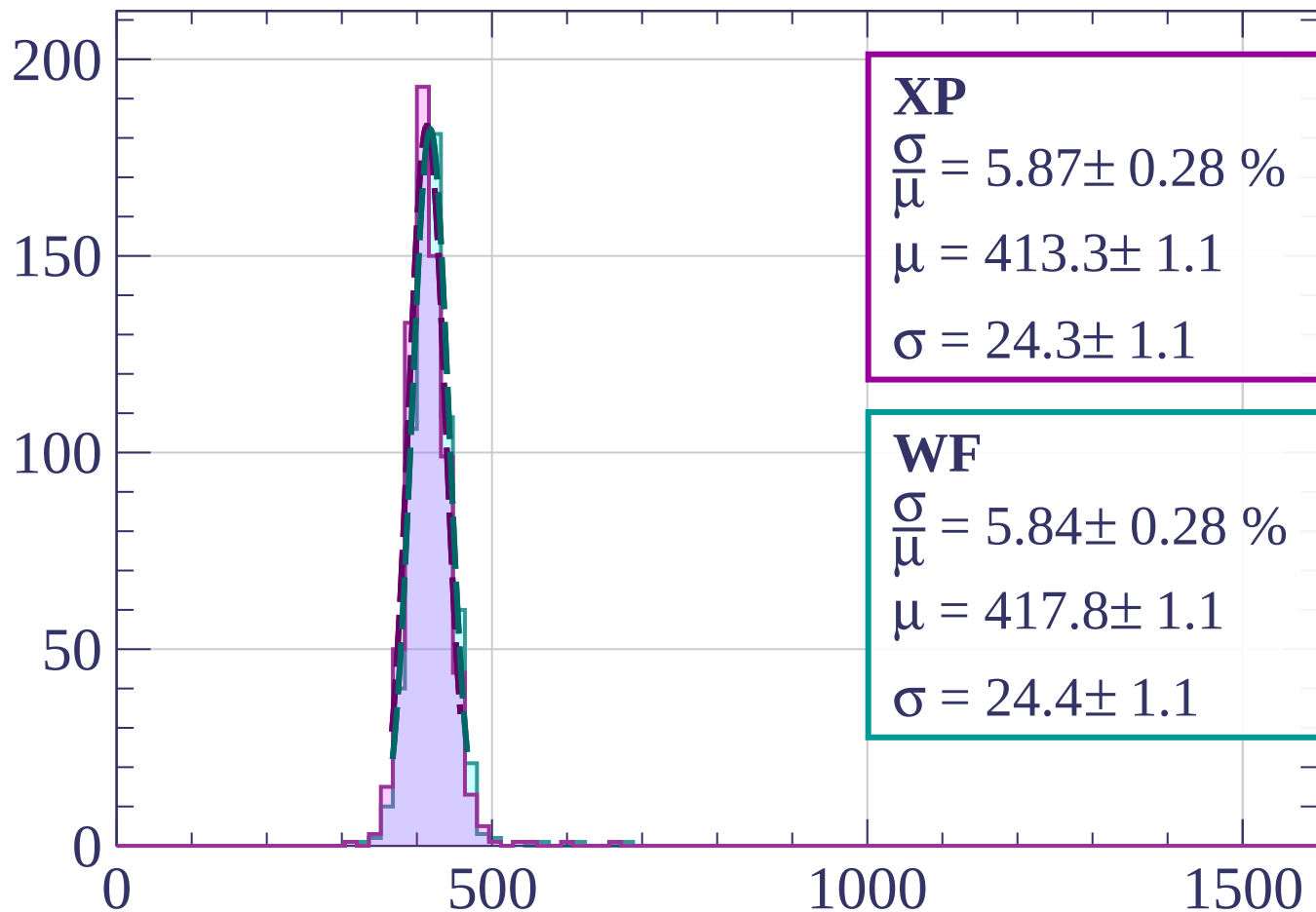
$$\mu = 416.0 \pm 1.0$$

$$\sigma = 23.4 \pm 0.9$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $6 < \phi < 9$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.87 \pm 0.28 \%$$

$$\mu = 413.3 \pm 1.1$$

$$\sigma = 24.3 \pm 1.1$$

**WF**

$$\frac{\sigma}{\mu} = 5.84 \pm 0.28 \%$$

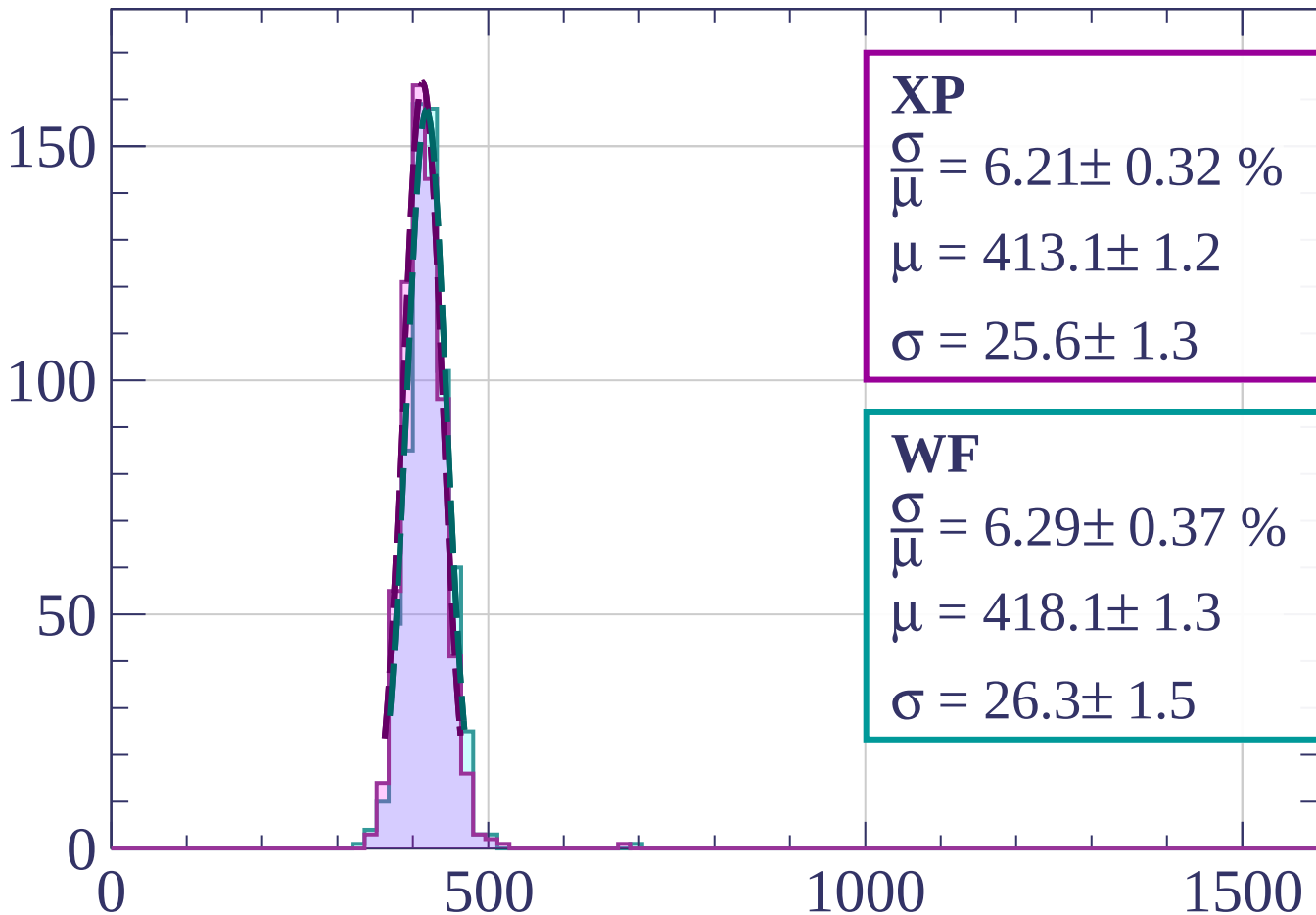
$$\mu = 417.8 \pm 1.1$$

$$\sigma = 24.4 \pm 1.1$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $9 < \phi < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.21 \pm 0.32 \%$$

$$\mu = 413.1 \pm 1.2$$

$$\sigma = 25.6 \pm 1.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.29 \pm 0.37 \%$$

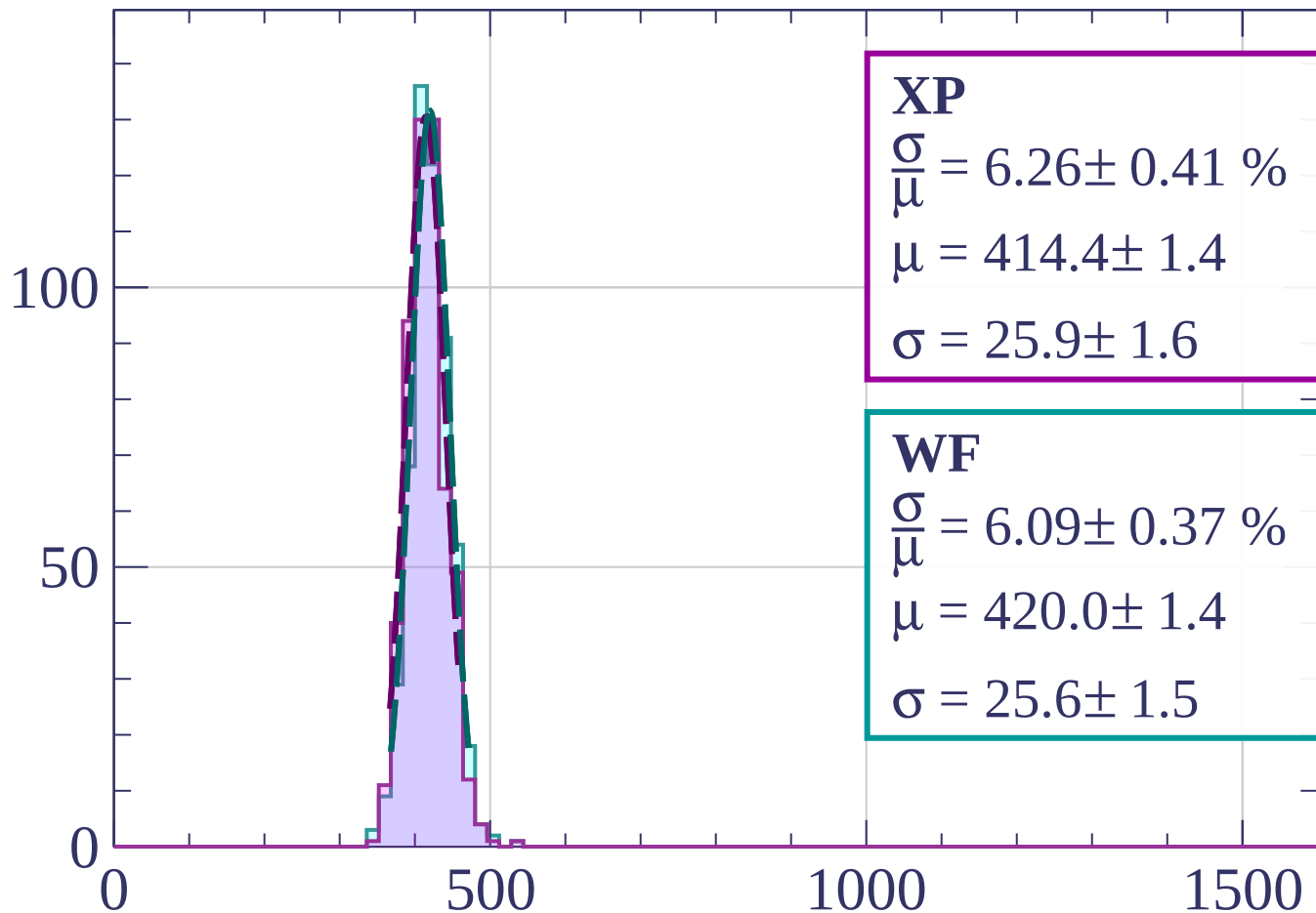
$$\mu = 418.1 \pm 1.3$$

$$\sigma = 26.3 \pm 1.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $12 < \phi < 15$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.26 \pm 0.41 \%$$

$$\mu = 414.4 \pm 1.4$$

$$\sigma = 25.9 \pm 1.6$$

**WF**

$$\frac{\sigma}{\mu} = 6.09 \pm 0.37 \%$$

$$\mu = 420.0 \pm 1.4$$

$$\sigma = 25.6 \pm 1.5$$

dE/dx [ADC counts/cm]

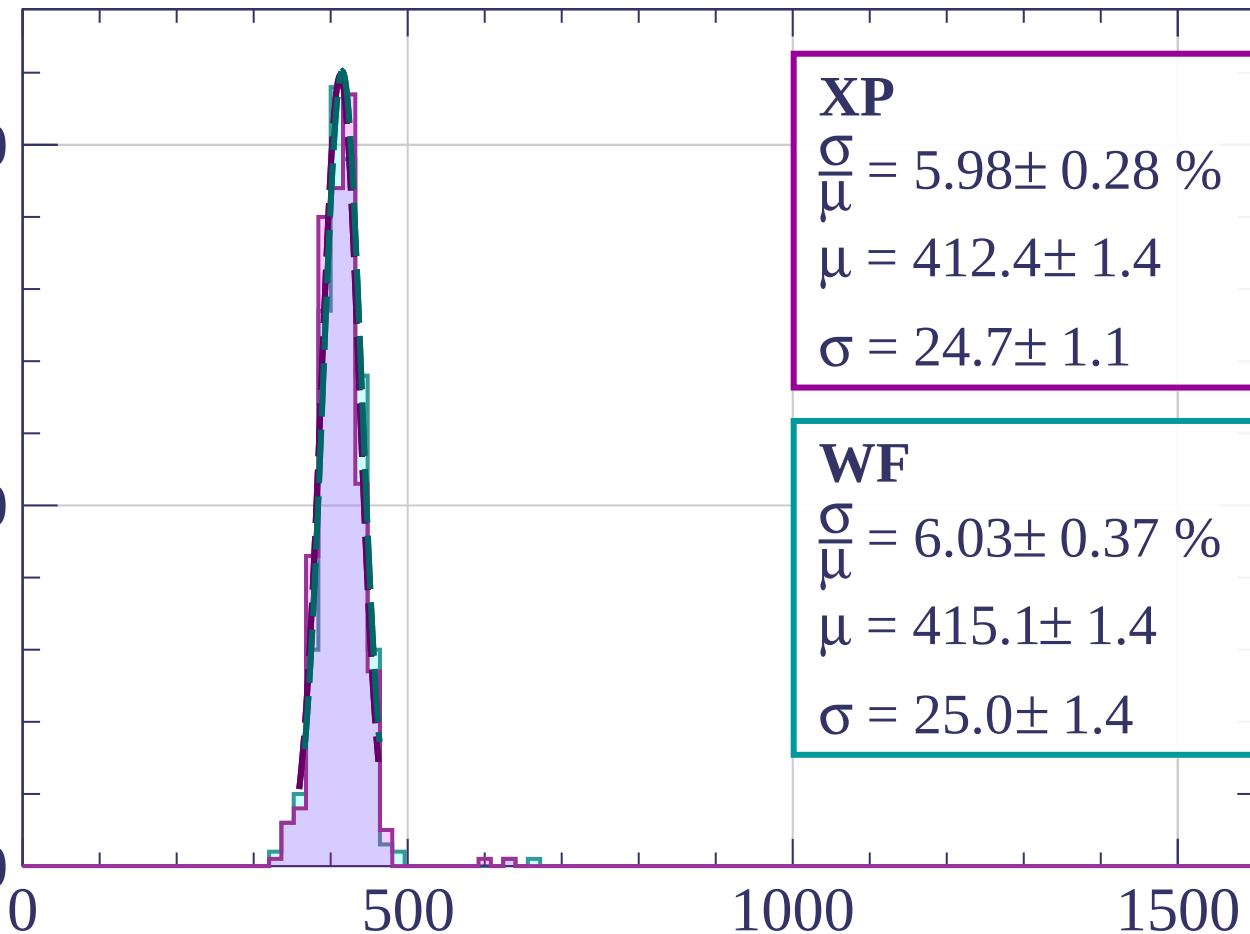
Energy loss |  $15 < \phi < 18$

Count

100

50

0

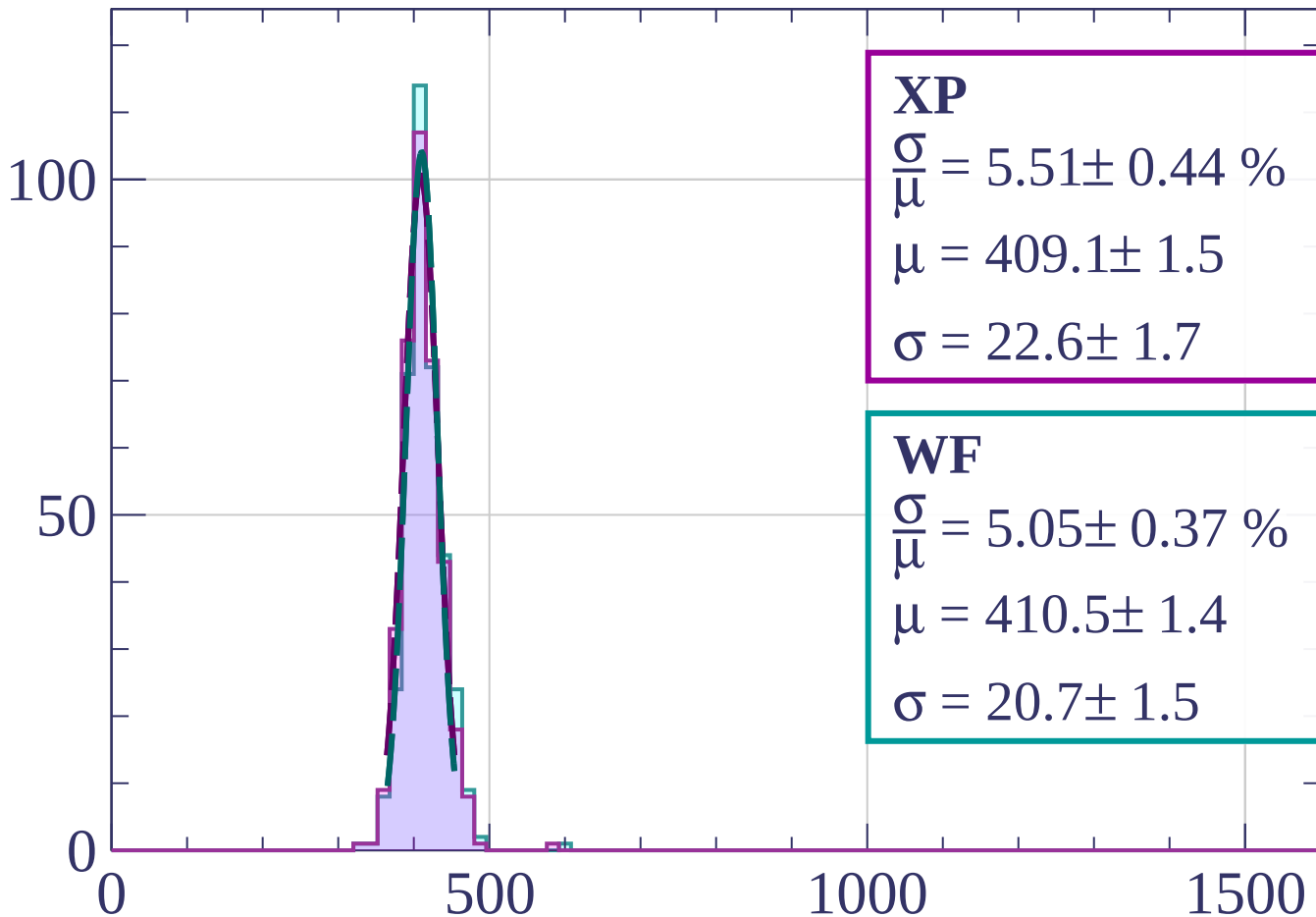


dE/dx [ADC counts/cm]



Energy loss |  $18 < \phi < 21$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.51 \pm 0.44 \%$$

$$\mu = 409.1 \pm 1.5$$

$$\sigma = 22.6 \pm 1.7$$

**WF**

$$\frac{\sigma}{\mu} = 5.05 \pm 0.37 \%$$

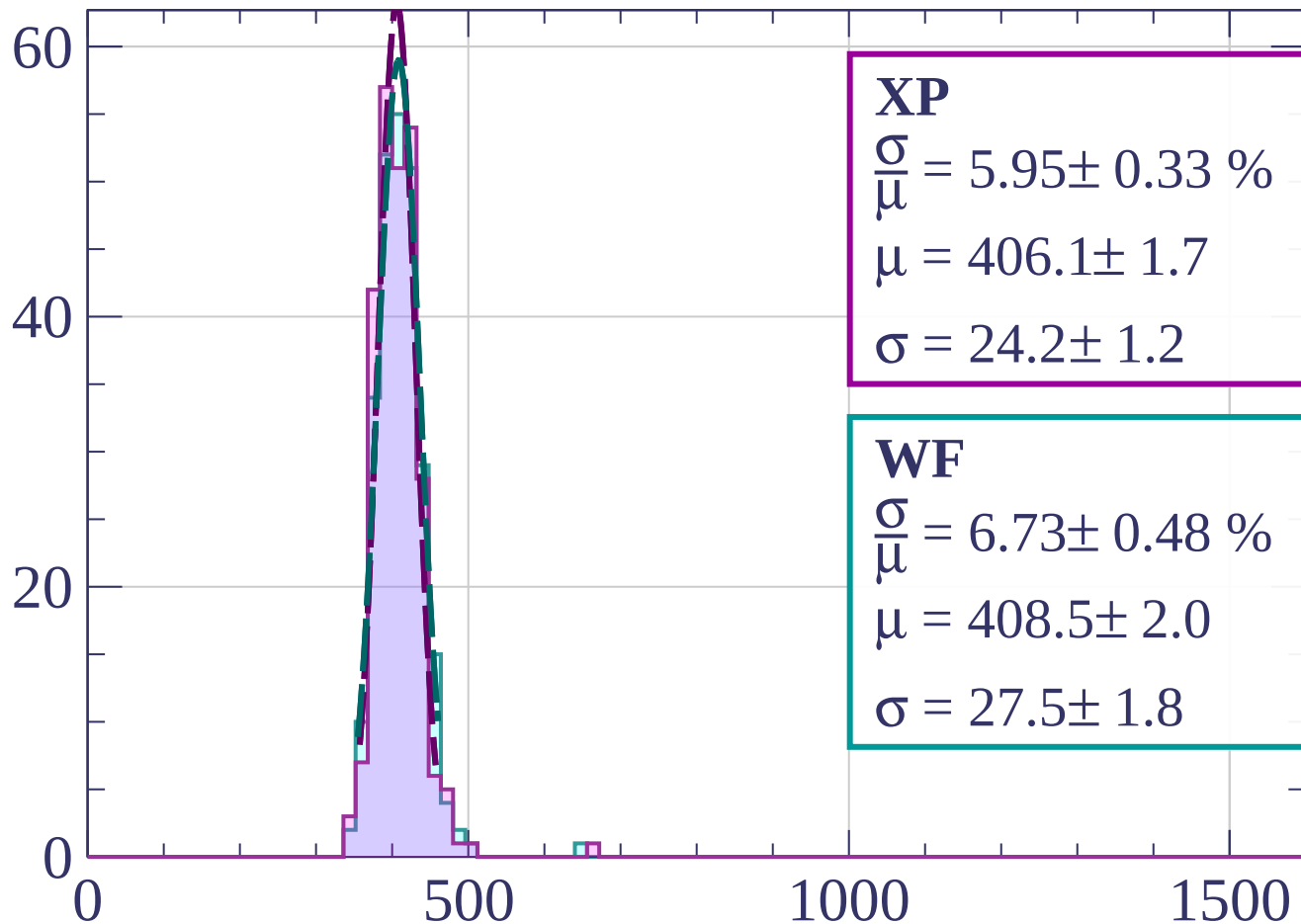
$$\mu = 410.5 \pm 1.4$$

$$\sigma = 20.7 \pm 1.5$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $21 < \phi < 24$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.95 \pm 0.33 \%$$

$$\mu = 406.1 \pm 1.7$$

$$\sigma = 24.2 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.73 \pm 0.48 \%$$

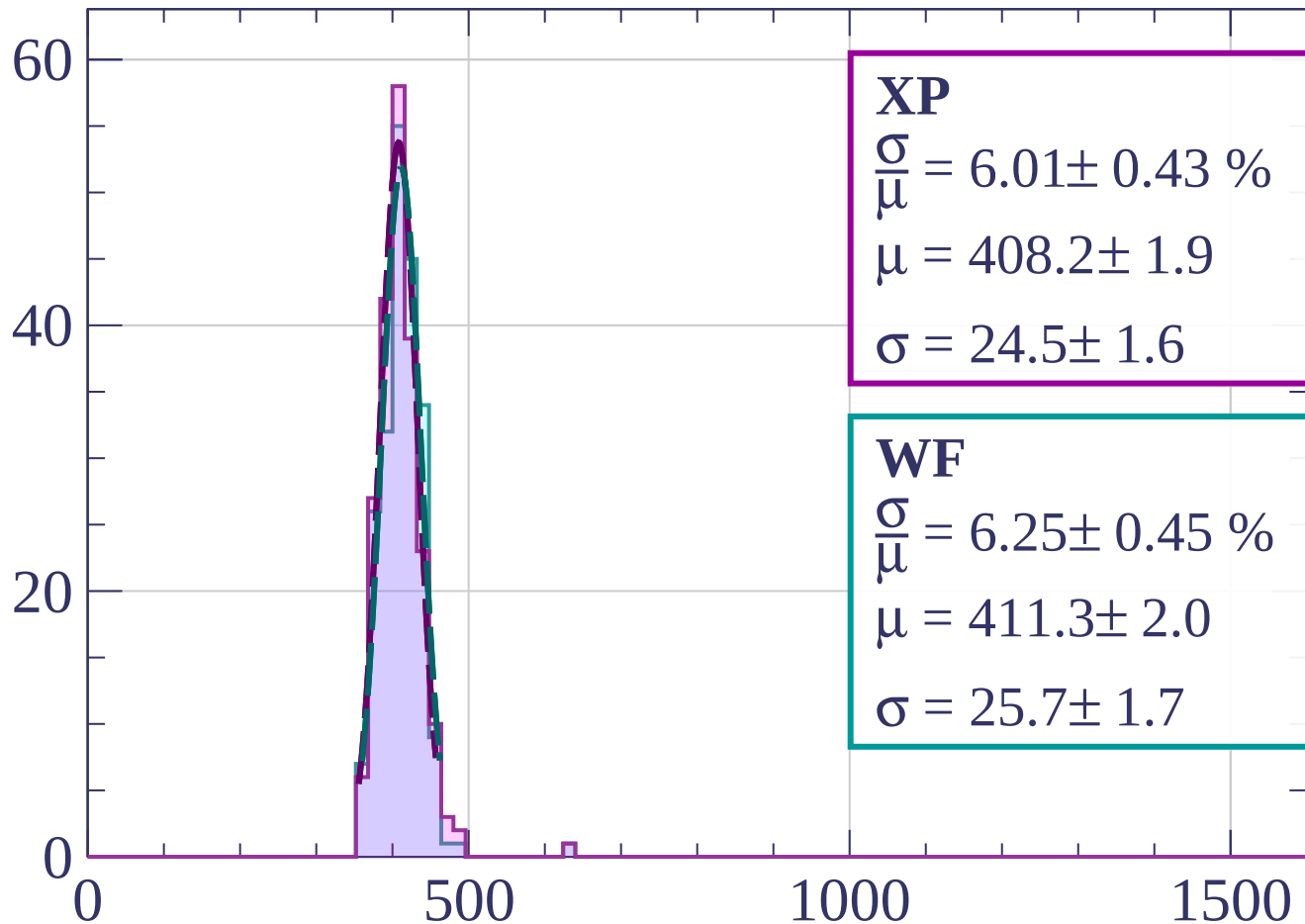
$$\mu = 408.5 \pm 2.0$$

$$\sigma = 27.5 \pm 1.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $24 < \phi < 27$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.01 \pm 0.43 \%$$

$$\mu = 408.2 \pm 1.9$$

$$\sigma = 24.5 \pm 1.6$$

**WF**

$$\frac{\sigma}{\mu} = 6.25 \pm 0.45 \%$$

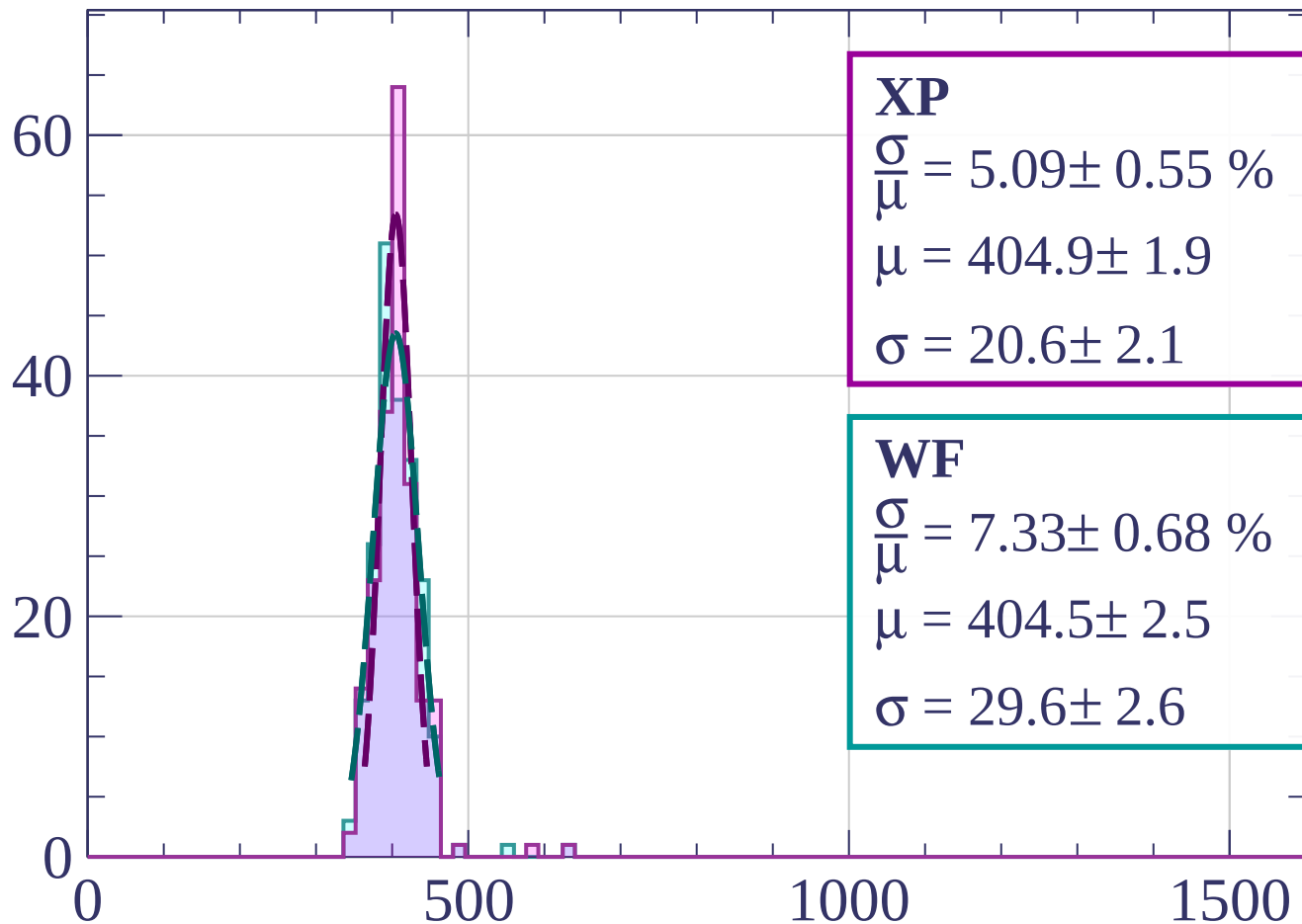
$$\mu = 411.3 \pm 2.0$$

$$\sigma = 25.7 \pm 1.7$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $27 < \phi < 30$

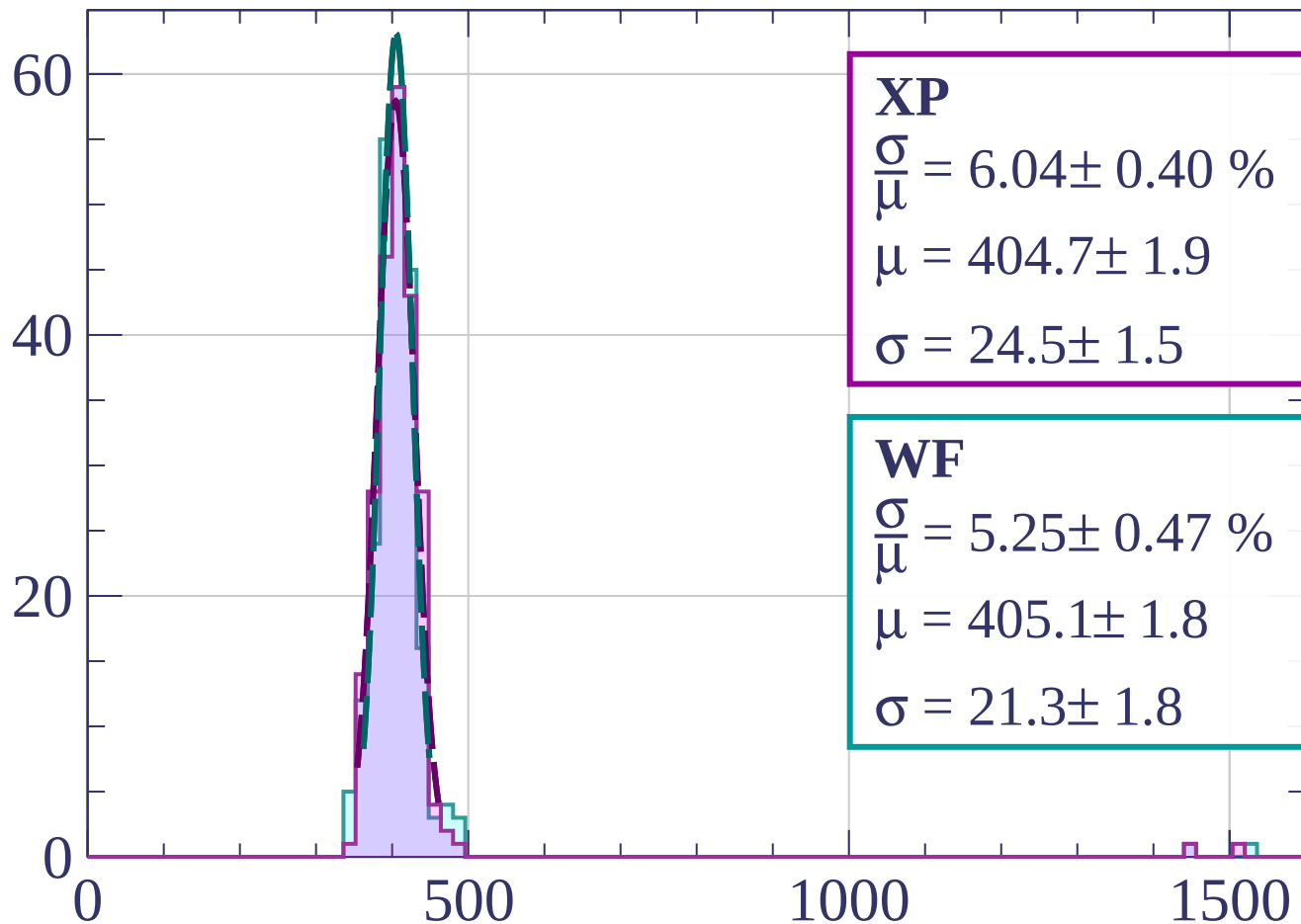
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $30 < \phi < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.04 \pm 0.40 \%$$

$$\mu = 404.7 \pm 1.9$$

$$\sigma = 24.5 \pm 1.5$$

**WF**

$$\frac{\sigma}{\mu} = 5.25 \pm 0.47 \%$$

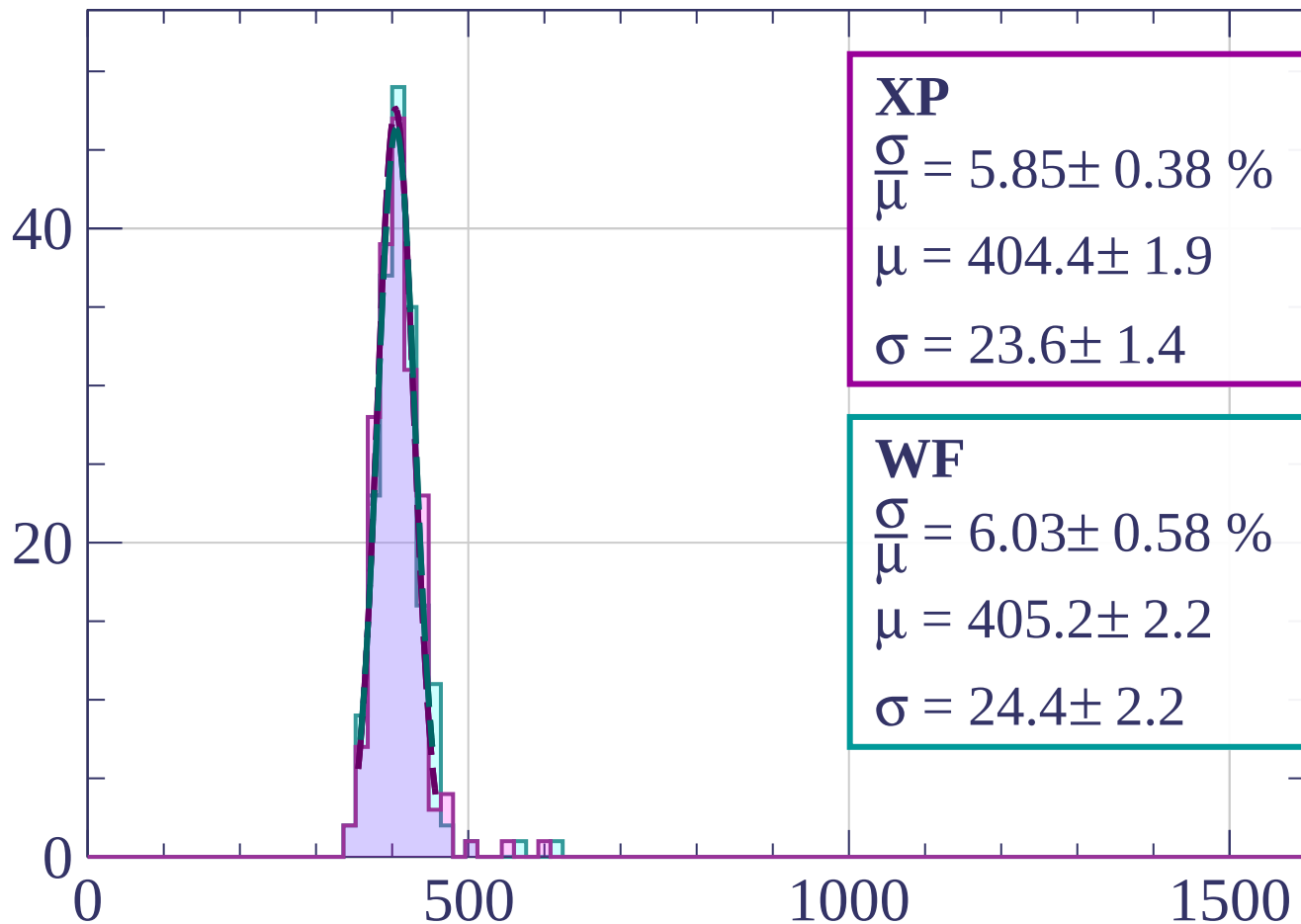
$$\mu = 405.1 \pm 1.8$$

$$\sigma = 21.3 \pm 1.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $33 < \phi < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.85 \pm 0.38 \%$$

$$\mu = 404.4 \pm 1.9$$

$$\sigma = 23.6 \pm 1.4$$

**WF**

$$\frac{\sigma}{\mu} = 6.03 \pm 0.58 \%$$

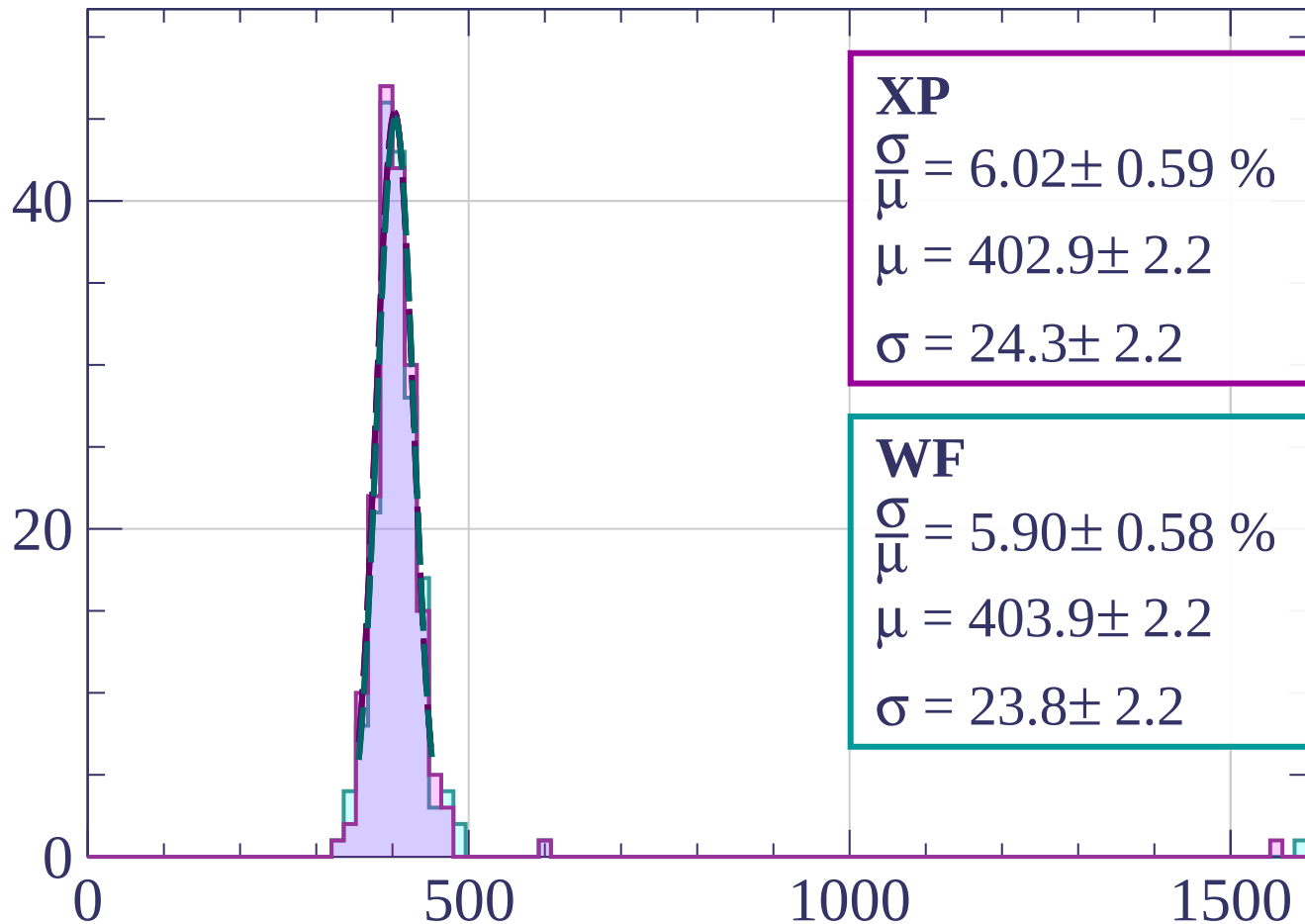
$$\mu = 405.2 \pm 2.2$$

$$\sigma = 24.4 \pm 2.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $36 < \phi < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.02 \pm 0.59 \%$$

$$\mu = 402.9 \pm 2.2$$

$$\sigma = 24.3 \pm 2.2$$

**WF**

$$\frac{\sigma}{\mu} = 5.90 \pm 0.58 \%$$

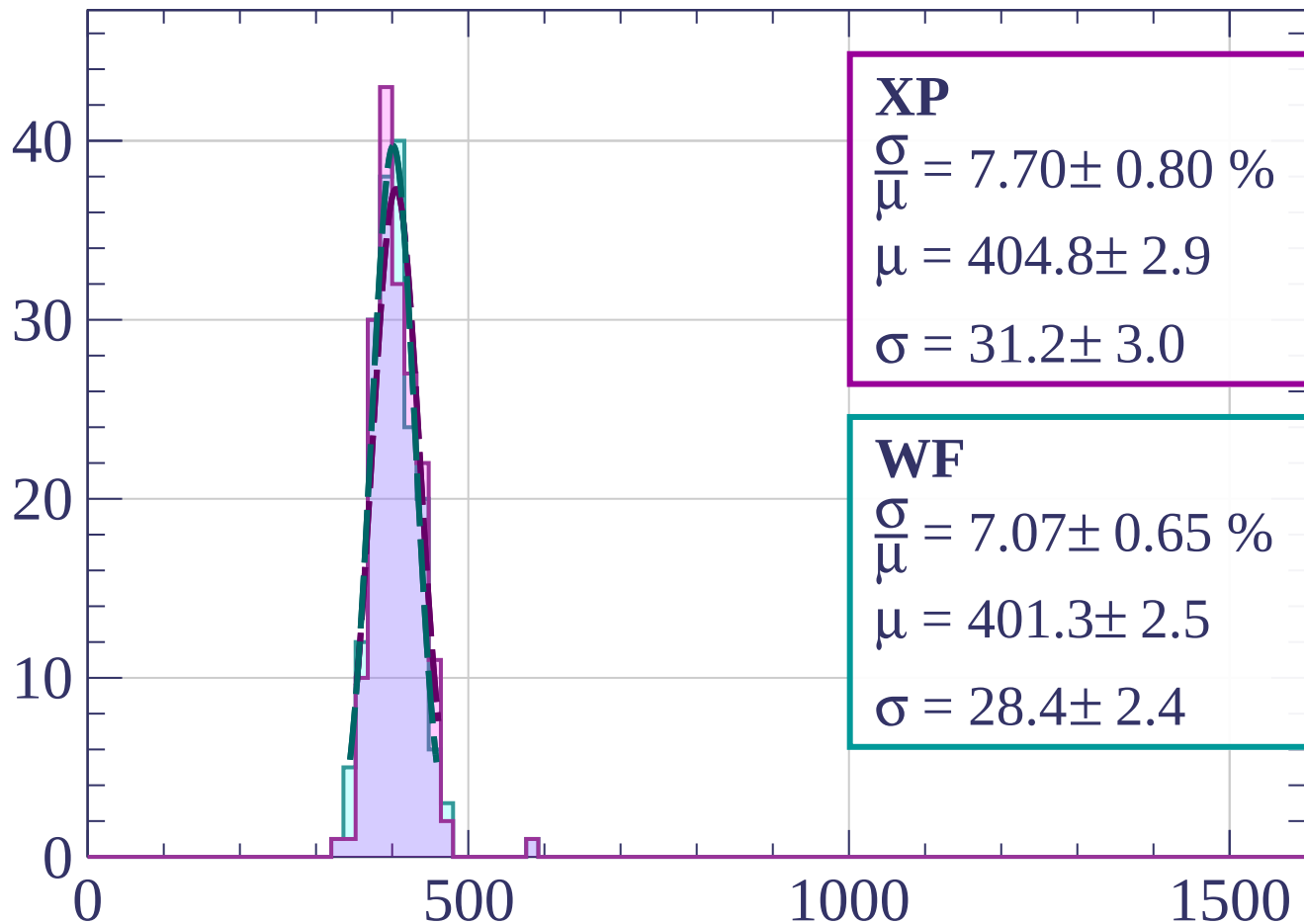
$$\mu = 403.9 \pm 2.2$$

$$\sigma = 23.8 \pm 2.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $39 < \phi < 42$

Count



**XP**

$$\frac{\sigma}{\mu} = 7.70 \pm 0.80 \%$$

$$\mu = 404.8 \pm 2.9$$

$$\sigma = 31.2 \pm 3.0$$

**WF**

$$\frac{\sigma}{\mu} = 7.07 \pm 0.65 \%$$

$$\mu = 401.3 \pm 2.5$$

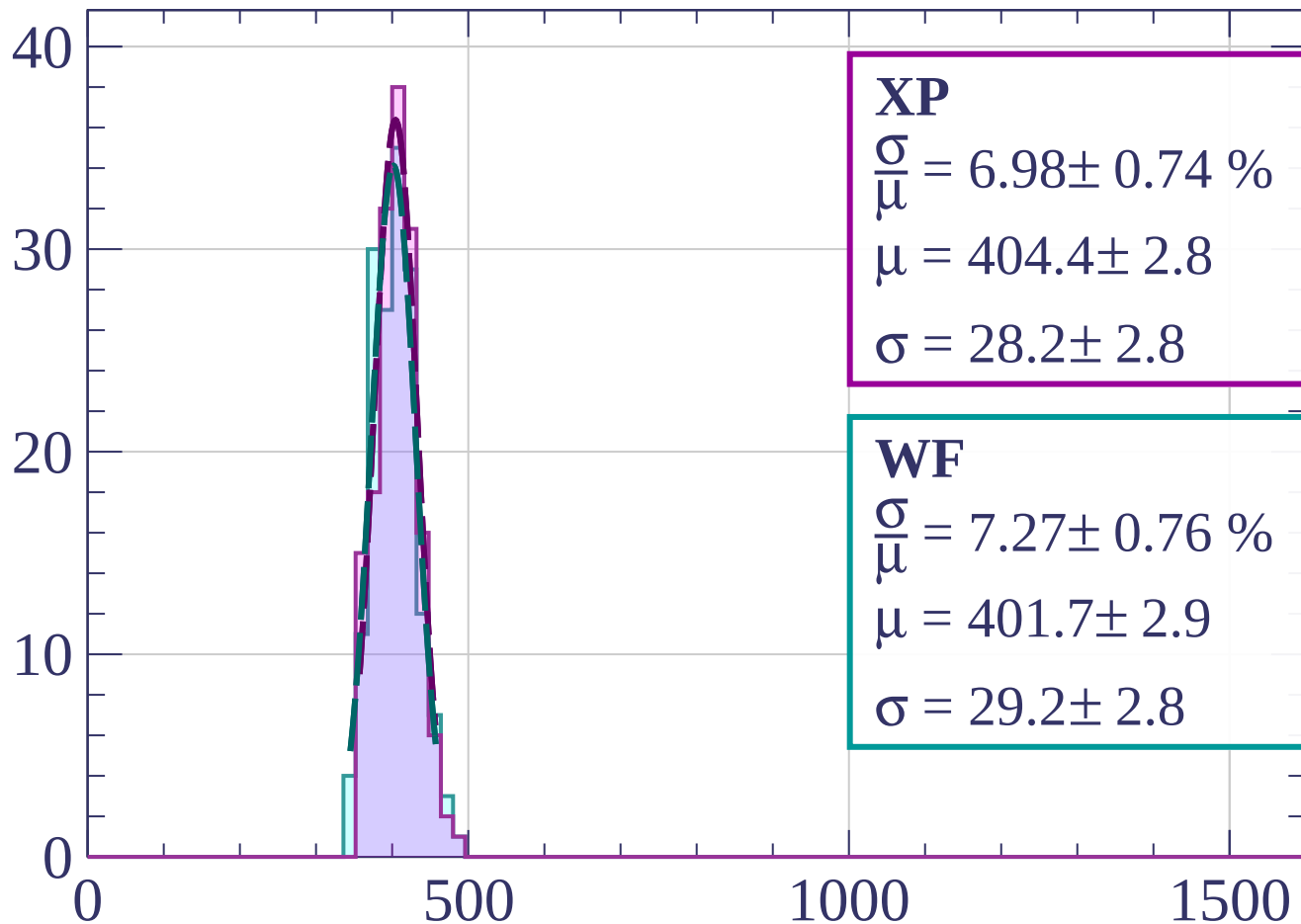
$$\sigma = 28.4 \pm 2.4$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $42 < \phi < 45$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.98 \pm 0.74 \%$$

$$\mu = 404.4 \pm 2.8$$

$$\sigma = 28.2 \pm 2.8$$

**WF**

$$\frac{\sigma}{\mu} = 7.27 \pm 0.76 \%$$

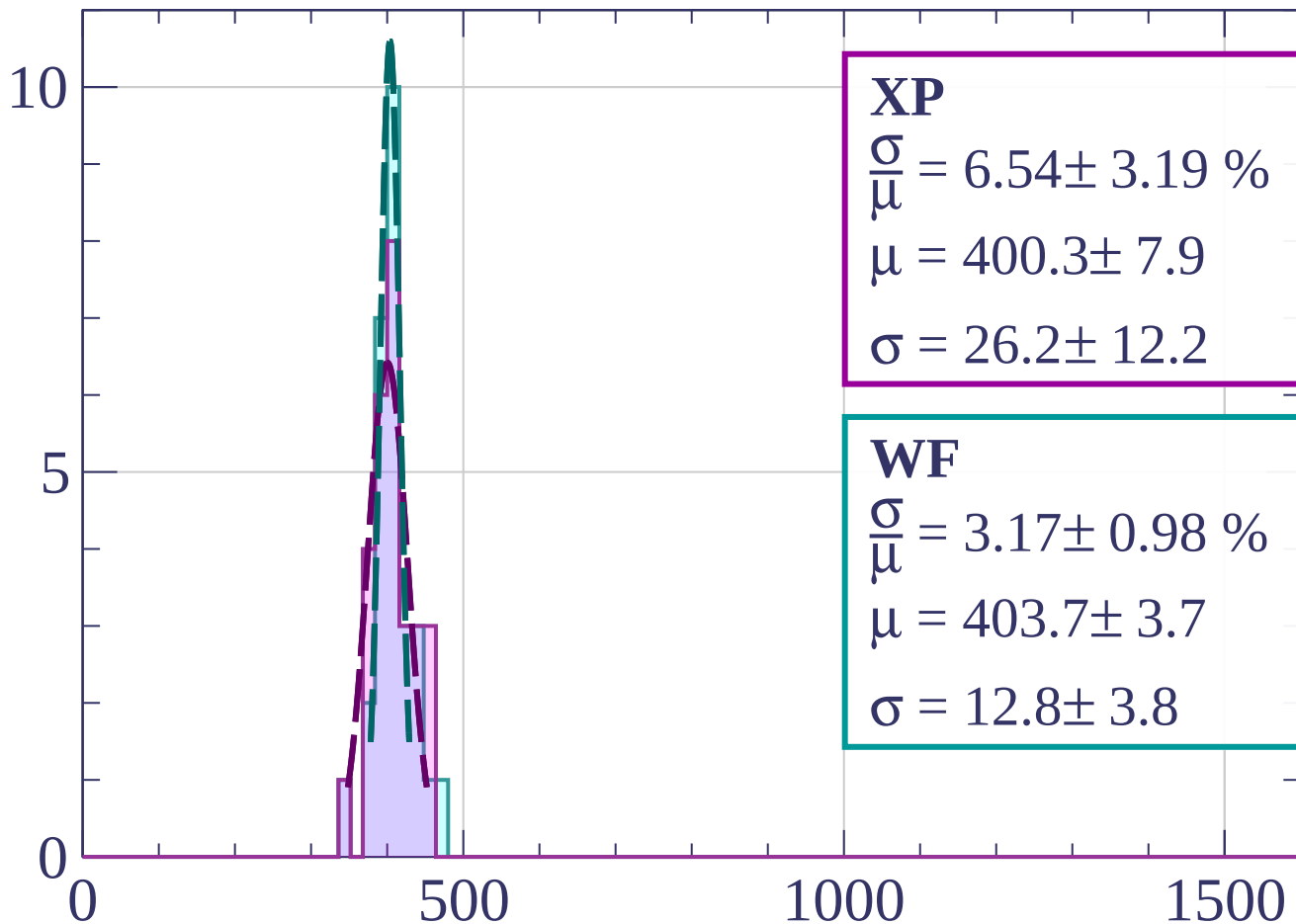
$$\mu = 401.7 \pm 2.9$$

$$\sigma = 29.2 \pm 2.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $45 < \phi < 48$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.54 \pm 3.19 \%$$

$$\mu = 400.3 \pm 7.9$$

$$\sigma = 26.2 \pm 12.2$$

**WF**

$$\frac{\sigma}{\mu} = 3.17 \pm 0.98 \%$$

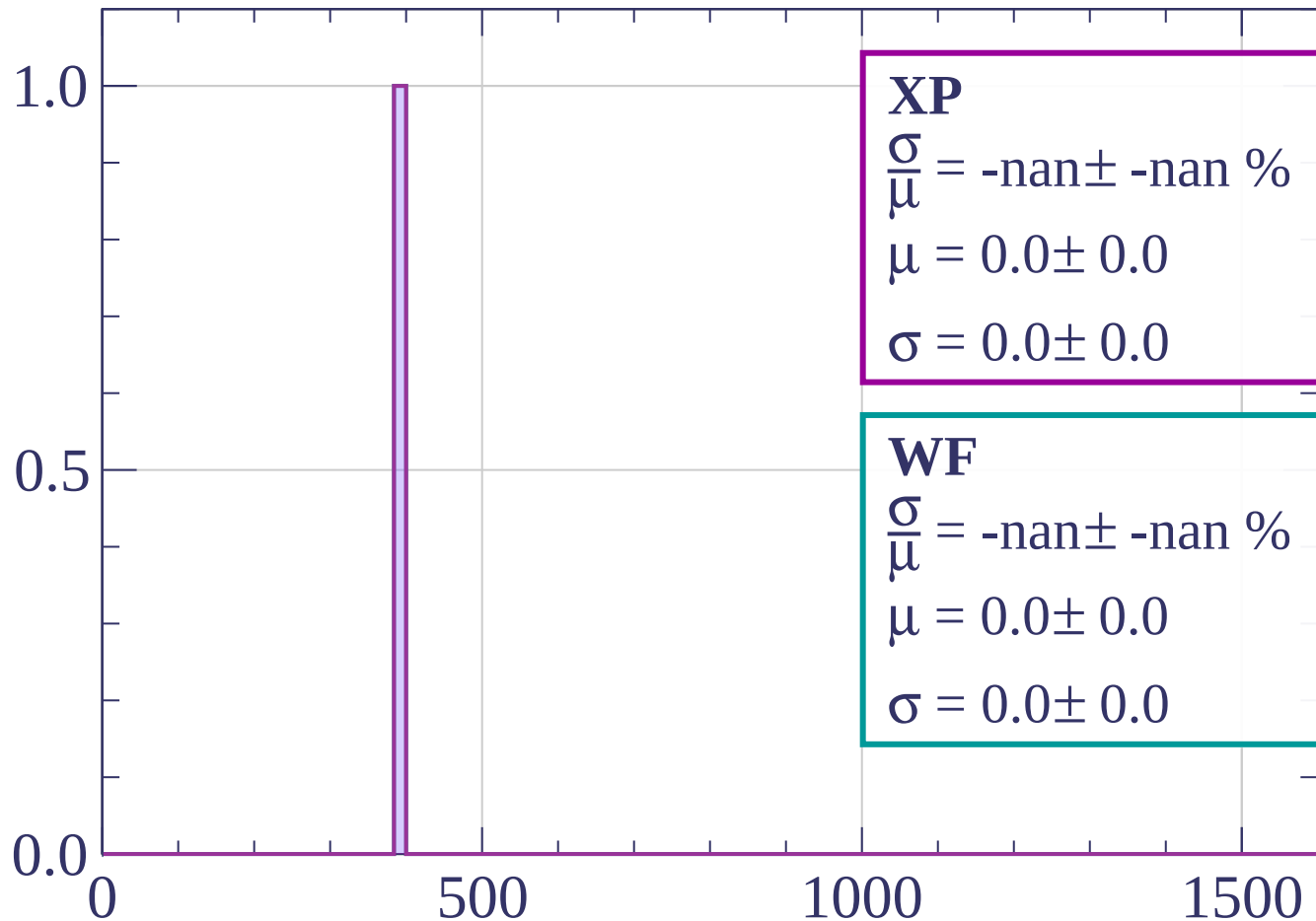
$$\mu = 403.7 \pm 3.7$$

$$\sigma = 12.8 \pm 3.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $48 < \phi < 51$

Count



**XP**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

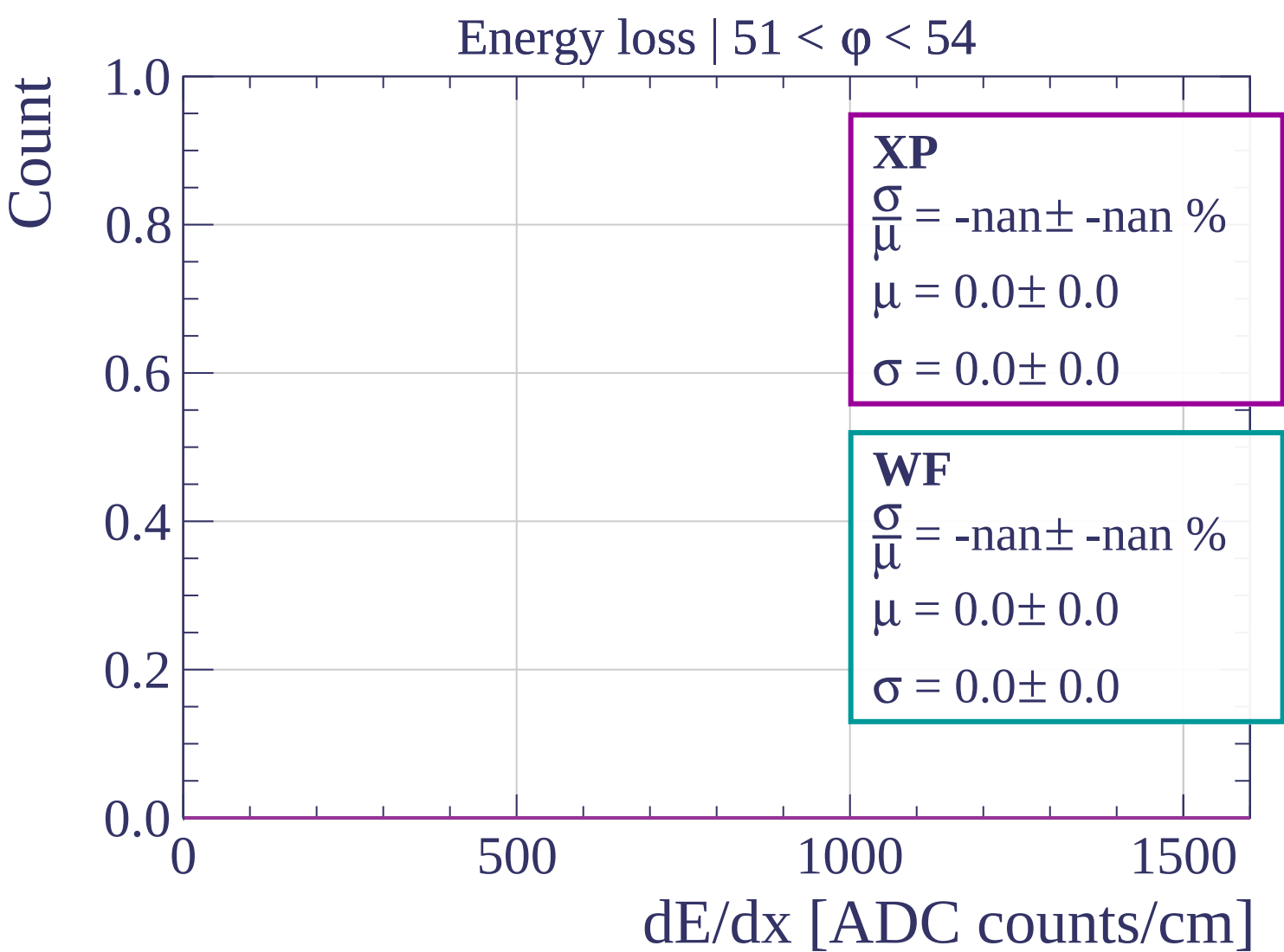
**WF**

$\frac{\sigma}{\mu} = \text{-nan} \pm \text{-nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

dE/dx [ADC counts/cm]



Count

Energy loss |  $54 < \phi < 57$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $57 < \phi < 60$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0  
0.8  
0.6  
0.4  
0.2  
0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $60 < \phi < 63$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

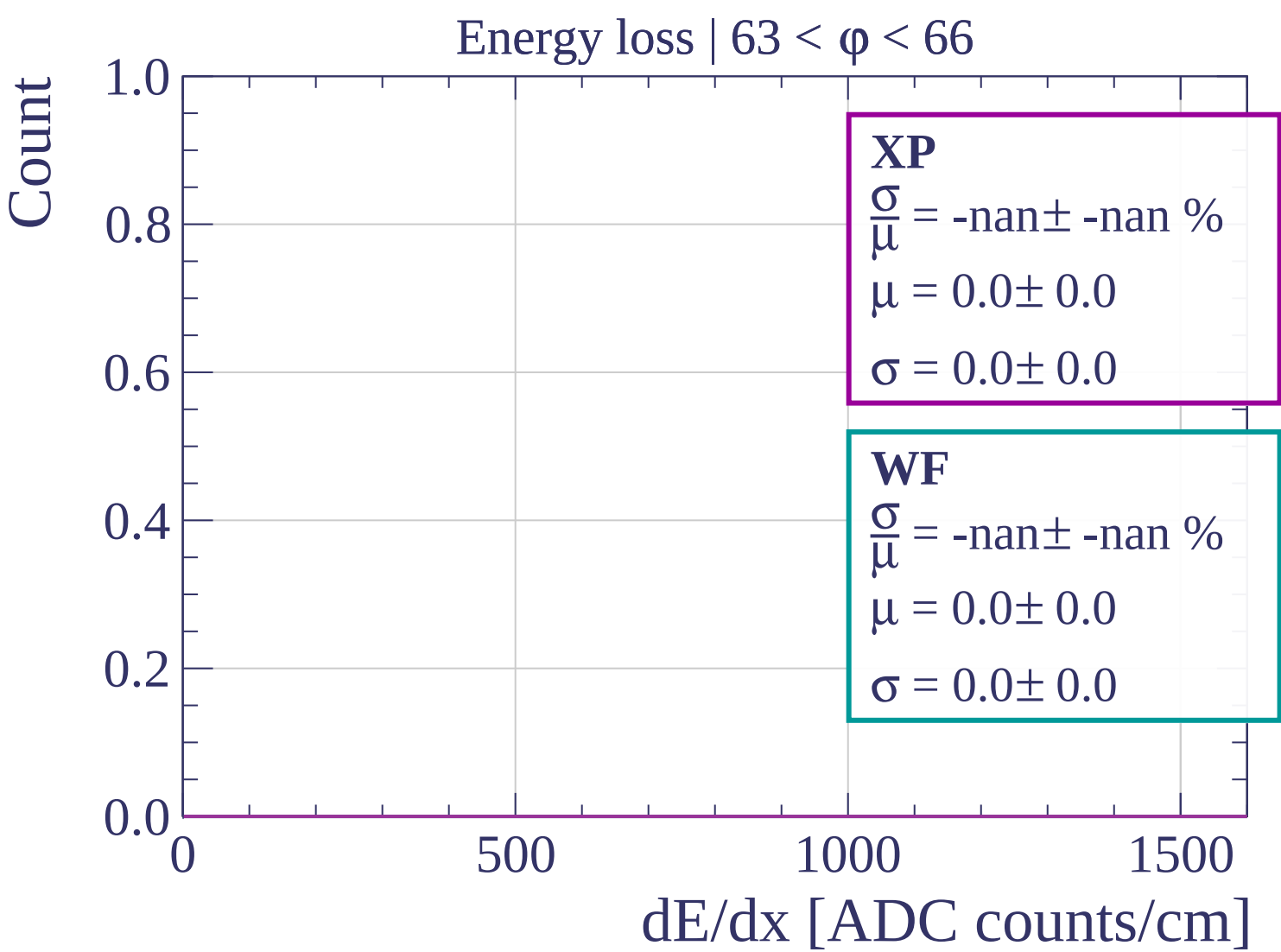
0

500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss |  $66 < \phi < 69$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $69 < \phi < 72$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $72 < \phi < 75$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

0

500

1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $75 < \phi < 78$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0

0.8

0.6

0.4

0.2

0.0

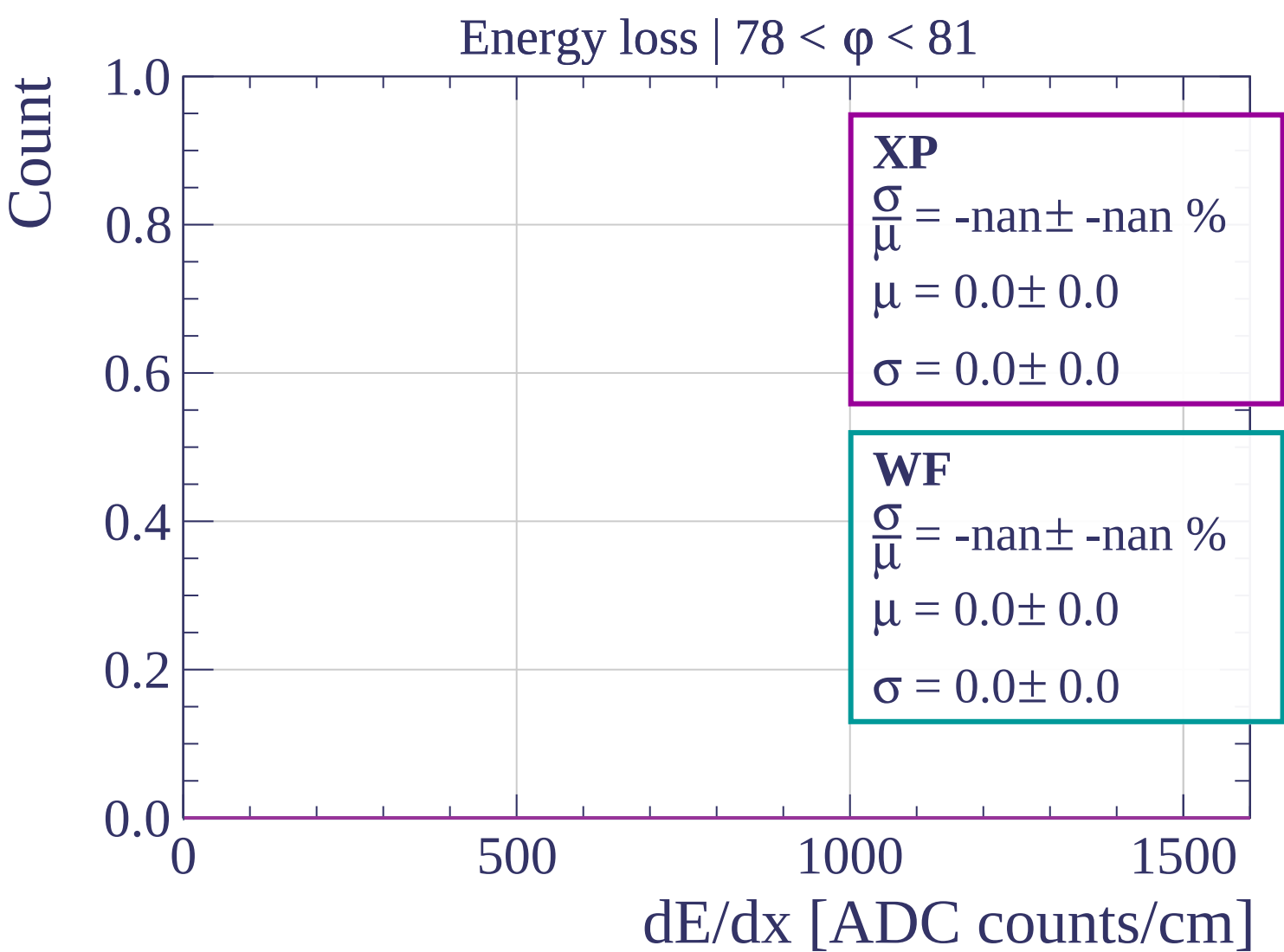
0

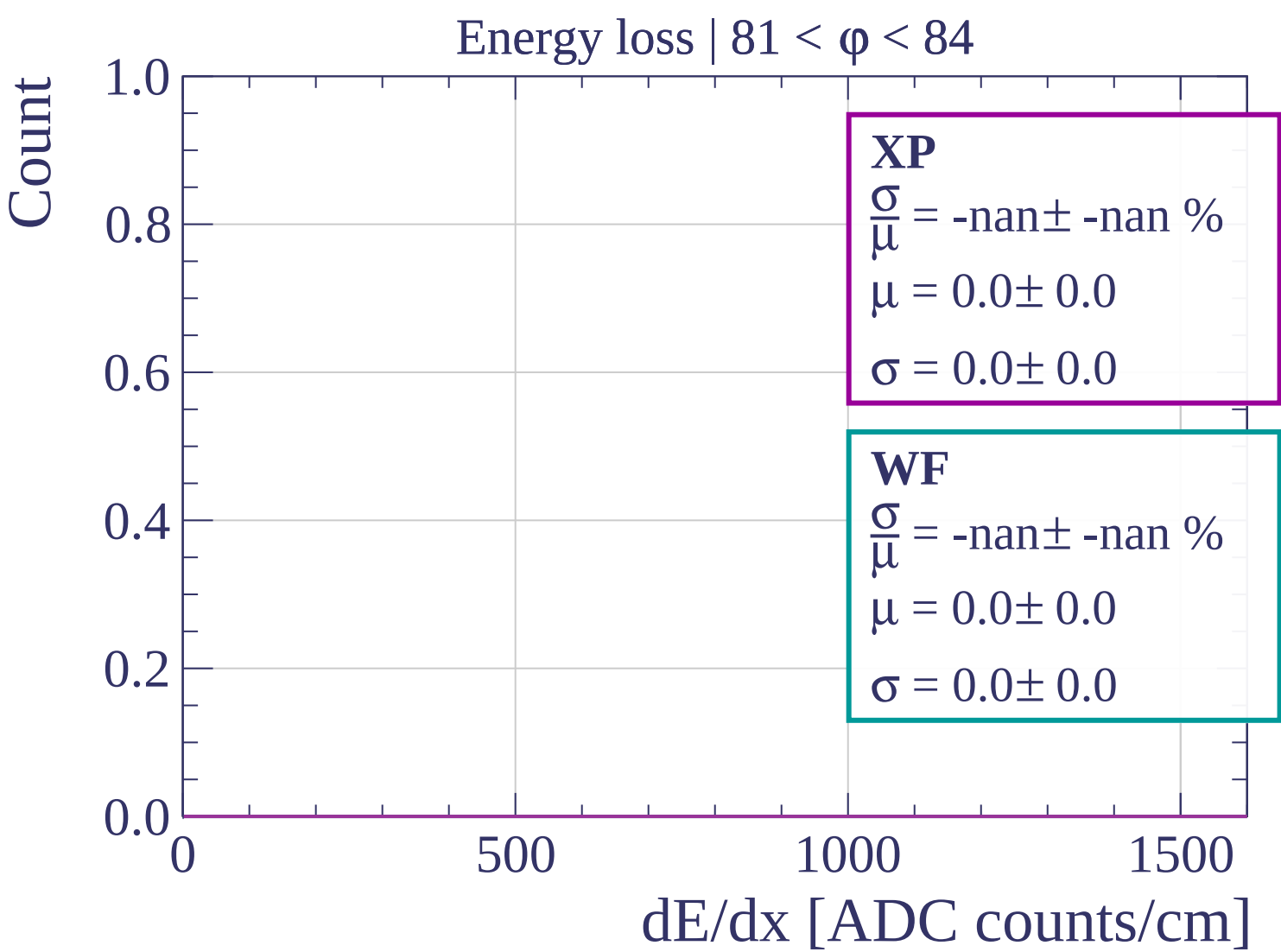
500

1000

1500

dE/dx [ADC counts/cm]





Count

Energy loss |  $84 < \phi < 87$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0  
0.8  
0.6  
0.4  
0.2  
0.0

0

500

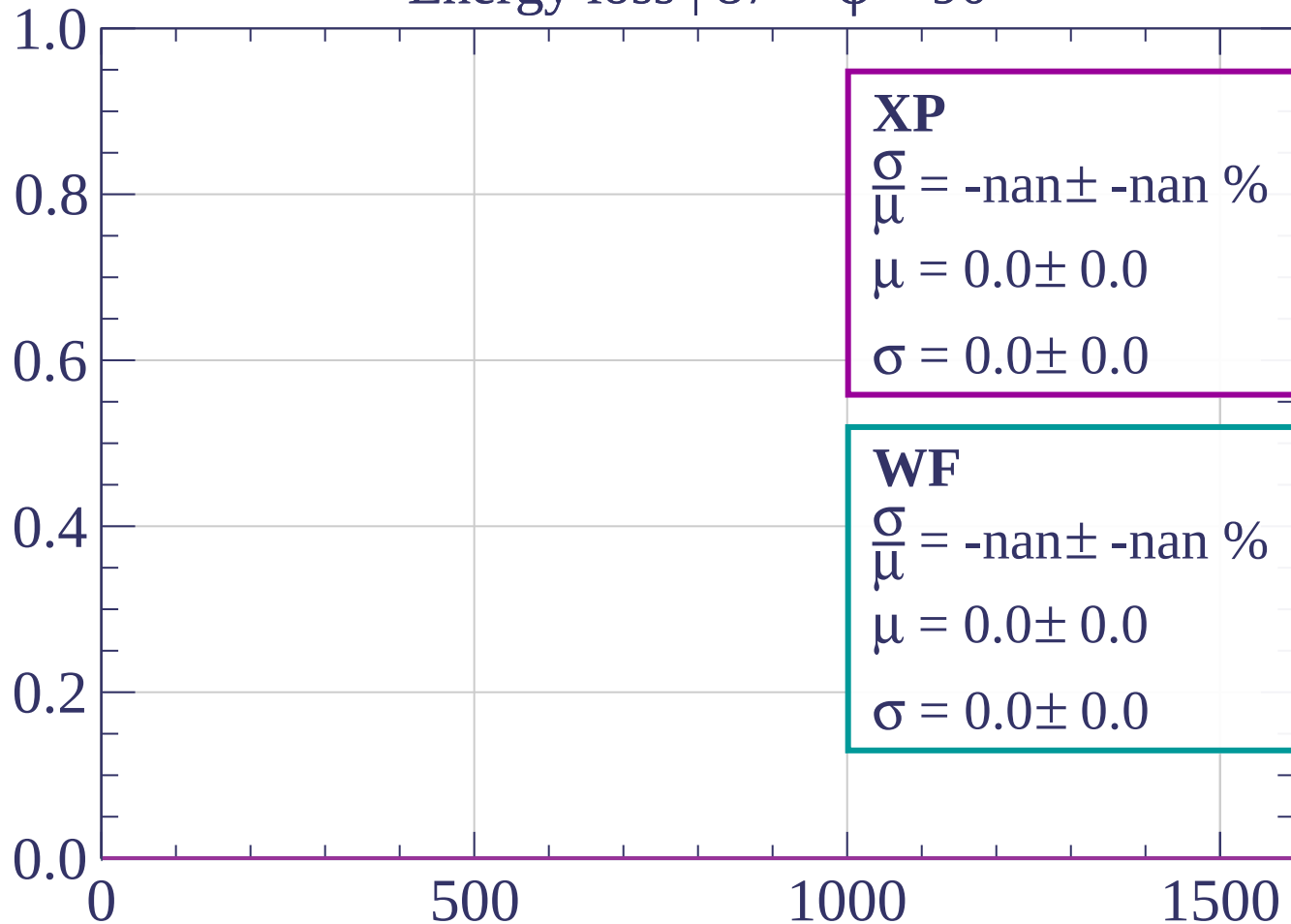
1000

1500

dE/dx [ADC counts/cm]

Count

Energy loss |  $87 < \phi < 90$

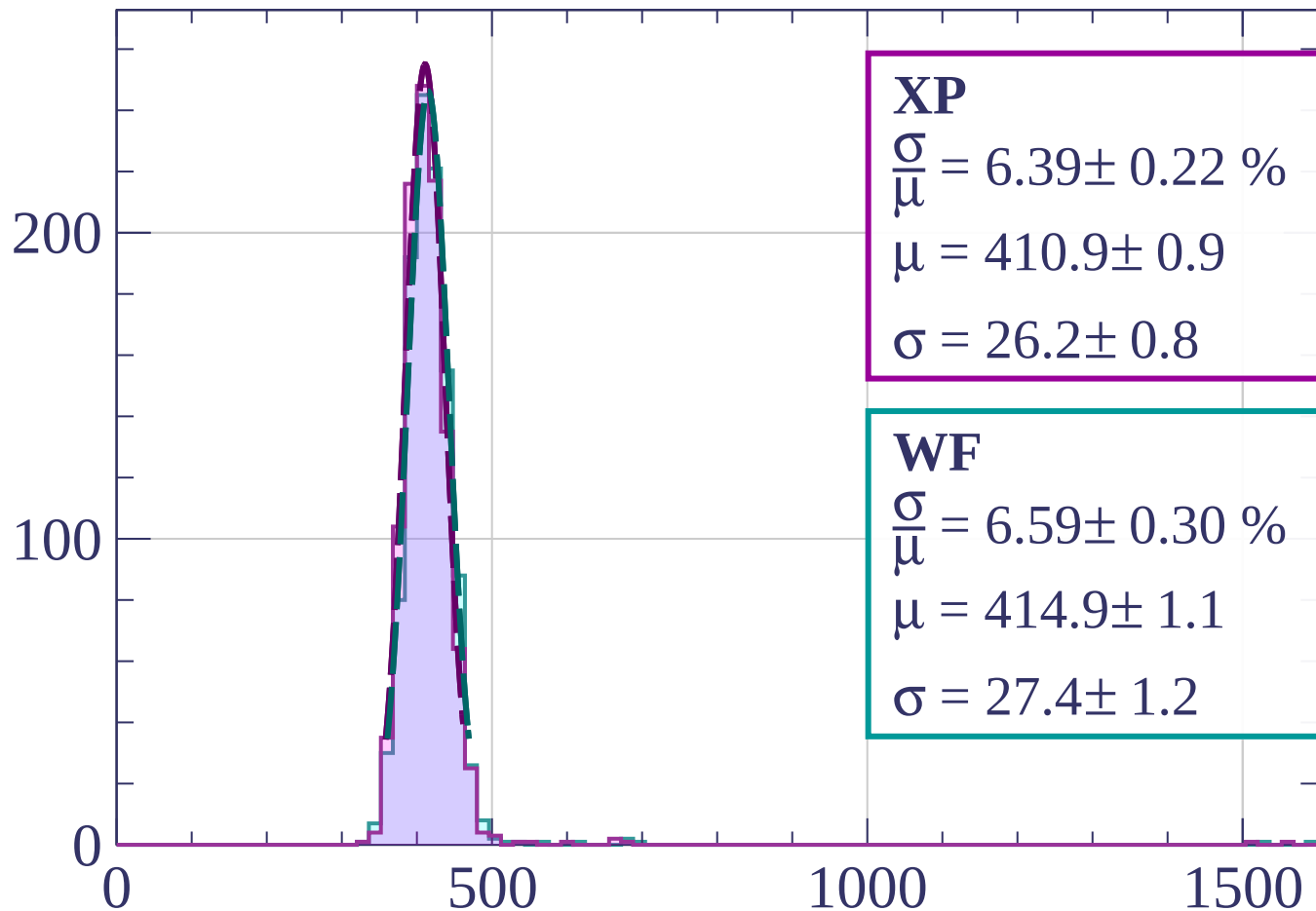


dE/dx [ADC counts/cm]



Energy loss |  $0 < \theta < 3$

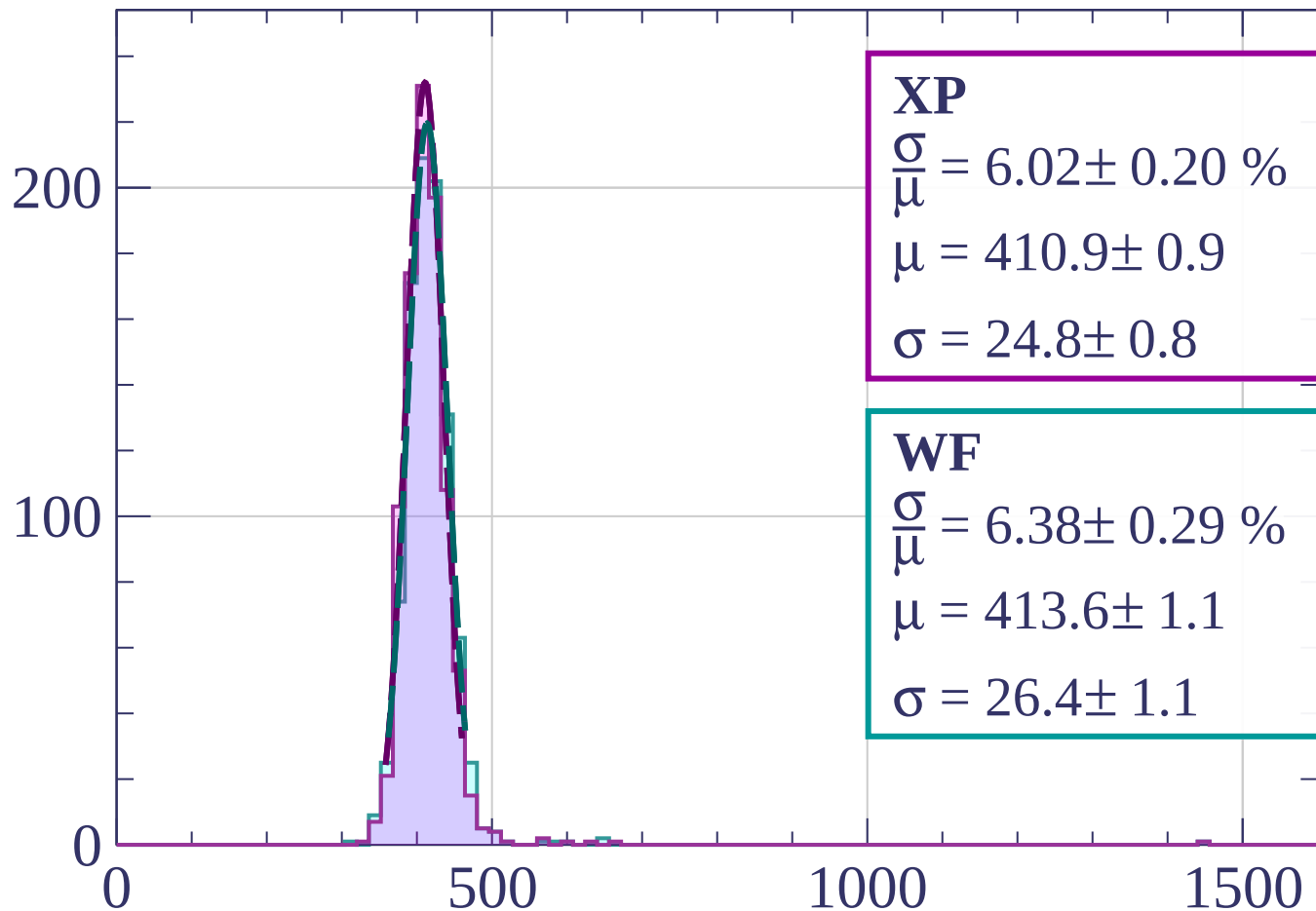
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $3 < \theta < 6$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.02 \pm 0.20 \%$$

$$\mu = 410.9 \pm 0.9$$

$$\sigma = 24.8 \pm 0.8$$

**WF**

$$\frac{\sigma}{\mu} = 6.38 \pm 0.29 \%$$

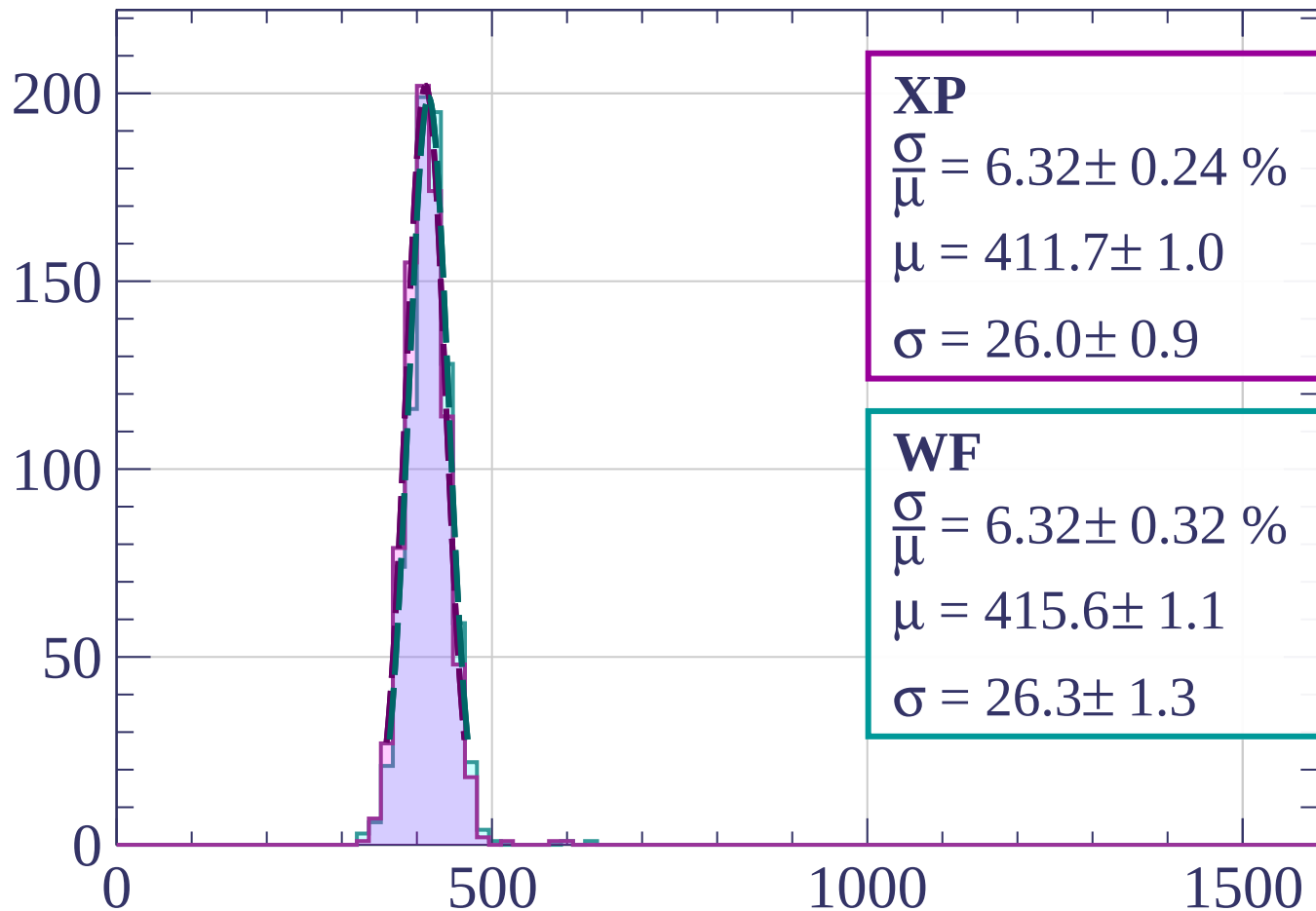
$$\mu = 413.6 \pm 1.1$$

$$\sigma = 26.4 \pm 1.1$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $6 < \theta < 9$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.32 \pm 0.24 \%$$

$$\mu = 411.7 \pm 1.0$$

$$\sigma = 26.0 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 6.32 \pm 0.32 \%$$

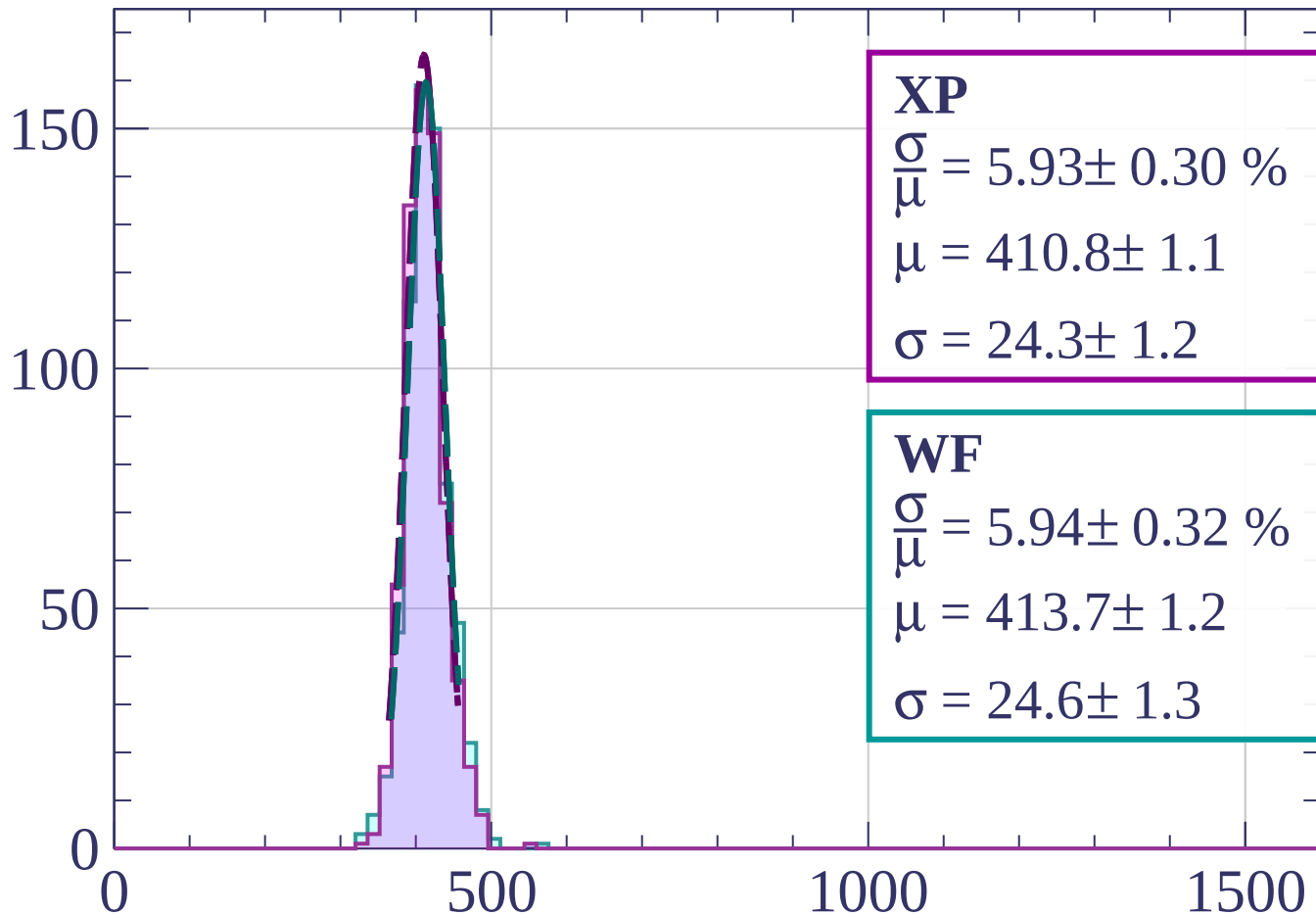
$$\mu = 415.6 \pm 1.1$$

$$\sigma = 26.3 \pm 1.3$$

dE/dx [ADC counts/cm]

Energy loss |  $9 < \theta < 12$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.93 \pm 0.30 \%$$

$$\mu = 410.8 \pm 1.1$$

$$\sigma = 24.3 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 5.94 \pm 0.32 \%$$

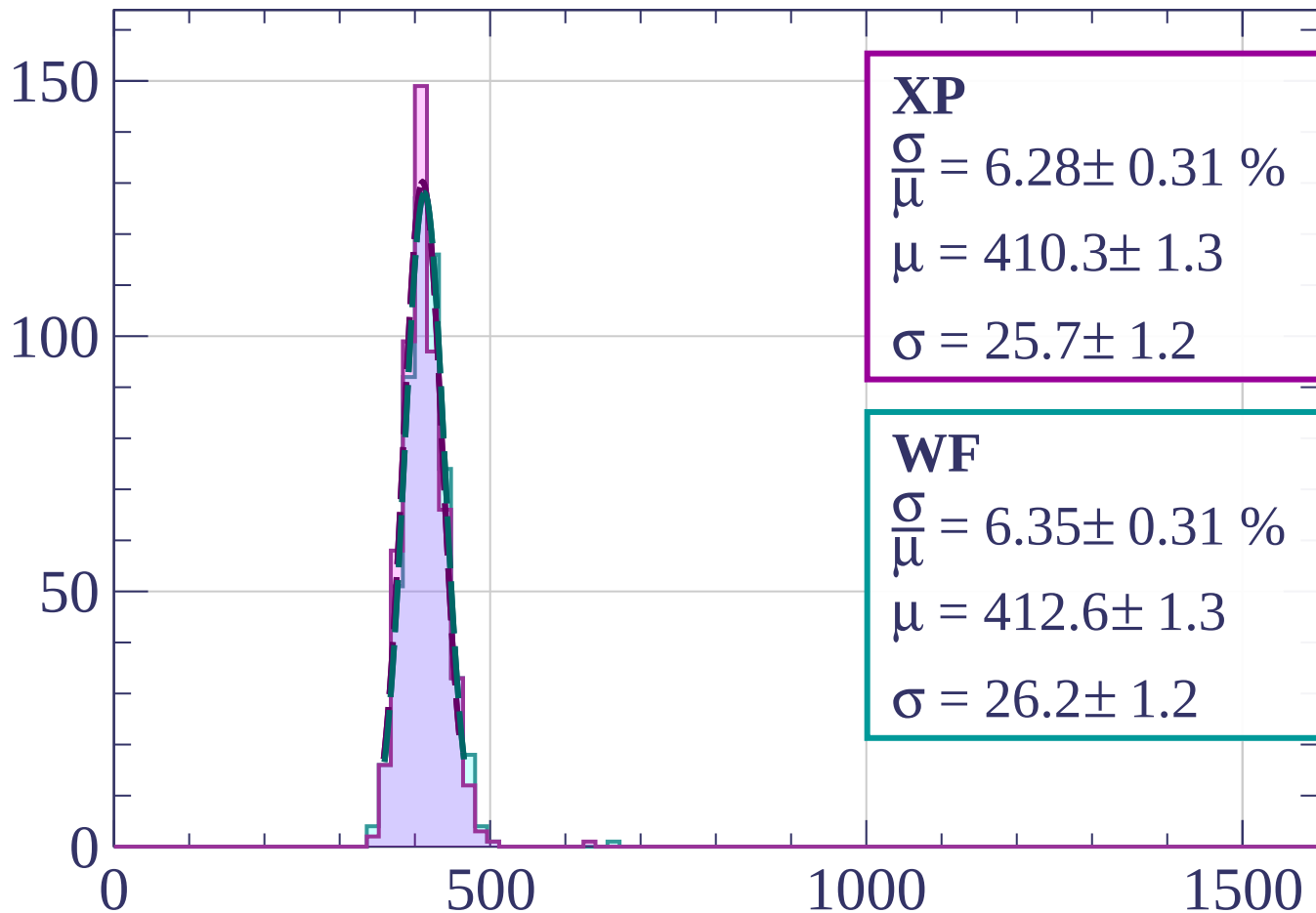
$$\mu = 413.7 \pm 1.2$$

$$\sigma = 24.6 \pm 1.3$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $12 < \theta < 15$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.28 \pm 0.31 \%$$

$$\mu = 410.3 \pm 1.3$$

$$\sigma = 25.7 \pm 1.2$$

**WF**

$$\frac{\sigma}{\mu} = 6.35 \pm 0.31 \%$$

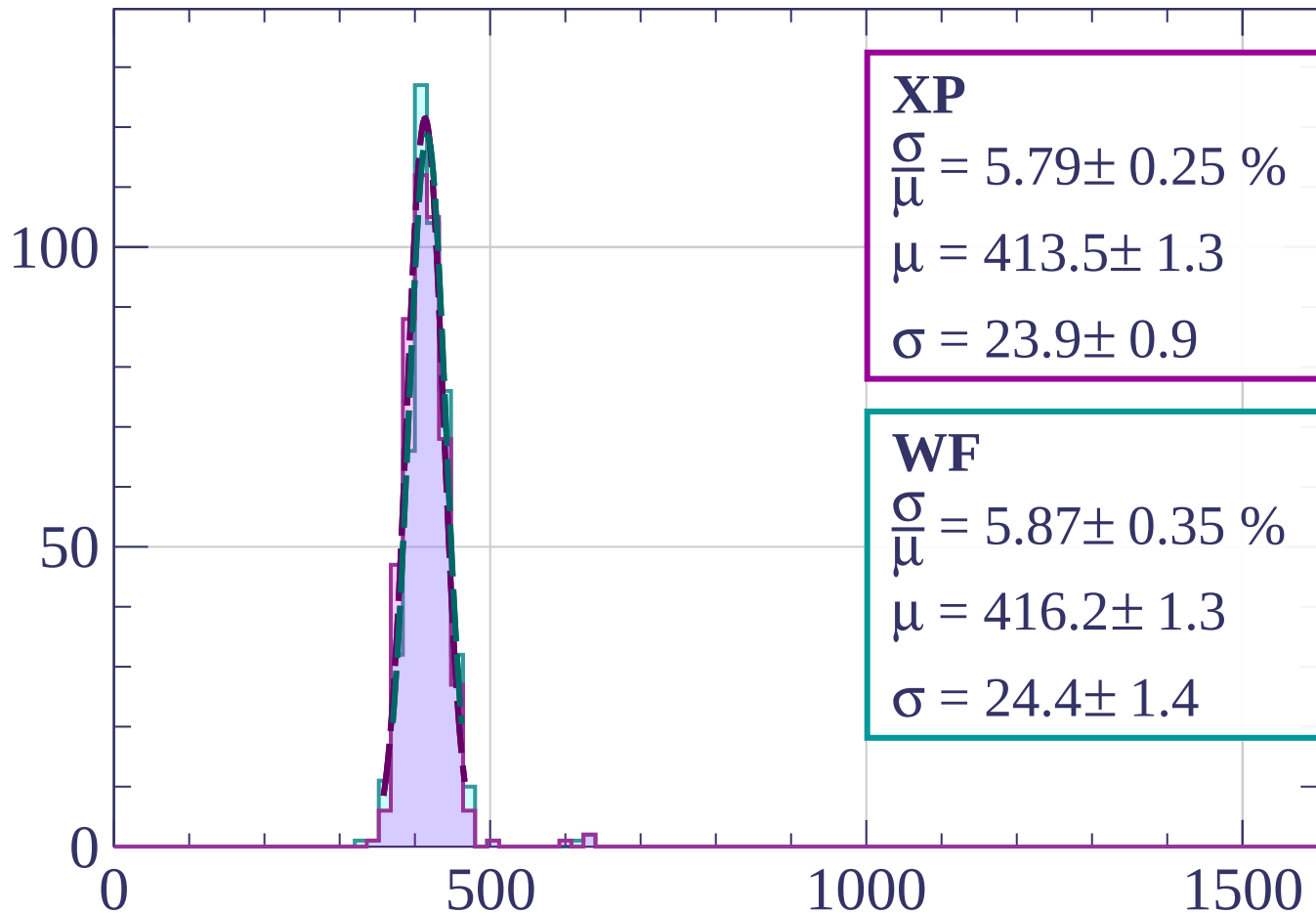
$$\mu = 412.6 \pm 1.3$$

$$\sigma = 26.2 \pm 1.2$$

dE/dx [ADC counts/cm]

Energy loss |  $15 < \theta < 18$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.79 \pm 0.25 \%$$

$$\mu = 413.5 \pm 1.3$$

$$\sigma = 23.9 \pm 0.9$$

**WF**

$$\frac{\sigma}{\mu} = 5.87 \pm 0.35 \%$$

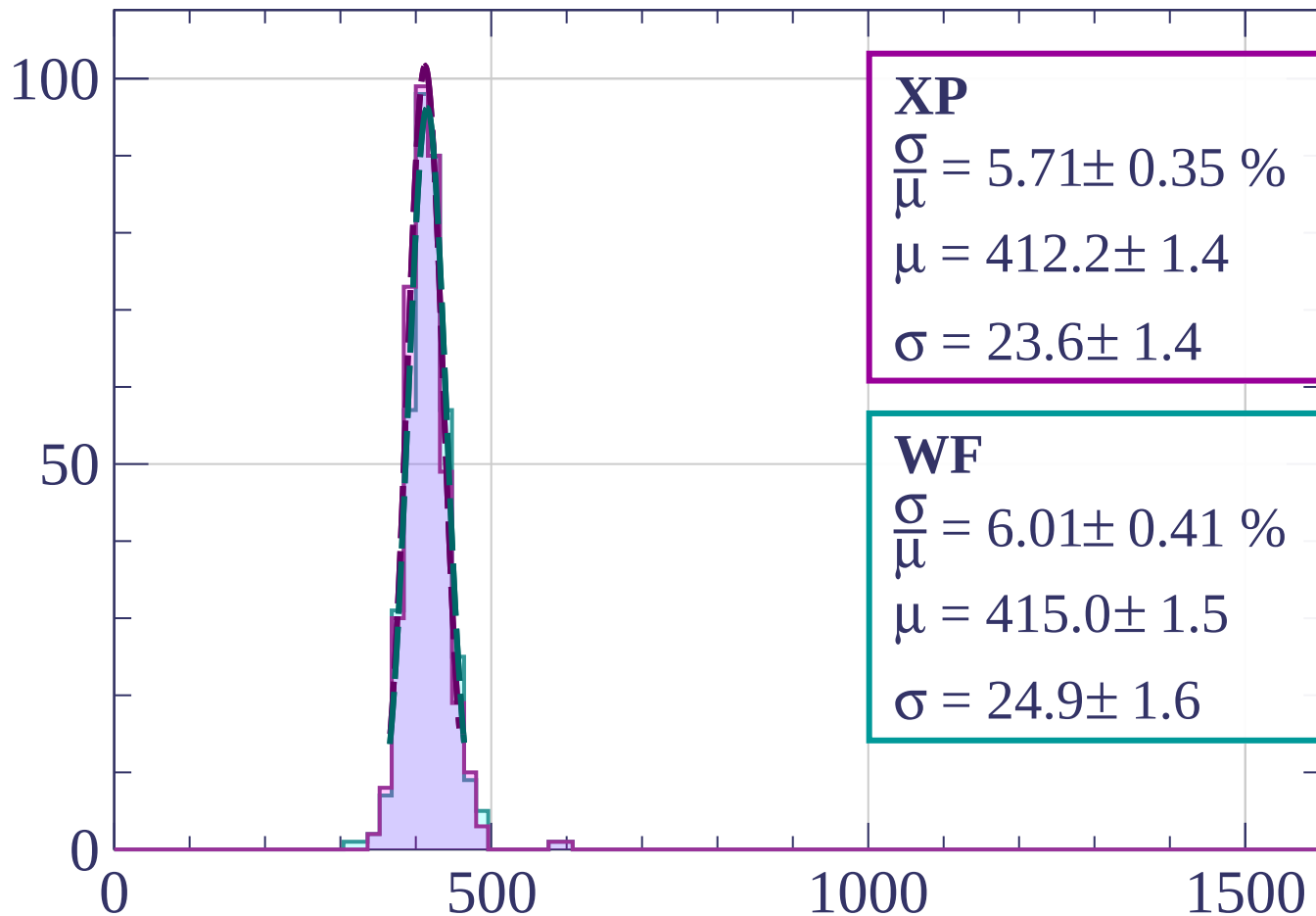
$$\mu = 416.2 \pm 1.3$$

$$\sigma = 24.4 \pm 1.4$$

dE/dx [ADC counts/cm]

Energy loss |  $18 < \theta < 21$

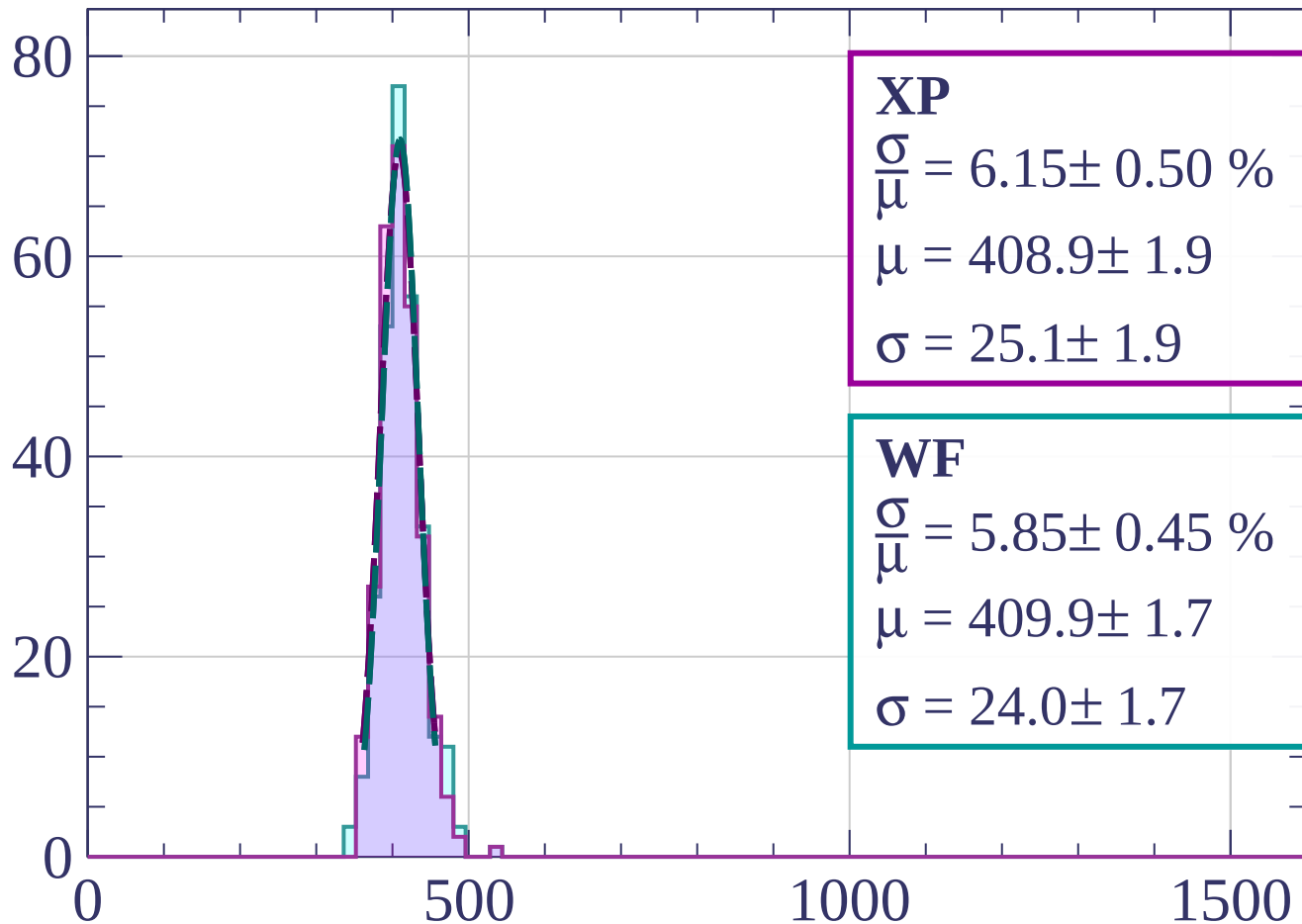
Count



$dE/dx$  [ADC counts/cm]

Energy loss |  $21 < \theta < 24$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.15 \pm 0.50 \%$$

$$\mu = 408.9 \pm 1.9$$

$$\sigma = 25.1 \pm 1.9$$

**WF**

$$\frac{\sigma}{\mu} = 5.85 \pm 0.45 \%$$

$$\mu = 409.9 \pm 1.7$$

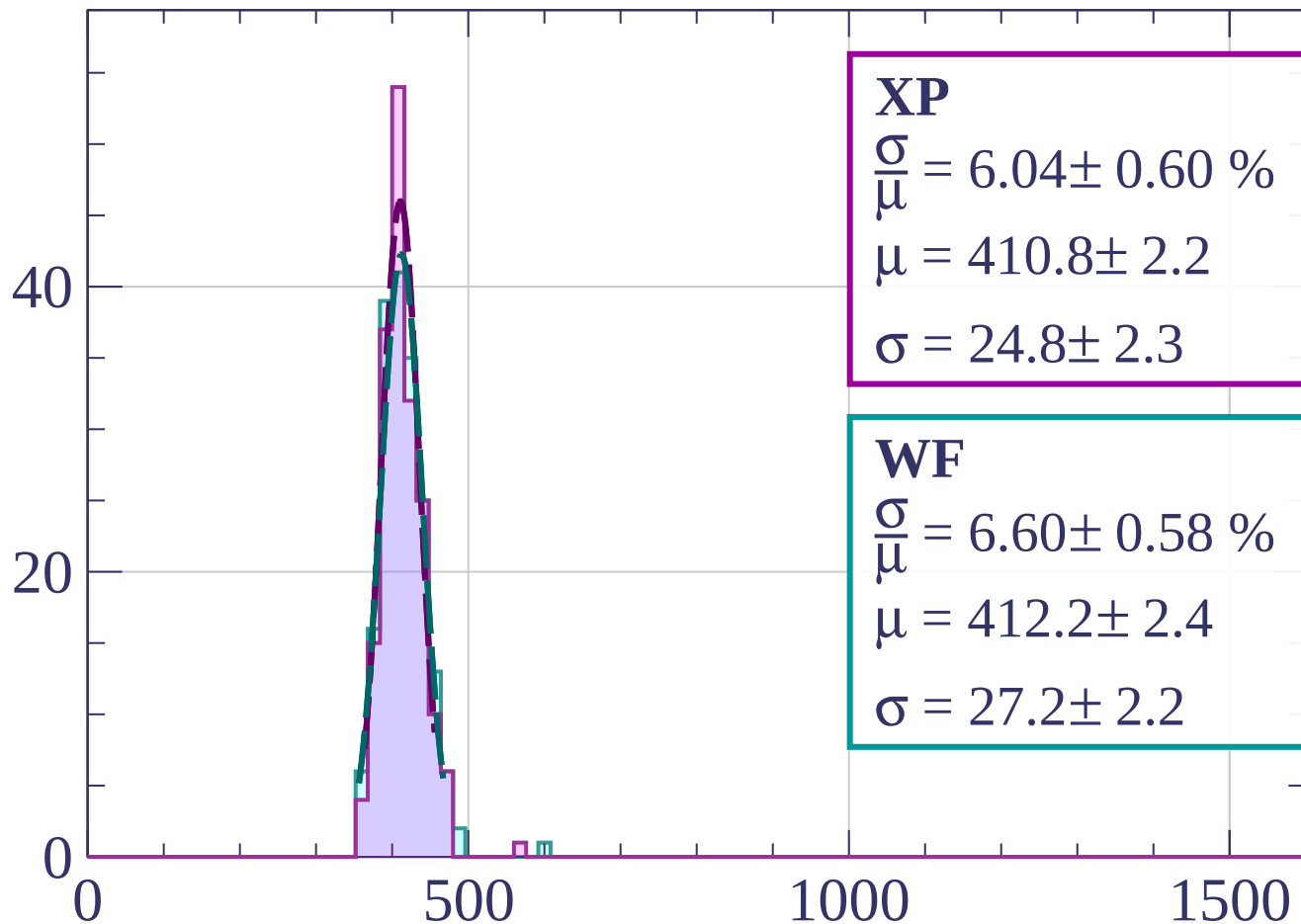
$$\sigma = 24.0 \pm 1.7$$

dE/dx [ADC counts/cm]



Energy loss |  $24 < \theta < 27$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.04 \pm 0.60 \%$$

$$\mu = 410.8 \pm 2.2$$

$$\sigma = 24.8 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 6.60 \pm 0.58 \%$$

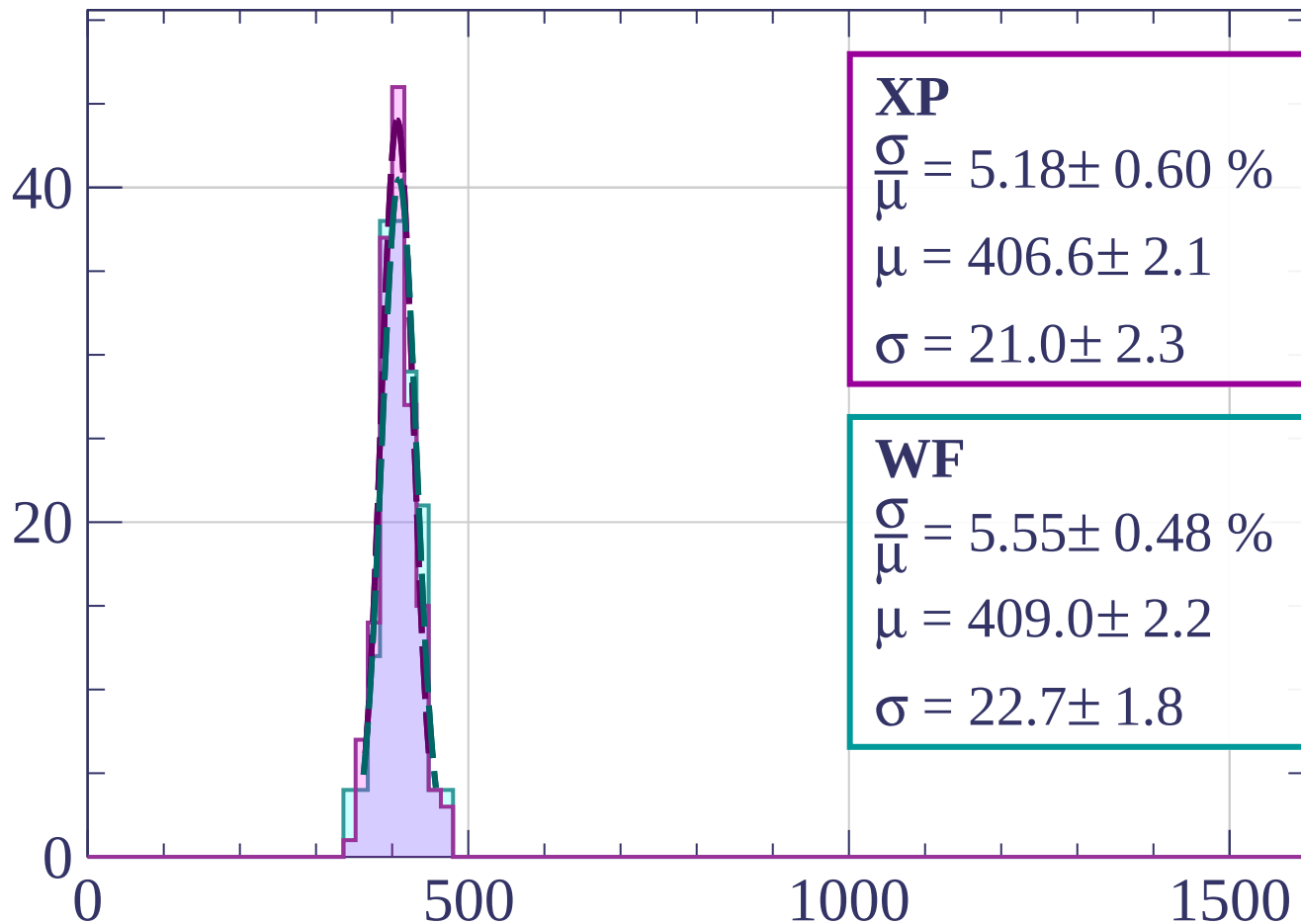
$$\mu = 412.2 \pm 2.4$$

$$\sigma = 27.2 \pm 2.2$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $27 < \theta < 30$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.18 \pm 0.60 \%$$

$$\mu = 406.6 \pm 2.1$$

$$\sigma = 21.0 \pm 2.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.55 \pm 0.48 \%$$

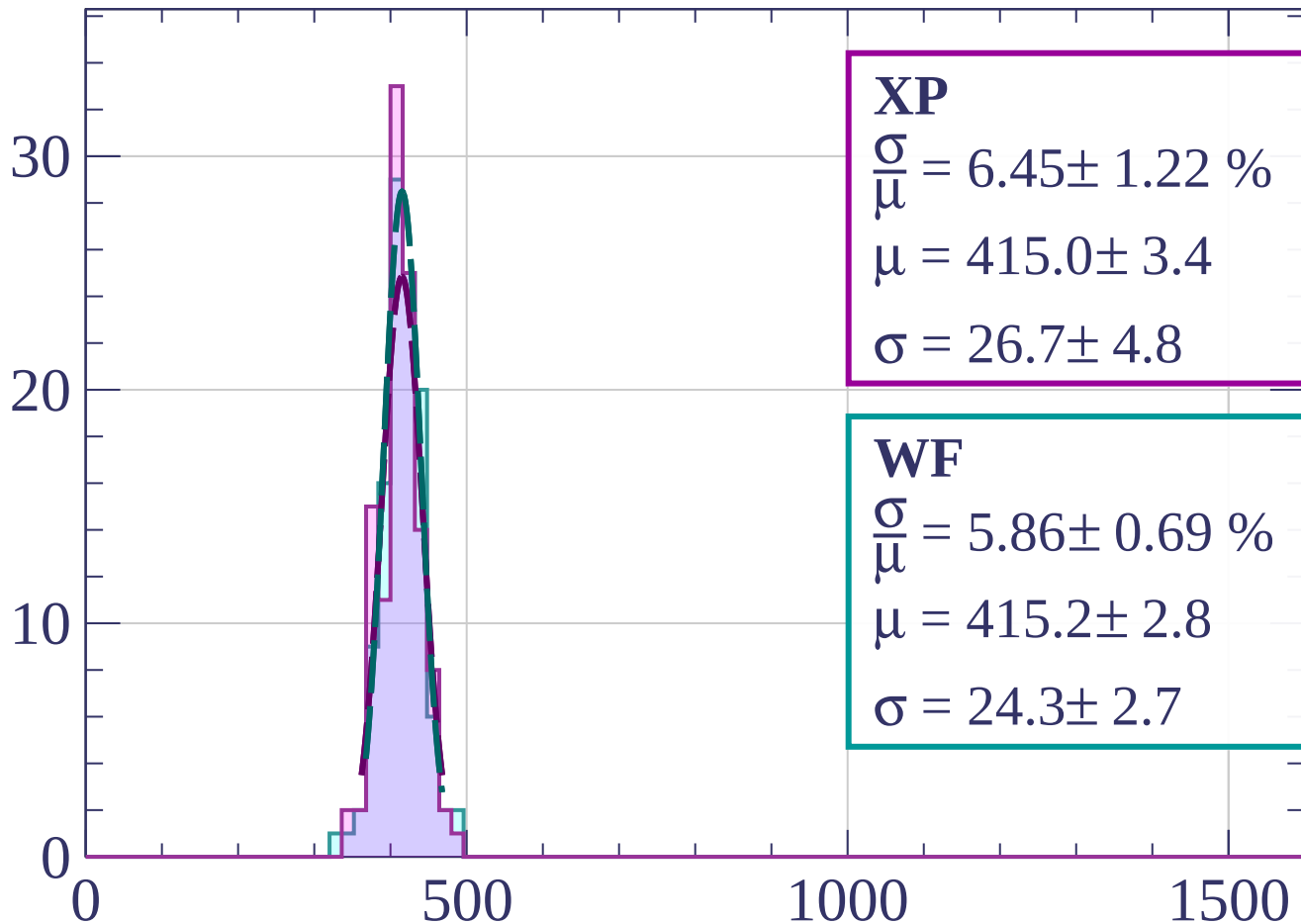
$$\mu = 409.0 \pm 2.2$$

$$\sigma = 22.7 \pm 1.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $30 < \theta < 33$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.45 \pm 1.22 \%$$

$$\mu = 415.0 \pm 3.4$$

$$\sigma = 26.7 \pm 4.8$$

**WF**

$$\frac{\sigma}{\mu} = 5.86 \pm 0.69 \%$$

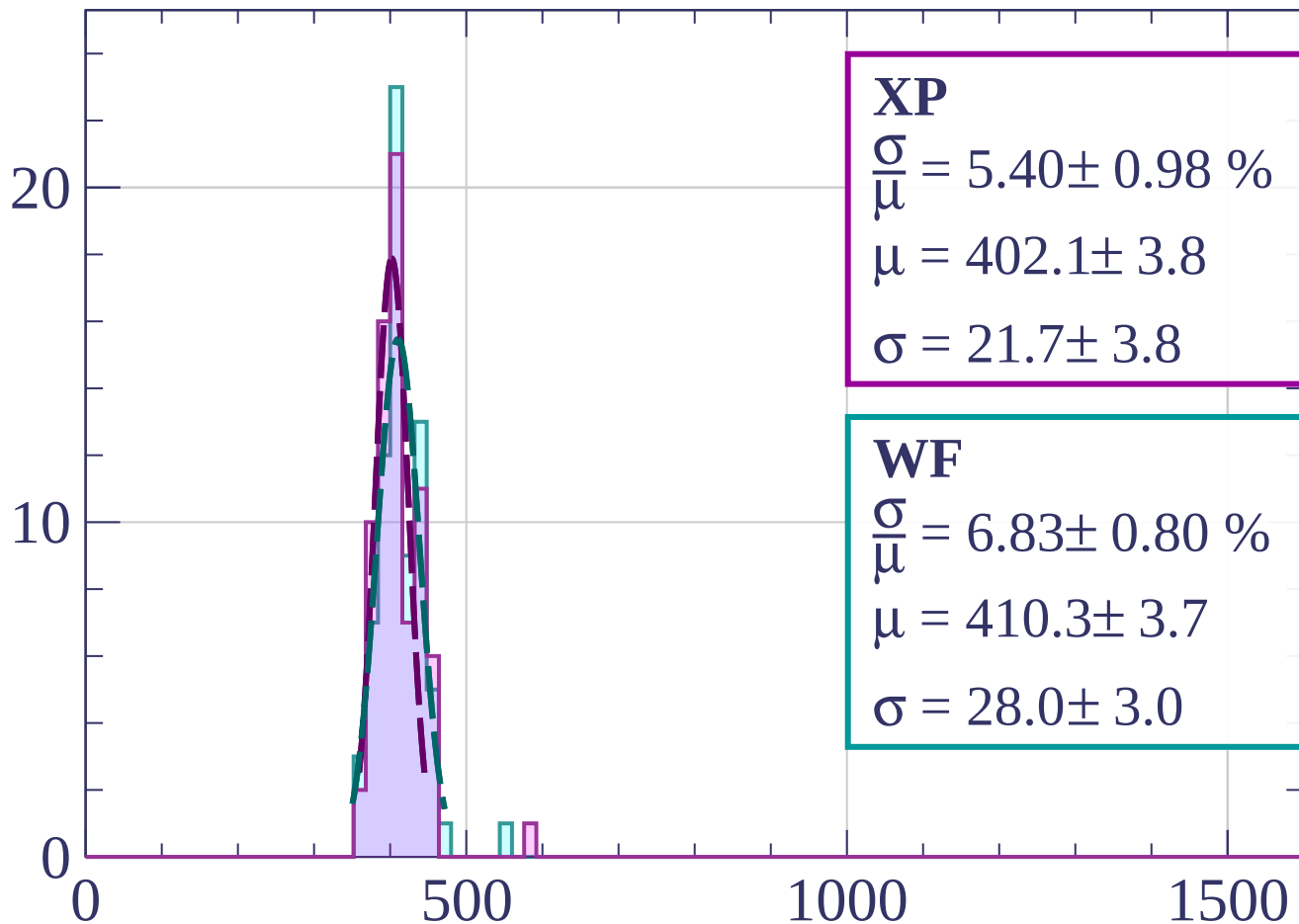
$$\mu = 415.2 \pm 2.8$$

$$\sigma = 24.3 \pm 2.7$$

dE/dx [ADC counts/cm]

Energy loss |  $33 < \theta < 36$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.40 \pm 0.98 \%$$

$$\mu = 402.1 \pm 3.8$$

$$\sigma = 21.7 \pm 3.8$$

**WF**

$$\frac{\sigma}{\mu} = 6.83 \pm 0.80 \%$$

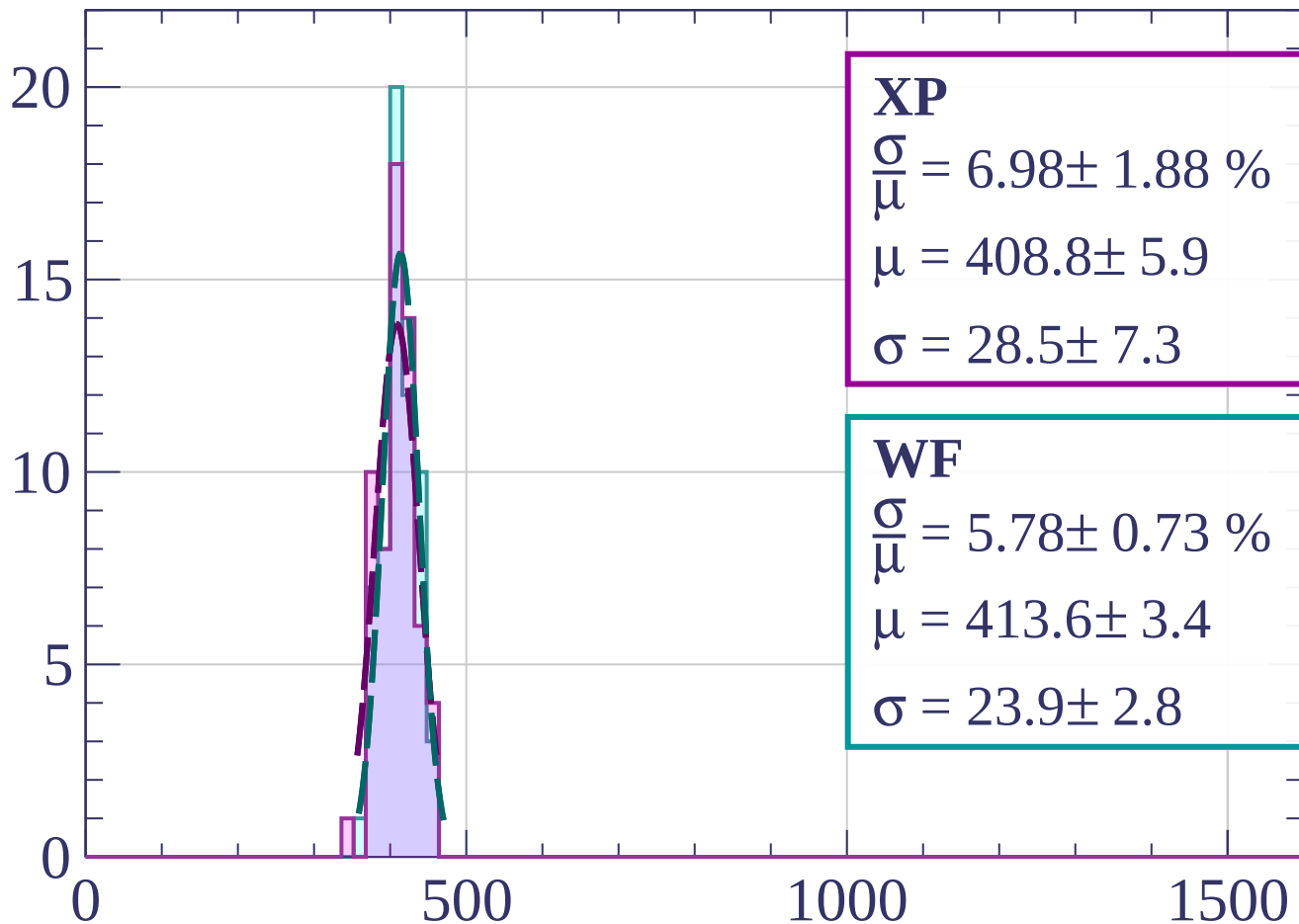
$$\mu = 410.3 \pm 3.7$$

$$\sigma = 28.0 \pm 3.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $36 < \theta < 39$

Count



**XP**

$$\frac{\sigma}{\mu} = 6.98 \pm 1.88 \%$$

$$\mu = 408.8 \pm 5.9$$

$$\sigma = 28.5 \pm 7.3$$

**WF**

$$\frac{\sigma}{\mu} = 5.78 \pm 0.73 \%$$

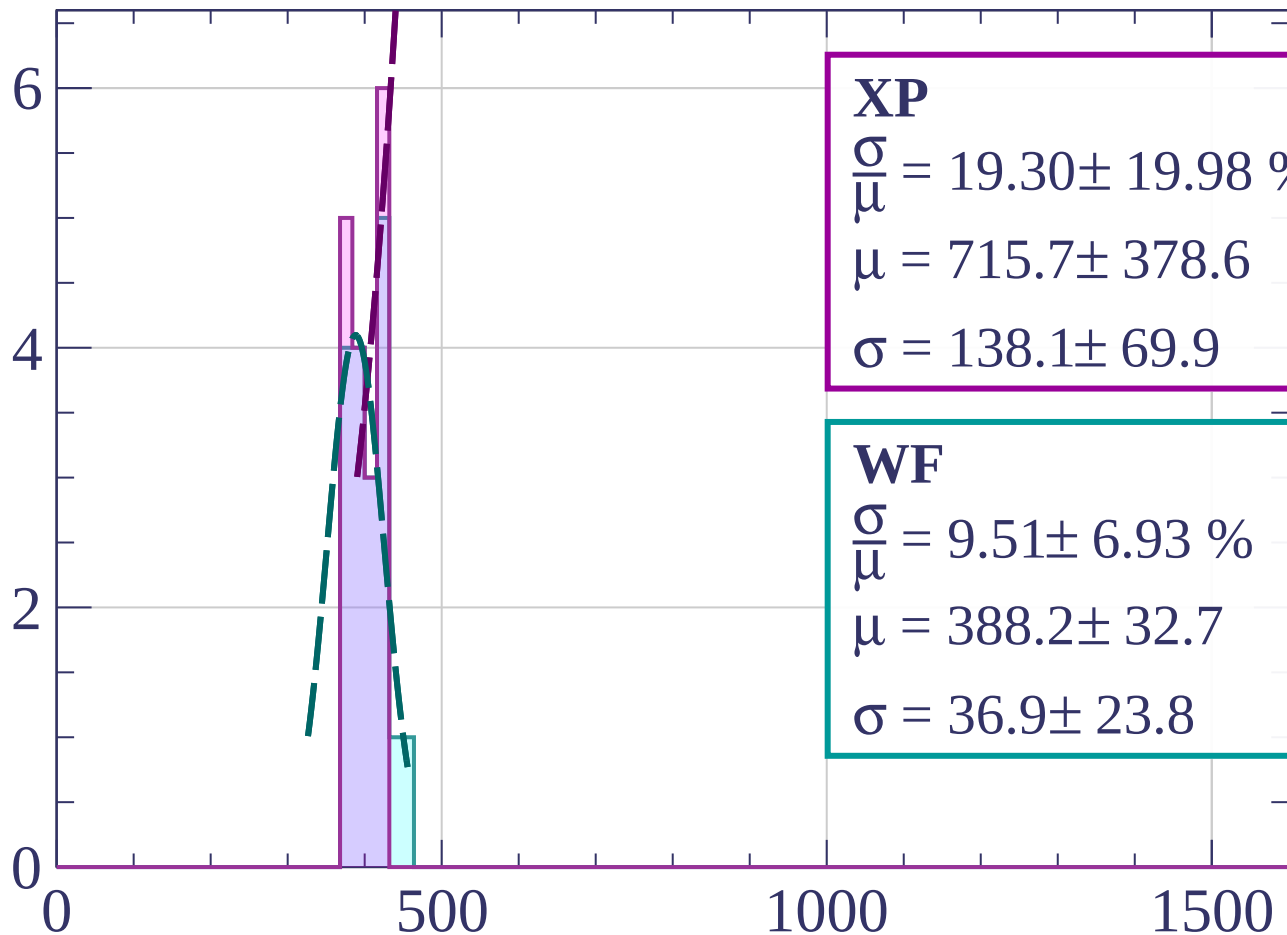
$$\mu = 413.6 \pm 3.4$$

$$\sigma = 23.9 \pm 2.8$$

$dE/dx$  [ADC counts/cm]

Count

Energy loss |  $39 < \theta < 42$



**XP**

$$\frac{\sigma}{\mu} = 19.30 \pm 19.98 \%$$

$$\mu = 715.7 \pm 378.6$$

$$\sigma = 138.1 \pm 69.9$$

**WF**

$$\frac{\sigma}{\mu} = 9.51 \pm 6.93 \%$$

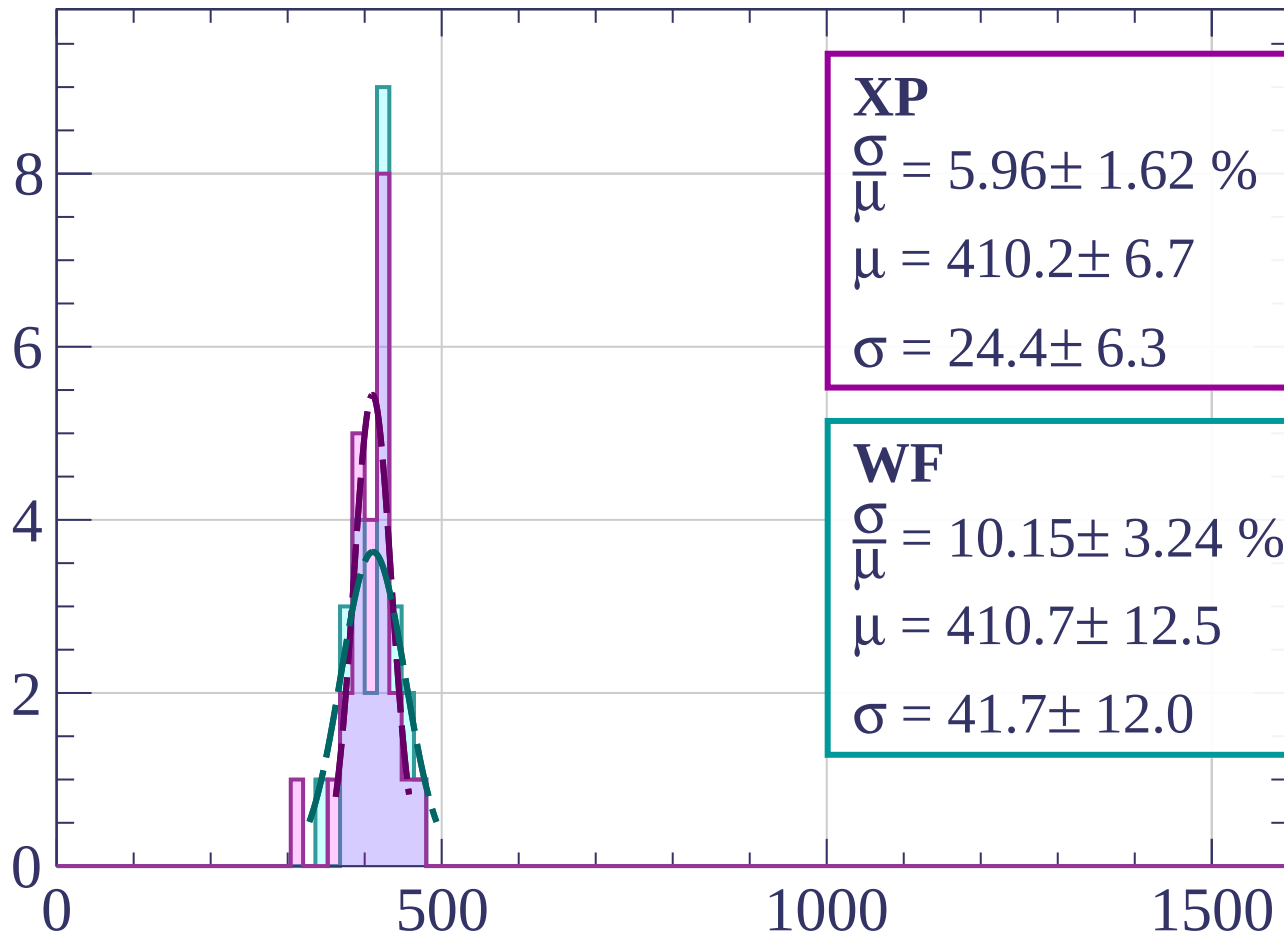
$$\mu = 388.2 \pm 32.7$$

$$\sigma = 36.9 \pm 23.8$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $42 < \theta < 45$

Count



**XP**

$$\frac{\sigma}{\mu} = 5.96 \pm 1.62 \%$$

$$\mu = 410.2 \pm 6.7$$

$$\sigma = 24.4 \pm 6.3$$

**WF**

$$\frac{\sigma}{\mu} = 10.15 \pm 3.24 \%$$

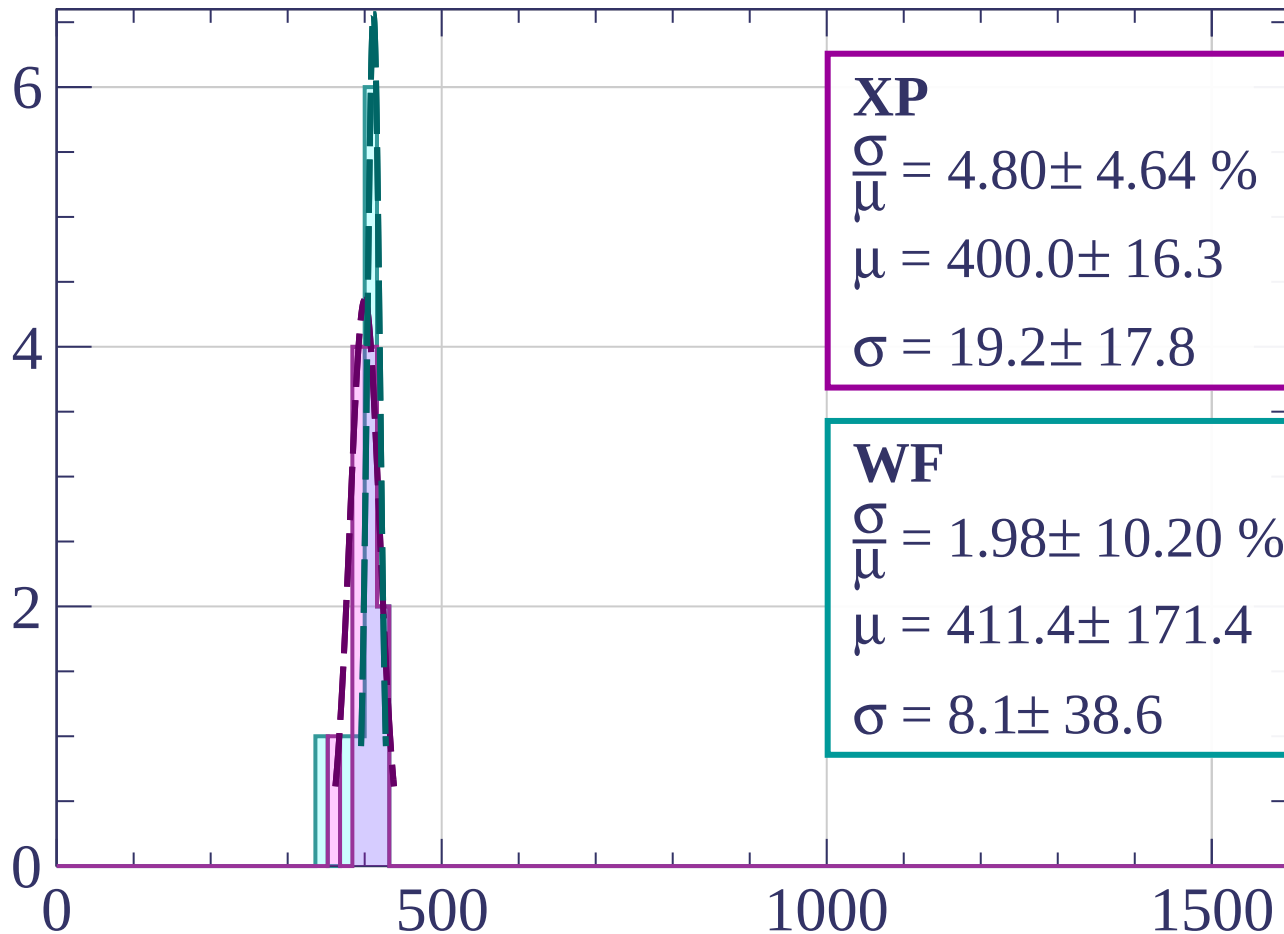
$$\mu = 410.7 \pm 12.5$$

$$\sigma = 41.7 \pm 12.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $45 < \theta < 48$

Count



**XP**

$$\frac{\sigma}{\mu} = 4.80 \pm 4.64 \%$$

$$\mu = 400.0 \pm 16.3$$

$$\sigma = 19.2 \pm 17.8$$

**WF**

$$\frac{\sigma}{\mu} = 1.98 \pm 10.20 \%$$

$$\mu = 411.4 \pm 171.4$$

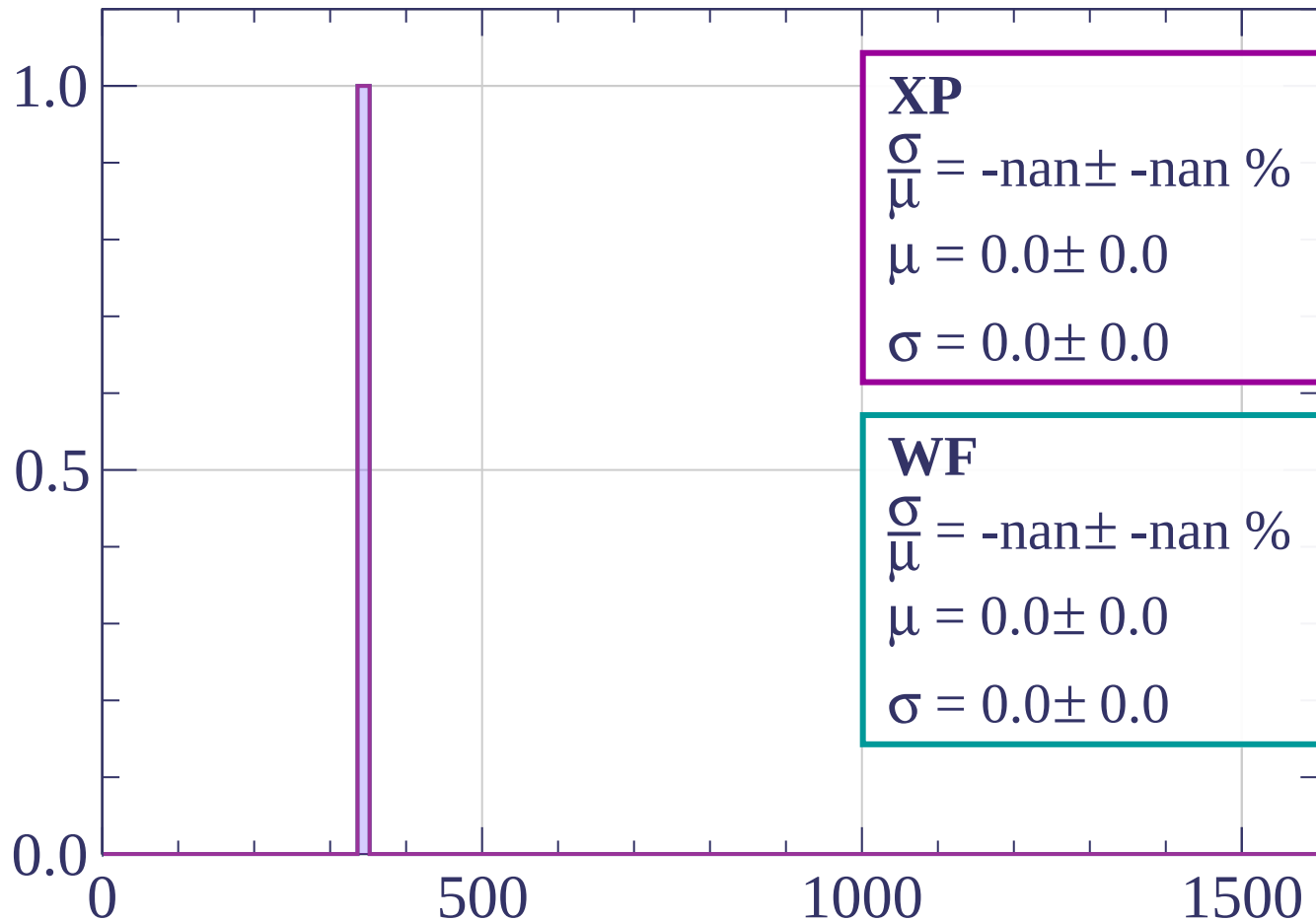
$$\sigma = 8.1 \pm 38.6$$

$dE/dx$  [ADC counts/cm]



Energy loss |  $48 < \theta < 51$

Count



**XP**

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

**WF**

$$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$$

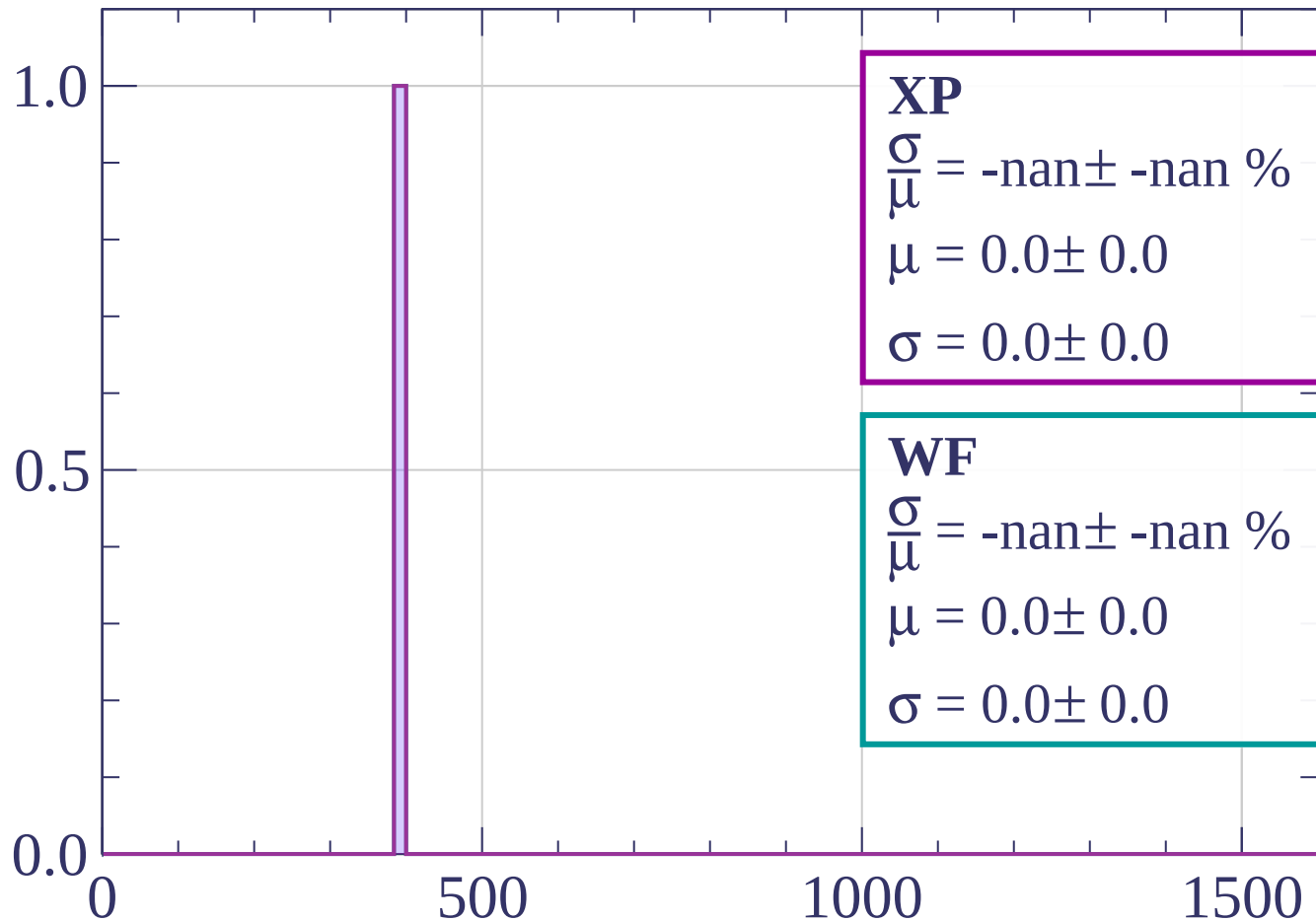
$$\mu = 0.0 \pm 0.0$$

$$\sigma = 0.0 \pm 0.0$$

$dE/dx$  [ADC counts/cm]

Energy loss |  $51 < \theta < 54$

Count



**XP**

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\bar{\mu}} = -\text{nan} \pm -\text{nan} \%$

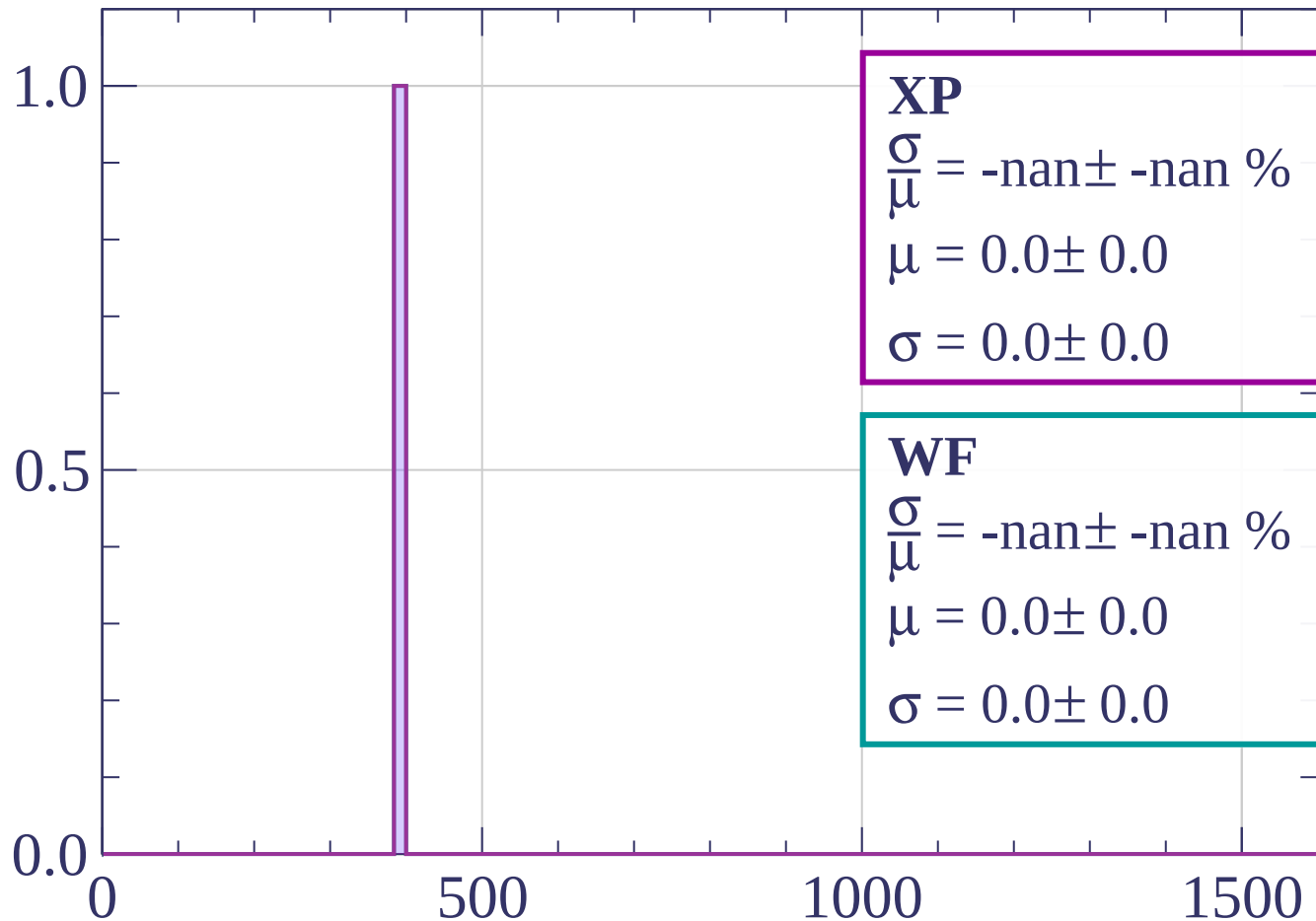
$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

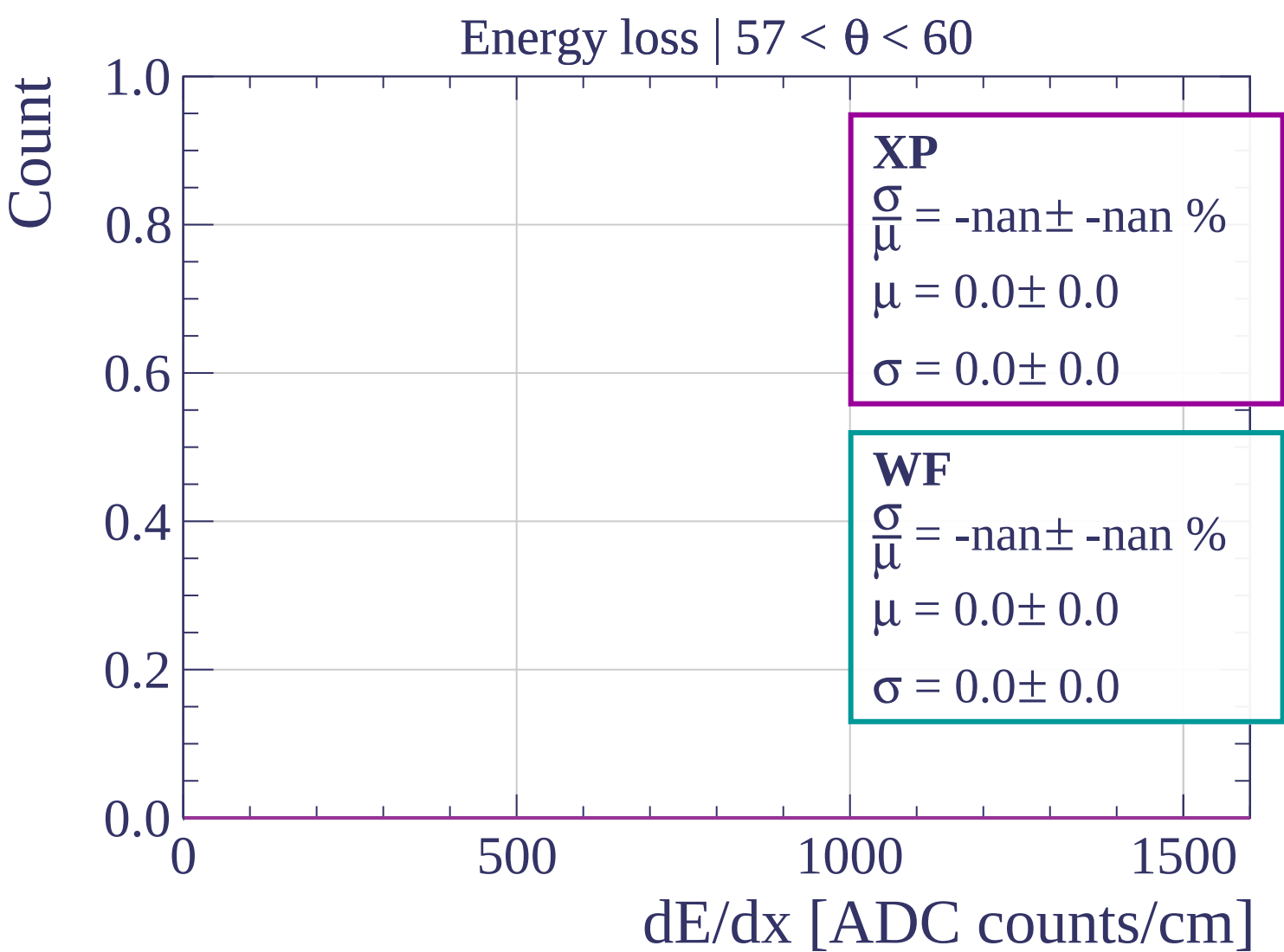
$dE/dx$  [ADC counts/cm]

Energy loss |  $54 < \theta < 57$

Count



$dE/dx$  [ADC counts/cm]



Count

Energy loss |  $60 < \theta < 63$

**XP**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

**WF**

$\frac{\sigma}{\mu} = -\text{nan} \pm -\text{nan} \%$

$\mu = 0.0 \pm 0.0$

$\sigma = 0.0 \pm 0.0$

1.0  
0.8  
0.6  
0.4  
0.2  
0.0

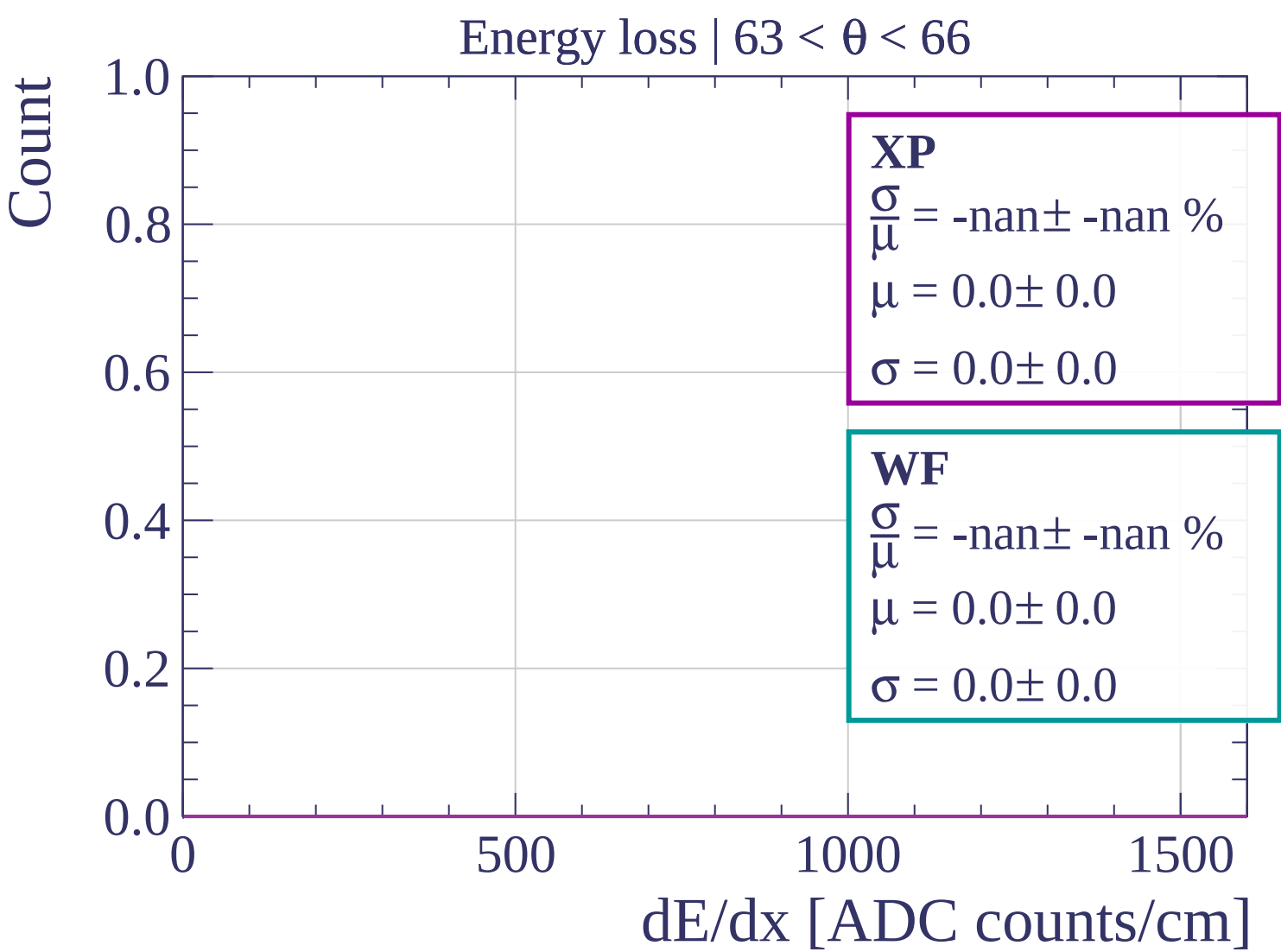
0

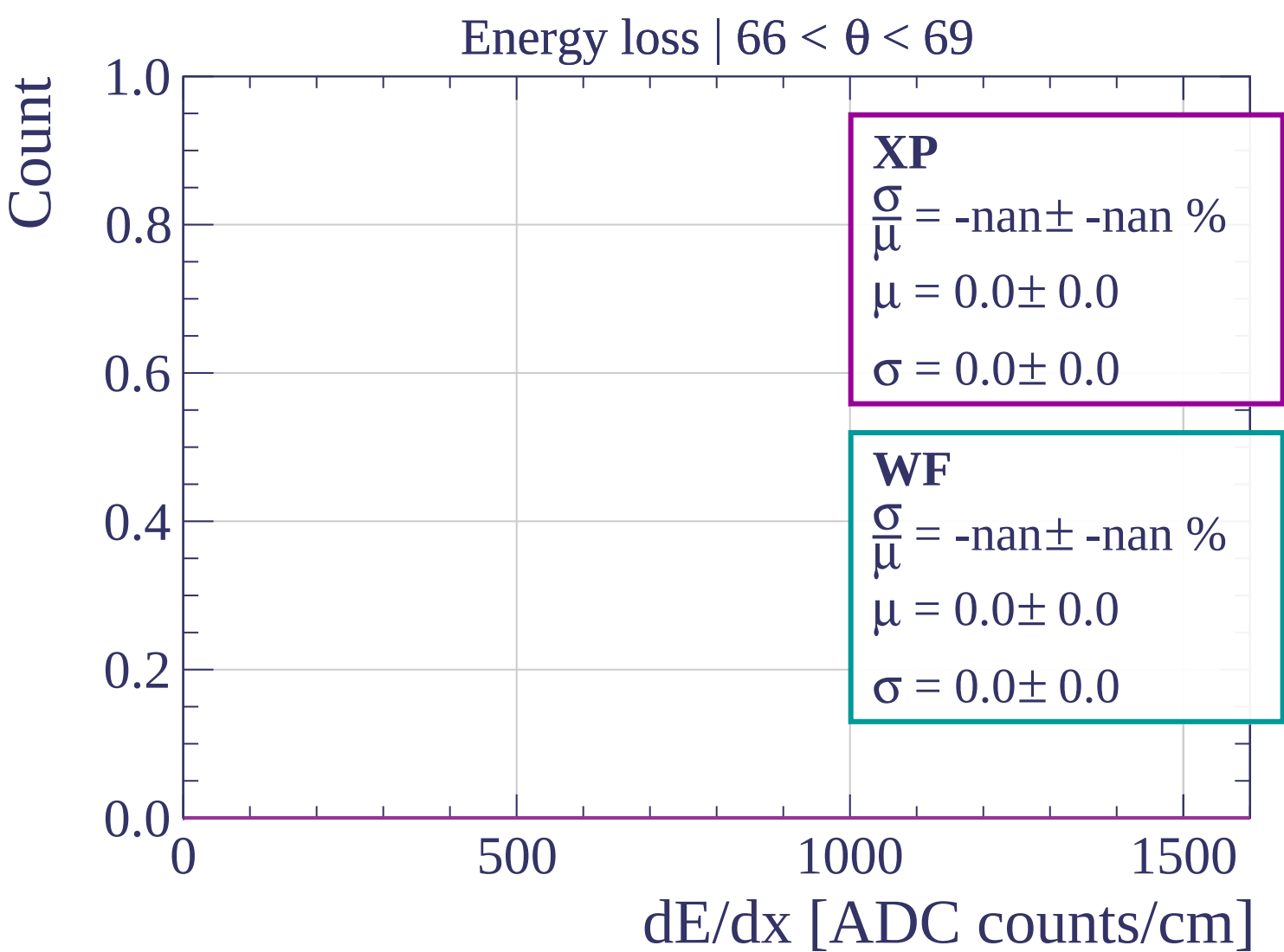
500

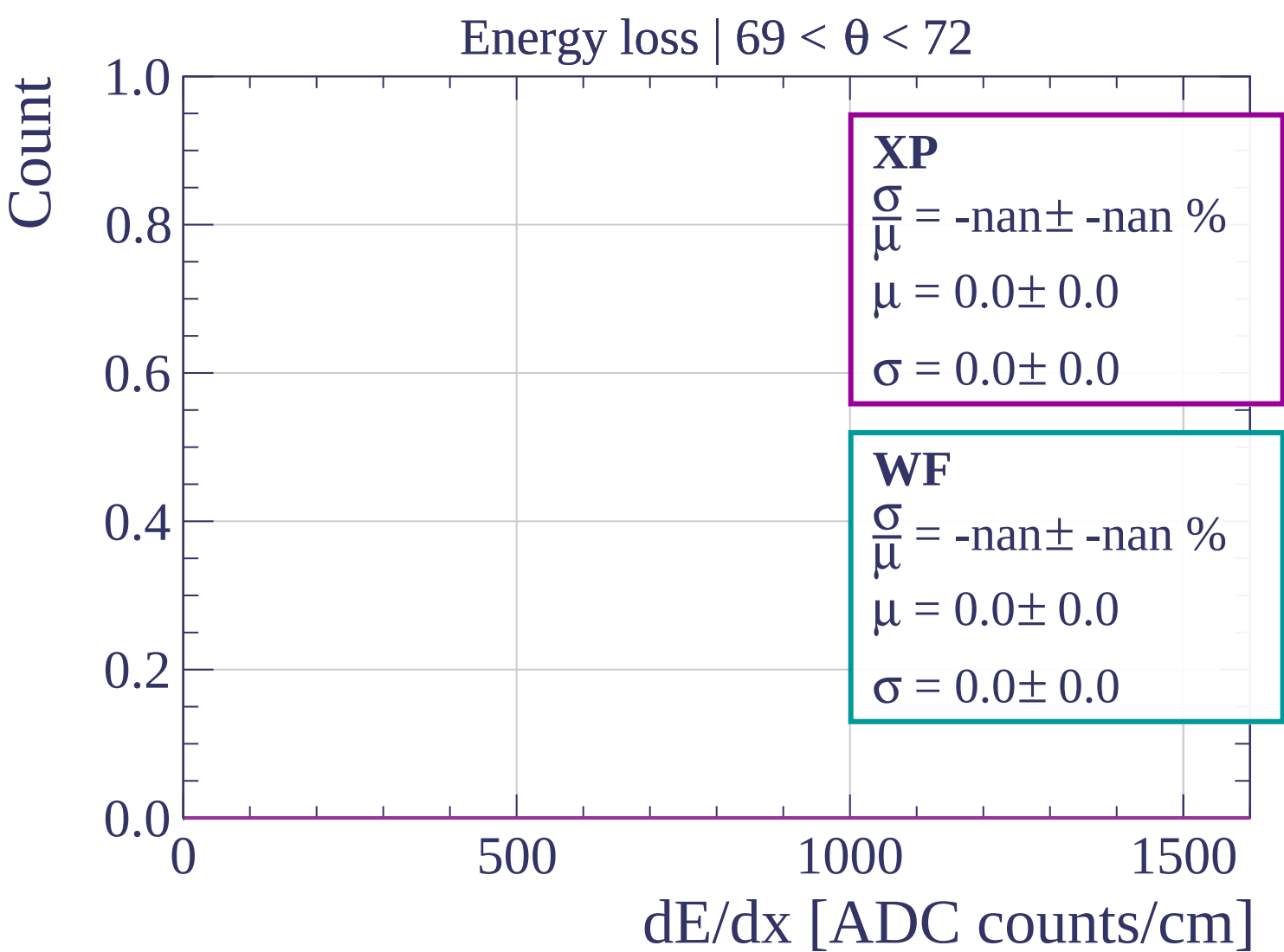
1000

1500

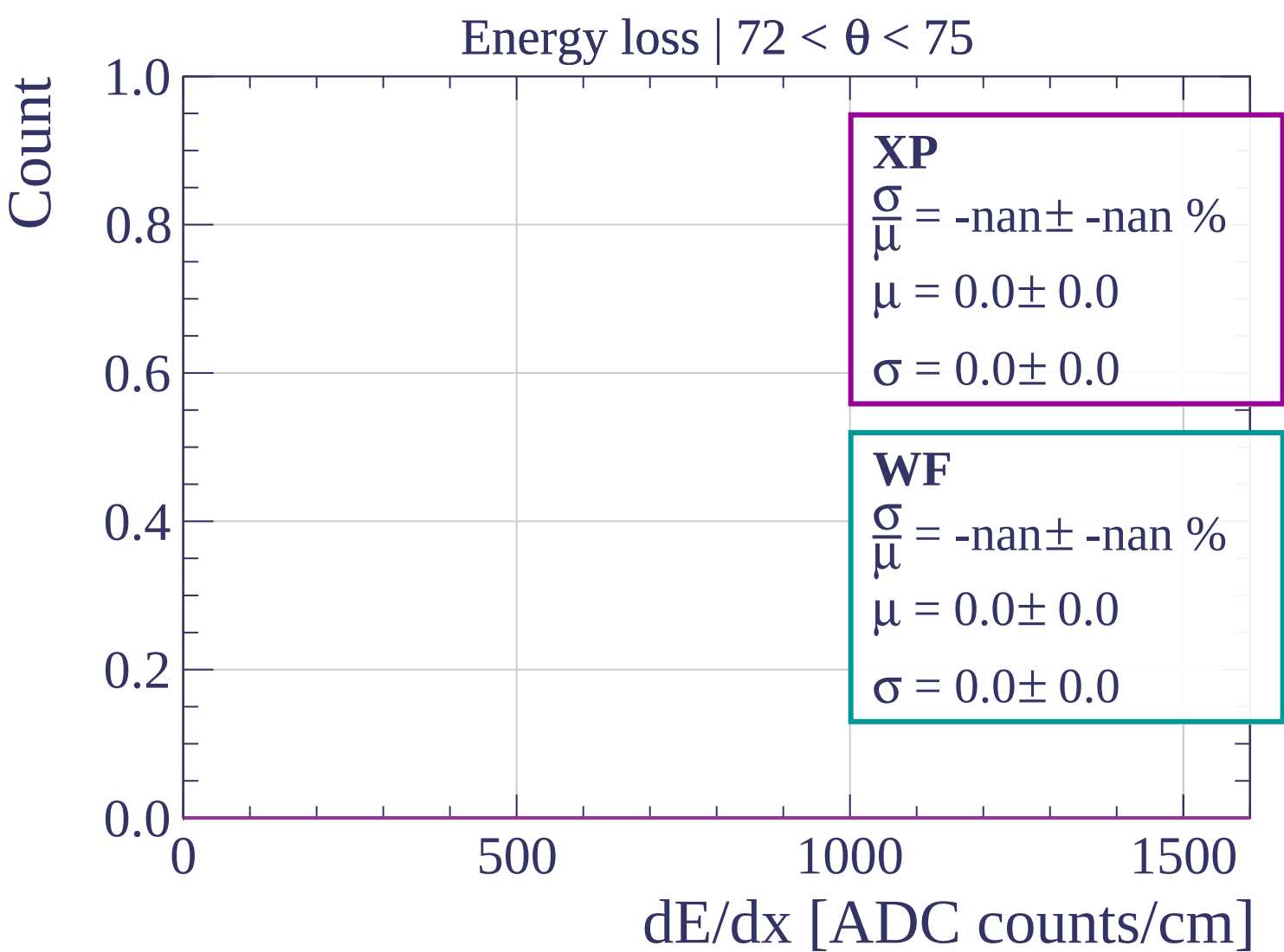
dE/dx [ADC counts/cm]

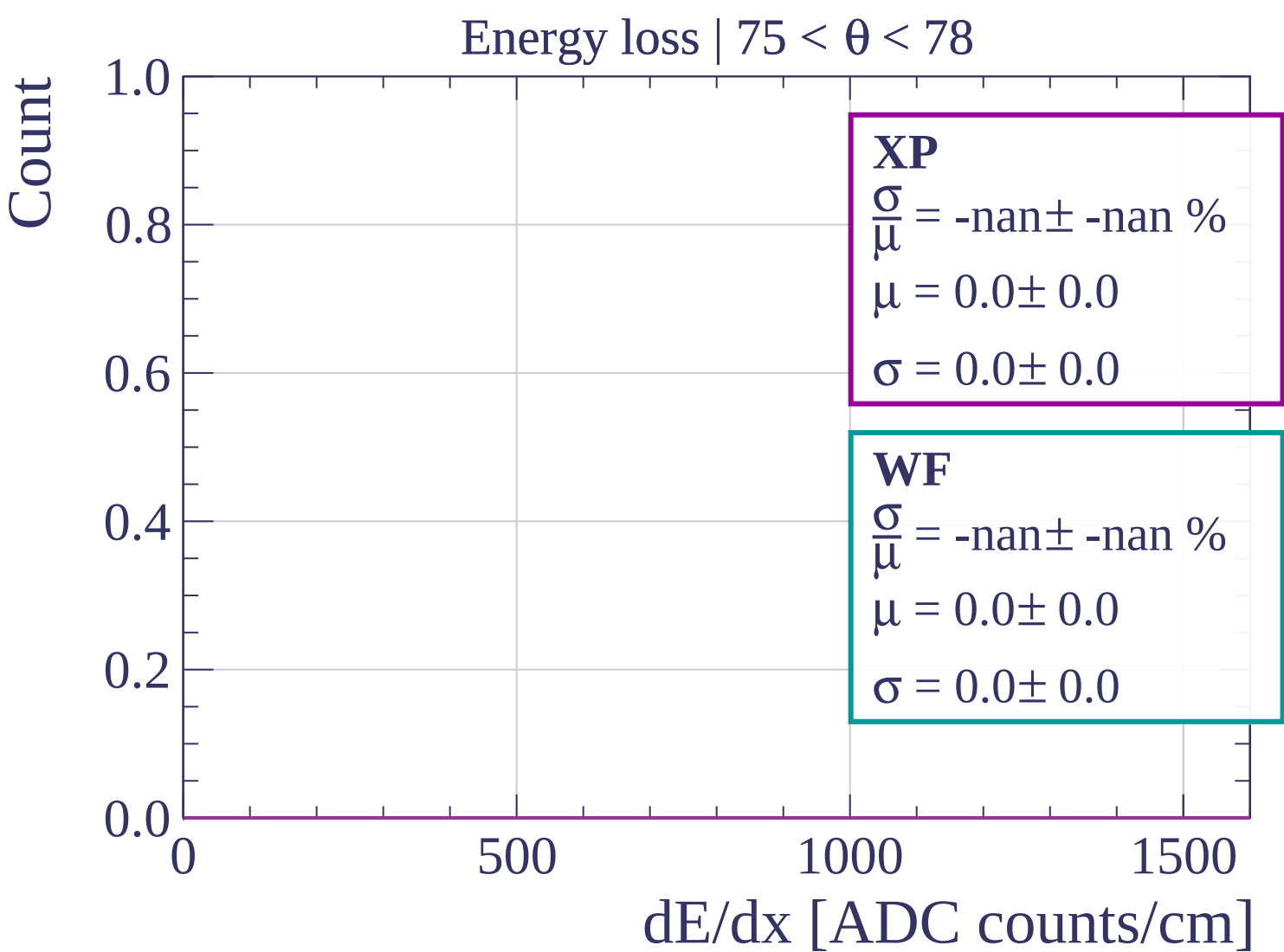


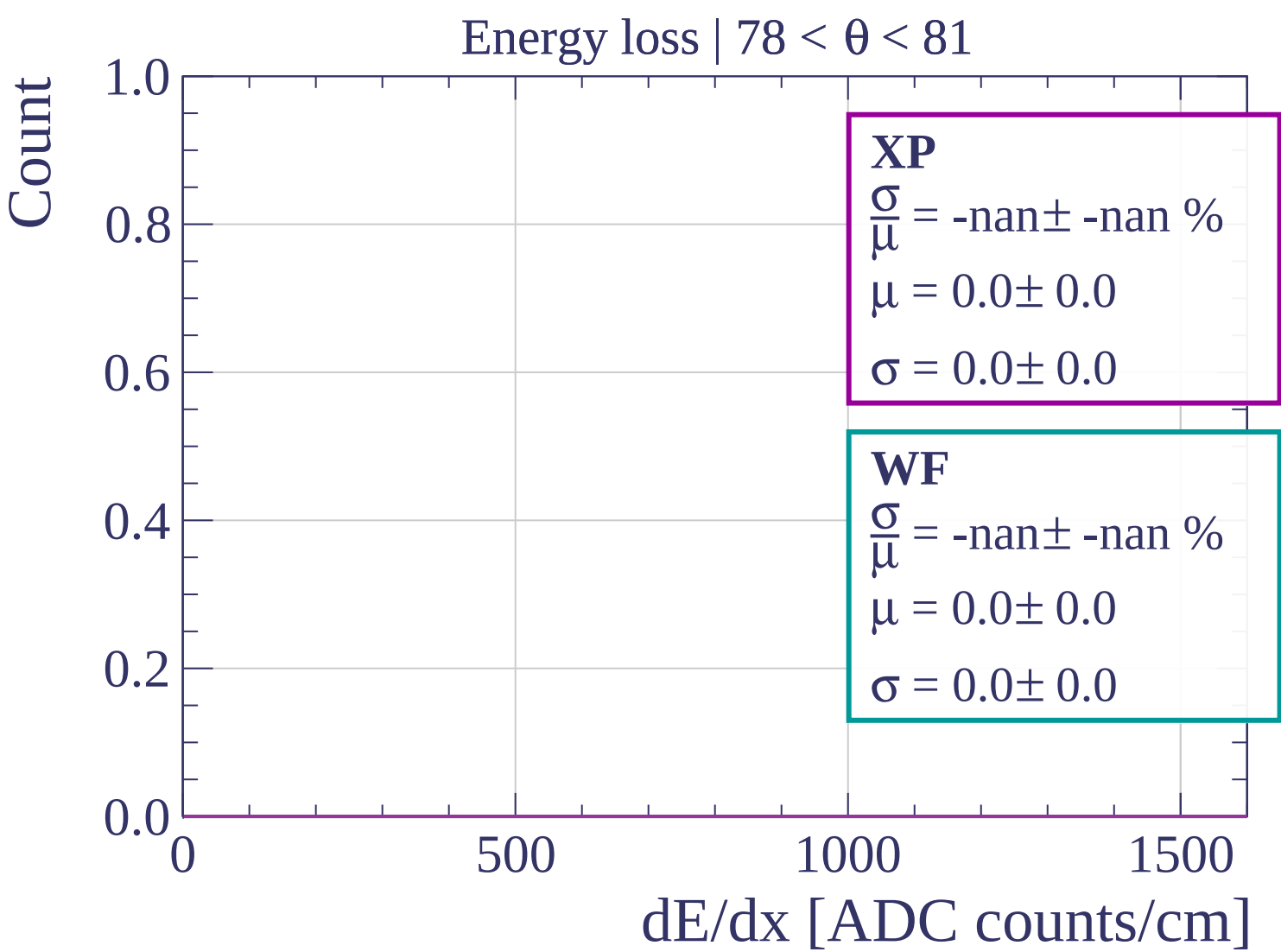


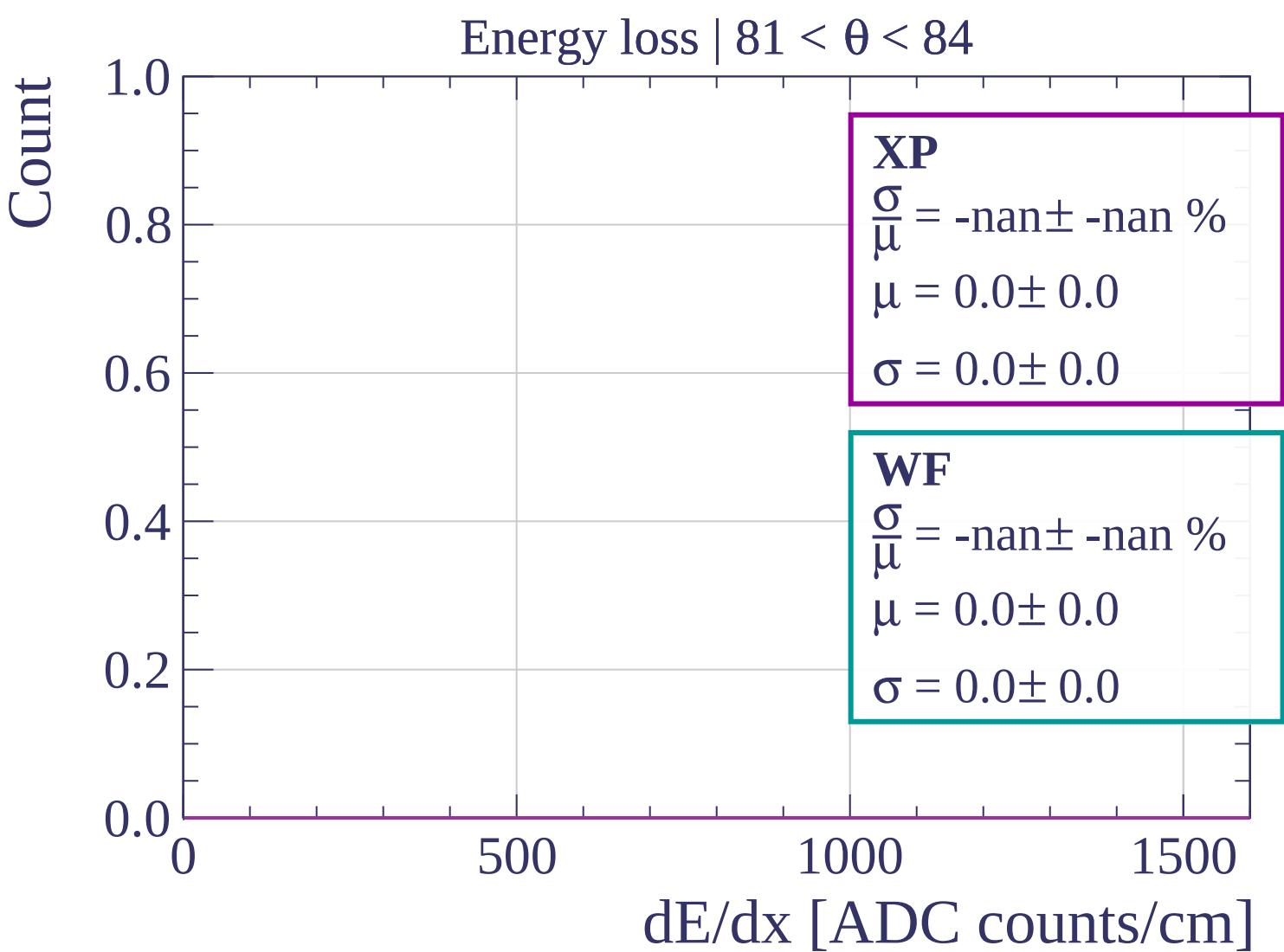


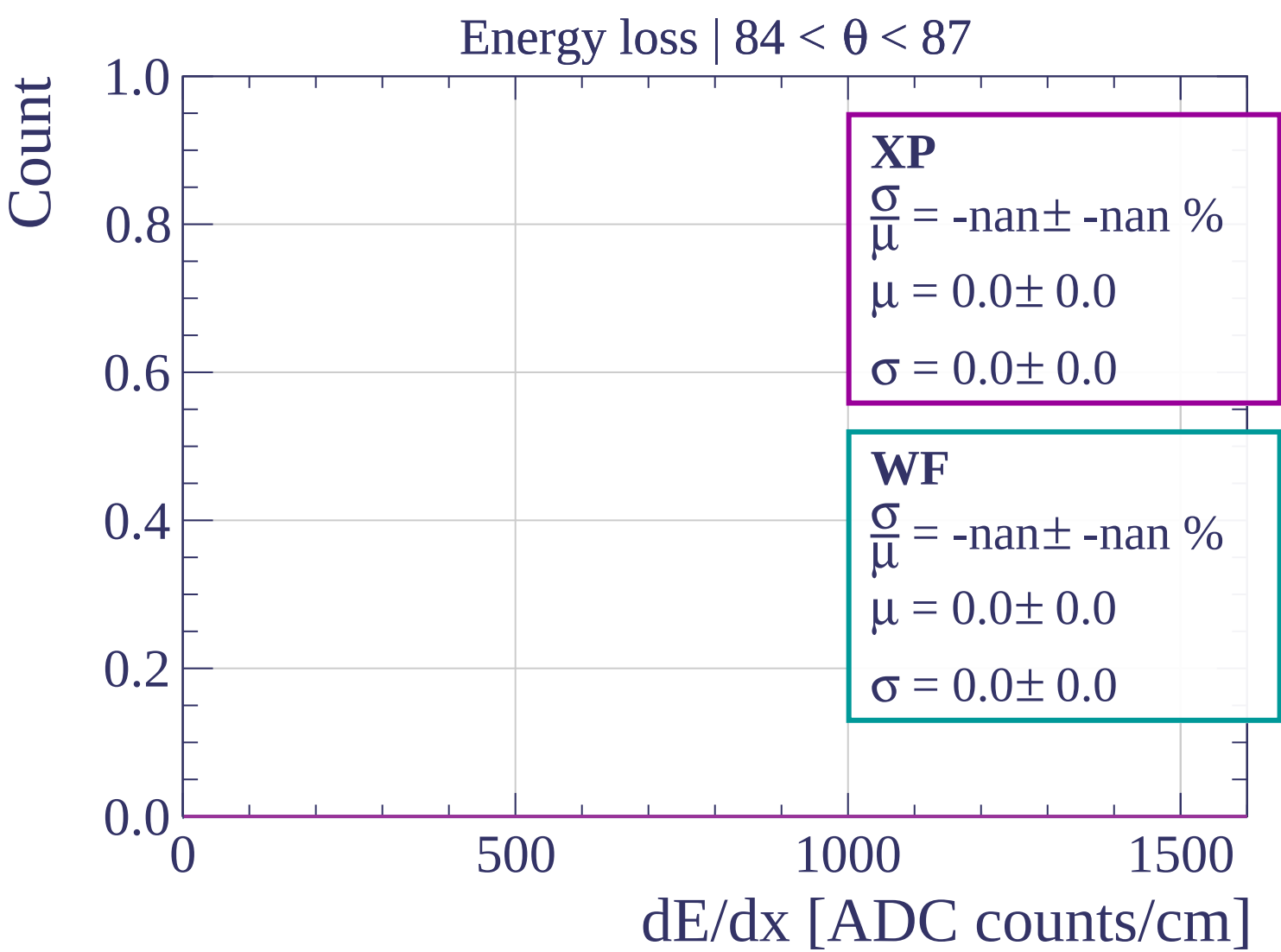


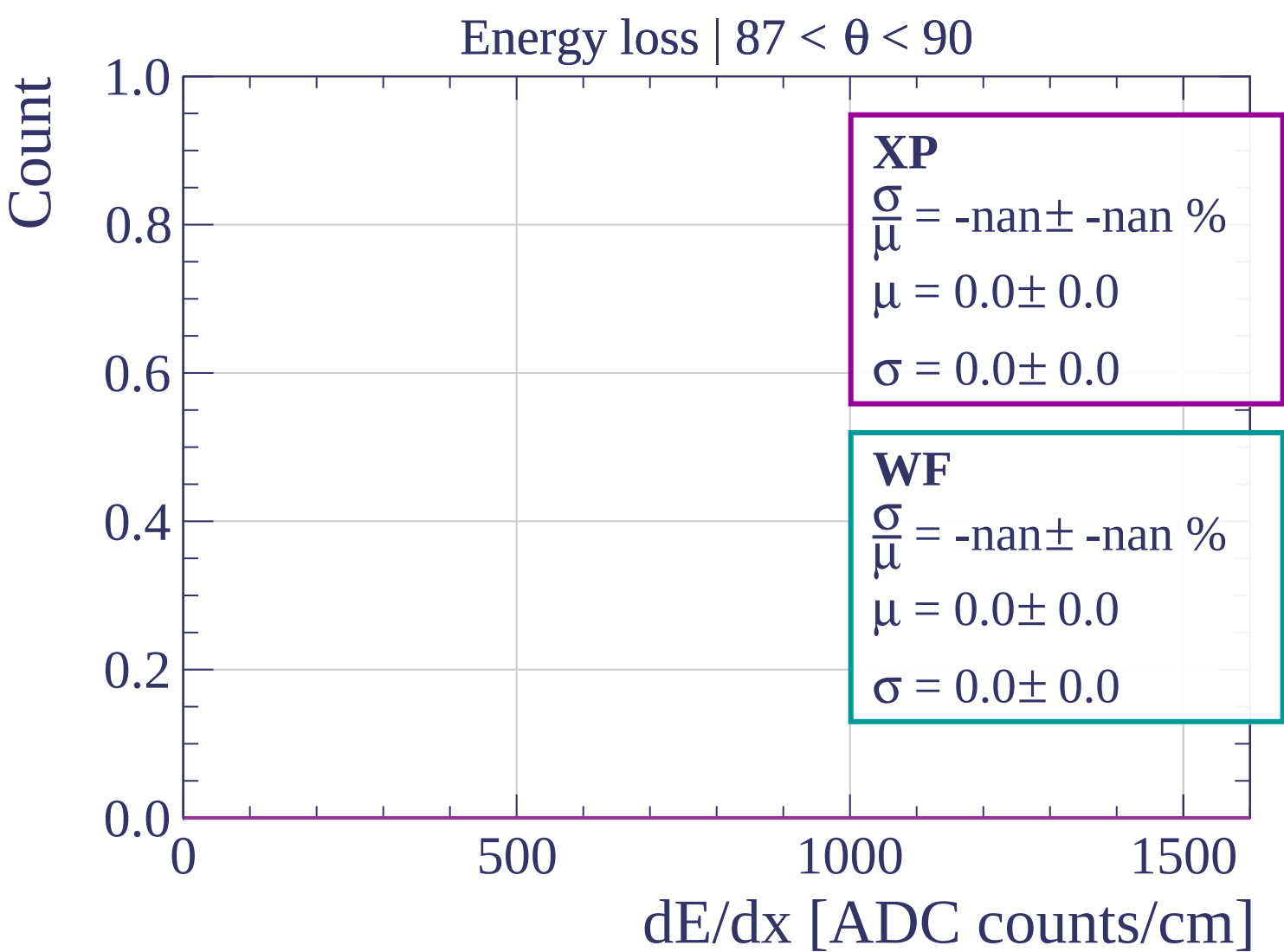




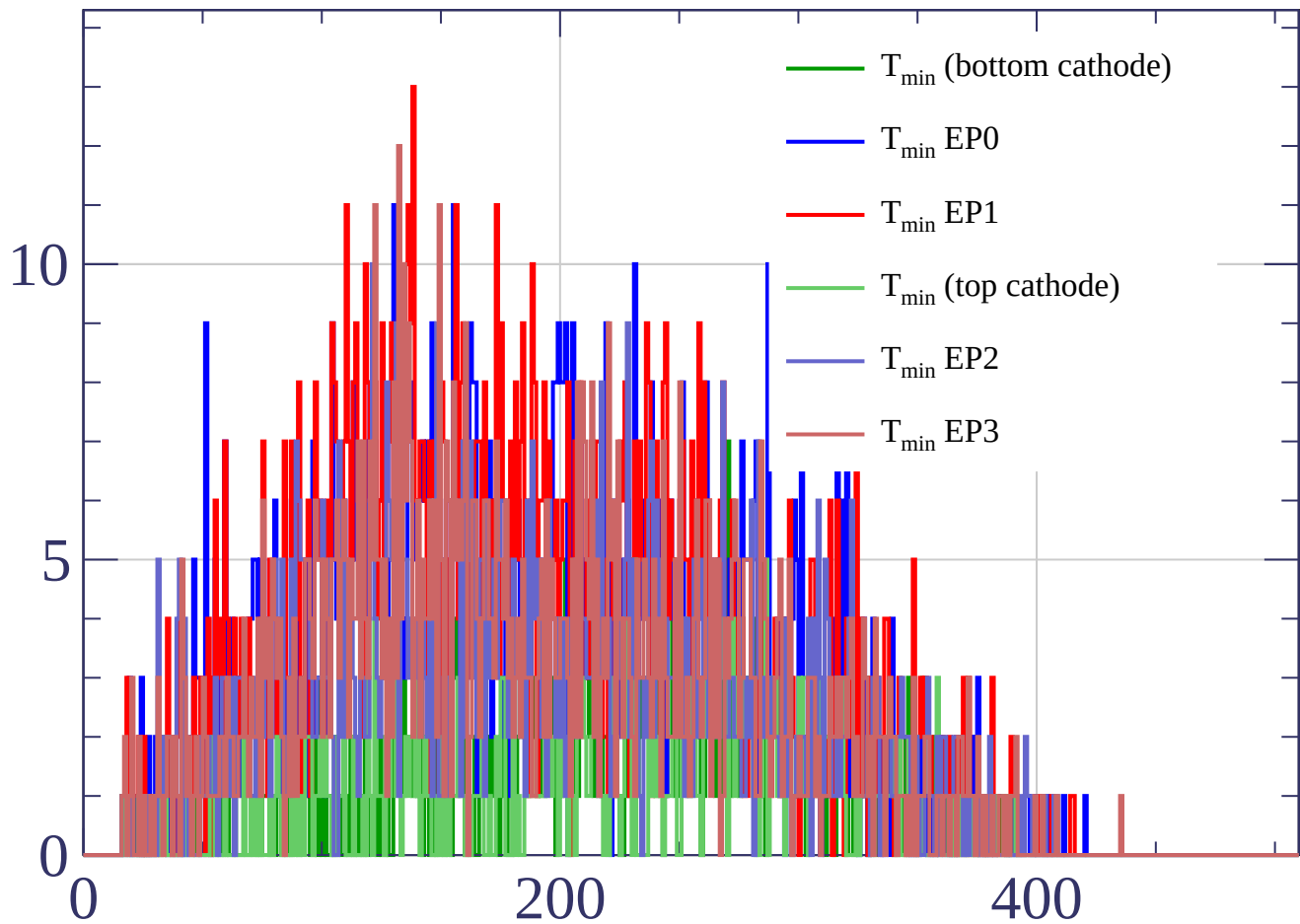








Count



Count

